BESMM4

Building and Engineering Standard Method of Measurement 4

Fourth Edition 2015



THE NIGERIAN INSTITUTE OF QUANTITY SURVEYORS

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FOREWORD

The review of BESMM3 was one of the items inherited by the 2013 – 2015 National Executive Council of The NIQS as it was initiated by the 2011 – 2013 National Executive Council. The Membership of the Ad-hoc Committee was reconstituted with the mandate to review the Building and Engineering Standard Method of Measurement 3 with a view to correcting all the errors, text or context i.e. typographical, inserting any omissions, removal of unnecessary features, rearrangement of contents and features and incorporating additional rules deemed necessary to enable the document serve its purpose more effectively.

In keeping with the Institute's policy of professional best practices at all times, the Committee reviewed the existing BESMM3 with other similar International Standard Methods of Measurement and came up with a complete updated document along the same principles of NRM2, CESMM4 and ISMM with some adjustments to suite the main contracting culture in the Nigerian Construction Industry. The final document was adopted as BESMM4 by The Nigerian Institute of Quantity Surveyors National Executive Council and its application in the Nigerian Construction Industry was generally accepted by the Council to take effect from January 2016.

Contractually, BESMM4 is neutral hence it is suitable for any Measure and Value Form of Contract. The introduction of Local Standards is highly encouraged as the use of British Standards is minimised in both Parts 1 and 2. The template for BESMM3 was retained in BESMM4 as the Ad-hoc Committee considered the template to be easily understandable.

BESMM4 provides fundamental guidance on the detailed measurement and description of Building, Engineering and Industrial Works for the purpose of obtaining a tender price, valuation for interim payments as well as a veritable basis for the valuation of variations for Building, Engineering and Industrial projects. The rules address all aspects of Bill of Quantities (BoQ) production, including setting out the information required from the employer and other construction consultants to enable a Bill of Quantities (BoQ) to be prepared, as well as dealing with the quantification of non-measurable work items and risks. Guidance is also provided on the content, structure and format of BoQ, as well as the benefits and uses of Bill of Quantities (BoQ).

While written mainly for the preparation of Bill of Quantities, Quantified Schedules of Works and Quantified Work Schedules, the BESMM4 will be invaluable when designing and developing standard or bespoke Schedules of Rates. These standard methods provide essential guidance to all those involved in the preparation of Bill of Quantities, as well as those who wish to be better informed about the purpose, use and benefits of Bill of Quantities.

I therefore on behalf of the NEC thank the Chairman, Mr. J.E. Abhulimhen and Members of the Committee who worked relentlessly to make the production of this document a reality. Special thanks also go to Members of my National Executive Council for their support in ensuring the preparation and production of this document remained a top priority right from inception of the Council up and including when the production was actualised.

Murtala M. Aliyu,FNIQS
NIQS President

INTRODUCTION

With the introduction of SMM 1 in 1988, the access to Standard Method of Measurement in the Nigerian Construction Industry for building works greatly improved. The essence of SMM1 was to provide a uniform basis of documentation for tendering purposes, to serve as a basis for fair valuation for interim payments as well as valuation of variations during the post contract of building projects and these have been effectively achieved over the years. The incorporation of Standard Method of Measurement of Civil Engineering Works in BESMM2 published in 2002 in line technological development further enhanced the document as an indispensable tool in the hands of Professionals in the Nigerian Construction Industry with particular reference the Professional Quantity Surveyors as well as Quantity Surveying Lecturers and Students. While BESMM2 was a radical departure from SMM1, BESMM3 was simply a corrected version of BESMM2.

The National Executive Council (NEC) of The Nigerian Institute of Quantity Surveyors (NIQS) in September, 2012 appointed an Ad-hoc Committee primarily charged with the responsibility of correcting all observed spelling and contextual errors in BESMM3 in addition to amending or incorporating additional rules deemed necessary to enable the document serve its purpose more effectively. The Committee in due response to the terms of reference, invited the generality of Members through their e-mail addresses to submit memoranda stating any desired change(s) including the rational for the proposed change(s).

The membership of the Ad-hoc Committee was reconstituted and the terms of reference was expanded in 2014 by the 2013-2015 NIQS NEC for the review of BESMM3 to incorporate the International Best Standard. The Committee in urgent need to come up with a generally accepted document that meets International Standard simply adopted the same procedure in the previous review by considering the review of BESMM3 along the same principles of the NRM2, CESMM4 and ISMM with some slight adjustments intended to suite the main contracting culture in the Nigerian Construction Industry. The BESMM4 differs significantly from the BESMM3 in some areas including Highways being replaced with Roads and Pavings, some areas that were previously left out in Part 2 of BESMM3 such as Piling, Piling Accessories, Masonry, Painting, Sewers, Water Main Renovation etc. for ease of application have been duly accommodated in Part 2 of BESMM 4, Classification Table in BESMM3 being subdivided into a brief description of work to be measured, unit of measurement and divisions 1-3 etc. Thus, the document will no doubt facilitate the measurement of Building, Civil and Industrial projects with ease.

The updated document was proposed and adopted as BESMM4 by the NIQS National Executive Council to replace BESMM3 with effect from January 2016.

The efforts of the Institute's Educators with particular reference to Dr. Babalola Olubola who took time and pains to edit the document did not go unnoticed. The effort of Malam Abdullahi Gimba in Part II is highly appreciated. Thanks to Isaac K. Thompson, Nike Orundami and Rashid Ali who repeatedly typed and retyped the document. Thanks to every Member of the Ad-hoc Committee with particular reference to the Vice President of the Institute, Mr. A. Ogunfidodo and the National Treasurer, Mr. Ruya T. Fadason without whose efforts the task would have been largely insurmountable. The efforts of the Secretariat with particular reference to the Executive Secretary for effectively tracking the production in order to ensure strict adherence to the timeline given to the Printer is highly appreciated. It is pertinent to briefly thank members of the National Executive Council for the rare privilege to serve The Institute once more.

The Ad-hoc Committee was made up of the following members:

1	Joseph E. Abhulimhen, BES, B. Bldg Tech. (Const. Mgt), M.Sc. (Const Mgt) FNIQS, MNIOB	Chairman
2	Olukayode S. Oyediran, BSc (QS), MSc (Const. Mgt), PhD, MNIQS	Secretary
3	Adegbemibo Ogunfidodo, BSc (QS), FNIQS, FAPM, MAACE	
4	K. T. Odusami, BSc, MSc, PhD, MNIQS, MRICS, MNIOB	
5	Alh Abba Bello, BSc (QS), MSc (Proj. Mgt), MSc (Const Mgt), PGDM, LLB, FNIQS, MACostE, MAPM	
6	Ray Agomoh, Dip (QS),PGD (Public Admin), PGD (Mgt), MBA (Proj. Mgt), FNIQS	
7	Ruya T. Fadason, Dip(QS) PGD (Bldg), MSc (QS), FNIQS, MAACE, MSAVE	

PREFACE TO THE 2008 EDITION OF BESMM3

In a review of the typographical and contextual errors obtained in BESMM2, as observed by the generality of Members, The NEC (National Executive Council of The NIQS) identified two alternative lines of action.

One was simply to correct the errors and reprint the document or the manual as a second impression.

The second was to revise the publication and issue a Third New Edition, in adopting the second option, The NEC set up an Ad-hoc Committee for the task.

The main Terms of Reference included the rearrangement of the measurement rules to ensure a more user friendly application of the document to the various types of construction activities in the Nigeria constructive ~~on~~ environment while maintaining the general philosophy of all the previous editions.

In partial response to this requirement, the document is now sub-divided into two main parts.

Aside from the General Rules, Part 1 contains the **Rules for Measurement for Buildings** and the like, while Part 2 is for **Civil and Industrial Engineering Installations** and the like.

Except in areas where it is obvious that the nature or circumstance of executing the work types in buildings and the like are radically different from similar work items in Civil and Industrial Installations, the Rules of Measurement are practically the same in both Parts 1 and 2. Consequently the rules in Part 2 in such cases are simply cross references to related items and sections in part 1.

This would no doubt ensure consistency and brevity in the presentation of data for the effective cost management of all types of project in Nigerian construction environment, I am aware that no book revision and is a minor task especially when it is a Committee affair. The completion of this assignment attests to the diligence and mention must be of J.O. Abhulimhen (A Past Chief Examiner), who went an extra mile to draw our attention to numerous small but crucial details.

The cooperation of the Acting Executive Secretary (Mrs, P. O Amobi) and other members of staff at the Secretariat were very helpful to the committee.

The contribution of Members of the NEC and especially the President Mr. F.O. Adetola, who took pains to review the type setting and other features, is also acknowledged.

The committee is also grateful to other Members of The Institute, too numerous to mention, who sent comments and suggestions on how and where to improve on the publication.

PREFACE TO 2002 EDITION OF BESMM2

The First Edition, then titled SMMBWS (Standard Method of Measurement of Building Works and Services) was produced in 1988, by an ad-hoc committee headed by Mr. J. O. Ayeni, FNIQS (now late).

In 1996, another Ad-hoc Committee under Mr. F. O. Adetola (FNIQS) corrected the errors in the First Edition which enabled the publication of the same edition with a second impression.

The rapid and wide spread computerisation of many economic and social activities during the last few decades has had a significant impact on the technological and management process of construction as well.

All the operators in the Nigeria construction environment were confronted with various challenges especially with the general belief that the cost of construction business could be significantly reduced by application of Information Technology (I.T.)

The NIQS, in various singular and joint research efforts with other operators in the public and private sectors, acknowledge that one area that required urgent application of IT was the process of contract data documentation.

In this regard, the harmonisation of the numerous Codes of Measurement, and presentation of the harmonised codes in a format that makes computers application effective in term of usage and cost of installation, was inevitable.

The task for revising the SMMBWS along the foregoing lines, was delegated to the Research and Development (R&D) Committee under the Chairmanship of Mr. E.C. Oforeh (was also the R&D Secretary in Council) in December 2000.

The Committee in discharging the assignment was guided by the philosophy behind the SMMBWS especially in relation to the peculiar procedure in the Nigerian Construction Industry.

It also relied on the model of the various existing codes which may have been adapted for computerisation of cost data processing, storage retrieval and exchange. Notable in this regard were SMM7 by the Royal Institution of Chartered Surveyors (RICS), CESMM3, issued by the Institution of Civil Engineers (ICE) and SMMIEC 1st Edition issue by the Joint Documentation Board of the RICS.

The rule for measuring certain aspect of construction common to both building and heavy engineering such as insitu concrete, steel work, piling engineering service installation in building were harmonized based largely on current common practice in the industry in Nigeria, largely contained in the SMMBWS.

The rule for measuring aspect of works which are strictly engineering in nature such as Geotechnical Process, Tunneling, Railway Tracks, Excavation for highways, Industrial Scaffolding are drawn largely from similar rules in the CESMM3 and SMMIEC and adapted to suit the peculiar format of the BESMM.

The only aspect where alternative rules have been allowed is with respect to preliminaries and general items. The fact that a document of this nature should even be more flexible in accommodating advances in technology and cost management procedure than all the rest from which it has been composed, was not omitted by the committee. However, it can only become the perfect document we desire, when due and constructive criticisms are offered.

For obvious reason, the title of the new document after the revision had to change. The name “BUILDING AND ENGINEERING STANDARD METHOD OF MEASUREMENT 2” (BESMM2) was approved by the Nigerian Executive Council of The Nigerian Institute of Quantity Surveyors.

Members of the R&D Sub-Committee were as follows:

1	Mr. E.C. Oforeh FNIQS	Committee Chairman	– Principal Partner in private practice and Chief Executive of a Construction Software Design and Development Firm. – Formerly Chief Q.S. in a frontline Building and Civil Engineering Company – Member of two previous Ad-hoc Committees on the SMMBWS.
2	Mr. H. Onukwube ANIQS	Member	– Senior Lecturer in Quantity Surveying University of Lagos – Principal Partner in private practice.
3	Mrs. O. Okusaga ANIQS	Member	– Chief Quantity Surveyor & Head of Department of Quantity Surveying in a Public Service Parastatal
4	Mr. G. O. Jagboro ANIQS	Member	– Principal Lecturer in Quantity Surveying, University of Ife
5	Mr. K. M. Tahir	Member	– Principal Partner in private practice
6	Mr. Martins Akumazi	Member	– Partner in private practice
7	Mr. Mohammed Ozama	Member	– Management staff of a leading Bank in Nigeria.
8	Mr. T. Ndah ANIQS	Committee Secretary	– Principal Partner in private practice

PREFACE TO THE 1996 REVISION OF SMM1

The First Edition of the Standard Method of Measurement was published in 1998 and became operative that same year.

The Council of the Nigerian Institute of Quantity Surveyors decided in 1995 that changes in Tendering Process and Procurement Methods in the Construction Industry should be effected in SMM.

This led to the setting up of the Presidential Ad-hoc Committee for the revision of the document by the President, Mr. S.E. Emiowele.

The Members of the Ad-hoc Committee are:

1	Mr. F.O. Adetola ARCIS, ANIQS	Chairman	Principal Partner in practice
2	Mr. E.C. Oforeh; ARCS, ANIQS	Member	Principal Partner in practice and Registrar of NIQS
3	Mr.Bisi Salau, FNIQS	Member	Principal Partner in private practice
4	Mrs. O. Obe ARICS, ANIQS	Member	Managing Partner in private practice (Co-opted)
5	Mr. O. Osisami ARICS, ANIQS	Member	Partner in private practice (Co-opted).

The Committee after various meetings saw the need to revise some of the Principles of Measurement and included some new principles peculiar to Nigerian Construction Industry in the document.

The Committee also proposed to have one unified Standard Method of Measurement for Building, Services, Structural and Heavy Engineering Works instead of the current 4 different Standard Methods of Measurement.

Due to the extent of the revision proposed, the Committee decided to re-print the SMM1 in its present format after minor amendments while approval of the Council is sought for the proposal.

PREFACE TO THE 1988 EDITION OF SMM 1

The building industry in Nigeria has come a long way. All along the World Wars, up to 1960, it has been British oriented both in design and operational modus.

Thereafter, various incursions were made from other areas of Europe and America. All these innovations and our inborn concepts metamorphosed into a new complexity which can rightly be regarded as Nigeria Building Industry Culture. This obviously necessitates a new approach to the measurement of building works.

The basic need for uniform method of measurement is to accord all tenderers a common basis for computing their prices. This in effect creates a congenial platform for comparison of quotations and makes easier, the settlement of Variations and Claims. The First Standard Method of Measurement evolved in 1922 by the RICS (Royal Institution of Chartered Surveyors of the United Kingdom) and subsequent revisions.

It is obvious that the nature or most reasonable mode of measurement may likely be constant. Hence, the possibility of semblance of this document to the hitherto existing types, such as the 6th Edition of Britain and those other Commonwealth countries like: Malaysia, New Zealand, Australia, Canada etc. This accounts for why we wrote the RICS whose document bears closet semblances to ours: and happily we received their encouragement. The present exercise has also taken into consideration, the situation in the Nigerian market, where the cost of labour is appreciably cheaper than the cost of materials which are very high.

Hence, less emphasis is placed on detailed measurement and has given opportunity for a choice where by such labours can be included in the major items of the trade or treated separately as desired. Similarly, the climatic condition which necessitate cooling rather than conserving heat.

It is the desired ambition to incorporate certain other cardinal features apparent in our design, viz: Mud- walling, Thatch-roofing and the likes. However, researches being carried out on the standardization of this by our governmental organization such as the Ministry of Science And Technology and some of its arms namely the Nigeria Building, Road Research Institute and Standard Organization of Nigeria are yet to be concluded. Therefore, future revisions of this document will certainly reflect such development.

It is necessary to mention that this document was rightfully demanded by the Federal Government of Nigeria. The Nigeria Institute of Quantity Surveyors took up the challenge and appointed an Ad-hoc Committee for the production of the Standard Method of Measurement. This Committee was inaugurated on 16th December, 1986 shortly after the official recognition of the Institute by Decree No. 31 of December 5 of the same year under the Presidency of Chief Ezuogo Isiadinso and composed of the following who participated actively in the production of the document.

1	Mr. J.O. Ayeni, FRICS, ANIQS	Chairman	Principal Partner in practice
2	Mr. E.C Oforeh, ARICS, ANIQS	Secretary	Senior Management Personnel in a front–line construction company
3	Mr. K.N Igbokwe, ANIQS	Ass. Secretary	Senior Quantity Surveyor in a front – line practice)
4	Mr. N.A. Kio, FRICS FNIQS		Principal Partner in private practice
5	Mr. A.O Nwanze, FRICS, FNIQS, PPNIQS		Former Director of Quantity Surveying, Federal Ministry of Works, and now Principal Partner in private practice
6	Mr. O.U Ikpa, ARICS, ANIQS		Principal Partner in practice

Consequent upon the initial draft, copies were sent out for comments from such organizations as:

Nigerian Institute of Architects, Nigerian Institute of Building; Federation of Building and Civil Engineering Contractors of Nigeria; Nigeria Society of Engineers; Reputable Building Contractors of Nigeria; Nigerian Society of Engineers; some Quantity Surveying firms in private practice, CASTLE; Associations of Professional Bodies of Nigeria; All Higher Institutions within the Federation, Ministry of Works and Housing. Their comments have been useful in production of this document.

The Committee welcomes contributions to further improvement of this document from all and sundry. Such contributions will guide the preparation of future editions.

The Institute cannot conclude this acknowledgement without mentioning the Executive Secretary, Mr. G.O. Moneke who did the entire proof reading of the draft with his staff.

J.O. AYENI
CHAIRMAN, AD-HOC COMMITTEE OF THE NIGERIAN INSTITUTE OF QUANTITY SURVEYORS.
For Drafting of the S.M.M

ACKNOWLEDGEMENT

The Nigerian Institute of Quantity Surveyors made extensive use of the New Rules of Measurement 2 published by the Royal Institution of Chartered Surveyors, CESMM4 published by the Institute of Civil Engineers hence, the Publishers are all singly and collectively acknowledged on behalf of The Nigerian Institute of Quantity Surveyors.

GENERAL RULES

GENERAL RULES

1.0 Part 1: General

1.1 Introduction

This Standard Method of Measurement provides a uniform basis for measuring construction works (building, civil and industrial engineering works) and embodies the essentials of good practice. Bill of Quantities shall fully describe and accurately represent the quantity and quality of the works to be carried out. More detailed information than is required by these rules shall be given where necessary in order to define the precise nature and extent of the work required. The rules apply to measurement of proposed and executed work.

1.2 Use of Tabulated Rules

Generally

1.2. 1 The rules in this document are set out in tables with a view to enhancing comprehensive description without difficulties. Each section of the rules comprises information (to be) provided, classification tables and supplementary rule. The tabulated rules are written in present tense.

1.2.2 Horizontal lines divide the classification table and supplementary rules into zones to which different rules apply.

1.3 Classification Tables

1.3.1 Within the classification table where a broken line is shown, the rules given above and below the broken line may be used as alternatives depending on the components of description.

1.3.2 In referring to columns in classification tables, the measurement unit column has been disregarded

1.3.3 The left-hand column of the classification table lists descriptive features commonly encountered in construction works. The next column lists further sub-groups into which each main group of items shall be divided and similarly the third column provides for further division.

1.3.4 Each item description shall identify the work with respect to one descriptive feature drawn from each of the first three columns in the classification tables and as many of the descriptive features in the fourth column as are applicable to the item. The general principle does not apply to Preliminaries in that it will be necessary to select as many descriptive features as appropriate from each column.

1.3.5 Where the abbreviation (nr) is given in the classification table, the quantity shall be stated in the item description.

1.4 Supplementary Rules

1.4.1 Within the supplementary rules, everything above the horizontal line, which is immediately below the classification table heading, is applicable throughout that table.

1.4.2 Measurement rules set out when work shall be measured and the method by which quantities shall be computed.

1.4.3 Definition rules, define the extent and limits of the work represented by a word or expression used in rules and the bills of quantities prepared in accordance with the rules

1.4.4 Coverage rules draw attention to particular incidental work which shall be deemed to be included in the appropriate items in a Bill of Quantities to the extent that such work is included in the tender documents. Where the coverage rules include materials, they shall be measured in the item descriptions.

1.4.5 The column headed Supplementary information contains rules governing the information which shall be given in addition to the information given as a result of the application of rule 2.6.

1.4.6 A separate item shall be given for any work which differs from other work with respect to any matter listed as supplementary information.

1.5 Quantities

1.5.1 Work shall be measured net as fixed in position except where otherwise stated in a measurement rule applicable to the work.

1.5.2 Dimensions used in calculating quantities shall be taken to the nearest 10mm (i.e. 5 mm and over shall be regarded as 10 mm and less than 5mm shall be disregarded)

1.5.3 Quantities measured in tonnes shall be given to two places of decimals. Other quantities shall be given to the nearest whole unit, except that any quantities less than one unit shall be given as one unit.

1.5.4 Unless otherwise stated, where minimum deductions for voids are dealt with in this document, they shall refer only to openings or wants which are within the boundaries of measured areas. Openings or wants which are at the boundaries of measured areas shall always be subject of deduction irrespective of size.

1.5.5 The requirement to measure separate items for widths not exceeding a stated limit, shall not apply where these widths are caused by voids.

1.6 Descriptions

1.6.1 Dimensions shall be stated in descriptions generally in the sequence length, width and height. Where ambiguity could arise, the dimensions shall be identified.

1.6.2 Information required by the application of rules 2.6 and 2.12 may be given in documents (e.g. drawings or specification) separate from the bills of quantities if a precise and unique cross reference is given in its place in the description of the item concerned.

1.6.3 Headings to groups of items in a Bill of Quantities shall be read as part of the description of the items to which the headings apply.

1.6.4 The use of a hyphen between two dimensions in this document or in a Bill of Quantities shall mean a range of dimensions exceeding the first dimension stated but not exceeding the second.

1.6.5 Each work section of a Bill of Quantities, shall begin with a description stating the nature and location of the work unless evident from the drawn information or other information required to be provided by these rules.

1.6.6 Unless otherwise specially stated in a Bill of Quantities or herein, the following shall be deemed to be included with all items:

a) Labour and all costs in connection with
b) Materials, goods and all costs in connection therewith.
c) Assembling, fitting and fixing materials and goods in position.
d) Plant and all costs in connection therewith
e) Waste of materials
f) Square cutting
g) Establishment charges, overhead charges and profit
h) Consumable stores including jointing materials (but excluding gaskets and bolts and nuts and washers).

1.6.7 A dimensioned description for an item in the Bill of Quantities shall define the item and state all the dimensions necessary to identify the shape and size of the item or the components.

1.7 Drawn information

1.7.1 Location drawings

a) Block Plan: shall identify the site and locate the outlines of the construction infrastructure in relation to a town or other context.

b) Site Plan: shall locate the position of the construction infrastructure in relation to setting out points, means of

access and general layout of the site

c) Plans, Sections and Elevations: shall show the position occupied by the various spaces in a building and the general construction and location of the principal elements.

1.7.2 Component drawings: shall show the information necessary for manufacture and assembly of components

1.7.3 Dimensioned diagrams shall show the spaces and shape of the work covered by an item and may be used in a Bill of Quantities in place of dimensioned description, but not in place of an item otherwise required to be measured.

1.7.4 Schedules which provide the required information shall be deemed to be drawings as required under these rules.

1.8 Catalogues or Standard Components

1.8.1 A precise and unique cross reference to a catalogue or to a standard specification may be given in an item description instead of the description required by rules 2.6 and 2.12 or instead of a component drawing.

1.9 Work of special types

1.9.1 Work of each of the following special types shall be separately identified:

a) Work on or in an existing building (see General Rule 1.3).
b) Work to be carried out and subsequently removed (other than temporary works)
c) Work outside the cartilage of the site.
d) Work carried out in or under water shall be so described stating whether canal, river, or sea water and where applicable the mean levels or high and low water levels at the peak rainy and dry seasons.
e) Work carried out under compressed air shall be so described stating the pressure and the method of entry and exit.

1.10 Fixing Base and Background

1.10.1 Method of fixing shall be measured and described where required by the rules of each Work Section. Where fixing through vulnerable materials is required to be identified, such materials are deemed to include those listed in rule 8.3(e).

1.10.2 Where the nature of the base is required to be identified, each type of base shall be identified separately.

1.10.3 Where the nature of the background is required to be identified, the item description shall state one of the following:

a) Timber, which shall be deemed to include manufactured building boards.
b) Masonry, which shall be deemed to include concrete, brick, block and stone.
c) Metal.
d) Vulnerable materials, which shall be deemed to include glass, marble,
e) Mosaic, tiled finishes and the like.

1.11 Composite items

1.11.1 Notwithstanding the requirement of Clause 4.2, work to be manufactured offsite may be combined into one item even though the rules require items to be measured separately, provided the items in question are all incorporated into the composite item off site. The Item description shall identify the resulting composite item and the item shall be deemed to include breaking down for transport and installation and subsequent assembly.

1.12 Procedure where the Drawn and Specification Information required by these rules is not available

1.12.1 Where work can be described and given in terms in accordance with these rules, but the quantity of work required cannot be accurately determined; an estimate of the quantity shall be given and identified as an approximate quantity.

1.12.2 Where work cannot be described and given as in items in accordance with these rules, it shall be given as Provisional Sum and identified as for either defined or undefined work as appropriate

1.12.3 A Provisional Sum for defined work is a sum provided for work which is not completely designed but for which the following information shall be provided:

a) The nature and construction of the work.
b) A statement of how and where the work is fixed to the building and what other work is to be fixed thereto.
c) A quantity or quantities which indicate the scope and extent of the work.
d) Any specific limitations and the like identified in Section A36.

1.12.4 Where Provisional Sums are given for defined work, the Contractor will be deemed to have made due allowance in programming, planning and pricing preliminaries. Any such allowance will only be subject to adjustment in those circumstances where a variation in respect of other work measured in detail in accordance with the rules, would give rise to adjustment.

1.12.5 A Provisional sum for undefined work is a sum provided for work where the information required in accordance with rule 10.3 cannot be given.

1.12.6 Where Provisional Sums are given for undefined work, the Contractor will be deemed to not to have made any allowance in programming, planning and pricing Preliminaries.

1.13 Work not covered

1.13.1 Rules of measurement adopted for work not covered by these rules, shall be stated in a Bill of Quantities. Such rules shall, as far as possible, conform to those given in this document for similar work.

1.14 Symbols and abbreviations and definitions

1.14.1 The following symbols and abbreviations are used in this method of measurement:

m	=	meter
m2 or m2	=	square meter
m3 or m3	=	cubic meter
mm	=	millimetre
nr or No	=	number
kg	=	kilogramme
kn	=	kilonewton
t	=	tonne
ha	=	hectare
hr	=	hour
p.c. sum	=	Prime Cost sum
prov. sum	=	Provisional sum
Sum	=	Lump sum
>	=	exceeding
>	=	equal to or exceeding
<	=	less than
<	=	not exceeding or equal to
%	=	percentage
-	=	hyphen (see rule 4.4)
FMW	=	Federal Ministry of Works
FMHUD	=	Federal Ministry of Housing and Urban Development
HFP	=	Health and Safety Policy

1.14.02 Cross references within the classification tables are given in the form:

work section number:	number from first column	number from second column	number from third column	number from fourth column

Example: 1a

1.5:6.1.1

Excavating and filling
Excavating
Bulk excavation (To reduce levels)
Maximum depth <2.00 m
Commencing level stated where > 0.25 m below existing ground level

Alternatively:

Cross references within the classification in columns 1-3 may be stated without the dot except in the case of numbers consisting of two digits (i.e. 10 and above) a slash (/) shall be used after the numbers of the first three columns before stating the number of the fourth column.

Example: 1b

1.11:22.1 Formwork for insitu concrete:
Walls;
Vertical;

Alternatively

Cross references within the classification columns, with or without the dots as stated in examples 1a and 1b above may be stated along with the item descriptions

Example 1c

A 1.11:22.1
Formwork for insitu concrete:
Walls; Vertical;

Alternatively

2d Where the production of bills of quantities with computer software demands the adoption of measured items referencing system that is different from the alternatives given above, the adopted measured items referencing system shall be explained in detail in the Preliminaries.

1.14.3 An asterisk with a cross reference represents all entries to the column in which it appears.

1.14.4 The digit, 0 without a cross reference represents no entries in the column in which it appears.

1.15 Work to existing buildings

1.15.1 Work to existing buildings shall be so described. Such work is defined as work on, or in, or immediately under work existing before the current project.

1.15.2 The additional rules for work to existing buildings are to be used in conjunction with the preceding rules in the appropriate Work Sections.

1.15.3 A description of the additional Preliminaries/General conditions which are pertinent to the work to the existing buildings shall be given, drawing attention to any specific requirements due to the nature of the work.

1.16 General definitions

1.16.1 Where the rules require work to be described as curved work with the radii stated, details shall be given of the curved work including if concave or convex, if conical or spherical if to more than one radius and shall state the radius or radii.

1.16.2 The radius stated shall be the mean radius measured to the centre line of the material unless otherwise stated.

1.17 Free issue materials for industrial installations and the like

1.17.1 Where materials are to be supplied as free issue, this shall be stated together with details of:

1) Point of delivery or collection and responsibility for handling together with a schedule of delivery of key details.
2) Materials test certificate and identification.
3) Methods of protection, storage and associated preparatory work such as degreasing and cleaning.
4) Conditions of steelwork and pipework ends
5) Responsibility for reconciliation and storage of materials whilst in the care of the contractor and return of surplus and scraps

1.18 Method-Related Charges

1.18.1 Method-related charges shall apply to measurements under Civil and Industrial Engineering Works Contract.

1.18.2 For the purposes of this section, the following words and expression shall have the meanings hereby assigned to them.

a) 'Method-Related Charges' means the sum for an item inserted in the Bill of Quantities by a tenderer in accordance with paragraph 16.3.

b) 'Time-Related Charge' means a Method-Related Charge for work the cost of which is to be considered as proportional to the length of time taken to execute the work.

c) 'Fixed Charge' means a Method-Related charge which is not a Time-Related Charge.

1.18.3 A tenderer may insert in the Bill of Quantities, such items for Method-Related Charges as he may decide to cover items of work relating to his intended method of executing the works, the costs of which are not to be considered as proportional to the quantities of the other items and for which he has not allowed in the rates and prices for the other items.

1.18.4 Where possible, the itemization of Method-Related Charges should follow the order of classification and other requirements set out in Section 2A of the work classification, distinguishing between Time-Related Charges and Fixed Charges. Method-Related Charges may be inserted to cover items of work other than those set out in Section 2A of this method of measurement.

1.18.5 Each item of Method-Related Charge inserted in the Bill of Quantities, shall be fully described so as to identify the resources to be used and the particular items of Permanent Works or Temporary works to which the item relates.

1.18.6 Insertion by the Contractor of an item for a Method-Related Charge in the Bill of Quantities when tendering, shall not bind him to adopt the method stated in the item description in executing the works.

1.18.7 Method-Related Charges shall not be subject to re-measurement.

1.19 Highlighting of some clauses of the supplementary rules

1.19.1 Clauses in the supplementary rules, especially the Measurement Rules and Supplementary Information which relate to work items with features having different properties that call for repeat or different measurement to distinguish the properties, are highlighted (to readily attract the attention of the Surveyor), by slightly shading over the rules.

1.19.2 Clauses in the supplementary rules, especially the Measurement Rules and Supplementary Information which relate to

work items with features or characteristics that could lead to:

1) Unwanted measurements, based on the prevailing circumstance.

2) Duplicated or repeat measurements, based on the prevailing circumstance are highlighted, to (readily attract the attention of the Surveyor), by slightly shading over the rules.

1.20 Definitions

Bill of Quantities (BoQ)	means a list of items giving detailed identifying descriptions and firm quantities of the work comprised in a contract.
Cost Breakdown Structure (CBS)	in the context of Bill of Quantities, represents the financial breakdown of a building project into cost targets for elements or work packages.
Cost Target	in the context of Bill of Quantities, means the total expenditure for an element or work package.
Credit	means a refund offered by the contractor to the employer in return for the benefit of taking ownership of materials, goods, items, mechanical and electrical plant and equipment, etc. arising from demolition or strip out works.
Daywork	means the method of valuing work on the basis of time spent by the contractor's Work people, the materials used and the plant employed.
Defined Provisional Sum	means a sum provided for work which is not completely designed but for which the following information shall be provided: 1. the nature and construction of the work; 2. a statement of how and where the work is fixed to the building and what other work is to be fixed thereto; 3. a quantity or quantities that indicate the scope and extent of the work; and 4. any specific limitations and the like identified.
Design Team	means architects, engineers and technology specialists responsible for the conceptual design aspects of a building, structure or facility and their development into drawings, specifications and instructions required for construction and associated processes. The design team is a part of the project team.
Director's Adjustment	means a reduction or addition to the tender price, derived by the contractor's estimating team, offered by the Director(s) of the contractor.
Employer	means the owner and/or the developer of the building/Engineering/Industrial Project; in some cases the ultimate user. The terms Senior Responsible Owner (SRO) and project sponsor are used by central civil government and the defense sector; being the representatives empowered to manage the building project building/Engineering/Industrial and make project- specific decisions. For the purpose of these measurement rules, the term 'employer' will also be used interchangeably with Senior Responsible Owner (SRO) or project sponsor.
Fixed Charge	is for work, the cost of which is to be considered independent of duration.
Health and Safety File	means a record of health and safety information for construction project. It is designed to identify key health and safety risks which may be encountered on site and must be kept up to date for the life of the building.
Health and Safety Management	means organised efforts and procedures for identifying workplace hazards and reducing accidents and exposure to harmful situations and substances.

Main Contractor	means the contractor responsible for the total construction and completion process of the building project.
Main Contract Preliminaries	are items that cannot be allocated to a specific element, sub-element or component. Main contract preliminaries include the main contractor's costs associated with management and staff, site establishment, temporary services, security, safety and environmental protection, control and protection, common user mechanical plant, common user temporary works, the maintenance of site records, completion and post-completion requirements, cleaning, fees and charges, sites services and insurances, bonds, guarantees and warranties. Main contractors' preliminaries exclude costs associated with subcontractors' or work package contractors' preliminaries.
Other Development/Project costs	means costs that are not necessarily directly associated with the cost of constructing the building, but form part of the total cost of the building project to the employer (e.g. land acquisition costs, fees for letting agents, marketing costs and contributions associated with planning permissions, compensation).
Overheads and Profit	means the contractor's costs associated with head office administration, proportioned to each building contact, plus the main contractor's return on capital investment.
Post Tender Estimate	means a cost estimate carried out after the evaluation of tenders to corroborate the funds required by the employer to complete the building project.
Pre-Tender Estimate	means a cost estimate prepared immediately before calling tenders for construction.
Preliminaries	see main contract preliminaries and work package contract preliminaries.
Provisional Quantity	means a quantity which cannot be accurately determined (i.e. an estimate of the quantity).
Provisional Sum	means a sum of money set aside to carry out work that cannot be described and given in quantified items in accordance with the tabulated rules of measurement. A provisional sum will be identified as either 'defined' or 'undefined' (see definitions of 'Defined provisional sums'.
Prime Cost Sum (PC Sum)	means a sum of money included in a unit rate to be expended on materials or goods from suppliers (e.g. ceramic wall tiles at ₦1560.00/m or door furniture at ₦1750.00/door). It is a supply only rate for materials or goods where the precise quality of those materials and goods are unknown. PC Sums exclude all costs associated with fixing or installation, all ancillary and sundry materials and goods required for the fixing or installation of the materials or goods, subcontractor's design fees, subcontractor's preliminaries, subcontractor's overheads and profit, Main Contractor's design fees, preliminaries and overheads and profit.
Residual Risk (or Retained Risk)	means the risks retained by the employer.
Statutory Undertaker	means organisations, such as water, gas, electricity and telecommunications companies that are authorised by statute to construct and operate public utility undertakings.
Sub-Contractor	means a contractor employed by the main contractor to undertake specific work within the building project/engineering/industrial; also known as specialist works, trade, work package, and labour only contractors.
Time-Related Charge	is for work, the cost of which is to be considered dependent on duration.

Undefined Provisional Sum	means a sum provided for work that is not completely designed, but for which the information required for a defined provisional sum cannot be provided.
Work Breakdown Structure (WBS)	in the context of Bill of Quantities, is used to sub-divide a building project into meaningful elements or work packages.
Work Package Contractor	a specialist contractor who undertakes particular identifiable aspects of work within the building project; e.g. ground works, cladding, mechanical engineering services, electrical engineering services, lifts, soft landscape works or labour only. Depending on the contract strategy, works contractors can be employed directly by the employer or by the main contractor.
Work Package Contract Preliminaries	are preliminaries that relate specifically to the work that is to be carried out by a work package contractor.

2.0 Part 2: Rules for Detailed Measurement of Building Works

2.1 Introduction

2.1.1 Part 2 of the rules describes the purpose and uses of BESMM4: Detailed Measurement for Building Works; describes the types of Bill of Quantities (BoQ); gives guidance on the preparation and composition of BoQ; and defines the information required to enable a BoQ to be prepared. Part 2 also sets out the rules of measurement of building items, and the rules for dealing with preliminaries, non-measurable works and contractor designed works, as well as risks, overheads and profit, and credits.

2.1.2 In addition, Part 2 of the rules deals with other aspects of BoQ production, including price fluctuations, Director's adjustments, day work and value added tax. Guidance is also provided on the codification of BoQ, the use of BoQ for cost control and cost management, and the analysis of a BoQ to provide cost data.

2.2 Purpose of Bill of Quantities

2.2.1 The primary purposes of a Bill of Quantities (BoQ), which becomes a contract document, are:

- to provide a co-ordinated list of items, together with their identifying descriptions and quantities, that comprise the works to enable contractors to prepare tenders efficiently and accurately; and
- when a contract has been entered into, to:
    i. provide a basis for the valuation of work executed for the purpose of making interim payments to the contractor; and
    ii. provide a basis for the valuation of varied work.

2.2.2 Essentially, a BoQ is a list of items with detailed identifying descriptions and quantities, which make up the component parts of a building, engineering or industrial facility.

2.3 Benefits of Bill of Quantities

2.3.1 Irrespective of what contract strategy is used, at some stage in the procurement process one party will need to quantify the extent of works to be executed; whether it be the employer's quantity surveyor/cost manager, the main contractor or the work package contractors for the purpose of obtaining a price for completing building, engineering or industrial works, valuing the extent of work complete for purposes of payment, valuing variations in the content or extent of works, or to support applications for tax or other financial incentives. Consequently, detailed measurement for the purpose of Bill of Quantities (BoQ) production is beneficial for a number of reasons:

i. it saves the cost and time of several contractors measuring the same design in order to calculate their bids for competition;
ii it provides a consistent basis for obtaining competitive bids;
iii it provides an extensive and clear statement of the work to be executed; it provides a very strong basis for budgetary control and accurate cost reporting of the contract (i.e. post contract cost control), including:
    - the preparation of cash flow forecasts,
    - a basis for valuing variations, and
    - a basis for the preparation of progress payments (i.e. interim payments).
iv it allows, when BoQ items are codified, reconciliation and any necessary transfers and adjustments to be made to the cost plan;
v when priced, it provides data to support claims for tax benefits (e.g. capital allowances and value added tax (VAT);
vi when priced, it provides data to support claims for grants; and
vii it provides one of the best sources of real-time cost data, which can be used for estimating the cost of future building, engineering and industrial projects (i.e. historic cost information as it provides a cost model in a single document.

2.4 Types of Bill of Quantities

2.4.1 The use of Bill of Quantities in support of a contract is the traditional and proven means of securing a lump-sum price for undertaking building, engineering and industrial projects. Bill of Quantities (BoQ) can be:

- firm (to obtain a lump-sum price for a fully designed project); or
- approximate (subject to re-measurement as built).

2.4.2 **Firm Bill of Quantities:**

2.4.2.1 The reliability of the tender price will increase in relation to the accuracy of the quantities provided (i.e. the more precisely the work is measured and described). In theory, were there no design changes, then a firm BoQ would provide a price at tender stage, which would equal the final cost. In practice there will be changes, and the BoQ provides a good basis for cost control, since the direct cost of change can be assessed with reference to the BoQ rates.

2.4.2.2 The firmer the BoQ the better it is as a means of financial control.

2.4.3 **Approximate Bill of Quantities:**

2.4.3.1 Approximate BoQ are used when there is insufficient detail to prepare firm BoQ or where it is decided by the employer that the time or cost of a firm BoQ is not warranted. Such contracts donor provide a lump-sum price, but rather tender price totals (i.e. a quantified schedule of rates), since the quantities are subject to re-measurement on completion by the quantity surveyor/cost manager. These contracts are usually subject to greater variation than lump sum contracts and therefore should only be used where time is a limiting factor or where there is great uncertainty in respect of certain elements, such as major excavation and earthworks.

2.4.3.2 The initial resource cost of an approximate BoQ is likely to be lower than for a firm BoQ, but the need for re-measurement invariably results in an overall higher resource cost.

2.4.3.3 Although the quantities are approximate, the descriptions of work items should be correct.

2.5 Preparation of Bill of Quantities

2.5.1 Bill of Quantities (BoQ) are produced at 'RIBA Work Stage G (Tender Documentation)' or as an intrinsic part of 'OGC Gateway 3C (Investment Decision) or at an agreed stage on a pre-contract programme on a project'. The basic requirements *include preparation and/or collation of tender documentation in sufficient detail to enable a tender or tenders to be obtained for the project.*

2.5.2 To enable the preparation of BoQ, the information resulting from RIB Work Stages E (Technical Design) and F (Production Information) will be required. The requirements of RIB Work Stages E and F, as described in the RIBA Outline Plan of Work, are as follows:

Preparation of technical design(s) and specifications, sufficient to co-ordinate components and elements of the project and information for statutory standards and construction safety.

F1 Preparation of production information in sufficient detail to enable a tender or tenders to be obtained.

F2 Application for statutory approvals. Preparation of further information for construction required under the building/civil engineering or industrial contract.

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2.5.3 The information and documents required for the preparation of BoQ are described in paragraph2.14 of these rules.

2.5.4 BoQ required for a lump sum contract based on firm or approximate quantities will normally be prepared by the employer's quantity surveyor/cost manager, whereas under a design and build contract, the employer's project team will prepare the 'employer's requirements' and the BoQ or quantified schedules of work will be prepared by either the main contractor or, more likely, the main contractor's work package contractors. The choice of who quantifies building, engineering and industrial works is solely down to the employer's preference of contract strategy (see Figure 2.1 below).

Figure 2.1: Responsibility for quantifying building works

Contract Strategy	Basis of 'Invitation Documents'	Preparation by
Traditional Lump Sum	(a) Firm BoQ --- (b) Approximate BoQ	Employer's Quantity Surveyor/Cost Manager
Design and Build	Employer's Requirements	Employers' Project Team (with compilation normally by the employer's Quantity Surveyor/Cost Manager). Note: Quantification of the Employer's requirements will be carried out by either the main contractor or work package contractors; who will prepare firm or approximate BoQ, or quantified Schedules of Work as appropriate.
Management	(a) Firm BoQ --- (b) Approximate BoQ	Employer's Quantity Surveyor/Cost Manager (or Main Contractor or Work Package Contractors if 'Invitation Documents' prepared by employer's Quantity Surveyor/Cost Manager or Main Contractor, respectively, are based on either application drawings' or un-quantified Schedule of Works' (i.e. un-quantified information).
Management (Design and Manage)	(a) Firm BoQ --- (b) Approximate BoQ	
Construction Management	(a) Firm BoQ --- (b) Approximate BoQ	

2.6 Composition of a Bill of Quantities

2.6.1 Bill of Quantities (BoQ) usually comprises the following sections:

- Form of Tender (including certificate of bona fide tender);
- Summary (or Main Summary);
- Preliminaries, comprising two sections as follows:
  i Information and requirements; and
  ii Pricing schedule;
- Measured work (incorporating contractor designed works);
- Risk;
- Provisional sums;
- Dayworks (Provisional); and
- Annexes.

2.6.2 **Form of Tender:**

2.6.2.1 This is a document that is used to record the main contractor's price for completing the building, engineering and industrial project (i.e. his tender price). If accepted by the employer, the tender price will become the 'contract sum'. The form of tender can be a separate document.

2.6.2.2 A separate 'certificate of bona fide tender', which is completed by the main contractor to confirm that he has not communicated his tender to other parties, is sometimes inserted after the form of tender. Alternatively, the employer's requirements for confirming that bona fide tender has been submitted by the main contractor can be incorporated in the form of tender.

2.6.3 **Summary (or Main Summary):**

2.6.3.1 The summary, sometimes called the main summary, is made at either the front or end of the Bill of Quantities and comprises a list of the bill that form the entire Bill of Quantities. The total price for each section of the Bill of Quantities (BoQ) is carried forward and inserted against the applicable item listed in the summary. For example, a summary for an elemental bill will address all or some of the following:

- Preliminaries;
- Measured works (including 'Contractor Designed Works':
  - Facilitating works
  - Substructure
  - Superstructure
  - Internal finishes
  - Fittings, furnishings and equipment
  - Services
  - Complete buildings
  - Works to existing buildings
  - External works
- Risks;
- Provisional Sums:
  - Defined
  - Undefined
- Works to be carried out by statutory undertakers;
- Overheads and profit;
- Credits (for materials arising from the works);
- Fixed price adjustment;
- Director's adjustment;
- Dayworks (Provisional);
- Total price (to Form of Tender).

2.6.3.2 Where the measured work has been divided into work sections, the work sections will be listed instead of elements.

2.6.3.3 At the end of the summary, provision is made to total the list to ascertain the total price and transfer the total to the form of tender, which, subject to verification and any necessary adjustments, will become the contract sum referred to by the conditions of contract.

2.6.4 **Preliminaries:**

2.6.4.1 Preliminaries address and communicate to the contractor items that are not directly related to any component, element, or work section (i.e. measured works).The information provided will enable the contractor to ascertain his or her price for, among other things, management of the building project, site establishment, security, safety, environmental protection and common user mechanical plant, as well as the employer's completion and post-completion requirements. Preliminaries are divided into two sections as follows:

2.6.4.2 **Information and Requirements**: The purpose of the information and requirements section is to describe the building, engineering or industrial project particulars; the drawings upon which the Bill of Quantities was based; the work in general; the site and any adjoining properties; the form of contract and any amendments and/or supplementary conditions to the form of contract; the employer's specific requirements; and any specific limitations or restrictions that might impact on the sequence and/or method of working. It also contains information on how to interpret the Bill of Quantities, including any special methods of measurement (i.e. where the method of measurement has deviated from BESMM4: Detailed measurement for building, engineering and industrial works).

2.6.4.3 **Pricing Schedule:** The pricing schedule is simply a schedule in which the contractor inserts preliminary costs relating to the employer's requirements and all contractor cost items (including management and staff, site accommodation, services and facilities, mechanical plant, and temporary works items). The quantification of preliminaries is dealt with in paragraph 2.7.

2.6.5 **Measured Work:**

2.6.5.1 This is the main part of the Bill of Quantities, which lists all the items of work to be undertaken. The quantities and descriptions of items should be determined in accordance with the tabulated rules of measurement in Part 3 (Tabulated rules for the measurement of building works) of these rules. Various methods can be used to present the measured work (i.e. bill breakdown structures – see paragraph A.1 at Appendix A: Guidance on the preparation of Bill of Quantities).

2.6.5.2 The rules relating to the quantification and description of measured work are given in paragraph 2.8.

2.6.6 **Risks:**

2.6.6.1 This section comprises a list of residual risks (i.e. unexpected expenditure arising from risks that materialise, for example, disposal of contaminated ground material), which the employer wishes to transfer to the contractor. The contractor is required to provide a lump-sum fixed price for taking, managing and dealing with the consequences of the identified risk should it materialise.

2.6.6.2 The rules relating to the quantification and description of risk are given in paragraph 2.10.

2.6.7 **Provisional Sums:**

2.6.7.1 Provisional sums are sums included for any items of work that are anticipated, but for which no firm design has been developed, including any sums listed for any items of work that are to be executed by a statutory undertaker.

2.6.7.2 This part of the BoQ, therefore, lists items of work that cannot be entirely foreseen or detailed accurately at the time tenders are invited (i.e. non-measurable items). Pre-determined sums of money are set against each item, determined by the quantity surveyor/cost manager, to cover their cost.

2.6.7.3 The rules relating to provisional sums are given in paragraph 2.9.1.

2.6.8 **Dayworks (Provisional):**

2.6.8.1 This section of the BoQ gives provision for the contractor to competitively tender rates and prices for works, which might be instructed to be carried out on a daywork basis.

2.6.8.2 The rules relating to time-charged work carried out on the basis of daywork rates are given in paragraph 2.13.3 below.

2.6.9 Annexes: The annexes comprise information referred to in the BoQ where it is not contained in, or to be issued as a separate document. Examples are: performance specifications (if not included in project specification); copies of quotations; and copies of communications with statutory undertakers. 2.7 Preliminaries.

2.7 Preliminaries

2.7.1 For the most part, preliminaries are the cost of administering a project and providing plant, site staff, facilities site-based services, and other items not included in the rates for measured works.

2.7.2 Preliminaries are divided into two categories:

1. Preliminaries (Main Contract)
2. Preliminaries (Works Package Contract).

2.7.3 Preliminaries (Main Contract) are divided into two discrete sections:

1. Information and Requirements; and
2. Pricing Schedule.

2.7.3.1 Information and Requirements:

This is the descriptive part of the preliminaries (Main Contract), which:

1. sets out the project particulars (e.g. the project title, the site address and the names and contact details of the employer and the employer's project team);

2. identifies the drawings upon which the Bill of Quantities was based;

3. depicts the boundary of the construction site;

4. provides information about existing buildings and existing mains services on or adjacent to the site, and about any existing records that will inform the main contractor of any known or potential hazards that need to be considered;

5. identifies known constraints and restrictions that might impact on the main contractor's methodology for constructing the building or buildings;

6. describes the building project in general;

7. specifies the standard form of contract, together with the contract particulars and any amendments and/or supplementary or special conditions to the standard form of contract, which is to be entered into by the contracting parties; as well as the employer's requirements in respect of insurances, parent company guarantees, performance bonds and collateral warranties;

8. explains the documents provided, their content and how they are to be used;

9. confirms the method of measurement (i.e. BESMM4: Detailed measurement for building, engineering and industrial works), how to interpret the Bill of Quantities, and any special methods of measurement (i.e. where the method of measurement has deviated from the specified rules);

10. sets out the employer's specific requirements in respect of:

    (a) management of the works by the main contractor, including progress reporting, programme management and cost reporting requirements;

    (b) quality standards to be achieved, and quality control requirements, by the main contractor, including inspection, testing and commissioning requirements;

    (c) security, safety and protection measures to be provided by the main contractor;

    (d) facilities, temporary works and services required by the employer;

    (e) specific requirements such as advertising, the provision of a marketing suite or a topping out event;

    (f) specific limitations on method, sequence, timing of the 'works' imposed by the employer, including out of normal hours working and phasing requirements;

    (g) operation and maintenance of the finished building, including operation and maintenance (O&M) manuals, familiarisation training, tools, and spare parts; and

    (h) post-completion requirements such as maintenance services.

Instructions to tendering contractors and other information relating to the tender process (e.g. information required to be submitted with tender submission, site visits, confidentiality, etc) form part of the invitation documents, but will not form part of the contract documents. Therefore, these should be addressed in a separate document (e.g. Conditions of tender).

2.7.3.2 Pricing schedule (quantification of main contractor's preliminaries):

It is not possible for the quantity surveyor/cost manager to quantify the main contractor's preliminaries. This is because it is for the contractor to interpret the information provided as part of the tender invitation documentation. From the information provided, the main contractor will ascertain his method of working and the resources required to complete the building project, as well as identify any other cost items that are to be recovered.

The preliminaries bill is therefore to include a pricing schedule that simply lists the headings under which the main contractor is to price his or her preliminaries items. Templates for preliminaries pricing schedules (condensed and expanded versions) are included in Appendices B and C, respectively, of these rules.

The pricing schedule is a simple a list of cost centres incorporated in the Bill of Quantities in which the main contractor will insert his charges relating to preliminaries.

The pricing schedule for main contractors' preliminaries is divided into two main cost centres:

1. employer's requirements; and

2. main contractor's cost items.

The items that comprise these two cost centres are listed and defined in Part B of Table 1 (Preliminaries (main contract)) at Part 3: Tabulated rules for measurement of these rules. Notwithstanding this, as part of the priced Bill of Quantities submitted by the main contractor, it is essential that the quantity surveyor/cost manager obtains a full and detailed breakdown that clearly identifies the items, shows how the price for each item has been calculated, and how the total price for preliminaries has been calculated.

Accordingly, as part of the conditions of tender, the quantity surveyor/cost manager should instruct the main contractor to return along with his or her tender a full and detailed breakdown that shows how the main contractor's total price for preliminaries has been calculated. It should be requested that the main contractor append this information to his priced Bill of Quantities. To ensure that the main contractor's detailed supporting calculations are presented in an easy-to-read and logical format, the main contractor should be instructed to ascertain the price for preliminaries in accordance with the rules of measurement for main contractor's preliminaries (refer to Part B (Pricing schedule) of Table 1 (Preliminaries (main contract)) at Part 3: Tabulated rules for measurement of these rules).

It is also essential that the quantity surveyor/cost manager makes it clear to the main contractor in the preliminaries bill and/or preliminaries pricing schedule that costs relating to items that are not specifically identified by the main contractor in his or her full and detailed breakdown will be deemed to have no cost implications or have been included elsewhere within his or her rates and prices.

2.7.4 **Preliminaries (Works Package Contract):**

2.7.4.1 **Information:**

The descriptive part of the work package contract preliminaries is prepared in the same way as the main contract preliminaries. The items to be considered when drafting the work package contracts are included in Part A of Table 1 (Preliminaries (works package contract) at Part 3: Tabulated rules for measurement of these rules.

2.7.4.2 **Pricing schedule:**

For the same reasons as for main contractor's preliminaries, the preliminaries bill for a works package shall comprise a pricing schedule that lists the headings under which the work package contractor will insert his charges relating to preliminaries items. It is a simple schedule comprising a list of cost centres.

The pricing schedule for work package contract preliminaries is divided into two main cost centres:

1. employer's requirements; and

2. work package contractor's cost items.

The items that comprise these two cost centres are listed and defined in Part B of Table 1 (Preliminaries (works package contract)) at Part 3: Tabulated rules for measurement of these rules.

Again, as part of the priced Bill of Quantities submitted by the work package contractor, it is essential that the work package contractor is instructed to provide a full and detailed breakdown that clearly identifies the items, shows how the price for each item has been calculated, and how the total price for preliminaries has been calculated.

2.8 Measurement Rules for Building, Engineering and Industrial Works

The rules for measuring and describing building, engineering and industrial items/components are set out in the tabulated rules of measurement rules for building works at Part 3 (Tabulated rules of measurement for works) of these rules.

2.9 Non-Measurement works

2.9.1 **Provisional Sums:**

2.9.1.1 Where building components/items cannot be measured and described in accordance with the tabulated rules of measurement they shall be given as a 'provisional sum' and identified as either 'defined work' or 'undefined work' as appropriate.

2.9.1.2 A provisional sum for defined work is a sum provided for work that is not completely designed but for which the following information shall be provided:

(a) the nature and construction of the work;
(b) a statement of how and where the work is fixed to the building and what other work is to be fixed thereto;
(c) a quantity or quantities which indicate the scope and extent of the work; and
(d) any specific limitations identified.

2.9.1.3 Where provisional sums are given for defined work, the contractor will be deemed to have made due allowance in his or her programming, planning and pricing preliminaries.

2.9.1.4 Where any aspect of the information required by paragraph 2.9.1.2 above cannot be given, work shall be described as an 'undefined' provisional sum. Where provisional sums are given for undefined work, the contractor will be deemed not to have made any allowance in programming, planning and pricing preliminaries.

2.9.1.5 Any provisional sum given for defined work that does not comprise the information required under 2.9.1.2 above shall be construed as a provisional sum for undefined work; irrespective that it was given in the BoQ as a provisional sum for defined work.

2.9.1.6 Provisional sums shall be exclusive of overheads and profit. Separate provision is to be made in the BoQ for overheads and profit (refer to paragraph 2.11).

2.9.2 **Contractor Designed Works:**

2.9.2.1 Contractor designed works include any works that require the contractor to undertake its design, whether directly or via a subcontractor. The employer shall be deemed responsible for works not clearly identified as contractor designed works.

2.9.2.2 Contractor designed work is sometimes referred to as the 'Contractor Designed Portion (CDP)'.

2.9.2.3 Where the contractor is required to take responsibility for the design of discrete parts of the building, such as piled foundations, windows, pre-cast concrete components, roof trusses and/or mechanical and electrical engineering services, the work items shall be identified as 'contractor designed works'.

2.9.2.4 The method of quantifying contractor designed work is dependent on the nature of the work.

2.9.2.5 Where contractor designed works can be measured and described in accordance with the tabulated rules of measurement (e.g. windows and pre-cast concrete components), the performance objectives or criteria that the contractor will be required to meet are to be clearly defined by way of a preamble to the work items that comprise the contractor designed works. Detailed documents defining the performance objectives and/or criteria to be met are to be incorporated as an annex to the Bill of Quantities (BoQ) and clearly cross- referenced in the preamble.

2.9.2.6 Where contractor designed works comprise a complete element or works package (e.g. the entire mechanical and electrical engineering services for the building), the works are to be measured and described as one or more item. The number of items is at the discretion of the quantity surveyor/cost manager but must be sufficient to provide an analysis of the price of the contractor designed works.

2.9.2.7 The quantity surveyor/cost manager is to obtain details of performance objectives and/or criteria from the relevant design consultant (refer to paragraph 2.14.3.2(2) below

2.9.3 **Risks**: The method of dealing with the employer's residual risks when preparing Bill of Quantities (BoQ) is addressed in paragraph 2.10.

2.9.4 **Works to be carried out by statutory undertakers**

2.9.4.1 Works that are required to be carried out by a statutory undertaker are to be given as a 'provisional sum', with the scope of works to be executed by the statutory undertaker described.

2.9.4.2 The contractor is to be deemed to have made due allowance in his programming, planning and pricing preliminaries for all general attendance on statutory undertakers.

2.9.4.3 Provisional sums for work to be carried out by statutory undertakers are to be exclusive of overheads and profit.

Separate provision shall be made in the BoQ for overheads and profit (refer to paragraph 2.11for further information).

2.10 Risks

2.10.1 **Risks generally:**

2.10.1.1 Every building engineering and industrial project involves risks and the proper management of risk saves time and money. Risks can occur at any point of a building, engineering and industrial project and it is essential that they are identified, assessed, monitored and controlled appropriately and effectively.

2.10.1.2 At the time of preparing a Bill of Quantities, a quantified schedule of works, or other quantity-based documents, whether for a complete building, engineering and industrial project or discrete works package, there will still be a number of risks remaining to be managed by the employer and his or her project team – this is called the employer's residual risk exposure (or residual risks).A risk response should only be decided after its possible causes and effects have been considered and fully understood. It will take the form of one or more of the following:

- risk transfer to the contractor;
- risk sharing by both employer and contractor; or
- risk retention by the employer.

2.10.1.3 Risks that can be designed out or avoided should have been addressed by this stage of the design development process. However, if time does not permit these risks to be designed out or properly dealt with, they will need to be dealt with using one of the above risk response strategies.

2.10.2 **Risk transfer to the contractor**:

2.10.2.1 The object of transferring risk is to pass the responsibility to another party able to control it better. If the risk materialises, the consequences are carried by the other party.

2.10.2.2 Whenever a risk is transferred there is usually a premium to be paid (effectively the contractor's valuation of the cost of the risk).Risk transfer will usually give the employer price/cost certainty for that aspect of the works. However, in return for price/cost certainty, the employer is required to pay the risk premium to the contractor, irrespective of whether the risk transferred does or does not materialize.

2.10.2.3 Risks that the contractor is required to manage, should they materialise, are to be fully described so that it is transparent what risk the contractor is required to manage, and what the extent of services and/or works the employer is paying for. Risks to be transferred to the contractor are to be listed in the BoQ under the heading 'schedule of construction risks'. A template for a 'schedule of construction risks' is in Appendix F of these rules.

2.10.2.4 The contractor will be deemed to have made due allowance in his risk allowances for programming, planning and pricing preliminaries.

2.10.2.5 Risk allowances inserted by the contractor shall be exclusive of overheads and profit. Separate provision should be made in the BoQ for overheads and profit (refer to paragraph 2.11 for further information).

2.10.3 **Risk sharing by both employer and contractor:**

2.10.3.1 Risk sharing occurs when a risk is not entirely transferred and some elements of it are retained by the employer. It is important that both the employer and the contractor know the value of the portion of the risk for which they are responsible. The objective should be to improve control and to reduce or limit the cost of the risk to the employer, should it materialise.

2.10.3.2 The method of dealing with risks that are to be shared by both employer and contractor will normally be dealt with using 'provisional quantities' with the pricing risk being taken by the contractor and the quantification risk being taken by the employer.

2.10.4 **Risk retention by the employer**

2.10.4.1 Where risks are to be retained by the employer, the applicable risk allowances included in the cost plan will be retained and managed by the employer or, if empowered by the employer, the project team.

2.10.4.2 Before deciding to retain a particular risk, the employer might wish to find out what the premium will be if the contractor were to be paid for resolving the risk should it materialise. The employer can then decide whether or not to

pay a premium for a defined scope of work. If the employer is content to pay a premium for transferring the risk, it is dealt with as a risk transfer in accordance with paragraph 2.10.2 above.

2.10.4.3 Risks retained by an employer are not necessarily controllable.

2.11 Overheads and profit

2.11.1 Provision shall be made in the Bill of Quantities (BoQ) for the contractor to apply their required percentage addition for overheads and profit on the following: (1) preliminaries; (2) measured work, including contractor designed works; (3) risk allowances; (4) work resulting from the expenditure of provisional sums (i.e. defined provisional sums, undefined provisional sums and works to be undertaken by statutory undertakers).

2.11.2 When required, overheads and profit can be treated as two separate cost items; namely, 'overheads' and 'profit'.

2.12 Credits

Credits are normally only applicable where the building project comprises the refurbishment or rehabilitation of an existing building, or demolition works. Provision for such provides the employer and opportunity to seek credits for old building material; components and items; mechanical and electric plant; and fittings, furnishings and equipment which arises from the stripping out or demolition works and for which the employer is content to pass ownership to the contractor for reuse.

2.12.2 Credits can be based on a pre-prepared list of items, which is incorporated in the BoQ and the contractor invited to insert the amount of credit he will give for each item. Alternatively, the contractor can be invited to list items for he is willing to offer a credit and the amount of credit he will give for each item.

2.12.3 A template for 'credits' is provided in Appendix F these rules.

2.13 Other Considerations

2.13.1 **Price fluctuations**:

2.13.1.1 The cost to the contractor of labour and materials etc. used in the works will alter during the contract period (i.e. they will be subject to price fluctuations).It might fall but, more usually, it will rise. The risk of fluctuating prices can be dealt with as follows:

1. contractor to price the risk (a fixed or firm price contract); or
2. allow provision for contractor to recover full or limited fluctuations on certain prices (a fluctuating price contract).

2.13.1.2 Most standard forms of contract conditions allow for either alternative to be used by providing clauses which may be included or deleted.

(1) **Fixed price contracts**:

These are contracts in which the price of labour, materials and plant is not subject to fluctuations. Fixed price contracts are sometimes referred to as 'fixed price lump sum contracts', 'firm price contracts', or 'firm price lump sum contracts'.

In the absence of any provision in the contract, or where the provision for recovering of price fluctuations has been deleted, the contractor will be required to take the risk (i.e. price the risk) of price fluctuations during the contract period. In order to cover himself, the contractor will make an estimate of the likely increase in costs and include this in his tender price.

Where there is no provision for recovering of price fluctuations, separate provision is to be incorporated in the Bill of Quantities for the contractor to tender his fixed price adjustment for pricing the risk. Such provision is to be referred to as either the 'main contractor's fixed price adjustment' or the 'work package contractor's fixed price adjustment', whichever is applicable.

When preparing Bill of Quantities, the quantity surveyor/cost manager shall ensure that no contract conditions relating to the recovery of price fluctuations exist.

(2) **Fluctuating price contracts:**

These are contracts in which adjustment is allowed for fluctuations in the prices of labour and materials etc. Various degrees of fluctuations are allowed under the provisions of standard contract conditions. The extent to which fluctuations are allowed will have a significant effect upon the contractor's tender price.

Where fluctuations are to be allowed, no provision for 'main contractor's fixed price adjustment' or the 'work package contractor's fixed price adjustment' is required.

2.13.2 **Director's adjustment:**

2.13.2.1 Separate provision is to be incorporated in the Bill of Quantities for the contractor to insert a 'director's adjustment'.

2.13.2.2 It is the responsibility of the contractor's directors, or other senior managers, to secure work for the company. Therefore, before submitting a tender price, the contractor's directors will undertake a commercial review of the project and the estimated price. This review might result in the contractor's directors requiring adjustment to the estimated price, referred to as a 'director's adjustment'. The Director's adjustment will include adjustments for commercial matters such as financing charges, cash flow, opportunities and competition. This is a sum added to or omitted from the estimated price to arrive at a tender price.

2.13.3 **Dayworks (provisional):**

2.13.3.1 Daywork is a method of valuing work on the basis of time spent by the contractor's employees, the materials used and the plant employed.

2.13.3.2 If required, a schedule of dayworks is to be incorporated in the Bill of Quantities (BoQ).The schedule of dayworks is to comprise a list of the various classifications of labour, estimates of the number of hours against each classification of labour, and estimated lump sums for materials and plant, for which daywork rates and percentage additions for overheads and profit are to be inserted by the contractor. A statement of the conditions under which the contractor will be paid for work executed on a daywork basis is to be given in either the preliminaries bill or schedule of dayworks.

2.13.3.3 The method of calculating labour time charge rates for work carried out in normal working hours (i.e. productive time) and work carried out outside of normal working hours (i.e. non-productive time) shall be defined in the schedule of dayworks. The definition of normal working hours shall be given in either the preliminaries bill or schedule of dayworks.

2.13.3.4 The total amount included for daywork by the contractor shall be omitted from the contract sum. The rates and percentage additions included in the BoQ shall be used to calculate the price of extra works instructed, which are authorised to be valued on a daywork basis.

Note: The total price derived from the schedule of dayworks can be included within or excluded from the contract sum. When included, it is to be treated as a provisional sum. When excluded, it shall be clearly stated that the rates, prices and percentage adjustments tendered are included in the contract.

2.13.4 **Value Added Tax (VAT):**

Value added tax (VAT) shall be excluded from Bill of Quantities (BoQ). Notwithstanding this, if required by the employer, provision for the contractor to provide a VAT assessment as part of his tender return can be incorporated in the form of tender.

2.14 Information Requirements for Measurement

2.14.1 The accuracy of Bill of Quantities (BoQ) is dependent on the quality of the information supplied to the quantity surveyor/cost manager by the employer, designers and other project team members; the more information provided, the more reliable the outcome will be. Where little or no information is provided, the quantity surveyor/cost manager will need to seek decisions from the employer as to how the uncertainty is to be managed and procured (refer to paragraph 2.11).

2.14.2 To enable the quantity surveyor/cost manager to prepare a BoQ, the information in paragraphs 2.14.3 to 2.15.6 inclusive (below) will be required.

2.14.3 **Specification:**

2.14.3.1 A specification defines what the employer wishes to buy and, consequently, what the contractor is expected to supply.

2.14.3.2 The two main types of specification used are:

1. Prescriptive specifications

2. Performance specifications.

(1) Prescriptive specifications:

This form of specification is required to enable a firm Bill of Quantities to be prepared. The function of a prescriptive specification is to prescribe the materials and workmanship required for a building project in as much detail as possible. Specific products and materials will be named, and the assembly of the building, engineering or industrial facility will be described and supported with drawn information and schedules (see paragraphs 2.14.4 and 2.14.5 below).Where materials are not named, reference will be made to published standards governing their composition (e.g. British standards or other country specific standards).

Where a prescriptive specification is used, the contractor will not carry any design responsibility.

(2) Performance specifications:

A performance specification describes the requirements of a product (e.g. windows), equipment (e.g. chillers plant), system or installation (e.g. mechanical and electrical installations) in terms of the performance objectives or criteria.

The main difference between a performance specification and a prescriptive specification is related to design responsibility. With a performance specification, the contractor is responsible for design development of the specification to meet the performance requirements.

The benefit to the employer is that design will not need to be advanced for performance-specified work before inviting tenders from contractors.

It should be noted that some standard forms of contract conditions do not include provision for contractor designed work. Therefore, care should be taken to ensure that the contract conditions used for the building, engineering or industrial project clearly transfer design responsibility for contractor designed work to the contractor (e.g.by incorporating supplementary contract conditions or other amendment to the standard forms of contract conditions used). Failure to do this will result in design responsibility remaining with the employer, though the employer did not undertake the design.

Usually, the quantity surveyor/cost manager will be faced with a combination of both types of specification, which needs to be organised in the BoQ (i.e. separated into measured works and non-measured works). Insufficient or poorly described information can mislead contractors, resulting in contract variations and potential time-related and/or cost-related claims.

2.14.4 **Drawn Information**

2.14.4.1 Drawn information is required to describe the assembly of the building, as well as any temporary works. Drawings shall be to a suitable scale.

2.14.4.2 Required drawn information shall include:

General arrangement (GA) drawings, comprising:

- a block plan: this shall identify the site and locates the outlines of the building works in relation to the town plan or other wider context;

- a site plan: this shall locate the position of the building works in relation to the setting out points, the means of access and the general layout of the site; and

- plans, sections and elevations: these shall show the position occupied by the various spaces in a building, engineering or industrial facility and the general construction and location of the principal elements and

components. The extent of elevations and sections shall be as appropriate to cover all major building, engineering or industrial facility zones. General arrangement drawings are sometimes called 'location' drawings.

- *Component drawings:* these shall show the information necessary for manufacture and assembly of a component; including key details/interfaces (e.g. interface between curtain walling system and structure, balconies and the like). Component drawings are sometimes called 'detail' or 'assembly' drawings.

- *Schematic drawings:* these show how something works and the relation between the parts (e.g. the wiring of an electrical system).

- *Record drawings:* these are a set of drawings that depict the actual as-built conditions of an existing building or structure, including mechanical and electrical engineering services installed. These are required for building, engineering and industrial projects involving the refurbishment or the demolition (partial or complete) of an existing building or structure. Record drawings are sometimes called 'existing', 'as-built' or 'as-installed' drawings.

2.14.4.3 Specific requirements for drawn information are further defined in the tabulated Rules of Measurement in Part 3 of these rules.

2.14.5 **Schedules:**

2.14.5.1 Schedules which provide the information required by the tabulated rules shall be deemed to be drawings. Schedules include:

- Room data sheets (including co-ordinated mechanical and electrical services engineering data sheets);
- Door schedules, including ironmongery;
- Window schedules, including ironmongery;
- Reinforcement (bar bending) schedules
- Landscaping and planting schedule (for internal and external works);
- Drainage schedules;
- Fittings, furnishings and equipment schedules;
- Luminaries schedules;
- Control schedules for mechanical and electrical engineering services;
- Primary mechanical and electrical plant and equipment schedules;
- Duties, outputs and sizes of primary mechanical and electrical plant and equipment;
- Builder's work in connection with mechanical and electrical engineering services;
- Other scheduled information necessary to specify the works.

The above list is not meant to be definitive or exhaustive, but simply a guide.

2.14.6 **Reports and other information:**

Reports and/or other information required for the preparation of Bill of Quantities (BoQ)will be dependent on the nature of the building, engineering or industrial project. However, such documents normally include some or all of the following:

- drawings showing the site boundary and all known site constraints and restrictions, including the means of access, restrictive covenants, party walls, rights of light issues and rights of access;

- a statement of, and drawing(s) detailing, phasing requirements;

- a statement of, and drawing(s) detailing, construction sequencing requirements;

- details of residual design development and construction risks (i.e. risk register or risk log);

- a schedule of gross external areas (GEA), gross internal floor areas (GIFA), net internal areas (NIA – i.e. usable

area for shops, supermarkets and offices) and site area (SA).

- site survey reports, including archaeological survey, ecological survey, invasive plant growth survey, etc.;
- details of wildlife, including protected species, protection measures;
- geotechnical and report(s) describing the intrusive ground and groundwater investigations completed, together with the results (e.g. the results of trial pits, auger holes, window samplers, boreholes, cone penetration tests (CPTs), and standard penetration tests (SPTs));
- environmental report(s) describing the sampling and analysis of soils, together with the results, giving information about the soil, ground water, and gases;
- details of any other facilitating works (e.g. soil stabilisation measures);
- remediation plan describing the method of dealing with contaminated materials and/or invasive plant growth (e.g. Japanese knotweed and giant hogweed), including requirements for post-remediation validation sampling;
- refurbishment and demolition survey reports, providing details of any asbestos containing materials (ACMs) and/or other hazardous materials, together with the scope of removal or encapsulation works required to be undertaken as part of the building works;
- details of any other facilitating works (e.g. soil stabilisation measures);
- details of any party wall awards or other agreements with adjoining owners and statutory undertakers, specifically detailing any requirements of the award to which the contractor should comply to ensure that the employer does not breach any agreement;
- temporary works methodology, drawings and sketches;
- details of which condition of contract is to be used for the building project;
- details of any planning conditions or information that the contractor is required to comply with;
- the employer's requirements in respect of insurances;
- the employer's requirements for the contractor to collect and report cost data to support claims for capital allowances, grants, value added tax (VAT) recovery and other tax incentives;
- the employer's policy documents, which the contractor will be required to comply with (e.g. site rules and regulations, environmental, corporate social responsibilities, and health and safety policies);
- hoarding requirements, including design where this is an employer's requirement; + details of the employer's post completion requirements (e.g. operation and maintenance of completed building (i.e. works and services)); and
- all other information necessary to construct the preliminaries bill for the building, engineering and industrial project or work package (refer to Work Section 1 (Preliminaries -Tabulated rules of measurement for building works) of these rules).

The above list is not meant to be definitive or exhaustive, but merely a guide. It is intended to be used by the quantity surveyor/cost manager to assist in identifying the types of reports and other information required to prepare a robust BoQ.

2.15 Cost Management/Control

2.15.1 The main purpose of a Bill of Quantities (BoQ) is to present a co-ordinated list of components/items, together with their identifying descriptions and quantities that encompass the building works so that the tendering contractors are able to prepare tenders efficiently and accurately. As well as assisting in ensuring parity of tendering. In addition, BoQ provide a vital tool, which can be used by the quantity surveyor/cost manager to manage and control the costs of the building projects. Cost management and control uses include:

i. Pre-tender estimates;

ii. Post tender estimates; Cost planning;
iii. Pricing variation; and
iv. Interim valuation and payment

2.15.2 **Pre-tender Estimates:**

Pre-tender estimates are prepared immediately before calling the first tenders for construction.

This is the final cost-check undertaken by the cost manager before tender bids for the building project, or any part the building project, are obtained. When a Bill of Quantities (BoQ) is the basis of obtaining a tender price, the pre-tender estimates will be based on the BoQ.

2.15.3 **Post-tender Estimate:**

A post-tender estimate is prepared at RIBA Work Stage H (tender action), after all the construction tenders have been received and evaluated. It is based on the outcome of any post-tender negotiations, including the resolution of any tender qualifications and tender price adjustments. The post-tender estimate will include the actual known construction costs and any residual risks. The aim of this estimate is to corroborate the funding level required by the employer to complete the building project, including cost updates of project and design team fees, as well as other development and project costs, where they form part of the costs being managed by the cost manager. When reporting the outcome of the tendering process to the employer, the quantity surveyor/cost manager should include a summary of the post- tender estimate(s).The post-tender estimate should be fairly accurate because the uncertainties of market conditions have been removed. Post-tender estimates are used as the control estimate during construction.

2.15.4 **Cost Planning:**

2.15.4.1 Cost planning is an iterative process, which is performed in steps of increasing detail as more design information becomes available. A cost plan provides both a work breakdown structure and a cost breakdown structure which can be used to redistribute works in elements to works packages for the purpose of procurement and cost control during the construction phase of the building project.

2.15.4.2 The third formal cost plan stage (completed at RIBA Work Stages E/F (Technical Design/Production Information) is based on technical designs, specifications and detailed information for construction. Formal cost plan 3 will provide the frame of reference for appraising tenders. It also provides the frame of reference for reconciling 'actual costs' against 'cost targets'. This is particularly beneficial where building works are being procured piecemeal (e.g. procuring discrete work packages as their design is completed).The proper use of the cost plan will allow the re-profiling of cost targets as necessary to ensure that the overall cost limit (i.e. the employer's authorised budget) is maintained by the project team.

2.15.5 **Pricing Variations:**

The rates in a priced Bill of Quantities (BoQ) provide a basis for the valuation of varied work. 'Pro-rata' and 'analogous' rates can also be ascertained from the base rates tendered to calculate the prices of other components not specifically described in the BoQ.

2.15.6 **Interim Valuations and Payment:**

2.15.6.1 Many building projects require interim payments to be paid to the contractor. This is in order to relieve the contractor of the burden of financing the whole of the building/engineering works until completion; works which may take many months or years to complete. Within each contract there will be clauses which set out the administrative rules under which the quantity surveyor/cost manager, architect (or contract administrator or project manager, quantity surveyor/cost manager) employer and contractor must operate. In many contracts, while the completion and calculation of the value is important, the method and procedure of the interim valuations and payment which the contractor receives is equally important.

2.15.6.2 A priced Bill of Quantities (BoQ) provides a comprehensive list of building, engineering and industrial components/items. Consequently, when a contract has been entered into, by assessing the building components/items in the BoQ, the priced BoQ can be used to ascertain periodic valuation of works properly executed in accordance with the provisions of the contract for the purpose of interim valuations and payment.

2.16 Analysis, Collection and Storage of Cost Data

2.16.1 Priced Bill of Quantities (BoQ) make available one of the best sources of real-time cost data, which can be used by quantity surveyors/cost managers to provide expert cost advice on the likely cost of future building, engineering and

industrial projects. Moreover, they afford a complete cost model in a single document.

2.16.2 The cost data provided in a BoQ can be retrieved, analysed, stored and reprocessed in various ways (e.g. as distinct rates, detailed elemental cost analyses, element unit rates (Naira), cost/m2 of gross internal floor area, and/or functional unit rates) for use in order of cost estimates and cost plans. It can also be used for benchmarking purposes.

2.16.3 **BESMM4**: Order of cost estimating and cost planning for capital building works can be used as a basis for measuring Element Unit Quantities (EUQ) for the purpose of preparing detailed cost analyses of building projects.

Part 3: Tabulated Rules of Measurement for Building Works

3.1 Introduction

3.1.1 Part 3 of the rules comprise:

(1) the information and requirements for main contractor's and work packages contractors' preliminaries, together with the rules for preparing the preliminaries pricing schedule; and

(2) the rules of measurement for building components/items.

3.1.2 The use of tabulated rules is also explained.

3.1.3 Bill of Quantities (BoQ) is to fully describe and accurately represent the quantity and quality of the works to be carried out. More details than is required by these rules should be given where necessary to define the precise nature and extent of the required work.

3.2 Use of tabulated rules of measurement for building works

3.2.1 General

3.2.1.1 The rules of measurement for building works are set out in tables. The tables are divided into two categories. namely those dealing with: measurement of building components/items

3.2.1.2 The rules are written in the present tense.

3.2.1.3 The symbol 'I' used between two or more units of measurement or within text, means 'or'.

3.2.1.4 Horizontal lines divide the tables and rules into zones to which different rules apply.

3.2.1.5 Where units of measurement or rules are separated by a broken line (------) this denotes a choice of units or choice of ways of measuring the work. The method chosen shall be the best to suit the particular situation.

3.2.1.6 The use of hyphen (-) or the phrase 'to' between two dimension in these tables or in a Bill of Quantities means a range of dimension exceeding the first dimension stated but not exceeding the second.

3.2.2 **Tables: Preliminaries**

3.2.2.1 Work Section I comprises the rules for describing and quantifying preliminaries. It is divided into two sub- sections as follow preliminaries (main contract); and preliminaries (work package contract).

3.2.2.2 Both sub-sections are sub-divided into parts as follows:

Part A: Information and requirements (i.e. dealing with the descriptive part of the preliminaries); and

Part B: Pricing schedule (i.e. provides the basis of a pricing document for preliminaries).

3.2.2.3 The tables for information and requirement for both preliminaries (main contract) and preliminaries (work package contract) are structured as follows:

(1) Information and requirements are specified under a number of headings, which are given above each table (i.e. project particulars; drawings and other documents; and the site and existing buildings);

(2) The left hand column (sub-heading 1) list the preliminaries items to be considered under each main heading,

(3) The second column (information requirements) lists the information which shall be included within the preliminaries descriptions; and

(4) The third column (information requirement/notes) lists supplementary information that might need to be included with the preliminaries descriptions, as well as providing additional guidance on the drafting of preliminaries statements.

(5) The fourth column (supplementary information/notes) lists supplementary information that might need to be

included within the preliminaries descriptions, as well as providing additional guidance on the drafting statements.

3.2.2.4 The pricing schedule tables for both preliminaries (main contract) and preliminaries (work package contract) are structured as follows:

(1) The contractor's pricing of preliminaries are captured under a number of headings, which are given above each table (i.e. projects particulars; drawings and other documents; and the site and existing buildings);

(2) The left hand column (component) lists the preliminaries items to be considered under each main heading;

(3) The second column (included/notes on pricing)lists the sub-items which form part of each main heading;

(4) The third column (information requirements) list the sub-items which form part of each item

(5) The fourth column (pricing method) stipulates if the component is a 'fixed charge', a 'time-related charge', or a combination of both; and

(6) The fifth column (excluded) describes the items excluded from components. Where exclusion is stated, cross reference to the appropriate component is given.

3.2.3 **Tables: Building Components/Items**

3.2.3.1 Work sections 2 to 40 comprise the rules of measurement for building components/items.

They are as follows:

No.	**Work Section:**
2	Off-site manufactured materials, components and buildings:
3	Demolitions;
4	Alterations, repairs and conservation;
5	Excavating and filling;
6	Ground remediation and soil stabilization;
7	Piling;
8	Underpinning;
9	Diaphragm walls and embedded retaining walls;
10	Crib walls, gabions and reinforced earth;
11	In-situ concrete works;
12	Precast/composite concrete;
13	Precast concrete;
14	Masonry;
15	Structural metalwork;
16	Carpentry;
17	Sheet roof covering;
18	Tile and slate roof and wall covering;
19	Waterproofing;
20	Proprietary linings and partitions;
21	Cladding and covering;
22	General joinery;
23	Windows, screens and lights;
24	Doors, shutters and hatches;
25	Stairs, walkways and balustrades;
26	Metalwork;
27	Glazing,
28	Floor, wall, ceiling and roof finishings;
29	Decoration;
30	Suspended ceilings;
31	Insulation, fire stopping and fire protection;
32	Furniture, fittings and equipment;
33	Drainage above ground;
34	Drainage below ground;
35	Site works;
36	Fencing;
37	Soft landscaping;

38 Mechanical services;
39 Electrical services; and
40 Transportation

3.2.3.2 Each table is structured as follows:

(1) The title of the work section is given in the headings;

(2) The first two rows set out the:

(a) drawn information required in respect of each work section to enable measurement and shall accompany the Bill of Quantities when issued,

(b) mandatory information that is to be provided in each work section,

(c) minimum information that shall be shown on the drawings or any other document that accompany each work section, and

(d) works and materials that not measured, but are deemed to be included in the building components/items measured in each work section;

(3) The left hand column (item) list the building components/items commonly encountered in building works;

(4) The left hand column (unit) lists the unit of measurement for building components/items;

(5) The third column (level 1) lists the information, including any dimension requirements, that shall be included in the description of the building components/items;

(6) The fourth column (level 2) lists the supporting information, including any additional dimension requirements, which shall be included in the description of the building components/items; and

(7) The fifth column (level 3) lists the further supporting information, including any additional dimension requirements, which shall be included in the description of the building components/items; and

(8) The sixth column (notes, comments and glossary) explains what work is deemed to be included in specific building components/items, clarifies the approach to quantification and description of building components/items, and contains definitions of specific terms and phrases used in connection with the building components/items.

The building components/items listed in tables comprises those commonly encountered in building/civil engineering and industrial works; the lists are not intended to be exhaustive.

3.3 Measurement Rules for Building Works

3.3.1 BoQ shall fully describe and accurately represent the quantity and quality of the works to be carried out. Where necessary, more detail than is required by these rules shall be given in order to define the precise nature and extent of the required work.

3.3.2 **Quantities:**

The rules for quantifying building components/items are as follows:

(1) Measurement and Billing:

(a) Measure work net as fixed in position unless otherwise stated.

(b) Net quantity measured shall be deemed to include all additional material required for laps, joints, seams and the like, as well as any waste material.

(c) Curved work shall be measured on the centre line of the material unless otherwise stated.

(d) Dimension shall be measured to the nearest 10mm, 5mm and over shall be regarded as 10mm and less that 5mm shall be disregarded.

(e) Except for quantities measured in tonnes (t), quantities shall be given to the nearest whole number. Quantities less than one unit shall be given as one unit; Quantities measured in tonnes (t) shall be given to two decimal places.

(2) Voids

(a) Unless otherwise stated, minimum deductions for voids refer only to openings or wants within the boundaries of the measured work.

(b) Always deduct openings or wants at the boundaries of measured areas, irrespective of size.

(c) Do not measure separate items for widths not exceeding a stated limit where these widths are caused by voids.

3.3.3 **Description:**

3.3.3.1 Each work section of a Bill of Quantities shall begin with a heading and a description stating the nature and location of the work.

3.3.3.2 Headings for groups of building components/items (i.e. components and sub-components) in a Bill of Quantities shall be read as part of the descriptions of the items to which the headings apply.

3.3.3.3 Descriptions shall state the building components/item being measured (taken from the first column of the tabulated rules) and include all level 1, 2 and 3 information (taken from the third, fourth and fifth columns respectively) applicable to that item. Where applicable, the relevant information from column five shall be included in the description.

3.3.3.4 Unless specifically stated otherwise in the Bill of Quantities or in these rules, descriptions for building components/items shall include the:

(1) type and quality of the material;

(2) critical dimension(s) of the materials(s).

(3) method of fixing, installing or incorporating the goods or materials into the work where not at the discretion of the contractor; and

(4) nature or type of background.

3.3.3.5 Where the nature or type of background is required to be identified, the building components/items description shall state one of the following;

(1) timber (the term includes all type of hard and soft building boards):

(2) plastics;

(3) masonry (the term includes brick, concrete, block, natural and reconstituted stone);

(4) metal, of any type;

(5) metal-faced timber or plastics; and

(6) vulnerable materials (the term includes glass, marble, mosaic, ceramics, tiled finishes, material finishes and the like).

3.3.3.6 Dimensions given as part of the descriptions shall be:

(1) stated in the sequence: length, width and height. White ambiguity could arise; the dimensions shall be identified in the description;

(2) the finished lengths, width and heights specified or shown on the drawings with no allowance made for overlaps, scarcements and the like.

3.3.3.7 Thicknesses given as part of the description shall be the finished thickness of the material after compaction and shall exclude the thickness of adhesives and bedding materials unless otherwise stated.

3.3.3.8 The use of hyphen (-) or the phrase 'to' between two dimensions in a description shall mean a range of dimensions exceeding the first dimension stated but not exceeding the second.

3.3.3.9 Where the rules require work to be described as 'curved' with the radius stated, details shall be given of the curved work, including if concave or convex, if conical or spherical, if to more than one radius, and shall state the radius or radii. The radius shall be the mean radius measured to the centre line of the material unless otherwise stated.

3.3.3.10 The information required by these rules may be given by a precise and unique cross-reference to another document (e.g. to a specification or to a catalogue).

3.3.3.11 Where other components and sub-components are referred to in other documents (e.g. a specification states that vinyl sheet flooring is to be laid on a plywood lining), each component and or sub-components shall be measured and described separately (i.e. both the vinyl sheet flooring and the plywood lining are to be measured as separate items).

3.3.3.12 Notwithstanding the requirements of paragraph 3.3.3.11 above, separate components or sun-components may be combined to form single composite building components/items. In such cases, the description of the composite building components/items shall clearly state what is included and how each component and or sub-component is to be incorporated. Any component, included and how each component and sub-components is to be incorporated. Any component, sub-component or other element of the work not clearly included in the description shall be deemed not to be included as part of the composite building components/items.

3.3.3.13 Unless specifically stated otherwise in the BoQ or in these rules, each building component/item shall be deemed to include the following:

(1) labour and all costs in connection therewith;
(2) materials and goods together with all costs in connection therewith;
(3) assembling, installing, erecting, fixing or fitting materials or goods in position;
(4) plant and all costs in connection therewith;
(5) waste of goods or materials;
(6) all rough and fair cutting unless specially stated otherwise;
(7) establishment charges, and cost of compliance with all legislation in connection with the work measured including health and safety, disposal of waste and the like.

3.3.4 **Work of Special Types:**

3.3.4.1 Work of each of the following types shall be separately identified. Work of special types includes:

(1) Work to existing buildings: such work is defined as work on, in or immediately under work existing before the current building project. Specific details pertaining to work carried out to an existing building shall be given at the start of each applicable work section.

(2) Work carried out and subsequently removed: specific details pertaining to work to be executed outside the cartilage of the site shall be given at the start of each applicable work section.

(3) Work outside the cartilage of the site: specific details pertaining to work to be executed outside the cartilage of the site shall be given at the start of each applicable work section.

(4) Work carried out in extraordinary conditions, including:

(a) In, on or under water, stating whether river, canal, lake or sea and, where applicable, stating the mean spring levels of high and low water;

(b) In tidal conditions;

(c) Underground, starting mean depth;

(d) In compressed air, stating the pressure and means of entry and exit; and

(e) In other types of extraordinary conditions.

3.3.4.2 Specific details pertaining to work carried out in each condition shall be given at the start of each applicable work section.

3.3.4.3 The additional rules for special types of work shall be read in conjunction with the rules in the appropriate work section.

3.3.4.4 Details of the additional preliminaries that are pertinent to the special types of work shall be given in the description, drawing attention to any specific requirements due to the nature of the work.

3.3.5 Measurable work not covered by the tabulated rules

3.3.5.1 Building components/items not covered by the tabulated rules shall, if possible, be measured by rules for similar types of work. Rules of measurement adopted for such building components/items shall be clearly stated and fully defined in either the preliminaries or in the Bill of Quantities (against the building components/items or items to which the rules relates). Such rules shall, as far as possible, conform to those given in the tabulated rules of measurement for similar building components.

3.3.5.2 Where it is not possible to derive the method of measurement from the tabulated rules, the rules chosen may be bespoken. In such cases, the rule or rules derived shall be reiterated in full either the preliminaries or in the Bill of Quantities (above the billing components/items or items to which the rule relates).

3.3.6 **Procedure where work cannot be quantified**

For the rules relating to work which cannot be quantified, refer to paragraph 2.9 (Non-measureable works) at Part 2 of these rules

3.3.7 **Procedure where exact type of product or component is not specified**

3.3.7.1 Where the exact type of product or components cannot be specified, an estimated price for the product or component shall be given in the description as a prime cost price (PC price). For example. ‘Allow the PC price of Nx per thousand delivered to site’, or ‘Allow Nx each delivered to site’.

3.3.7.2 Unless specifically stated otherwise in the Bill of Quantities or in these rules, the contractor shall be deemed to have allowed for all items listed in paragraph 3.3.3.13 in his or her priced rate for each building component/item incorporating a PC price.

3.3.7.3 PC prices shall exclude any allowance for the main contractor’s overhead and profit, which are dealt with separately.

3.3.8 **Procedure where quantity of work cannot be accurately determined**

3.3.8.1 Where work can be described and given in items in accordance with the tabulated rules of measurement but the quantity of work cannot be accurately determined, an estimate of the quantity shall be given and identified as ‘provisional quantity’.

3.3.8.2 Work items identified as a ‘provisional quantity’ shall be subject to remeasurement when they have been completed. The ‘approximate quantity’ shall be substituted by the ‘firm quantity’ measured, and the total price for that item adjusted to reflect the change in quantity. Where the variance between the ‘provisional quantity’ and the ‘firm quantity’ measured is less than 20 per cent, the rate tendered by the contractor shall not be subject to review. Where the variance is significant (i.e. 20 per cent or more), the rate can be reviewed to ensure that the rate is fair and reasonable to both the employer and contractor.

SECTION 1 - MEASUREMENT OF BUILDING WORKS

1.0 SECTION 1 - MEASUREMENT OF BUILDING WORKS

1.1 PRELIMINARIES

Part A: Information and Requirements

1.1.1A Project Particulars

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
1 Project Particulars	**1** Name of project	Short project title to be stated	
	2 Nature of project	Short description to be stated	
	3 Location of project	Full postal address to be stated	
	4 Length of contract	Period, in weeks to be stated	**1** Where to be stated by the contractor, insert 'To be confirmed'.
	5 Names and addresses of Employers and Consultants	Function (e.g. architect), name of organisation, address point of contact, telephone number and email address to be stated. To include: **1** Employer. **2** Project sponsor (e.g. employer's internal project manager). **3** Project Manager (if applicable). **4** Person empowered by the contract to act on behalf of the employer (person's title to be given (e.g. contract administrator). **5** Quantity Surveyor. **6** Consultants (separately identified). **7** Clerk of works (if required by the employer).	

PRELIMINARIES

1.1.2A Drawings

	SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
	2 Drawings	**1** List of drawings from which the bill of quantities was prepared	Drawing number including revision status, drawing titles and author to be stated	**1** Exceptions to be stated.
	3 Other documents	**1** Pre-construction information.	Explain how pre-construction information is dealt (i.e. within the preliminaries bill or as a separate document). Cross-reference to pre-construction information document if separate document.	
		2 List of drawings and other documents relating to the contract but not included in the tender documents.	**1** Provide a list of drawings and other documents relating to the contract but not included in the tender documents, which may be seen by the contractor during the tender period. **2** Document title, reference, revision, date of issue and author to be stated **3** Details of where documents can be seen to be stated.	
1.1.3A	**The Site/Existing buildings** **1** The site		**1** Description of the site **2** Reference to drawing, or drawings, showing the site boundaries and contractor's working area(s).	
	2 Existing buildings on or adjacent to the site		Description of existing building on or adjacent to the site	
	3 Surrounding land/building uses	**1** Address.	**1** Sub heading 2 to identify land or building **2** Use or activities carried out on the land or in building(s).	

The Site/Existing Buildings (cont'd)

	SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
	4 Existing mains/services	**1** On the site. **2** Adjacent to the site.	List drawings and other applicable information	
	5 Soils and ground water **6** Site investigation		Site information provided and where included in documentation (e.g. 'Annex B of the BoQ' as a separate document) Cross reference as necessary.	
	7 Health and safety file	**1** Health and safety file **2** Other documents	Availability for inspection Arrangement for inspection	
	8 Health and safety hazard		Details of hazards that are or may be present at the site.	Where information provided in the pre-construction information, cross-reference accordingly
	9 Access to the site **10** Parking		**1** Description. **2** Limitations Details of employer's requirement in respect of parking and payment of fees and charges in connection of parking including parking bay and parking meter suspension.	
	11 Use of the site		Details of any limitations	
	12 Site visits		**1** Purpose. **2** Arrangements for site visit.	Include where information not given in the 'conditions of tender'
1.1.4A	**Description of the works**			
	1 The work		Description of the Works.	
	2 Preparatory work by others		Description of the works that will be carried out by others under a separate contract before the start of work on site for this contract.	
	3 Work by others concurrent with the contract		Description of the Works	
	4 Completion work by others			

PRELIMINARIES

1.1.5A The Contract Conditions

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
1 Condition of contract	**1** Form of contract title to be stated	**1** Full title of the standard or bespoke form of contract, including edition, revision and standard amendments applicable. **2** Schedule of clause/condition headings in the standard form of contract (see note (1)). **3** Reference to any amendments to clauses/conditions to standard form of contract (see note (1)). **4** Reference to any supplementary or special clauses/conditions to standard form of contract. **5** Insertions relating to articles of agreement articles, recitals and contract particulars or abstract of particulars (see note (1)). **6** Employer's insurance responsibility **7** Employer's requirements in respect of performance bonds. **8** Employer's requirements in respect of parent company guarantees. **9** Employer's requirements in respect of collateral warranties.	**1** Neither a schedule of clause/condition headings in the standard form of contract nor reference to any supplementary or special clauses/conditions to standard form of contract or details of insertions relating to articles of agreement, articles, recitals and contract particulars or abstract of particulars is required where addressed through a schedule of amendments to the standard conditions of contract. Notwithstanding this, reference to the schedule of amendments is to be made in this section of the preliminaries bill **2** Where bespoke, or uncommon, forms of contract are used – a copy is to be appended to the bill of quantities or included as part of the tender.

PRELIMINARIES

1.1.5 Employer's Requirements: Provision, Content and Use of Documents

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
1 Definitions and interpretations	**1** Definitions	Explain how to interpret key words, terms, phrases and synonyms used in the preliminaries and specification.	
	2 Communication	Definition and format of communications and	
	3 Products	Definition and/or meaning	
	4 Site equipment		
	5 Drawings		
	6 Contractor's choice **7** Contractor's designed work		
	8 Submit proposals		
	9 Terms used in specification	Definitions and/or meaning of key words, terms, phrases and synonyms used in the specification	
	10 Manufacturer and product references.	**1** Definition of terms **2** Version of manufacturer's technical literature applicable to tender and contract (e.g. current on the date of invitation to tender).	
	11 Substitution of products	**1** Definition and/or meaning of substitute/alternative products **2** Process for acceptance and rejection of substitute/alternative products.	
	12 Cross-references	Explain method of cross referencing used.	
	13 Referenced documents	Order of precedence of referenced documents	
	14 Equivalent products	Definition and/or meaning	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Definitions and interpretations (Cont'd)	**15** Substitution of standards	**1** Definition and/or meaning **2** Process for acceptance and rejection of substitute standards	
	16 Currency of documents	Version of published documents, including revisions and amendments, applicable to tender and contract (e.g. current on the date of invitation to tender).	
	17 Product sizes	General definition of product sizes Exceptions to general definition.	Products to be specified by their co-ordinating size. Exceptions to this rule to be stated
2 Documents provided on behalf of employer	**1** Additional copies of drawings and documents	Describe procedure	
	2 Dimensions	Explain ownership of scaled dimensions	
	3 Measured quantities	Explain precedence of measured quantities	
	4 The specification	**1** Reference the specification, or specifications, the preliminaries. **2** Explain the method used to cross reference specification clauses on or in other tender/contract documents.	
	5 Divergence from the statutory requirements	Method for dealing with divergence from the statutory requirements should they occur	
	6 Employer's policy documents	Requirements in respect of compliance with the employer's policies. Examples include: - environmental. - sustainability. - corporate social responsibilities (CSR). - health and safety.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
3 Documents provided by the contractor, subcontractors and suppliers	**1** Design information	**1** General requirements, including design management and programming requirements. **2** Specific requirements in respect of design documents and information **3** Format. **4** Number of copies. **5** Submission requirements.	
	2 Production information	**1** General requirements in respect of product information **2** Format. **3** Number of copies. **4** Submission requirements.	
	3 As-built/as-installed drawings and information	**1** General requirements. **2** Submission requirements. **3** Number of copies.	
	4 Technical literature.	**1** Literature to be maintained. **2** Requirements for literature to be available on site.	
	5 Maintenance instructions and guarantees	**1** Information requirements. **2** Format. **3** Number of copies. **4** Submission requirements. **5** Storage and information management. **6** Requirements in respect of emergency and/or out of normal working call-out services, including requirements for contact details and extent of cover.	**S2** Requirements relating to 'management information systems' to be stated under storage and information management requirements.

1.1.7A **Employer's Requirements: Management of the Works**

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
	6 Energy rating calculations	**1** Information requirements. **2** Number of copies. **3** Submission requirements.	
	7 Environmental assessment information	**1** Scheme type **2** Environmental targets (in respect of site activities and the works). **3** Information requirements. **4** Format. **5** Number of copies. **6** Submission requirements.	
	8 Documents required before Practical Completion	Specific requirements.	
4 Document and data interchange	**1** Electronic data interchange (EDI).	**1** Types and classes of communication **2** Parties communication between **3** Requirements.	
1 Employer's requirements – generally	**1** Supervision	Specific requirements	
	2 Considerate constructors scheme	**1** Registration requirements **2** Contact details **3** Compliance requirements.	
	3 Insurance	**1** Documentary evidence required. **2** When required.	
	4 Insurance claims	**1** Requirements for notifying events **2** Requirements for indemnifying the employer	
	5 Climatic conditions	Records to be maintained by contractor	
	6 Ownership of materials/products arising from the works.	Requirements in respect of ownership, and removal from site.	**S1** Where to become the property of the contractor; consider seeking credit for materials/products arising from the works

1.1.7A Employer's Requirements: Management of the Works (cont'd)

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
2 Programme / Progress	**1** Programme	**1** Format and content **2** Exclusions. **3** Document control requirements **4** Submission requirements	
	2 Revised programme	Specific requirements for re-profiling and reissuing programme.	
	3 Commencement of work	Notice period to be given before the commencement of work on site	
	4 Monitoring progress	**1** Employer's specific requirements in respect of reporting and avoiding potential delay **2** List and description of key performance indicators (KPIs) to be maintained by contractor **3** Requirements for reporting against KPIs. **4** Actions of contractor if KPIs not achieved	
	5 Notification of compensation event(s)	Employer's specific requirements in respect of contractor notifying events that compensate potential delay.	

Employer's Requirements: Management of the Works (cont'd)

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Programme/progress (Cont'd)	**6** Project progress meetings	**1** General requirements **2** Proposed agenda **3** Frequency. **4** Location **5** Accommodation availability. **6** Attendees. **7** Chairperson	
	7 Contractor's progress report	**1** Form and content of report **2** Method of presentation **3** Submission requirements.	
	8 Contractor's site meetings.	General requirements	
	9 Photographic records	**1** Image format. **2** Frequency. **3** Number of locations **4** Number of images from each location **5** Other requirements.	
	10 Early possession	Employer's specific requirements for early possession and taking over parts of the works before practical completion.	
	11 Notice of completion	**1** General requirements. **2** Minimum period of notice to be given by contractor to be stated	
	12 Extensions of time	**1** Requirements in respect of notification by the contractor. **2** Submission requirements.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
3 Cost control	**1** Cash flow forecast	**1** Basis of cash flow forecast **2** Frequency. **3** Submission requirements.	
	2 Removal/replacement of existing work	**1** Location. **2** Extent **3** Requirements in respect of execution.	
	3 Proposed instructions	**1** Requirements in respect of estimates **2** Content of estimates. **3** Other requirements.	
	4 Measurement of covered work	**1** General requirements **2** Notice period to be given by contractor before covering works which are to be measured	
	5 Daywork vouchers	**1** Notice period to be given by contractor before commencement of work to be carried out on a daywork basis. **2** Submission requirements	
	6 Interim applications, valuations and payments	**1** Process for agreeing interim payments and payment dates. **2** Content and submission requirements for contractor's interim applications for payment **3** Employer's specific invoicing requirements	

1.1.8A Employer's Requirements: Quality Standards and Quality Control

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
	7 Payment for products not incorporated into the works.	Information/evidence of freedom from title required from contractor in respect of products stored on-site before payment is to be considered by employer	
	8 Payment for products stored off-site.	Information/evidence of freedom from title required from contractor in respect of products stored off-site before payment is to be considered by employer	
	9 Labour and equipment returns	**1** Records to be maintained by the contractor. **2** Content of records. **3** Submission of records	
1 Standards of products and executions	**1** Incomplete information	**1** Requirements for dealing with products/materials that are not fully specified. **2** Status of works where the extent is not fully documented. **3** Status of omissions or errors in description and/or quantity.	
	2 Workmanship skills	Requirements in respect of: **1** Appropriateness of contractor's operatives. **2** Registration schemes to which contractor's operatives should belong **3** Evidence of scheme registration **4** Other requirements.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Standards of products and executions (Cont′d)	**3** Quality of products	Requirements in respect of: **1** Using new and recycled products. **2** Supply of products. **3** Tolerances. **4** Deterioration prevention. **5** Other requirements.	
	4 Quality of execution	Requirements in respect of: **1** Fixing, application and installation of products, including alignment.	
		2 Colour batching. **3** Dimensions. **4** Finished work. **5** Location and fixing of products. **6** Other requirements.	
	5 Compliance	Requirements in respect of: **1** Compliance with proprietary specifications. **2** Compliance with performance specifications. **3** Other requirements.	
	6 Inspections.	Requirements for inspecting products and work executed.	
	7 Related work	Requirements in respect of co-ordinating workmanship of trades.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Standards of products and executions (Cont'd)	**8** Manufacturer's recommendations/ instructions.	**1** Requirements in respect of compliance. **2** Version of manufacturer's recommendations/instructions applicable to tender and contract (e.g. current on the date of invitation to tender).	
	9 Water for the works	General requirements	
2 Samples/Approvals	**1** Samples.	Requirements in respect of samples of products, work	
	2 Approval of product samples	**1** General requirements (including programming requirements) for submission of, inspection of, and tests on samples. **2** Definition of approval in context of samples **3** Retention of complying samples, including storage requirements **4** Other requirements.	
	3 Approval of work executed/mock-ups.	**1** General requirements (including programming requirements) for submission of, inspection of, tests on work executed and mock-ups **2** Definition of approval in context of work executed and mock-ups.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Samples/Approvals (Cont'd)		**3** Retention of complying work executed and mock-ups, including storage requirements. **4** Other requirements.	
3 Accuracy/Setting out	**1** Accuracy of instruments **2** Setting out.	General requirements	
	3 Appearance and fit	Requirements in respect of tolerances and dimensions.	
	4 Critical dimensions	Details of critical dimensions.	
	5 Levels of structural floors	Maximum tolerances for designed levels to be stated	
	6 Record drawings	**1** Requirements in respect of recording details of grid lines, setting-out stations, benchmarks and profiles **2** Information retention requirements **3** Submission requirements **4** Other requirements.	
4 Services	**1** Services regulations **2** Water regulations/byelaws notification	General requirements	
	3 Water regulations/byelaws contractor's certificate	**1** Content of certificate **2** Submission requirements **3** Other requirements.	
	4 Electrical installation certificate	**1** Submission requirements **2** Other requirements.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Services (Cont'd)	**5** Gas, oil and solid fuel appliance installation certificate.	**1** Content of certificate. **2** Submission requirements. **3** Other requirements	
	6 Mechanical and electrical services	**1** Requirements in respect of final tests and commissioning. **2** Requirements in respect of Building Regulations notice. **3** Other requirements.	
5 Supervision/inspection/defective work	**1** Supervision.	**1** General requirements. **2** Notice period for replacement of contractor's person in charge by contractor	
	2 Co-ordination of mechanical and electrical engineering services.	**1** General requirements. **2** Requirements for documentary evidence of contractor's staff.	
	3 Overtime working	General requirements	
	4 Defects in existing work	**1** Process for dealing with undocumented defects. **2** Process for dealing with documented remedial work.	
	5 Access for inspection	General requirements	
	6 Tests and inspections	**1** Timing requirements. **2** Records required. **3** Submission requirements. **4** Other requirements.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Supervision/inspection/defective work (Cont'd)	**7** Air permeability	**1** Method. **2** Performance requirements. **3** Submission requirements (results). **4** Other requirements.	
	8 Continuity of thermal insulation	**1** General requirements. **2** Content of reports. **3** Submission requirements. **4** Number of copies. **5** Other requirements.	
	9 Resistance to passage of sound.	**1** Method. **2** Compliance requirements. **3** Submission requirements. **4** Other requirements.	
	10 Energy performance certificate.	**1** General requirements. **2** Format of certificate and report. **3** Submission requirements. **4** Other requirements.	
	11 Proposals for rectification of defective products/executions.	General requirements.	
	12 Measures to establish acceptability.		
	13 Quality control.	**1** Procedural requirements. **2** Records required. **3** Content of records. **4** Other requirements.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
6 Work at or after completion	**1** Work before completion.	**1** General requirements. **2** Cleaning requirements, including cleaning materials and methods of cleaning. **3** Requirements for rectifying minor faults. **4** Requirements in respect of moving parts. **5** Other requirements.	
	2 Security at completion.	**1** General requirements. **2** Requirements in respect of keys	
	3 Making good/rectification of defects	**1** Access arrangements. **2** Notice periods. **3** Completion requirements.	
	4 Highway/sewer adoption.	**1** Description of work to be adopted. **2** Requirements in respect of work for adoption.	

PRELIMINARIES

1.1.9A Employer's Requirements: Security, Safety and Protection

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
1 Security/health and safety	**1** Pre-construction information	Location of information	
	2 Execution hazards.	**1** Management requirements for common hazards **2** Details of significant hazards incorporated in the design of the project.	
	3 Product hazards.	**1** Requirements in respect of hazardous substances. **2** Management requirements for common hazards. **3** Details of significant hazards in specified construction materials.	
	4 Construction phase health and safety plan	**1** General requirements. **2** Content. **3** Submission requirements.	
	5 Security.	**1** Requirements for protecting the site, the works, products, materials, and existing buildings affected by the works, from damage and theft. **2** Requirements for preventing unauthorised access to the site, the works, and adjoining property. **3** Description of any special security requirements.	
	6 Stability	Requirements for maintaining the stability and structural integrity of the works and adjoining property during the contract.	
Security/Health and Safety (Cont'd)	**7** Occupied premises.	**1** Extent existing buildings will be occupied and/or used during the contract. **2** Method of working.	

Employer's Requirements: Security, Safety and Protection (cont'd)

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Security/health and safety (Cont'd)	Occupied premises (Cont'd).	**3** Requirements in respect of overtime working.	
	8 Passes.	**1** Details of controlled areas. **2** Procedures for obtaining and returning passes	
	9 Occupier's rules and regulations	**1** General requirements. **2** Location of rules and regulations **3** Arrangements for inspection	
	10 Use of mobile telephones	Requirements relating to use of mobile telephones on-site.	
	11 Employer's representatives site visits	**1** Safety requirements **2** Protective clothing and/or equipment requirements	
	12 Working precautions/restrictions	**1** Details of hazardous areas. **2** Permit to work requirements.	
2 Protection against	**1** Explosives	**1** Use. **2** Details of restrictions.	
	2 Noise consent by local authority.	**1** General requirements. **2** Special requirements	
	3 Noise control **4** Pollution control		
	5 Pesticides.	**1** Use. **2** Details of restrictions. **3** Disposal requirements **4** Operatives competency requirements	
	6 Nuisance.	**1** General requirements. **2** Special requirements	
	7 Asbestos containing materials (ACM)s		
	8 Antiquities.		
	9 Fire prevention.	**1** General requirements. **2** Details of standards to which contractor is to comply.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Protection against (Cont'd)	**10** Smoking on-site.	**1** General requirements. **2** Special requirements	
	11 Burning on-site		
	12 Moisture.	**1** Requirements for preventing wetness and dampness. **2** Requirements for drying out.	
	13 Infected timber/contaminated	General requirements.	
	14 Waste.	**1** Definition of waste. **2** General requirements, including minimizing waste, cavities in the construction removing waste and excluding waste from voids and cavities in the construction. **3** Requirements in respect of 'site waste management plans' (SWMP) **4** Documentation requirements. **5** Details of key performance indicator (KPI) data to be provided **6** Information and documentation submission requirements	
	15 Electromagnetic interference.	General requirements	
	16 Laser equipment.		
	17 Power actuated fixing systems.		
	18 Invasive species	**1** General requirements for the prevention of invasive species (e.g. plants and animals). **2** Details of any special precautions required. **3** Requirements in respect of discovery and reporting.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
3 Protection	**1** Existing services	**1** Requirements in respect of notifying services authorities, statutory undertakers and/or adjoining adjacent owners. **2** Requirements for identification of existing services. **3** Requirements should damage occur to existing services. **4** Other requirements.	
	2 Roads and footpaths	**1** General requirements. **2** Requirements should damage occur to existing roads and/or footpaths. **3** Other requirements.	
	3 Existing topsoil and subsoil	**1** Requirements for preventing over compaction of existing topsoil and subsoil. **2** Details of protective measures.	
	4 Retained trees, shrubs and grassed	**1** Details of protective measures. **2** Requirements should damage occur to retained trees, shrubs and grassed areas. **3** Other requirements.	
	5 Areas of retained trees	Details of protective measures.	
	6 Wildlife species and habitats.	**1** General requirements. **2** Details of protective measures. **3** Other requirements.	

1.1.10A Employers Requirements: Specific Limitations on Method/Sequence and Timing

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
	7 Existing features.	**1** Details of protective measures. **2** Special requirements.	
	8 Existing work.	**1** Details of protective measures. **2** Requirements for removing and replacing existing work.	
	9 Building interiors	Details of protective measures.	
	10 Existing furniture, fittings and equipment	**1** Details of protective measures. **2** Extent of removal work to be carried out by the employer	
	11 Especially valuable and vulnerable items	**1** Details of protective measures. **2** Extent of removal work to be carried out by the employer	
	12 Adjoining property	Permission requirements.	
	13 Adjoining property restrictions	**1** Precautions to be taken by the contractor. **2** Consequence of damage.	
	14 Existing structures	General requirements	
	15 Materials for recycling and/or reuse	**1** Sorting and damage prevention requirements. **2** Storage requirements.	

PRELIMINARIES

Employers Requirements: Specific Limitations on Method/Sequence and Timing (cont'd)

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
1 General		Statement explaining that limitations described in this section of the preliminaries bill are supplementary to limitations described or implicit in information given in other sections or on the drawings.	
2 Design constraints		Details of any design constraints.	
3 Method and sequence of work		Specific limitations relating to method and sequence of working, including phasing requirements to be included in the programme.	
4 Use or disposal of materials found		Specific limitations.	
5 Working hours		**1** Definition of work hours. **2** Definition of normal working hours.	
6 Completion of any section or part of the works	**1** Employer requirements for possession	**1** General requirements. **2** Special requirements	
	2 Remainder of the works	Requirements in respect of: **1** Provision of services. **2** Fire precautions. **3** Means of escape and safe access. **4** Other requirements	

1.1.11A **Employer's Requirements: Site Accommodation/Facilities/Temporary**

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
1 Generally	**1** Spoil heaps, temporary works and services	Requirements in respect of the sitting of spoil heaps and the maintenance, alteration, movement and removal of temporary works.	
2 Site accommodation	**1** Room for meetings.	**1** Specific requirements, including furniture and equipment, to be stated **2** Cleaning and maintenance requirements	
	2 Site offices		
	3 Off-site offices/room for meetings	**1** Specific requirements, including furniture and equipment. **2** Preparatory works, including painting, decoration and applied finishings (e.g. carpet). **3** Cleaning and maintenance requirements	
	4 Sanitary accommodation	Specific requirements, including furniture and equipment.	
	5 Use of accommodation/land not included in the site	**1** Identification of accommodation and or land that may be used by the contractor for the duration of the contract without charge **2** Limitations/restrictions on use. **3** Requirements for temporary adaptations. **4** Cleaning and maintenance requirements. **5** Accommodation/land use. **6** Location of accommodation/land. **7** Reinstatement requirements on vacation of accommodation/land	
	6 Car parking	Specific requirements for the provision of car parking	

Employer's Requirements: Site Accommodation/ Facilities/Temporary Works/Services (cont'd)

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
3 Services and facilities	**1** Lighting	Specific requirements for the provision of temporary lighting for finishing work and inspection.	
	2 Lighting and power	**1** Requirements in respect of: a) Use of employer's mains supply. b) Responsibility for continuity of supply **2** Metering requirements. **3** Location of supply point. **4** Available capacity, frequency, phase and current type **5** Conditions/restrictions imposed on the contractor	
	3 Water	**1** Requirements in respect of: a) Use of employer's mains supply. b) Responsibility for continuity of supply **2** Metering requirements. **3** Location of supply point. **4** Available capacity, frequency, phase and current type **5** Conditions/restrictions imposed on the contractor	
	4 Contractor's on-site telephones	**1** Date to be installed. **2** Responsibility for paying installation and all rental charges, including paying the cost of calls. **3** Requirements for disseminating telephone number. **4** Other requirements	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Services and facilities (Cont'd)	**5** Mobile telephones	Requirements in respect of: **1** Contractor's staff required to be provided with mobile telephones. **2** Responsibility for paying installation and all rental charges, including paying the cost of calls. **3** Requirements for disseminating telephone numbers. **4** Other requirements	
	6 Telephones	**1** System requirements **2** Date to be installed.	
	7 Fax installation	**3** Responsibility for paying installation and all rental charges, including paying the cost of calls. **4** Other requirements.	
	8 Computers.	**1** System requirements, including computers, software, printers, cables and consumables. **2** Responsibility for paying the cost of consumables **3** Date to be installed. **4** Other requirements.	
	9 Email and Internet facilities	**1** System requirements. **2** Date to be installed. **3** Responsibility for paying installation and all rental charges. **4** Other requirements.	

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Services and Facilities (Cont'd)	**10** Photocopier	Employer's requirements.	
	11 Temperature and humidity	Levels to be maintained by the contractor.	
	12 Use of permanent heating systems	**1** Confirmation or otherwise that the contractor is permitted to use permanent heating systems for drying out the works, services and controlling temperature and humidity levels **2** Requirements for operation, maintenance and remedial work **3** Requirements for contractor to arrange supervision of use by subcontractor and indemnification of subcontractor. **4** Other requirements.	
	13 Beneficial use of permanent installed systems	Details of services systems, including lifts and sanitary installations, which can be used by the contractor to complete the works.	
	14 Meter readings	Requirements for obtaining meter readings.	
	15 Thermometers	Requirements for providing maximum and minimum thermometers for measuring atmospheric shade temperature.	
	16 Surveying equipment	Requirements for providing surveying equipment.	
	17 Personal protective equipment	Specific requirements for those acting on behalf of the employer.	
	18 Other requirements	Specific requirements.	

1.1.12A Employer's Requirements: Operation/Maintenance of Finished Building

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
4 Temporary works	**1** Roads, hard standings and footpaths	Requirements in respect of permanent roads, hard standings and footpaths on the site, including restrictions on use and protective and/or remedial measures.	
	2 Temporary works	Specific requirements (e.g. fences, hoardings, screens and roofs).	
	3 Temporary protection measures to existing trees/vegetation.	**1** Location of temporary protection (by reference to drawing) **2** Standards of protective barriers and any other applicable physical protection measures. **3** Design details of physical protection measures (by reference to drawing). **4** Areas of structural landscaping to be protected from construction operations. **5** Requirements for maintaining integrity of protection for the duration of the works. **6** Requirements for removing protection on completion of the works.	
	4 Name boards. **5** Advertising.	Specific requirements.	
	6 Other requirements		
1 Operation and Maintenance Manual	**1** Generally	**1** Purpose. **2** Scope. **3** Responsibility for preparation. **4** Information to be provided by others. **5** Review process. **6** Number of copies.	The operation and maintenance information, the health and safety file, and all other information can be combined as a single document. In this case, the document can be referred to as the 'building manual'.

1.1.12A Employer's Requirements: Operation/Maintenance of Finished Building

SUB-HEADING 1	SUB-HEADING 2	INFORMATION REQUIREMENTS	SUPPLEMENTARY INFORMATION/NOTES
Operation and Maintenance Manual (Cont'd)	Generally (Cont'd)	**7** Latest date for submission of final manual. **8** As-built/as-installed drawings: a) format and standard b) number of copies.	
	2 Content.	Specific requirements.	
2 Health and Safety file	**1** Generally	**1** Purpose. **2** Scope. **3** Responsibility for preparation. **4** Information to be provided by others. **5** Review process. **6** Number of copies. **7** Latest date for submission of final manual.	
	2 Content.	Specific requirements.	
3 Web-based information management system		Specific requirements.	
4 Presentation of documents	**1** Operation and maintenance manual	Specific requirements in respect of format and presentation.	
	2 Health and safety file.		
	3 Other requirements	Specific requirements.	
5 Other employer specific requirements	**1** Maintenance services	Specific requirements for post completion maintenance (planned and reactive).	
	2 Information for commissioning services **3** Training. **4** Spare parts. **5** Tools.	Specific requirements.	
6 Other information		Specific requirements.	

1.1B Part B: Pricing schedule

1.1.1 Employer's requirements

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1 Site accommodation	Site accommodation for the employer and the employer's representatives where separate from main contractor's site accommodation, including: – site offices – sanitary accommodation – welfare facilities – foundations to site accommodation – temporary drainage to accommodation – temporary services – intruder alarms Type and extent of accommodation to be provided to be stated; with each type separately quantified.			Site accommodation, furniture and equipment, telecommunication and IT systems for the employer and the employer representatives where an integral part of the main contractor's site accommodation (included in section 1.2: in contractor's cost items, as appropriate).
	1 Bringing to site and installing, including all temporary drainage, services and intruder alarms.	Item	Fixed charge	
	2 Adaptations/alterations during works. **3** Dismantling and removing from site, including rectifying any damage.			
	4 Maintaining.	Weeks	Time-related charge	
	5 Cleaning.			
	6 Charges.			

Part B: Pricing Schedule (cont'd)
Employer's Requirements (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Site Accommodation (Cont'd)	**7** Off-site rented temporary accommodation			
	8 Rectifying damage to off-site rented temporary accommodation	Item	Fixed charge	
2 Furniture and Equipment	Furniture and equipment for the employer and the employer's representatives where separate from main contractor's site accommodation. For example, desks, chairs, meeting table and chairs, cupboards, kettles, coffee maker, photocopier and consumables.			
	1 Bringing to site and installing.	Item	Fixed charge	
	2 Cleaning.	Weeks .	Time-related charge.	
	3 Charges			
	4 Dismantling and removing from site.	Item	Fixed charge	
3 Telecommunications and IT systems	Telecommunication and IT systems for the employer and the employer's representatives where separate from main contractor's site accommodation, including telephones, fax machines, photocopier, computers, printers and consumables.			
	1 Purchase charges.	nr	Fixed charge.	
	2 Hire charges.	week	Time-related charge.	
	3 Consumables.			

1.1.1.02 Site Records

	COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
	1 Site records	**1** Operation and maintenance manuals	item .	Fixed charge	
		2 Compilation of health and safety file			
	2 Web-based information management system	**1** Provision of system.			
		2 Uploading of data.			
		3 Training of building user's staff in the operation of the web-based management system.			
1.1.1.03	**Completion and post-completion requirements**				
	1 Handover requirements	**1** Training of building user's staff in the operation and maintenance of the building engineering services systems.	item	Fixed charge	
		2 Provision of spare parts for maintenance of building engineering services			
		3 Provision of tools and portable indicating instruments for the operation and maintenance of building engineering services systems			

Part B: Pricing Schedule (cont'd)

Employer's Requirements (cont'd)

1.1.1.02 Site Records

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
2 Operation and maintenance services	**1** Operation and maintenance of building engineering services installations, mechanical plant and equipment and the like during the defects liability period, period for rectifying defects, maintenance period or other specified period (i.e. additional services that are normally required by the contract).	week	Time-related charge	
3 Landscape management services	**1** Maintenance of internal and external planting.			

Part B: Pricing Schedule (cont'd)

1.1.2 Main Contractor's Cost Items

1.1.2.1 Management and Staff

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1 Project-specific management and staff	Main contractor's project specific management and staff such as: **1** Project manager/director **2** Construction manager. **3** Supervisors, including works/trade package managers, building services engineering managers/co-ordinators and off-site production managers **4** Health and safety manager/officers. **5** Commissioning manager (building engineering services). **6** Planning/programming manager and staff. **7** Senior/managing quantity surveyor **8** Project/package quantity surveyors **9** Procurement manager **10** Design manager **11** Project engineers **12** Environmental manager. **13** Temporary works design engineers **14** Materials management staff (e.g. storeman). **15** Administrative staff, including secretary, document controllers, finance clerks and the like. **16** Other management and staff	week (number of staff by number of man hours per week by number of weeks)	Time-related charge	**1** Security staff (included in Section 1.2.4: Security).

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
2 Visiting Management and Staff	**1** Managing Director, Regional Director, Operations Director, Commercial Director and the like.			**1** Visiting management and staff for which an allowance has been made within the main contractor's overheads
	2 Quality Manager.			
	3 Contracts/Commercial Manager			
	4 Health and Safety manager			
	5 Environmental Manager/Consultant.			
	6 Other visiting management and staff			
3 Extraordinary support costs	**1** Legal advice costs (i.e. solicitors)	Item	Fixed charge	**1** Extraordinary support costs for which an allowance has been made within the main contractor's overheads.
	2 Recruitment costs.			
	3 Team building costs.			
	4 Other extraordinary support costs.			
	5 Day transport	week (number of days per week by number of weeks)	Time-related charge	
	6 Personnel transport (i.e. transportation of work operatives to site).			

Part B: Pricing Schedule (cont'd)

Main Contractor's Cost Items (cont'd)

Management and Staff (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Extraordinary support costs (Cont'd)	**7** Temporary living accommodation (e.g. long/medium term accommodation costs).	week (number of staff by number of man hours per week by number of weeks)		
	8 Subsistence payments.			
	9 Out of normal hours working, including non-productive overtime allowances			
4 Staff travel	Costs associated with off-site visits such as:			
	1 Visits to employer's and consultants' offices	nr (number of occasions)	Fixed charge.	
	2 Visits to subcontractors' offices/works			
	3 Overseas visits.			
	4 Accommodation charges and overnight expenses.			

1.1.2.2 Site Establishment

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1 Site accommodation	Main contractor's and common user temporary site accommodation such as: – offices. –conference/meeting rooms. – canteens and kitchens. – drying rooms – toilets and washrooms – first aid room			**1** Employer's accommodation, where not an integral part of the main contractor's site accommodation (included in section 1.1.1:Site accommodation). **2** Temporary bases, foundations and provision of drainage and services to temporary site accommodation (included in component 1.2.2.2: Temporary works in connection with site establishment). **3** Service provider's charges for temporary services (included in section 1.2.12: Fees and charges). **4** Rates for temporary services (included in section 1.2.12: Fees and charges).
	– laboratories	item	Fixed charge	
	– workshops. – secure stores. – compounds, including containers for material storage. – security control room. – stairs and office staging. Type and extent of accommodation to be provided to be stated, with each type separately quantified.			
	1 Purchase charges.	item	Fixed charge	
	2 Hire charges.	week	Time-related charge	
	3 Employer's accommodation, where an integral part of the main contractor's site accommodation.			
	4 Delivery of temporary site accommodation to site, erection, construction and removal.	item	Fixed charge	

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Site accommodation (Cont'd)	**5** Temporary accommodation made available by the employer	week	Time-related charge	
	6 Intruder alarms	item	Fixed charge	
	7 Land/property rental where site accommodation located off-site	week	Time-related charge	
	8 Alterations and adaptations to site accommodation, including partitioning, doors, painting and decorating, and the like.	item	Fixed charge	
	9 Relocation and alterations of temporary accommodation during construction stage. **10** Reinstating temporary site accommodation to original condition prior to removal from site. **11** Removal of site accommodation and temporary works in connection with site accommodation.			
2 Temporary works in connection with site establishment	**1** Temporary bases and foundations for site accommodation, including maintenance and reinstatement of existing surfaces on completion of the works.	m^2	**1** Fixed charge. **2** Time-related charge.	**1** Provision of temporary services to site establishment (included in section 1.2.3: Temporary services). **2** Provision of temporary drainage to site establishment (included in section 1.2.3: Temporary services).

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
	2 Connections to temporary service, including maintenance and removal on completion of the works.	nr		
	3 Connections to temporary drainage, in including maintenance and removal on completion of the works.	nr		**3** Hoardings, fans, fencing and the like to site boundaries and to form site compounds (included in section 1.2.4: Security (Hoardings, fences and gates).
	4 Temporary site roads, paths and pavings (including on-site car parking), including reinstatement of existing surfaces on completion of the works.	m		
	5 Temporary surface water drainage to temporary site roads, paths and pavements, including maintenance and removal on completion of the works.	m		
3 Furniture and Equipment	**1** Workstations for staff, including maintenance.	nr	**1** Fixed charge **2** Time-related charge.	**1** Telephone and fax installations (included in section 1.2.3:Temporary services). **2** Computers and IT associated equipment (included in component 1.2.2.4: IT systems).

Part B: Pricing Schedule (cont'd)

Main Contractor's Cost Items

Site Establishment (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Furniture and Equipment (Cont'd)	**2** General office furniture, including maintenance.	item		
	3 Conference/Meeting Room furniture, including maintenance.			
	4 Photocopiers, including purchase/rental, maintenance and other running costs.			
	5 Canteen furniture, including maintenance.			
	6 Canteen equipment, including purchase/rental, maintenance and other running costs.			
	7 Floor coverings, including maintenance.			
	8 Water dispensers, including purchase/rental, maintenance and other running costs.			
	9 Heaters, including maintenance of heaters.			
	10 Other office equipment, including maintenance.			
	11 Removal of furniture and equipment.			
	12 Maintenance of furniture and floor covering.			

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
4 IT systems	**1** Computer hardware, including purchase/rental, installation, initial set up, maintenance and running costs, such as: – desktop computers and laptop – CAD stations. – server and network equipment. – printers and plotters. – other computer system hardware.	item	**1** Fixed charge **2** Time-related charge.	Computer and printer consumables (included in component 1.1.2.5: Consumables and services). Document management, including electronic data management systems (EDMS) (included in component 1.1.2.6: Brought in services).
	2 Software and software licenses.	item		
	3 Modem lines, modems and connections (i.e. email and internet capability).			
	4 WAN lines and connections (if on WAN).	item		
	5 Line rental charges.	week	Time-related charge.	
	6 Internet/website addresses.	nr	Fixed charge	
	7 Internet service provider (ISP)			
	8 Line calls charges.	week	Time-related charge.	
	9 IT support and maintenance.			
5 Consumables and services	**1** Stationery	week	Time-related charge.	
	2 Computer and printer consumables (e.g. ink cartridges).			

Part B: Pricing Schedule (cont'd)

Main Contractor's Cost Items

Site Establishment (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Consumables and services (Cont'd)	**3** Postage.			
	4 Courier charges.			
	5 Tea, coffee, water bottles and the like.			
	6 First aid consumables.			
	7 Photocopier consumables (e.g. paper and toners).			
	8 Fax consumables (e.g. paper and toners).			
	9 Drawing printer consumables (e.g. ink cartridges			
6 Brought-in services	Services outsourced by the main contractor such as:	week	Time-related charge.	
	1 Catering.			
	2 Equipment maintenance.			
	3 Document management, including management information systems and electronic data management systems (EDMS).			
	4 Printing (purchasing), including reports and drawings.			
	5 Staff transport.			
	6 Off-site parking charges.			
	7 Meeting room facilities.			
	8 Photographic services.			
	9 Others.			

Part B: Pricing Schedule (cont'd)

Main Contractor's Cost Items (cont'd)

Temporary Services (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
7 Sundries	**1** Main contractor's signboards	Item	Time-related charge.	
	2 Safety and information notice boards.			
	3 Fire points.			
	4 Shelters.			
	5 Tool stores.			
	6 Crane signage.			
	7 Employer's composite signboards			
Temporary Services				
1 Temporary water supply	**1** Temporary connections	nr	**1** Fixed charge	
	2 Distribution equipment, installation and adaptations.	item	**2** Time-related charge.	
	3 Meter Charges	week		
2 Temporary gas supply	**1** Gas connection	nr	**1** Fixed charge	
	2 Distribution equipment, installation and adaptations.	item	**2** Time-related charge.	
	3 Charges.	week		
	4 Bottled gas			
3 Temporary electricity supply	**1** Temporary connections	nr	Fixed charge	
	2 Temporary electrical supply for tower cranes	item		
	3 Charges – power consumption for site establishment	item	Time-related charge.	
	4 Charges – power consumption for the works.			

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
	5 Distribution equipment, installation and adaptations	item	Fixed charge	
	6 Attendance	nr (number of man hours per week by number of weeks)	Time-related charge.	
	7 Uninterrupted power supply (UPS).	item	Fixed charge	
	8 Temporary substation modifications			
4 Temporary Telecommunication Systems	**1** Landlines (including connection and rental charges), including: – telephone and fax lines – ISDN lines	item	**1** Fixed charge **2** Time-related charge	**1** Fax consumables (included in component 1.1.2.5: Consumables and Services).
	2 Telephone and facsimile equipment (including connection and rental – PABX equipment. – handsets, including purchase or rental. – fax machines, including purchase or rental. – installation of equipment. – maintenance of equipment.			

Part B: Pricing Schedule (cont'd)

Main Contractor's Cost Items (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Temporary Telecommunication Systems (Cont'd)	**3** Mobile (cellular) phones, including: – mobile phones, including purchase or rental and connection charges. – spare batteries. – mobile phone charges.			
	4 Telephone charges, including: – telephone call charges. – fax charges. – fax and telephone consumables			
	5 Radios (including purchase or rental charges), including: – base set. – handsets and chargers. – repairs and maintenance. – licenses. – spare batteries.			
5 Temporary drainage	**1** Temporary mains	item	**1** Fixed charge	
	2 Septic tanks.	nr	**2** Time-related charge	
	3 On-site treatment plant.	item		
	4 Holding tanks.	nr		
	5 Sewage pumping.			
	6 Distribution pipework, etc.	item		
	7 Drainage installation and adaptations.			
	8 Disposal charges (i.e. rates).	week	Time-related charge	
	9 Disposal costs (i.e. tanker charges).			

Part B: Pricing Schedule (cont'd)

Main Contractor's Cost Items (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1 Security Staff	**1** Security guards (day and night).	nr (number of staff by number of man hours per week by number of weeks).	Time-related charge	**1** Security staff accommodation (included in section 1.1.2: Site establishment).
	2 Watchmen (day and night).			
2 Security Equipment	**1** Site pass issue equipment, including maintenance and removal	item	**1** Fixed charge	
	2 Site pass consumables		**2** Time-related charge	
	3 CCTV surveillance installation, including maintenance and removal.			
	4 Temporary vehicle control barriers, including maintenance and removal	nr		
3 Hoardings, Fences and Gates	**1** Perimeter hoardings and fencing and the like to site boundaries and to form site compounds.	m	**1** Fixed charge	
	2 Access gates, including frames and ironmongery	nr	**2** Time-related charge	
	3 Painting of hoardings, fencing, gates, and the like	m		
	4 Temporary doors	nr		
	5 Modification to line of hoarding and fencing during construction.			

Main Contractor's Cost Items (cont'd)

1.1.2.05 Safety and Environmental Protection

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Hoardings, fences and gates (Cont'd)	**6** Dismantling and removal of hoarding, fencing, gates, and the like.	m		
1 Safety programme	Works required to satisfy requirements of Statutory Regulations: **1** Health and safety manager/officers.	nr (number of staff by number of man hours per week by number of weeks).	Time-related charge	**1** Health and safety manager/officers (included in section 1.2.1: Management and staff). **2** Welfare facilities (included in section 1.2.2: Site establishment).
	2 Safety audits, including safety audits carried out by external consultant.	nr	**1** Fixed charge **2** Time-related charge	
	3 Staff safety training.	item		
	4 Site safety incentive scheme.			
	5 Notices and information to neighbours.			
	6 Personal protective equipment (PPE), including for employer and consultants.	nr (sets)		
	7 PPE for multi-service gangs.			
	8 Fire points.	nr		
	9 Temporary fire alarms. **10** Fire extinguishers			
	11 Statutory safety signage	Item		
	12 Nurse	nr (number of staff by number of man hours per week by number of weeks).	Time-related charge	
	13 Traffic marshals.			

Safety and Environmental Protection (Cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
2 Barriers and Safety Scaffolding	**1** Guard rails and edge protection (e.g. to edges of suspended slabs and roofs).	Item	**1** Fixed charge	
	2 Temporary staircase balustrades (i.e. to new staircases during construction).		**2** Time-related charge	
	3 Lift shaft protection. **4** Protection to holes and openings in ground floor slabs, suspended slabs and the like.			
	5 Debris netting/plastic sheeting			
	6 Fan protection	Item		
	7 Scaffold inspections	nr		
	8 Hoist run-offs.	item		
	9 Protective walkways.			
	10 Other safety measures			
3 Environmental Protection Measures	**1** Control of pollution.	Item	**1** Fixed charge **2** Time-related	
	2 Residual control of noise **3** Environmental monitoring.			
	4 Environmental manager/ consultant	nr (number of staff by number of man hours per week by number	Time-related charge	
	5 Environmental audits, including safety audits carried out by external consultant.	nr	**1** Fixed charge **2** Time-related	

1.1.2.06 Control and Protection

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1 Survey, Inspections and Monitoring	**1** Surveys **2** Topographical survey. **3** Non-employer dilapidation survey. **4** Structural/dilapidations survey adjoining buildings. **5** Environmental surveys. **6** Movement monitoring. **7** Maintenance and inspection costs.	Item	**1** Fixed charge **2** Time-related charge	**1** Environmental monitoring (included in section 2.2.5: Safety and environmental protection).
2 Setting out	**1** Setting out primary grids **2** Grid transfers and level checks **3** Maintenance of grids. **4** Take over control and independent checks (i.e. on change of subcontractors). **5** Instruments for setting out	Item	**1** Fixed charge **2** Time-related charge	
3 Protection of works	**1** Protection of finished works to project handover **2** Protection of stairs, balustrades and the like works to project handover. **3** Protection of fittings and furnishings works to project handover. **4** Protection of entrance doors and frames works to project handover. **5** Protection of lift cars and doors works to project handover.	Item	**1** Fixed charge **2** Time-related charge	

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Protection of works (Cont'd)	**6** Protection of specifically vulnerable products to project handover. **7** Protection of all sundry items.			
4 Samples	**1** Provision of samples **2** Provision of sample room **3** Mock-ups and sample panels **4** Testing of samples/mock-ups, including testing fees **5** On-site laboratory equipment **6** Mock-ups of prefabricated units (e.g. residential units, student accommodation units, hotel accommodation and the like).	Item	**1** Fixed charge **2** Time-related charge	
5 Environmental Control of Building	**1** Dry out building **2** Temporary heating/cooling **3** Temporary waterproofing, including over roofs **4** Temporary enclosures.	Item	**1** Fixed charge **2** Time-related charge	

Mechanical Plant

	COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1.1.2.07	**Mechanical plant**				
	1 Generally	Common user mechanical plant and equipment used in construction operations.			Plant and equipment used for specific construction operations, such as: **1** Earthmoving plant. **2** Piling plant. **3** Paving and surfacing plant. **4** Wheel spinners, and road sweepers (included in section 1.2.11: Cleaning). **5** Access scaffolding (included in section
	2 Tower cranes	Type of craneage to be provided shall be stated; with each type separately quantified. **1** Hire charges	week	Time-related charge	**1** Temporary electrical supply to tower crane (included in section 1.2.3:Temporary services).
		2 Crane operator	week (number of staff by number of man hours per week by number of weeks).		
		3 Overtime for crane and operator.			
		4 Piles for tower crane bases, including maintenance removal	nr	**1** Fixed charge	
		5 Temporary bases for tower cranes, including anchors. maintenance; removal and reinstatement on completion (size, in m2, to be stated).		**2** Time-related charge	

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Tower cranes (Cont'd)	**6** Ties.	week	Time-related charge	
	7 Connections to temporary electrical supply.	nr	Fixed charge	
	8 Bring to site, erection, test and commission			
	9 Periodic safety checks/inspections	week	Time-related charge	
	10 Dismantling and removing from site	nr	Fixed charge	
	11 Other costs specific to tower crane such as: – chain pack and sundries – relief operator – banksman – man cage.	Item	**1** Fixed charge **2** Time-related charge	
	12 Temporary voids in building structure for craneage, hoists and the like including filling voids after removal.	nr	Fixed charge	

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
3 Mobile cranes	Type of craneage to be provided shall be stated; with each type separately quantified. **1** Mobile crane hire charges, including driver/operator charges	week	**1** Fixed charge **2** Time-related charge	
	2 Attendance	nr (number of man hours per visit by number of visits).		
	3 Other costs specific to mobile crane	item		
4 Hoists	Type of hoist to be provided shall be stated; with each type separately quantified. **1** Goods and passenger hoists, including protection cages and embedment frames.	week	Time-related charge	**1** Temporary services to hoist installations (included in section 1.2.3:Temporary services).
	2 Hoist bases	nr	**1** Fixed charge **2** Time-related	
	3 Bringing to site, erecting, testing and commissioning	nr	Fixed charge	
	4 Dismantling and removing from site.			

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Hoists (Cont'd)	**5** Protection systems	Item	**1** Fixed charge **2** Time-related charge	
	6 Hoist operator, including overtime	week (number of staff by number of man hours per week by number of weeks)	Time-related charge	
	7 Beam hoists	Item	**1** Fixed charge **2** Time-related charge	
	8 Periodic safety checks/inspections	month	Time-related charge	
	9 Other costs specific to temporary hoist installations	Item	**1** Fixed charge **2** Time-related charge	
5 Access plant	**1** Forklifts	week	**1** Fixed charge **2** Time-related charge	
	2 Scissor lifts.			
	3 Loading platforms.			
	4 Maintenance of mechanical access equipment			
	5 Other costs specific to mechanical access equipment	Item		

Part B: Pricing Schedule (cont'd)

Main Contractor's Cost Items (cont'd)

Mechanical Plant (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
6 Concrete plant	**1** Concrete plant	week	**1** Fixed charge **2** Time-related charge	**1** Temporary service to concrete plant (included in section 1.2.3:Temporary services).
	2 Plant operator	week (number of staff by number of man hours per week by number of weeks)	Time-related charge	
	3 Overtime for plant and operator.			
	4 Bases for concrete plant	nr	**1** Fixed charge **2** Time-related charge	
	5 Power connections, including cabling and statutory undertaker's charges for temporary connection to their supply.	nr	**1** Fixed charge **2** Time-related charge	
	6 Bring to site, erection, test and commission	nr	Fixed charge	
	7 Maintenance of concrete plant	week	Time-related charge	
	8 Dismantling and removing from site	nr	Fixed charge	
7 Other plant	**1** Small plant and tools	week	Time-related charge	

1.1.2.08 Temporary Works

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1 Access scaffolding	Common user access scaffolding (type of access scaffolding to be specified): – access scaffolding to elevations, lift shafts and the like, including: fans and mesh screens. – structural scaffolding (e.g. to party walls). – birdcage scaffolding. – cantilever access scaffolding. – staircase platforms. – primary loading platforms. – travelling access platforms. **1** Bringing to site, erecting and initial safety checks	nr	Fixed charge	**1** Scaffolding specific to works packages (included in appropriate element or sub-element). **2** Scaffold inspections (included in sub-element 1.2.5: Safety and environmental protection).
	2 Hire charges	week	Time-related charge	
	3 Altering and adapting during construction	nr	Fixed charge	
	4 Dismantling and removing from site			
2 Temporary works	Common user temporary works: – support scaffolding and propping. – crash decks. – temporary protection to existing trees and/or vegetation. – floodlights			**1** Temporary works design (included in section 1.1.1: Management and staff). **2** Temporary bases, drainage and services to site accommodation (included in section 1.2.2: Site establishment).

	COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
		1 Bringing to site, erecting and initial safety checks	nr	Fixed charge	**3** Temporary roads, paths and pavement, including on-site car parking (included in section 1.2.2: Site establishment (i.e. Builder's work in connection with site accommodation)). **4** Hoardings, fans, fencing and the like to site boundaries and to form site compounds (included in section 1.2.4: Security (hoardings, fences and gates).
		2 Hire charges	week	Time-related charge	
		3 Altering and adapting during construction			
		4 Dismantling and removing from site	nr	Fixed charge	
1.1.2.09	**Site records**				
	1 Site records	Unless otherwise indicated, costs associated with the following shall be deemed to be included in management and staff costs: **1** Photography: – camera purchase. – consumables. – printing and presentation	Item	**1** Fixed charge **2** Time-related charge	
		2 Works records: – progress reporting. – site setting out drawings. – condition surveys and reports. – operation and maintenance manuals. – as-built/installed drawings and schedules. – compilation of health and safety file (if required).			

Main Contractor's Cost Items (Cont'd)

1.1.2.10 Completion and Post-Completion Requirements

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1 Testing and commissioning plan	Costs associated with the following shall be deemed to be included in section 1.2.1: Management and staff costs: **1** Preparation of Commissioning Plan	Item	**1** Fixed charge **2** Time-related charge	**1** Testing and commissioning of services.
2 Handover	Unless otherwise indicated, costs associated with the following shall be deemed to be included in section 1.2.1: Management and staff costs: **1** Preparation of Handover Plan.			
	2 Training of building user's staff in the operation and maintenance of the building engineering services systems			
	3 Provision of spare parts for maintenance of building engineering services			
	4 Provision of tools and portable indicating instruments for the operation and maintenance of building engineering services systems.			
	5 Pre-completion inspections.			
	6 Final inspections.			

Part B: Pricing Schedule (cont'd)

Main Contractor's Cost Items (Cont'd)

	COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
	3 Post-Completion Services	**1** Supervisory staff (employer/tenant care).	week (number of staff by number of man hours per week by number of weeks)	Time-related charge	
		2 Handyman.			
		3 Minor materials and sundry items **4** Insurances.	Item	Fixed charge	
		5 Other post-construction staff	week (number of staff by number of man hours per week by number of weeks)	Time-related charge	
1.1.2.11	**Cleaning**				
	1 Site Tidy	**1** Cleaning site accommodation – internal, including cleaning telephone handsets, other office furniture and equipment and window cleaning.	week	Time-related charge	
		2 Periodic maintenance of site accommodation, including redecoration (internal and external).			
		3 Waste management, including rubbish disposal (including compactor visits; skips and waste bins; roll-off, roll-on waste bins) and other disposal			

	COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
		4 Pest control.			
	2 Maintenance of Roads, Paths and Pavings	**1** Maintenance of temporary site roads, paths, and pavements	week	Time-related charge	
		2 Maintenance of public and private roads, including wheel spinners and road sweepers.			
	3 Building clean	**1** Final builder's clean	Item	Fixed charge	
1.1.2.12	**Fees and charges**				
	1 Fees	**1** Building control fees, where not paid by the employer **2** Oversailing fees, where not paid by the employer. **3** Any other fees operated by local authority). **4** Scheme registration fees or similar fees, where not paid by the employer.	Item	**1** Fixed charge **2** Time-related charge	**1** Building control fees, where paid by the employer. **2** Oversailing fees, where paid by the employer. **3** Scheme registration fees or similar fees, where paid by the employer.
	2 Charges	**1** Rates on temporary accommodation	week	Time-related charge	**1** Statutory undertaker's charges in connection with permanent services to the building.
		2 Licenses in connection with hoardings, scaffolding, gantries and the like	Item	**1** Fixed charge	**2** Statutory undertaker's charges in connection with temporary services.
		3 Licenses in connection with crossovers, parking permits, parking bay suspensions and the like.		**2** Time-related charge	

1.1.2.13 Site services

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
1 Temporary works	**1** Temporary works that are not specific to an element	item/nr/m/m^2/m^3	**1** Fixed charge **2** Time-related charge	**1** Temporary screens in connection with minor demolition works and alteration works. **2** Supports to small openings cut into existing walls or after removal of internal walls or the like in connection with minor demolition works and alteration works. **3** Temporary or semi-permanent support for unstable structures or facades – façade retention works (i.e. structures not to be demolished).
2 Multi-service gang	**1** Ganger **2** Labour. **3** Fork lift driver. **4** Service gang plant and transport.	week (number of staff by number of man hours per week by number of weeks)	Time-related charge	

1.1.2.14

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
Insurance, bonds, guarantees and warranties				
1 Works insurance	**1** Contractor's 'all risks' (CAR) **2** Contractor's plant and equipment insurance **3** Temporary buildings insurance. **4** Terrorism insurance. **5** Other insurances in connection with the works. **6** Insurance premium tax (IPT). **7** Allowance for recovery of all or part of insurance premium excess.	Item	**1** Fixed charge **2** Time-related charge	
2 Public liability insurance	**1** Non-negligence insurance **2** Professional indemnity insurance **3** Insurance premium tax (IPT) **4** Allowance for recovery of all or part of insurance premium excess	Item	**1** Fixed charge **2** Time-related charge	
3 Employer's (main contractor's) liability insurance	**1** Management and staff, including administrative staff. **2** Works operatives **3** Insurance premium tax (IPT) **4** Allowance for recovery of all or part of insurance premium excess	Item	**1** Fixed charge **2** Time-related charge	

Main Contractor's Cost Items (cont'd)
Insurance, Bonds, Guarantees and Warranties (cont'd)

COMPONENT	INCLUDED/NOTES ON PRICING	UNIT	PRICING METHOD	EXCLUDED
4 Other insurances	**1** Employer's loss of liquidated damages **2** Latent defects cover **3** Motor vehicles. **4** Other insurances. **5** Insurance premium tax (IPT) **6** Allowance for recovery of all or part of insurance premium excess	Item	**1** Fixed charge **2** Time-related charge	
5 Bonds	**1** Tender bonds (if applicable) **2** Performance bonds	Item	**1** Fixed charge **2** Time-related	
6 Guarantees	**1** Parent company guarantees **2** Product guarantees, insurance backed guarantees	Item	**1** Fixed charge **2** Time-related	
7 Warranties	**1** Collateral warranties **2** Funder's warranties **3** Purchaser's and tenant's warranties. **4** Other warranties.	Item	**1** Fixed charge **2** Time-related charge	

1.2 OFF-SITE MANUFACTURED MATERIALS, COMPONENTS OR BUILDINGS

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Site plans. **2** Plans. **3** Sections. **4** Elevations. **5** Installation details **B Mandatory Information** **1** Kind and quality of materials **2** Method of fixing or installing. **3** Connecting to other work and services **4** Special finishes **C Minimum information that must be shown on the drawings that accompany this section of measurement include:-** **1** Major dimensions of component, structure or unit. **2** Location of component, structure or unit							These are deemed to be included: **C1** All factory applied finishes **C2** Transport from factory to site. **C3** Offloading and storing on site **C4** Setting, hoisting and placing in final position **C5** All connection and joint materials **C6** All service connections	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Component	nr	**1** Overall dimensions	**1** Description of component	**1** Method of fixing/installing. **2** Height above structural ground floor level. **3** Services connections		**D1** These are prefabricated proprietary components that are not adequately covered by the other Work Sections in this document.		

OFF-SITE MANUFACTURED MATERIALS, COMPONENTS OR BUILDINGS (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Prefabricated structures	nr	Overall dimensions	**1** Roofs. **2** External walls. **3** Internal walls/ partitions. **4** Floors. **5** Stairs. **6** Bridges. **7** Masts. **8** Other, type stated.			**D1** These are complete or substantially complete building elements of proprietary construction, largely prefabricated.	**C7** The fixing of items supplied only as part of the proprietary package is included here. Other work not forming part of the proprietary package is measured separately in the appropriate Work Section.	
3 Prefabricated building units	nr		**1** Toilet/ bathroom units **2** Sound proof rooms **3** Cold rooms. **4** Spray booths **5** Kiosks. **6** Other, type stated.		**M1** These are complete or substantially complete room units, usually of proprietary construction, for incorporation into buildings, structures or siteworks. The list is not exhaustive.			

OFF-SITE MANUFACTURED MATERIALS, COMPONENTS OR BUILDINGS (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
4 Prefabricated buildings	nr		**1** Description of building			**D2** These are complete or substantially complete building superstructures of proprietary construction, largely prefabricated.	**C8** The fixing of items supplied only as part of the proprietary package is included here. Other work not forming part of the proprietary package is measured separately in the appropriate Work Section	

1.3 DEMOLITIONS

Demolitions

Shoring, Facade Retention and Temporary Works

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Location drawings **2** Drawings of existing structures to show full extent of demolition. **B Mandatory information to be provided.** **1** Brief description and size of structure to be demolished **2** Extent of parts of structure to be temporarily retained **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Lowest level of demolition. **2** Extent of any temporary works not at the discretion of the contractor **3** Position and extent of any temporary screens, roofs and the like.			The following are deemed to be included:- **C1** All temporary works unless stated otherwise **C2** Temporarily diverting, maintaining or sealing off existing services **C3** Disposal of all debris unless stated otherwise **C4** Method of demolition unless stated otherwise. **C5** All temporary support left to the discretion of the contractor. **C6** Clearing away all temporary works.	**S1** Any limitations due to presence of toxic or hazardous materials. **S2** Asbestos surveys

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C7 Disposing of rainwater **C8** Making good all work disturbed	
Item or Work to be measured		**First Division**	**Second Division**	**Third Division**				
1 Demolitions	item	**1** All structures **2** Individual structures **3** Parts of structures	**1** Description of building(s) or parts of building(s). **2** Lowest levels to which structure(s) are to be demolished	**1** Limitations on disposal of materials **2** Any material to be retained for re-use. **3** Materials remaining the property of the employer				**S3** Contractor should be advised to inspect the structure(s) to be demolished
2 Temporary support of structures, roads and the like		**1** Parts of existing building to be retained.	**1** Description of building(s) or parts of building(s) or road or other surface or feature to be retained.			**D1** Support is for parts of the structure that must be retained. It does not mean any type of support required as incidental to the demolitions.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Temporary support of structures, roads and the like (Cont'd)	item	**2** Adjoining buildings not forming part of the works. **3** Roads and other surfaces to be retained **4** Any other existing feature to be retained	**2** Type of shoring. **3** Length of exposed edge of surface to be retained and average height(s).					
3 Temporary works	m^2	**1** Roofs. **2** Screens. **3** Floors. **4** Roads	**1** Weatherproof. **2** Watertight. **3** Dustproof. **4** Fireproof.	**1** Method of construction if not at the discretion of the contractor. **2** Maintaining: duration stated	**M1** In order to ensure the full extent and scope of this work the surveyor may need to provide additional information if not readily ascertained from the drawings.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Temporary works (Cont'd)			**5** Any other requirement: type stated	**3** Adapting during course of works **4** Clearing away. **5** Disposing of rainwater: details stated. **6** Providing openings: details stated				
4 Decontamination	item	**1** Removal of toxic/hazardous materials. **2** Decontamination of existing premises. **3** Infestation removal /eradication. **4** Fungus removal/ eradication.	**1** Scope and location of work. **2** Type of contamination /infestation/ fungus to be treated.	**1** Preparatory works	**M2** (Excluding decontamination of existing ground: this work is measured in Work Section 1.5: Excavating and Filling.)	**D2** Sufficient information must be given to fully describe the cause of the contamination or infestation together with such information as is appropriate to allow the contractor to fully treat the condition described.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Recycling	item	**1** Detailed description of type of material to be recycled and any limitations imposed by employer or local authority.	**1** To be collected by local authority. **2** To be transported to recycling depot, details and location stated.					

1.4 ALTERATIONS, REPAIRS AND CONSERVATION

Alteration Work to Existing Buildings

Repairs/Cleaning/Renovating and Conserving

Decontamination

Re-Cycling

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this Section of Measurement **1** Location drawings **2** Drawings of existing structures to show full extent of demolition. **B Mandatory information to be provided.** **1** Description of operations where not left to discretion of the contractor **2** Details of all materials to be set aside for subsequent re-use including means of storage. **3** Any restrictions on method, sequence and/or timing of the works. **4** Any restrictions on methods of storage of materials to be re-used or to remain property of the employer. **5** Compliance with all regulations relating to the handling, transport and disposal of hazardous waste materials **C Minimum information that must be shown on the drawings that accompany this Section of Measurement.** **1** Scope and location of work relative to existing structures.	**M1** The rules within this Section apply to works to existing buildings as defined in clause 3.3.4.1 of the Measurement Rules for Building Works. **M2** Inserting new work includes re-fixing or re-using removed materials.	**D1** All materials arising from these works become the property of the Contractor unless otherwise stated	**C1** Details of all materials to remain the property of the employer including means of storage. **C2** All temporary works including Shoring and scaffolding incidental to the work excluding those listed below **C3** Making good all work disturbed. **C4** Extending and making good existing finishes **C5** Disposal of all waste materials	**S1** Specific location of each item of work relative to the existing building. **S2** Any restrictions on method or location of disposal of waste. **S3** Asbestos surveys. **S4** The rules within this section do not apply to temporary works except as those listed in Rule 24 of this section.

ALTERATIONS, REPAIRS AND CONSERVATION (cont'd)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C6 All work and materials incidental to the items of alteration. **C7** Materials required for bonding new work to existing.	
1 Works of alteration	item	**1** Dimensioned description sufficient to identify extent and location of work.	**1** Extent, nature and scope of work described including type and thickness of existing structure.				**C8** Details must be given of all work involved in each item including method of operation where not at the discretion of the contractor	

ALTERATIONS, REPAIRS AND CONSERVATION (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Removing	item m^2 m nr	**1** Fittings and fixtures **2** Plumbing items or installations. **3** Electrical items or installations. **4** Finishes. **5** Coverings. **6** Pavings.	**1** Details sufficient for identification stated **2** Approximate area or size of area of each type of finish, covering or paving				**C9** Disconnecting and, if required, subsequent re-connection of plumbing and electrical or other services installations is deemed included.	**S5** The surveyor shall choose the unit most suitable for the type of work being removed.
3 Cutting or forming openings **4** Cutting or forming recesses **5** Cutting back **6** Filling in openings **7** Filling in recesses	item m^2 m nr	**1** Dimensioned description **2** Type and thickness of existing structure. **3** Method of performing the work if not left to discretion of the contractor.	**1** Re-use of existing materials stated. **2** Type and size(s) of new materials stated.		**M3** Details given of new work are to be the equivalent of those details required by the rules for the measurement of the same in other Work Sections. The surveyor shall choose the unit most suitable for the type of work being cut or filled.			

ALTERATIONS, REPAIRS AND CONSERVATION

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Removing existing and replacing	m^2	**1** Thickness stated.	**1** Brickwork.	**1** Treatment of exposed sound surface(s) stated.	**M5** The unit of measurement shall be left to the discretion of the surveyor but shall reflect the size and extent of the work.	**D2** Making good is deemed to be to match existing unless described otherwise	**C10** Formwork and any other form of temporary support is deemed included.	
	m	**2** Width and thickness stated.	**2** Concrete.					
		3 Length, width and thickness stated.	**3** Stonework. **4** Timber.	**2** Treatment of exposed reinforcement or other material stated.				
	nr		**5** Glass.					
	item	**4** Dimensioned description	**6** Other, type stated.	**3** Making good with new materials other than to match existing to be described.				
				4 Bonding new to existing.				
9 Preparing existing structures for connection or attachment of new work	nr	**1** Dimensioned description	**1** Description sufficient to determine scope and location.	**1** Nature of existing structure to receive new work			**C11** This will include preparing structural steel sections for attachment to new steel framing and the like.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
10 Repairing	m^2	**1** Thickness stated.	**1** Nature of surface to be repaired stated.	**1** Method of repairing stated.	**M6** The unit of measurement shall be left to the discretion of the surveyor but shall reflect the size and extent of the work.			
	m	**2** Width and thickness stated						
	nr	**3** Length, width and thickness stated.						
	item	**4** Dimensioned description						
11 Repointing joints	m	**1** Nature of existing joint. **2** Type of pointing.	**1** Materials required stated	**1** Width and depth stated.			**C12** Removal of existing joint material and preparation of exposed surfaces deemed included.	
12 Repointing	m^2	**1** Thickness stated.	**1** Nature of existing surface.	**1** Composition and mix of mortar and/or other joint material(s).	**M7** Areas each less than 1m2 to be enumerated.		**C13** Linear items would include reveals, wall ends and the like.	
	m	**2** Width and thickness stated.	**2** Size of existing components.		**M8** The unit of measurement shall be left to the discretion of the surveyor but shall reflect the size and extent of the work.		**C14** Types of surface would include brickwork, blockwork, stonework and the like.	
	nr	**3** Length, width and thickness stated.	**3** Bond of existing joints.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Repointing (Cont'd)			**4** Type of new pointing. **5** Width and depth of raking out of existing joints					
13 Resin or cement impregnation/ injection	m^2	**1** Thickness stated.	**1** Method of impregnation or injection stated.	**1** Centres or spacings of drilling holes.	**M9** Overall removal of finishes prior to this work would be measured elsewhere.		**C15** Work is deemed to include making good holes and finishes on completion.	
	m	**2** Width and thickness stated.						
	nr	**3** Length, width and thickness stated.	**2** Nature of existing material.	**2** Localised removal of finishes.				
	item	**4** Dimensioned description	**3** Nature of existing finish where applicable. **4** Thickness or depth of treatment.		**M10** The unit of measurement shall be left to the discretion of the surveyor but shall reflect the size and extent of the work.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
14 Inserting new walls	m^2 nr	**1** Size and type of new ties	**1** Method of insertion. **2** Nature and thickness of outer skin. **3** Nature of existing finish where applicable.	**1** Centres or spacings of drilling holes. **2** Localised removal of finishes.	**M11** Overall removal of finishes prior to this work would be measure		**C16** Work is deemed to include making good holes and finishes on completion.	
15 Re-dressing existing flashings and the like.	m nr	**1** Girth and thickness stated. **2** Length, width and thickness stated.	**1** Dimensioned description of flashing. **2** Description of new profile.	**1** Raking out existing joint. **2** Repointing with new material: method and type of pointing material stated.	**M12** The girths and lengths stated are net. No allowance to be made for additional materials required for labours.		**C17** Removal, cleaning, re-shaping, trimming and re-fixing existing flashing is deemed included.	
16 Damp-proof course renewal **17** Damp-proof course insertion	m	**1** Method of renewal or insertion stated. **2** Nature and thickness of existing wall.	**1** Centres or spacings of drilling holes. **2** Localised removal of finishes.	**1** Chemical. **2** Injection mortar. **3** Electro osmosis. **4** Other mechanical methods.	**M13** Overall removal of finishes prior to this work would be measured elsewhere.		**C18** Work is deemed to include making good holes and finishes on completion.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Damp-proof course insertion		**3** Nature of existing finishes where applicable **4** Thickness or depth of treatment						
18 Cleaning surfaces	m^2	**1** Over 500mm wide.	**1** Nature of surface to be treated stated.	**1** Treatment material stated.	**M14** The unit of measurement shall be left to the discretion of the surveyor. **M15** Any repair and remedial works required to the surface prior to treatment to be measured separately.			
19 Removing stains	m	**2** Not exceeding 500mm wide. width stated.	**2** Required finished	**2** Method of treating stated				
20 Artificial weathering	nr	**3** Length and width stated.						
	item	**4** Dimensioned description.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
21 Renovating	m^2	**1** Thickness stated.	**1** Brickwork.	**1** Details and nature of the renovation stated.	**M16** The unit of measurement shall be left to the discretion of the surveyor but shall reflect the size and extent of the work.			
	m	**2** Width and thickness	**2** Concrete.	**2** Materials required stated.				
	nr	**3** Length, width and thickness stated.	**3** Stonework.	**3** Method of renovation stated where not at the discretion of the contractor.				
	item	**4** Dimensioned description	**4** Timber.					
			5 Other, type stated.					
22 Conserving	m^2	**1** Thickness stated.	**1** Brickwork.	**1** Details and nature of the renovation stated.	**M17** The unit of measurement shall be left to the discretion of the surveyor but shall reflect the size and extent of the			
	m	**2** Width and thickness stated	**2** Concrete.	**2** Materials required stated.				
	nr	**3** Length, width and thickness	**3** Stonework.	**3** Method of renovation stated where not at the discretion of the contractor.				
	item	**4** Dimensioned description	**4** Timber.					
			5 Other, type stated.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
23 Decontamination	item	**1** Removal of toxic/hazardous materials. **2** Decontamination of existing premises. **3** Infestation removal /eradication. **4** Fungus removal/ eradication.	**1** Scope and location of work. **2** Type of contamination /infestation/fungus to be treated .	**1** Preparatory works	**M18** (Excluding decontamination of existing ground: this work is measured in Work Section 1.5: Excavating and Filling.)	**D3** Sufficient information must be given to fully describe the cause of the contamination or infestation together with such information as is appropriate to allow the contractor to fully treat the condition described.		
24 Temporary works	m^2	**1** Roofs. **2** Screens. **3** Floors.	**1** Weatherproof. **2** Watertight. **3** Dustproof. **4** Fireproof. **5** Any other requirement: type stated	**1** Method of construction if not at the discretion of the contractor. **2** Maintaining: duration stated **3** Adapting during course of works **4** Clearing away.	**M19** In order to ensure the full extent and scope of this work the surveyor may need to provide additional information if not readily ascertained from the drawings.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Temporary works (Cont'd)				**5** Disposing of rainwater: details stated. **6** Providing openings: details stated				
25 Recycling	item	**1** Detailed description of type of material to be recycled and any limitations imposed by employer or local authority.	**1** To be collected by local authority. **2** To be transported to recycling depot, details and location stated.					

1.5 EXCAVATING AND FILLING

Site clearance/preparation Excavations
Disposal
Fillings
Membranes

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Site plan(s) showing all major excavations. **2** Locations of spoil heaps if not left to the discretion of the contractor. **3** Existing site survey **B Mandatory information to be provided.** **1** Ground water level(s) and date(s) established **2** Starting level of each type of excavation **3** Levels of rock where applicable **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Original and proposed ground levels **2** Any item(s) that must remain on site during the works and be protected from damage. **3** Any item(s) adjacent to site that may impact on the works **4** Details of trial pits or boreholes including their location. **5** Details of live over or underground services including their location. **6** Pile sizes and their locations where applicable	**M1** Location of works in accordance with clause 3.3.3.1 of the Measurement Rules for Building Works. **M2** Support to faces of excavation unless not at the discretion of the contractor **M3** Excavation and filling for temporary works unless not at the discretion of the contractor. **M4** Levelling, grading, trimming and compacting surfaces exposed by the site excavations.	**D1** Disposal of all surface water. **D2** Ground water levels subject to periodic changes due to tidal or similar effects are to be so described giving the mean high and low water levels. **D3** The ground water level must be re-established at the time each excavation is carried out and is defined as the post-contract ground water level	**C1** Nature of any known hazardous contamination on the site or in the ground including any restrictions on disposal of surface or ground water. **C2** Quantities relate only to the depths and dimensions shown on the drawings and overdig is not included. This applies to associated Extra over items and backfilling.	**S1** Date of existing site survey **S2** Ground conditions including anticipated stability of excavations. **S3** Working space. **S4** Curved work. **S5** Multiple handling of excavated materials on site unless specified.

EXCAVATING AND FILLING (cont'd)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
	M5 The quantities given for excavation and filling are the bulk before excavating or the net void to be filled. No allowance is made for subsequent variations to bulk or for the extra space taken up by working space or earthwork support unless the type of backfill is not left to the discretion of the contractor. This also applies to the quantities given for any subsequent Extra over items listed in Rule 7.		**C3** All excavated material is deemed to be inert unless described otherwise	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Preliminary sitework	item nr	**1** Locating underground services. **2** Trial pits to locate existing services or determine ground conditions.	**1** Maximum depth and type of service stated.	**1** Specified location(s) stated **2** Means of locating service if not left to discretion of contractor.				
	nr	**3** Boreholes to determine ground conditions.	**1** Diameter. **2** Maximum depth stated	**1** Destination of core samples stated **2** Type and extent of report required				
2 Removing trees **3** Removing tree stumps	nr	**1** Girth 500-1500mm **2** Girth 1,500-3,000mm **3** Girth over 3,000 mm, stated in 1,500mm stages.	**1** Filling material stated		**M6** Tree girths are measured at a height of 1.00 m above original ground level **M7** Stump girths are measured at the top		**C4** Removing trees is deemed to include removing the stump and roots unless otherwise stated.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Removing tree stumps (Cont'd)							**C5** This work is deemed to include: grubbing up roots, disposal off site of all material arising and filling void	
4 Site clearance	m^2	**1** Clearing site vegetation and other growth and dispose off site	**1** Description sufficient to identify scope and location of work				**C6** All growth includes trees and tree stumps less than 500mm girth, unless specifically designated to remain. **C7** The removal of invasive vegetation such as Japanese knotweed or the like should be specifically mentioned in the description	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Site preparation	m^2	**1** Lifting turf for Preservation thickness stated	**1** Method and location of preservation stated					
	m^2	**2** Remove topsoil: depth stated						
	m^2	**3** Remove hard surface paving: thickness stated	**1** Destination stated **2** Type of paving stated	**1** Method of breaking up if not left to discretion of contractor	**M8** Excludes any hardcore beds below the pavings. **M9** Removal of the hardcore is treated as reduced level excavation.			
	nr	**4** Remove specific items	**1** Description sufficient to identify size and location of each item			**D4** This excludes all but the simplest of building structures whose demolition is covered in Work Section 1.3: Demolitions.	**C8** Any existing items on site not specifically designated to remain including all types of rubbish such as abandoned cars, fridges and the like.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Site preparation (Cont'd)							**C9** Removal of any associated foundations, fixings, supports, fastenings and the like is deemed included.	
6 Excavation, commencing level stated if not original ground level	m^3	**1** Bulk excavation.	**1** Not exceeding 2m deep. **2** Over 2m not exceeding 4m deep. **3** And thereafter in stages of 2m.	**1** Details of obstructions in ground to be stated.	**M10** Bulk excavation includes excavating to reduce levels or to form basements, pools, ponds or the like. For clarity each type of excavation may be measured and described separately. **M11** Obstructions will be piles, manholes and the like that must remain undisturbed.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Excavation (Cont'd)	m^3	**2** Foundation excavation			**M12** Foundation excavation includes excavating for strip and pad foundations, pile caps and all other types of foundations			**S6** For clarity each type of excavation may be measured and described separately
7 Extra over any types of excavation irrespective of depth	m^3	**1** Excavating in.	**1** Hazardous material, details stated. **2** Non-hazardous material, details stated. **3** Below ground water level. **4** In running water. **5** Unstable ground.		**M13** Hazardous materials are any material that require special precautions taken when handling, transporting or disposing. **M14** Unstable ground is running silt,	**D5** Running water is a spring, stream or river.	**C10** Ground water is any water encountered below the established water table level. It does not include water from underground streams, broken drains or culverts or water arising from surface flooding	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Extra over any types of excavation irrespective of depth (Cont'd)	m^3	**2** Breaking up.	**1** Rock **2** Reinforced concrete **3** Concrete **4** Masonry or stonework		**M15** Breaking up hard surface pavings is measured in accordance with rule 5.3 of this Section.	**D6** Rock is any hard material which is of such size or location that it can only be removed by wedges, rock hammer, special plant or explosives. **D7** A boulder = $5m^3$ in volume or one that can be lifted out in the bucket of an excavator will not constitute rock.	**C11** Trimming exposed faces of rock is deemed included	**S7** Degraded or friable rock that can be scraped out by the excavator bucket does not constitute rock
	m	**3** Excavating alongside existing underground services	**1** Type, size and depth of service stated.	**1** Nature of precaution	**M16** These items are measured where there is a risk of the existing service being affected by the excavation process. The method of protection is left to the discretion of the contractor. **M17** If in doubt the surveyor must measure an item giving the nature of the live service			
	nr	**4** Excavating across existing underground services						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Support to face(s) of excavation where not at the discretion of the contractor	m^2	**1** Maximum depth stated.	**1** Location stated	**1** Method of forming support where not left to discretion of the contractor.	**M18** This work shall only be measured where it has been specifically specified in the contract documents or if the contractor has been instructed by the contract administrator to provide the support during the course of the works.			
9 Disposal	item	**1** Ground water.	**1** Depth below original ground level stated **2** Polluted water described if known.	**1** If the post-contract water level differs from the pre-contract level the measurements must be revised	**M19** The method and place of discharge left to the discretion of the contractor unless stated otherwise			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Disposal (Cont'd)		**2** Excavated material off site	**1** Destination if not at the contractor's discretion **2** Hazardous material. **3** Non-hazardous material where it requires to be disposed to a specific location		**M20** Irrespective of where excavated material originates			
10 Retaining excavated	m^3	**1** Top soil. **2** All other excavated material.	**1** To temporary spoil heaps. **2** Average distance to spoil heap stated	**1** Specified handling: details stated		**D8** If the distance to spoil heap is not stated the location is left to the discretion of the contractor		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
11 Filling obtained from excavated material	m^2	**1** Final thickness of filling not exceeding 500mm deep, finished thickness stated **2** Final thickness of filling exceeding 500mm deep	**1** Source, distance, destination and method stated **2** Maximum or average depth of layers stated	**1** Treatment of material prior to depositing in final location	**M21** The thickness stated will be that after compaction **M22** Destinations will comprise general areas to make up levels, backfilling foundations, landscaping areas, planter beds and the like	**D9** Source will be either direct from excavations or from temporary spoil heaps	**C12** This includes topsoil and any other material arising from the excavations that have been specified to remain on site. **C13** Compacting layers and surfaces are deemed included irrespective of depth and number of layers.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
12 Imported filling	m^3	**1** Blinding bed not exceeding 50mm thick, finished thickness stated	**1** Level, to falls, cross falls or cambers, stated. **2** Sloping not exceeding 15° from horizontal.	**1** Destination stated. Maximum or average depth of layers stated.	**M23** The thickness stated will be that after compaction		**C14** All types of surface treatments are deemed included. **C15** Compacting layers and surfaces are deemed included irrespective of depth and number of layers.	
	m^3	**2** Beds over 50mm thick but not exceeding 500mm deep, finished thickness stated.						
	m^3	**3** Beds exceeding 500mm deep.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
13 Geotextile fabric **14** Radon barrier **15** Methane barrier **16** Damp proof membrane **17** Ground movement protection boards **18** Any other fabric, membrane or board: type stated **19** Ground stabilisation meshes and the like, type stated	m	**1** Not exceeding 500mm wide, thickness or gauge stated	**1** Horizontal. **2** Sloping. **3** Vertical	**1** Protective fleeces or boards, type stated. **2** Method of anchoring stated			**C16** All turn-ups, turndowns, laps and joints deemed included. **C17** Forming holes deemed included	
	m^2	Over 500mm wide, thickness or gauge stated						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
20 Cutting off tops of piles irrespective of length	nr	**1** Size stated					**C18** Cutting off tops of piles is deemed to include preparation and integration of reinforcement into pile cap or ground beam and disposal of all debris.	

1.6 GROUND REMEDIATION AND SOIL STABILISATION

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this Section Of Measurement **1** Site plan(s) showing location of work(s). **B Mandatory information to be provided.** **1** Ground conditions. **2** Ground water level(s) and date(s) established **3** Starting level of each type of excavation **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Original and proposed ground levels **2** Any item(s) that must remain on site during the works and be protected from damage. **3** Any item(s) adjacent to site that may impact on the works **4** Details of trial pits or boreholes including their location. **5** Details of live over or underground services including their location. **6** Pile sizes and their locations where applicable			The following are deemed to be included: **C1** Disposal off site of all surplus excavated material. **C2** Disposal of all surface water. **C3** Support to faces of excavation unless not at the discretion of the contractor. **C4** Working space **C5** Excavation and filling for temporary works unless not at the discretion of the contractor.	**S1** The ground water level must be re-established at the time each excavation is carried out and is defined as the post-contract ground water level. **S2** Ground water levels subject to periodic changes due to tidal or similar effects are to be so described giving the mean high and low water levels. **S3** Nature of any known non-hazardous or hazardous contamination on the site or in the ground.

1.6 GROUND REMEDIATION AND SOIL STABILISATION (cont'd)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C6 Levelling, grading, trimming and compacting surfaces exposed by the excavations **C7** Curved work. **C8** Multiple handling of excavated materials on site unless specified.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Site dewatering	item	**1** Area of site to be dewatered. **2** Maximum depth of	**1** Method of disposal of water stated if not at discretion of contractor.	**1** Pre-contract water level. **2** Level to which ground water must be lowered to and maintained at.	**M1** Each type of remedial work shall be accompanied by a full description of the proposed works including limits on the extent, the proximity of adjoining building, restrictions on method, sequence and timing.			

GROUND REMEDIATION AND SOIL STABILISATION (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Sterilisation	m^3	**1** Maximum depth of ground to be treated	**1** Method of sterilisation stated.					
3 Chemical neutralising	m^3	**1** Maximum depth of ground to be treated	**1** Method of neutralisation stated.					
4 Freezing	m^3	**1** Maximum depth of ground to be treated	**1** Method of freezing stated. **2** Duration of freezing stated if not left to discretion of contractor	**1** Duration may be stated as a period of time or to a point within the contract programme such as 'completion of foundation				
5 Ground gas venting	m^2	**1** Type of gas to be vented.	**1** Method of collection stated **2** Method of disposal stated.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
6 Soil nailing	m^2	**1** Area of site to be treated	**1** Length and diameter of nails, details stated **2** Spacing of nails. **3** Method of grouting.			**D1** Soil nailing is a technique used for constructing excavations and stabilising slopes by reinforcing the ground insitu with relatively small, fully bonded inclusions, usually steel bars		
7 Ground anchors	nr	**1** Diameter and length of borehole.	**1** Details stated.					
8 Pressure grouting/ground permeation	m^2	**1** Area and depth of site to be treated.	**1** Details stated.					
9 Compacting	m^2	**1** Area of site to be treated. **2** Weight of compactor	**1** Details stated.					
10 Stabilising soil insitu by incorporating cement with a rotavator	m^2	**1** Area of site to be treated. **2** Grams/m^2 of cement.	**1** Details stated.					

1.7 PILING

Interlocking
Bored Piling
Driven
Piling
Vibro-Compacted Stone Piling

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Site plan showing site boundary and any adjacent buildings or features that might affect or be affected by the piling. **2** General piling layout. **B Mandatory information to be provided.** **1** Types of piles. **2** Ground water level(s) and date(s) established **3** Nature of the ground. **4** Commencing levels from which work is expected to begin. **5** Details of compaction. **6** Limitations on headroom. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Positions of piles. **2** Positions of existing buildings adjacent to the site. **3** Position of existing services.			The following are deemed to be **C1** Features that might affect piling include rivers, canals, tidal waters, flood areas and the like sufficiently close to the works that their situation may affect the piling process. **C2** Temporary containment of spoil. **C3** Any concrete placed in excess of the designed completed length **C4** Backfilling empty bores.	**S1** Kind and quality of materials. **S2** Types of tests. **S3** Type of grout. **S4** Irregular ground levels must be stated.

PILING (cont'd)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C5 All pre-boring. **C6** Re-positioning piling plant during the works **C7** Maintaining all piling plant.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Interlocking sheet piles	m^2	**1** Total driven area, maximum length stated	**1** Section modulus and cross section size, or section reference stated	**1** Removal to be stated if not left to discretion of contractor.			**C8** The cost of extraction is	
2 Bored piles **3** Driven piles **4** Others, type stated	m	**1** Type stated. **2** Nominal size or diameter stated. **3** Total bored or driven length: maximum length stated. **4** Total concreted length. **5** Total number	**1** Contiguous/secant piling shall be so described. **2** Permanent casings shall be stated. **3** Raking, inclination stated	**1** Reinforcement to precast concrete piles.	**M1** Lengths are measured along the axes of the piles from the commencing level to the bottom of the pile.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Vibro-compacted piles	m	**1** Nominal diameter. **2** Total number.						
6 Vibro-compacted trench fill	m	**1** Nominal width and depth stated						
7 Extra over piling	nr	**1** Enlarged bases. **2** Enlarged heads.	**1** Type of piling stated. **2** Diameter or size of enlargement stated.				**C9** This work is deemed to include everything necessary to form the enlargement including disposal of additional spoil.	
	m	**3** Pile extensions.	**1** Total number stated. **2** Total concreted length.				**10** Preparing head of pile to receive extension is deemed included.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Reinforcement to Insitu concrete piles	t	**1** Nominal size and type of bars stated.					**C11** Reinforcement is deemed to include tying wire and spacers plus links and binders that are incorporated at the discretion of the contractor. **C12** Types of bars include plain, deformed and helical.	
9 Breaking through obstructions	hr	**1** Rig standing.			**M2** Only measured where the obstruction is encountered above the founding stratum of the pile.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
10 Disposal of excavated materials	m^3	**1** Off site. **2** On site.	**1** Hazardous material. **2** Non-hazardous material where it requires to be disposed to a specific location. **3** Destination of spoil if not at the contractor's discretion.		**M3** The volume calculated is the nominal cross section by the pile lengths		**C13** The volume of enlarged heads and bases is to be included.	
11 Delays	hr	**1** Rig standing			**M4** Delays are measured only where specifically instructed		**C14** Delays are deemed to include associated labour, plants and overhead	
12 Test	nr	**1** Details stated	**1** Timing stated					

1.8 UNDERPINNING

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Location drawings. **2** Detailed section(s). **B Mandatory information to be provided.** **1** Limit of length of work to be carried out in one operation. **2** Maximum number of sections to be carried out at any one time. **3** Ground conditions including anticipated stability of excavations **4** Ground water level(s) and date(s) established **5** starting level of each type of excavation **Minimum information that must be shown on the drawings that accompany this section ofmeasurement.** **1** Extent and method of work. **2** Details of existing structure to be underpinned.	**M1** If the work of underpinning is extensive the various elements of this work may be measured separately in accordance with the rules of the relevant trades or Work Sections, stating that the work is 'in underpinning'.		The following are deemed to be included: **C1** Temporary support of existing structures. **C2** Excavation and disposal. **C3** Earthwork support. **C4** Preliminary trenching. **C5** All working space. **C6** Disposal of ground and surface water. **C7** Cutting awa y existing foundations and footings and disposal. **C8** Preparing the underside of existing work. **C9** Backfilling. **C10** Surface	**S1** Nature of any known hazardous contamination on the site or in the ground.

UNDERPINNING

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C11 All new work associated with the process of underpinning. **C12** All making good if specified.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Underpinning	m m nr	**1** Foundations. **2** Walls. **3** Bases	**1** Brief description of work stating depth, maximum width and method of underpinning. **2** Curved work	**1** Method, if not left to discretion of contractor, may be given by reference to drawing(s).				
2 Concrete	m^3				**M2** Measured in accordance with the rules of the appropriate trade or Work Section stating that the work is 'in underpinning'.			
3 Formwork	m^2							
4 Reinforcement	t							
5 Brickwork or blockwork	m^2							
6 Tanking	m^2							

1.9 DIAPHRAGM WALLS AND EMBEDDED RETAINING WALLS

<table>
<tr><th colspan="5">INFORMATION PROVIDED</th><th>MEASUREMENT RULES</th><th>DEFINITION RULES</th><th>COVERAGE RULES</th><th>SUPPLEMENTARY INFORMATION</th></tr>
<tr><td colspan="5">A Drawings that must accompany this Section of Measurement

1 Location drawings.
2 Site plans showing site boundary and any adjacent buildings or features that might affect or be affected by the construction of the diaphragm walls.

B Mandatory information to be provided.
1 Ground conditions including anticipated stability of excavations.
2 Ground water level(s) and date(s) established
3 Limit of length of work to be carried out in one operation.
4 Maximum number of sections to be carried out at any one time.
5 Starting level of each type of excavation

C Minimum information that must be shown on the drawings that accompany this section of measurement.
1 Extent of work.</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">The following are deemed to be included:

C1 Excavation and disposal.
C2 Earthwork support.
C3 Preliminary trenching.
C4 All working space.
C5 Disposal of ground and surface water.
C6 Backfilling.
C7 Surface treatments.</td><td rowspan="2">S1 Nature of any known hazardous contamination on the site or in the ground.

S2 Irregular ground shall be so described.</td></tr>
<tr><td>Item or work to be measured</td><td>Unit</td><td>First Division</td><td>Second Division</td><td>Third Division</td></tr>
<tr><td>1 Walls, thickness stated</td><td>m^2</td><td>1 Commencing levels of excavation.
2 Maximum depth of
3 Finished top level of concrete if different from commencing level of excavation.</td><td>1 Details and method of construction stated.
2 Type of concrete.
3 Details of reinforcement.</td><td>1 Details of support fluid.</td><td></td><td></td><td></td><td></td></tr>
</table>

DIAPHRAGM WALLS AND EMBEDDED RETAINING WALLS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Extra over excavation and disposal	m^3	**1** Breaking out hard materials. **2** Excavating in hazardous material.						
	m^2	Breaking up hard surface pavings, thickness stated.						
3 Joints	m	**1** Dimensioned description	**1** Method of forming.	**1** Vertical. **2** Horizontal. **3** Raking. **4** Curved: radius stated.				
4 Trimming and cleaning exposed faces	m^2	**1** Details, stated						
5 Delays	hr	**1** Authorised standing time.			**M1** Delays are measured only where specifically instructed		**C8** Delays are deemed to include associated labour, plants and overhead	
6 Tests	item	**1** Details stated						

1.10 CRIB WALLS, GABION AND REINFORCED EARTH

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Plans showing scope and location of each type of work. **B Mandatory information to be provided.** **1** Kind and quality of materials **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Original ground levels. **2** Finished ground levels. **3** Condition of ground.		**D1** Final excavation means any minor trimming of earth surfaces required during each installation. All reduced level and foundation excavations, foundations and backfill are measured in accordance with the rules of Work Section 5: Excavating and filling	The following are deemed to be included: **C1** Final excavation associated with each installation. **C2** Disposal of any excavated material including hazardous material. **C3** Earthwork support. **C4** Preparing surfaces to receive each installation. **C5** Disposal of surface water.	**S1** Kind and quality of materials

CRIB WALLS, GABION AND REINFORCED EARTH (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Crib walls	m^2	**1** Thickness stated	**1** Vertical **2** Battering **3** Curved on plan: radius stated		**M1** No deductions made for voids = $1.00m^2$	**D2** Area is measured on front face.	**C6** Dowels, pins, granular infill, compacting fill, special units, providing manufacturers certificates, building in pipes and forming openings = $1.00m^2$ deemed included.	
2 Extra for	m	**1** Ends. **2** Corners irrespective of angle						

CRIB WALLS, GABION AND REINFORCED EARTH (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
3 Gabion baskets walls	m^2	**1** Basket size stated. **2** Gauge of basket wire stated. **3** Width of the base of the wall.	**1** Vertical **2** Battering: rate stated. **3** Sloping **4** Curved on plan: radius stated **5** Stepped on face	**1** Type of fill to baskets described. **2** Specified treatment of basket fill described **3** Specified handling details stated	**M2** Area is measured on front face	**D3** Filling is deemed to include compaction and overfilling of fill material.	**C7** Gabion baskets are deemed to include assembling, tying, fixing, bracing and tying lids	
4 Earth reinforcement	m^2	**1** Mesh **2** Fabric	**1** Horizontal **2** Sloping **3** Curved on plan : radius stated	**1** Method of anchoring **2** Minimum laps stated **3** Final facing, details stated	**M2** Area measured in contact with base and excludes laps **M3** No deductions made for voids = $1.00m^2$		**C4** Assembling, tying, fixing, stacking and tensioning deemed included.	

1.11 INSITU CONCRETE WORKS

Insitu Concrete
Surface finishes to Insitu Concrete
Formwork
Reinforcement
Designed Joints in Insitu Concrete
Accessories Cast in to Insitu Concrete
Insitu Concrete Sundries

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. 1 General arrangement drawings **B Mandatory information to be provided.** **1** Kind, quality and size of materials **2** Details of tests of materials **3** Details of tests of finished work **4** Limitations on method sequence, speed or size of pouring **5** Method of compaction **6** Method of curing **7** Details of watertightness **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Relative position of all members. **2** The size of members. **3** The thickness of slabs. **4** The permissible loads in relation to casting times.	**M1** Deductions are not made for reinforcement, steel sections, cast in accessories, voids = $0.05m^3$ except voids in troughed and coffered slabs.		**C1** Concrete is deemed finished as struck from basic finish formwork. **C2** No allowance in volume to be made for deflection of formwork. **C3** Concrete is deemed cast into formwork unless otherwise described. **C4** All top surfaces and soffits are deemed to finish horizontal unless otherwise stated.	**S1** Work in substructures, superstructures or external works to be stated in headings or descriptions. **S2** Watertight work shall be so described. **S3** This applies to concrete laid on ribbed metal decking as well as other types of formwork.

INSITU CONCRETE WORKS (cont'd)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C5 All top surfaces are deemed finished tamped. **C6** Concrete volume is measured net.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
Plain Insitu concrete Reinforced Insitu concrete Fibre reinforced Insitu concrete Sprayed Insitu concrete								
1 Mass concrete	m^3	**1** Any thickness	**1** In filling voids. **2** In trench filling. **3** In any other situation: details stated.	**1** Poured on or against earth or unblinded hardcore.	**M2** The volumes of each type of mass concrete work may be aggregated or given separately.	**D1** Mass concrete is any unreinforced bulk concrete not measured elsewhere.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Horizontal work	m^3	**1** = 300 thick. **2** > 300 thick.	**1** In blinding. **2** In structures	**1** Poured on or against earth or unblinded hardcore. **2** Reinforced = 5%	**M3** The volumes of each type of horizontal work may be aggregated or given separately. **M4** Work laid in bays shall be so described giving average area of bays.		**C7** Horizontal work includes blinding, beds, foundations, pile caps, column bases, ground beams, slabs, coffered and troughed slabs, landings, beams, attached beams, beam casings, shear heads, upstands whose height is = than three times their width, kerbs, copings	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
3 Sloping work $< 15^0$ **4** Sloping work = 15^0	m^3	**1** = 300 thick. **2** > 300 thick.	**1** In blinding. **2** In structures **3** In staircases.	**1** Poured on or against earth or unblinded hardcore. **2** Reinforced = 5%	**M5** The volumes of each type of sloping work may be aggregated or given separately **M6** Work laid in bays shall be so described giving average area of bays.		**C8** Sloping work includes blinding, beds, slabs, steps and staircases, kerbs, copings. **C9** Includes any attached beams, upstands, shear heads or similar.	
5 Vertical work	m^3	**1** = 300 thick. **2** > 300 thick.	**1** In structures	**1** Reinforced = 5%	**M7** The volumes of each type of vertical work may be aggregated or given separately		**C10** Vertical work includes columns, attached columns, column casings, walls, retaining walls, filling to hollow walls, parapets or upstand beams where height is greater than three times the width.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
6 Sundry Insitu concrete work	m m^3	**1** Work = 300 wide or thick. **2** Work > 300 wide or thick.	**1** Horizontal. **2** Sloping. **3** Vertical	Reinforced = 5%			**C11** Includes work such as backsills, machine and plant bases and the like.	
7 Sprayed Insitu concrete	m^2	**1** Thickness stated.	**1** Slabs. **2** Walls. **3** Beams. **4** Columns	**1** Tops. **2** Soffits. **3** Curved. **1** Curved.	**M8** The method of application and finish to be stated in the description			
Surface finishes to Insitu concrete								
8 Trowelling **9** Power floating **10** Hacking **11** Grinding **12** Any other surface treatment not left to discretion of the contractor	m^2	**1** To top surfaces. **2** To faces. **3** To soffits	**1** Sloping. **2** Falls. **3** Crossfalls	**1** Application of surface hardeners, sealers, dust proofers, waterproofers, carborundum grains or the like shall be so described.				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Formwork		**1** Plain formwork **2** Special finish formwork.			**M9** Curved work shall be described stating the radii. **M10** No deductions shall be made for voids= 5.00m^2 **M11** Top formwork is measured for sloping surfaces that are > 15° or where otherwise specifically required	**D2** Permanent formwork or formwork left in shall be so described.	**C12** All kickers except to walls shall be deemed included. **C13** All square, raking and curved cutting deemed included. **C14** All holes, boxings, recesses, rebates, chamfers, nibs, channels and the like are deemed included.	**S4** Plain finish shall be left to discretion of the contractor.
13 Sides of foundations and bases **14** Edges of horizontal	m m^2	**1** = 500 high: width stated. **2** > 500 high.						
15 Soffits of horizontal work **16** Soffits of troughed or waffled horizontal work, details described	m^2	**1** For concrete = 300 thick. **2** For concrete 300 to 450 thick. **3** For concrete > 450 thick.	**1** Propping = 3m high. **2** Propping over 3m but not exceeding 4.5m high. **3** And thereafter in 1.5m stages				**C16** Includes suspended slabs and stair landings.	**S6** Through propping to be described if not left to discretion of contractor.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Formwork (Cont'd)								
17 Sides and soffits of isolated beams	m^2	**1** Regular: shape stated.				**D4** Irregular shape is any shape other than square or rectangular	**C17** Includes concrete casings to steel beams and columns.	**S7** Shape is deemed regular unless described as otherwise.
18 Sides and soffits of attached beams		**2** Irregular shaped, dimensioned description or diagram.						
19 Sides of upstand beams								
20 Sides of isolated columns, nr stated								
21 Sides of attached columns								
22 Faces of walls and other vertical work	m^2	**1** Vertical. **2** Battered one face. **3** Battered both faces.	**1** Rate of batter to be stated					**S8** Work to single sides shall be so described.
23 Extra over	nr	**1** Openings for doors or the like: thickness of wall stated.	**1** = $5.00m^{2.}$ **2** $5.00m^2$ to $10.00m^2$. **3** > $10.00m^2$				**C18** All additional labour and material needed to form the opening is deemed included.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
24 Wall ends, soffits and steps in walls	m m^2	**1** = 500 wide: width stated. **2** > 500 wide.					**C19** Excludes ends and soffits of walls created by the formation of an opening. These are deemed included in the item for forming the opening.	
25 Soffits of sloping work	m^2	**1** Sloping one way. **2** Sloping two ways.					**C20** This includes work to soffits of slabs, ramps, steps, staircases and the like	
26 Staircase strings and the	m	**1** Maximum width stated						
27 Staircase risers and the like	m	**1** Vertical: width stated. **2** Undercut: width stated.						
28 Sloping top surfaces	m^2	**1** = 15°. **2** >15°.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Formwork (cont'd) **29** Steps in top surfaces **30** Steps in soffits	m	**1** = 500 high: width stated. **2** > 500 high.						
31 Complex shapes	nr	**1** Dimensioned description or diagram.	**1** Propping = 3m high. **2** Propping over 3m but not exceeding 4.5m high. **3** And thereafter in 1.5m stages.					
32 Wall kickers	m	**1** Plain. **2** Suspended.			**M12** Formwork to wall kickers is measured along the centre line of the wall and is deemed to include both sides			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Reinforcement								
33 Mild steel bars **34** High yield steel bars	t	**1** Nominal size stated	**1** Straight. **2** Bent. **3** Curved. **4** Links.	**1** Bars exceeding 12m long: length stated. **2** Deformed. **3** Bending restrictions			**C21** Forming hooks, tying wire, spacers, cutting, and bending is deemed included. **C22** Chairs and connectors are deemed included unless not at discretion of contractor.	
35 Accessories not at the discretion of the contractor	nr	**1** Nominal size stated	**1** Chairs or stools **2** Connectors.					
Reinforcement (Cont'd)								
36 Pre/Post-tensioned members	nr	**1** Dimensioned description. **2** Nominal size stated **3** Method of tensioning stated.	**1** Composite construction described.	**1** Sleeves, tendons, fittings and grouting described.	**M13** Post-tensioning is measured by the number of tendons in identical members.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
37 Mesh	m^2	**1** Weight per m^2 stated. **2** Fabric reference stated **3** Minimum laps stated.	**1** Bent. **2** Strips in one width, width stated.		**M14** Voids = 1.00m2 in area not deducted.		**C23** Laps, tying wires, all cutting, bending, spacers, stools, chairs and other supports deemed included. **C24** Bent fabric is deemed to include fabric that is wrapped around steel members.	
Designed joints in Insitu concrete					**M15** Details of primers, cleaners, fillers, waterstops, backing strips, reinforcement, ties, sealants, the method of application, preparation and the like shall be stated in the description			**S9** Joints located at the discretion of the contractor are not measured.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
38 Plain **39** Formed **40** Cut	m	**1** Dimensioned description. **2** total depth stated.	**1** Horizontal. **2** Vertical. **3** Curved, radius stated			**D5** Plain joints are those that do not require formwork	**C25** Formed joints are deemed to include formwork. **C26** All preparation, cleaners, primers and sealers are deemed included.	
Accessories cast into Insitu concrete								
41 Type or proprietary reference stated	m2 - - - m - - - nr	**1** Dimensioned description.	**1** If linear or superficial quantities are used the description must include any appropriate spacing dimensions.				**C27** Cast-in accessories include anchor bolts, anchor boxes, fixing bolts, dowels, column guards, isolated glass blocks and any other ancillary item that is specified to be cast in as the concrete work proceeds.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C28 Cast-in accessories exclude, tying wire, distance blocks, spacers, chairs, structural steel members, hollow blocks, filler blocks, void formers, permanent formwork, joints, all components around which concrete is cast but which are not fixed in position by the contractor **C29** Formwork or other temporary means of support to exposed edges and the like is deemed included	
Insitu concrete sundries								
42 Grouting	nr	**1** Dimensioned description	**1** Stanchion bases. **2** Grillages.					
43 Filling mortices or holes	nr							
44 Filling chases	m							

1.12 PRECAST/COMPOSITE CONCRETE

Precast/Composite Concrete Walls, Partitions and Panels

Precast/Composite Concrete Decking and Flooring

Other Precast/Composite Concrete Work

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. 1 General arrangement drawings **2** Specific drawings or details relating to precast works. **B Mandatory information to be provided. 1** Sizes and spacing of planks and blocks **2** Method of compaction and curing **3** Bedding and fixing. **4** Surface finishes **5** Kind and quality of reinforcement or pre/post-tensioning, spacing and stresses **6** Finish of exposed surfaces. **7** Details of tests of materials. **8** Details of tests of finished work **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** The relative position of concrete members. **2** Size of members. **3** The thickness of slabs and panels. **4** The permissible loads.			The following works and materials are deemed included: **a** Moulds and formwork. **b** Reinforcement. **c** Bedding. **d** Fixings. **e** Temporar y support. **f** Cast-in accessories. **g** Pre-tensioning or pre-stressing. **h** Filled ends. **j** All grouting. **k** Margins = 500mm wide	**S1** Kind and quality of materials. **S2** Kind, quality and mix of concrete.

PRECAST/COMPOSITE CONCRETE

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Composite concrete work	m^2	**1** Nature of work described. **2** Composition of work described. **3** Thickness stated.	**1** Horizontal. **2** Sloping = 15° **3** Sloping > 15°. **4** Vertical.	**1** Reinforcement: details stated. **2** Post-tensioning: number of tendons and details stated. **3** Cast-in accessories: details stated.	**M1** Margins greater than 500mm wide are measured as ordinary Insitu concrete slabs. **M2** No deduction is made for voids = 1.00m^2.	**D1** The thickness stated is the combined thickness of both precast and in-situ work.		**S3** Will apply to panels, slabs, walls, partitions, decking.
2 Designed joints	m	**1** Dimensioned description.	**1** Type and sizes of filling and sealant stated.					**S4** Joints formed at the discretion of the contractor are not measured.
3 Holding down or tie straps	nr	**1** Dimensioned description or proprietary reference number.	**1** Material type stated. **2** Protective coating stated	**1** Method of fixing.				

1.13 PRECAST CONCRETE

Precast Concrete Frame Structures
Precast Concrete Sills, Lintels, Copings and Other Units
Precast Concrete Panel Cladding
Precast Concrete Slabs
Precast Concrete Walls and Partitions
Precast Concrete Decking
Precast Concrete Rooflights and Pavement Lights
Precast Composite Concrete Work

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** General arrangement drawings **2** Specific drawings or details relating to precast works. **B Mandatory information to be provided.** **1** Method of compaction and curing **2** Bedding and fixing **3** Surface finishes **4** Kind and quality of reinforcement or pre/post-tensioning, spacing and stresses **5** Finish of exposed surfaces. **6** Details of tests of materials. **7** Details of tests of finished work **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** The relative position of concrete members. **2** The thickness or size of members. **3** The thickness of slabs and panels. **4** The permissible loads. **5** Full details of anchorages, ducts, sheathing and vents.			**C1** The following works and materials are deemed included: **(a)** Moulds and formwork. **(b)** Reinforcement. **(c)** Bedding. **(d)** Fixings. **(e)** Temporary support. **(f)** Cast-in accessories. **(g)** Pre-tensioning or pre-stressing. **(h)** Filled ends. **(j)** All grouting. **(k)** Margins $\leq$ 500 wide **(l)** Angles and fair ends.	**S1** Kind and quality of materials.

PRECAST CONCRETE

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Precast concrete goods	nr	**1** Dimensioned description or dimensioned diagram.	**1** Reinforcement: details stated. **2** Post-tensioning: number of tendons and details stated. **3** Cast-in accessories: details stated.	**1** Stoolings, number stated.				**S2** Will apply to sills, lintels, padstones, staircases, landings, panels, partitions, columns, beams, structural frames and other precast features
	m	**2** Dimensioned description stating number of pieces.	**4** Sloping not exceeding 15°. **5** Sloping exceeding 15°.					**S3** Will apply to frame members, copings and the like.
	m^2	**3** Dimensioned description stating thickness.		**2** Span where relevant.				**S4** Will apply to panels, slabs, walls, partitions, decking and the like.

PRECAST CONCRETE

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Rooflights **3** Pavement lights **4** Vertical panel lights	m^2	**1** Dimensioned description, Number stated	**1** Sizes and extent of reinforcement stated.		**M1** Isolated glass lenses are measured as accessories cast into in situ concrete.		**C2** Roof, pavement and panel lights are deemed to include moulds, formwork, temporary propping, reinforcement, bedding and glass lenses.	
	nr	**2** Dimensioned description.						
5 Designed joints	m	**1** Dimensioned description.	**1** Type and sizes of filling and sealant stated.					**S5** Joints formed at the discretion of the contractor are not measured.
6 Holding down or tie straps	nr	**1** Dimensioned description or proprietary reference number	**1** Material type stated. **2** Protective coating stated	**1** Method of fixing.				

1.14 MASONRY

Brick/Block Walling, Glass Block Walling, Natural Stone Rubble Walling and Dressings, Natural Stone Ashlar Walling And Dressings

Artificial/Cast Stone Walling and Dressings

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Plans of each floor level **2** Principal sections showing floor to floor height **3** External elevations **4** Any other major masonry work **B Mandatory information to be provided** **1** Bonding to other work **2** Raking or curved work so described **3** Radius of curved work **4** Type of finish/facing to each side **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Major horizontal and vertical dimensions **2** Types of materials used to construct the walls and other structures. **3** Position of main structural frame members.	**M1** All walling is measured on the centre line irrespective of construction. **M2** No deductions will be made for voids or built in items whose cross sectional area is less than $0.50m^2$. **M3** Deductions for string courses, lintels, sills, plates, and the like, are measured as regards height to the extent only of full brick or block courses displaced and as regards depth to the extent of only of full half brick beds displaced	**D1** Thicknesses stated are nominal **D2** Face is any work in brick or block finished fair **D3** Work is deemed vertical unless otherwise described **D4** Walls include skins of hollow walls	The following are deemed included in brick and blockwork: **C1** All rough and fair cutting **C2** All ends and angles, either formed or proprietary **C3** Extra materials for curved work **C4** Forming rough and fair grooves, throats, mortices, chases, rebates, and holes, stops, mitres and the like labours. **C5** Raking out joints to form a key **C6** Raking out joints to insert flashings.	**S1** Kind, quality and size of brick/block/stone units. **S2** Type of finish/facing to each side. **S3** Type of bond **S4** Composition and mix of mortar **S5** Type of pointing **S6** Method of cutting where not at the discretion of the Contractor

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
					M4 All wall dimensions exclude applied finishes		**C7** Labour in eaves filling **C8** Labour in returns, ends and angles. **C9** Centering **C10** Overhead work **C11** Bonding ends of walls to other work **C12** All extra materials required for bonding **C13** Additional material in laps **C14** Preparation of all surfaces to receive subsequent applications.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Walls; overall thickness stated **2** Diaphragm walls; overall thickness stated, spacing and thickness of ribs stated	m^2	**1** Brickwork **2** Blockwork **4** Natural stone **5** Cast stone **6** Other: type stated	**1** Skins of hollow walls **2** Battered **3** Tapered; one side or both sides **4** Built against other work	**1** Method of forming	**M5** The description shall describe the type of construction of the masonry such as rubble or ashlar work and height of coursing.	**D5** Battering walls are sloping walls with parallel sides **D6** Thickness stated for tapering walls is mean	**C15** Brickwork and blockwork bonded to another material is deemed to include extra material for bonding	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
			5 Used as formwork		**M6** Walls are measured on the centre line of the material unless otherwise stated. **M7** The radius of curved work is taken from the centre line. **M8** No deductions to be made for voids or built in items whose cross sectional area equals to or less than 0.50m^2	**D7** Thickness stated for diaphragm walls is total thickness of both skins and cavity void.		
3 Vaulting; thickness and type stated	m^2							
4 Isolated piers; isolated casing; chimney	m	**1** Dimensioned description or dimensioned	**1** Vertical **2** Battered **3** Curved: radius stated		**M9** No deductions to be made for flues and the like equal to or less than 0.10m^2 in cross sectional area	**D8** These are such when their length on plan is equal to or less than four times their thickness except where caused by opening.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Attached Projections	m		**1** Vertical **2** Raking **3** Horizontal **4** Curved: radius stated			**D9** Attached projections are attached piers whose length on plan is equal to or less than four times their thickness plus plinths, oversailing courses and the like		
6 Arches (number	m	**1** Height of face, width of soffit and shape of arch stated.				**D10** The length is the mean girth or length on face		
7 Bands; dimensioned description	m	**1** Flush **2** Sunk depth of set back stated **3** Projecting depth of set forward stated	**1** Vertical **2** Raking **3** Horizontal **4** Curved: radius stated	**1** Entirely of stretchers **2** Entirely of headers **3** Alternate header/ **4** Other bond		**D11** Bands are brick on edge bands, brick-on-end bands, basket pattern bands, moulded or splayed cappings, moulded string courses, moulded cornices and the like.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Flues **9** Flue linings **10** Filling around flues	m	**1** Dimensioned description	**1** Method of forming.					
11 Extra over walls for perimeters and abutments, details stated	m	**1** Dimensioned description.	**1** Method of forming. **2** Closing cavities, additional ties, insulation and all other associated work is deemed included.			**D12** This will include work forming eaves, copings, kerbs, quoins, ends and the like.		
12 Extra over walls for opening perimeters, details stated	m					**D13** This will include work forming sills, jambs, reveals, cavity closers, architraves, lintels, mullions, transoms, thresholds, steps and the like.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
13 Special purpose blocks	nr	**1** Dimensioned description	**1** Function stated		**M10** Descriptions of stones are given as the smallest block from which each item can be obtained having regard in the case of natural stone to the plane in which the stone is required to be laid with relation to its quarry bed. The dimensions are taken over one mortar bed and one mortar joint			
14 Forming cavity	m^2	**1** Width and method of forming	**1** Type and spacing of ties.					
15 Cavity insulation	m^2	**1** Type and thickness	**2** Method of installing or fixing.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
16 Damp-proof courses ≤ 300mm wide **17** Damp-proof courses > 300mm wide **18** Pre-formed cavity trays	m m^2 m	**1** Gauge or thickness. **2** Number of layers **3** Composition and mix of bedding materials.	**1** Vertical **2** Raking **3** Horizontal **4** Curved, mean radius on face stated **5** Stepped				**C16** Damp-proof courses are deemed to include: **(a)** forming laps, ends and angles. **(b)** pointing exposed edges **(c)** bonding to damp-proof membrane and the like.	
19 Joint reinforcement	m	**1** Width stated						
20 Fillets	m	**1** Width and thickness stated		**1** Weather **2** Angle **3** Other				
21 Pointing	m	**1** Width and depth of joint to be pointed		**1** Type of finish		**D14** This relates to the pointing of joints, flashing, frames and the likes.		
22 Joints	m	**1** Width and thickness stated		**1** Type of filler, sealant, pointing, method of application and preparation.	**M11** Joints are only measured where their composition and position is designed			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Joints (Cont'd)				**2** Ties, centres stated **3** Channels				
23 Wedging and pinning	m	**1** Width and thickness stated						
24 Creasing	m	**1** Width stated **2** Number of courses						
25 Proprietary and individual spot items.	nr	**1** Dimensioned description or dimensioned diagram.	**1** Proprietary reference or catalogue number where applicable.	**1** Method of forming, building in or fixing.			**C17** All of these items are deemed to include all necessary forming of openings or pockets, liners, cavity closers, damp-proof courses, fixings and fastenings and builder's work in connection with any associated mechanical or electrical connections.	**S7** The list is not exhaustive. Any type of individual shaped piece should be enumerated and described either by a dimensioned description or a dimensioned diagram or by reference to the specification or trade brochure.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Proprietary and individual spot items (Cont'd)							**C18** This will include items such as windposts, head restraint channels, steel lintels, wall end ties, wall end bonding channels, cappings, chimney pots, finials, boiler seats, soot doors, plinths, steps, winders, landings, bases, key blocks, air bricks, ventilator gratings, flue blocks, fire backs and sides, grates, carved bricks, blocks or stones, quoin stones, jamb stones, hearths, weep- hole formers and the like.	

1.15 STRUCTURAL METALWORK

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** All general arrangement drawing, Plans, Sections and Elevations. **2** Drawings specific to the work **B Mandatory information to be provided** **1** Specification describing fabrication, welding, testing, erection and everything else necessary to complete the installation. **2** Surface preparation prior to application of any surface treatment or finish. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Position of the work in relation to the proposed structure and any existing structure that the new work is being connected to. **2** The types and sizes of all structural members and their positions in relation to each other **3** Details of connections or of the reactions, moments and axial loads at connection points.							**C1** The mass of framing includes all components. **C2** No allowance is made for the mass of welds, bolts, nuts, washers, rivets and protective coatings. **C3** Permanent erection is deemed to include all specified operations subsequent to fabrication including delivery to site.	**S1** Types and grade of materials including steel and steel to steel bolts.
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Framed members, framing and fabrication	t	**1** Lengths not exceeding 1.00m **2** Lengths over 1.00 but not exceeding 9.00m **3** Lengths exceeding 9.00m	**1** Weight $\leq$ 25 kg/m **2** Weight 25 - 50 kg/m **3** Weight 50 - 100 kg/m **4** And so on in increments of 50kg/m	**1** Columns **2** Beams **3** Rafters **4** Bracings **5** Purlins and cladding rails **6** Grillages **7** Trusses **8** Plate girders **9** Framing to doors and windows	**M1** Hollow, shape stated **M2** Built-up work; details of construction stated **M3** Compound fabrication, details of construction stated.		**C4** Temporary bracing or support work and subsequent removal is deemed to be included	**S1** Castellated **S2** Tapered **S3** Curved **S4** Cambered **S5** Cellular **S6** Secondary **S7** steelwork Temporary bracing or support work

STRUCTURAL METALWORK (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Framed members, framing and fabrication (Cont'd)				**10** Trimmers to roofs and walls. **11** Rafter and column stay s number stated				
2 Framed members, permanent erection on site.	t			**1** Crane rails	**M4** Details and centres of fixing clips and resilient pads stated.			
				1 Wires, cables, rods, ties and bars				
3 Isolated structural members, fabrication	t	**1** Plain member, use stated	**1** Dimensioned description and weight per lin m			**D1** An isolated structural member is the one not integrally connected to other structural members.		
4 Isolated structural members permanent erection on site	nr	**2** Built-up member, use stated						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Allowance for fittings	t	**1** Calculated weight **2** Percentage allowance; percentage stated.	**1** To framed members. **2** To isolated members.			**D2** Fitting are components that allow members to be joined together or are other brackets, supports and the like that are supplied and attached, either on or off site, to the main loadbearing frame by the structural metal contractor.		
6 Cold rolled purlins, cladding rails and the like	m	**1** Type and method of fixing stated **2** Size or proprietary reference stated.	**1** Purlins and cladding rails **2** Sag rods **3** Stays **4** Others.	**1** Method of fixing stated.				
7 Extra over for	nr	**1** Forming cranks	**1** Dimension of member stated					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Profiled metal decking, type and/or profile stated	m^2	**1** Height not exceeding 5.00m **2** Height exceeding 5.00m but not exceeding 10.00m **3** And so on in 5.00m increments.	**1** Shear studs, size and spacing stated.	**1** Method of fixing	**M5** This work is only measured here when it forms part of the structural steel package otherwise it would be measured in accordance with the rules for permanent formwork. **M6** Heights are always measured from finished floor level unless otherwise stated. **M7** The area of metal decking is calculated as the finished area			
9 Extra over for	m	**1** Edge trims; size, girth, profile or proprietary reference	**1** Straight **2** Raking **3** Curved, radius stated					
	m	**2** Curved cutting						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
10 Holding down bolts or assemblies	nr	**1** Type and diameter stated.	**1** Anchor plates, frames, members, tubes, cones and any other associated accessory stated	**1** Supply only				
11 Special bolts	nr	**1** Type and diameter stated		**1** Background if other than structural steel.		**D3** Special bolts are all bolts and fasteners other than grade 4.6 black bolts and holding down assemblies.		
12 Connections to existing steel and other members or structures	nr	**1** Details stated			**M8** Labours on non-steel structures are measured elsewhere		**C5** All labours on new and existing steel are deemed included.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
13 Trial erection	t	**1** Details and location stated			**M9** Only measured when not at discretion of contractor. **M10** The information given in the description must include items such as the number of erectable pieces and number of site welds. This information may be given by reference to drawing(s).			
14 Filing hollow sections	item	**1** Water **2** Concrete **3** Other material type stated	**1** Details stated					
15 Surface treatments	m^2	**1** Galvanising **2** Sprayed coating **3** Painting **4** Other treatment; type stated	**1** On site. **2** Off site	**1** Preparation described **2** Number of coat stated **3** Thickness of coat(s) stated **4** Fire rating stated **5** Finish stated			**C6** All preparation is deemed included.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
16 Isolated protective coatings	nr	**1** Approximate size or area stated.	**1** Type of protective coating stated **2** Preparation described.	**1** On site. **2** Off site				
17 Testing		**1** Load tests **2** Fire protection test **3** Other tests.	**1** Details stated.					

1.16 CARPENTRY

Timber framing and First Fixings,
Timber, Metal and Plastic Boarding, Sheeting, Decking, Casting and Linings
Metal and Plastic Accessories

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** All general arrangement plans, sections and elevations. **2** Drawings specific to the work **B Mandatory information to be provided 1** Grade of timber **2** Spacing of battens and grounds **3** Nature of base **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Layout and spacing of timbers. **2** Types of materials used. **3** Position of timbers in relation to adjacent structures.			**C1** Works in this section are deemed to include: **a** All sizes nominal unless otherwise stated. **b** All timbers are sawn unless otherwise stated. **c** All work fabricated, assembled and erected on site unless otherwise stated. **d** All work fixed by nails unless otherwise stated. **e** Holes in timber for bolts and all other fixings. **f** All labours on timber. **g** All webs, gussets and the like on trusses and portals.	**S1** Kind quality and size of materials **S2** Type of preservative treatment. **S3** Type of protective coating **S4** Method of fixing where not at the discretion of the contractor. **S5** Method of jointing or construction where not at the discretion of the contractor. **S6** Spacing of battens and grounds.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Primary or structural timbers	m	**1** Nominal size stated.	**1** Rafters and associated timbers. **2** Purlins **3** Wall plates **4** Roof and floor joists **5** Beams **6** Posts or columns **7** Partition and wall members **8** Strutting	**1** Selection and protection for subsequent treatment **2** Matching grains or colours **3** Limits on planning margins **4** Fixing centres stated where not at the discretion of the contractor. **5** Type and size of bolts and other fixings and fastenings **6** Background or nature of base. **7** Fixing through vulnerable materials.	**M1** Strutting is measured through the structural members being stiffened.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
				8 Surface finish or treatment applied as part of the manufacturing process. **9** Length> 6.00 in one continuous length.				
2 Engineered or prefabricated members/items	nr	**1** Finished dimension description	**1** Roof trusses **2** Portal frame **3** Trusses **4** Wall panels **5** Beams **6** Joists **7** Posts or columns	**1** Manufacturer's reference **2** Length> 6.00 in one continuous length.				
3 Backing and other first fix timbers	m	**1** Nominal dimensioned description of each member.	**1** Grounds **2** Battens					
	m^2	**2** Nominal dimensioned description of each member.	**1** Framed grounds, battens and bracketing	**1** Centres, each way stated				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
4 Boarding, flooring, sheeting, decking, casings, linings, sarking, fascias, barge boards, soffits and the like	m m^2	**1** Not exceeding 600mm wide; finished width and thickness stated. **2** Over 600mm wide; finished thickness stated	**1** Horizontal **2** Sloping **3** Vertical **4** Curved; radius stated **5** Soffit or ceiling **6** Other shape described	**1** Finish stated unless sawn **2** Types of joints where not at the discretion of the contractor **3** Profile				**S7** The location of the work shall be given such as external walls, internal walls and attached piers, isolated columns, floors, ceilings and attached beams, isolated beams, roofs, eaves, verges, tops and cheeks of dormers and the like.
5 Ornamental ends of timber members	nr	**1** Size and detail stated						
6 Metal fixing, fastenings and fittings	nr	**1** Dimensioned description **2** Proprietary reference or catalogue number where applicable.	**1** Ties. **2** Rods. **3** Brackets. **4** Straps. **5** Shoes. **6** Bolts. **7** Joist hangers. **8** Other: details stated.	**1** Method of fixing stated **2** Background stated if not	**M2** The length of bolt is measured over the head		**C2** Bolts includes heads, nuts and washers. **C3** Work is deemed to include all labours in fabricating and fixing including drilling holes in the fitting, the timber and the background.	

1.17 SHEET ROOF COVERING

Bituminous felts
Plastic sheets
Sheet metals
Rigid boards with factory applied sheet covering
Any other type of sheet roof covering

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Plans of each roof **2** Principal sections of roof **3** External elevations. **B Mandatory information to beprovided** **1** Thickness, gauge, weight and temper **2** Method of fixing **3** Details and position of laps, drips, welts, beads, rolls, joints, upstands and downstands **4** Bonding to other work **5** Radius of curved work **6** Type(s) and spacing of joints. **7** Type(s) and spacing of seams **8** Special finishes **C Minimum information that must be shown on the drawings that accompany this section ofmeasurement.** **1** Major horizontal and vertical dimensions. **2** Types of materials used to construct the roof and its structures **3** Position of main structural frame members **4** Height of work above ground level stating whether working platformlevel or finished **5** All labours and dressings.	**M1** The areas and lengths measured are net in contact with base	**D1** Examples of roof coverings measured in accordance with the rules of this section are: sheet lead, aluminium, copper, zinc, stainless steel, fibre etc. This list is not exhaustive.	**C1** Works and materials deemed included: **a** Underlay in contact with the covering. **b** All rough and fair cutting and waste. **c** Extra material required for bonding. **d** Extra material in laps and dressings. **e** Extra material for curved work. **f** All ends, angles and intersections either formed or proprietary. **g** Forming all rough and fair grooves, throats, mortices, chases, rebates, holes, stops, mitres and the like labours.	**S1** Kind and quality of materials.

SHEET ROOF COVERING (cont'd)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							h Raking out joints to insert flashings. **j** Labours in returns, ends and angles. **k** Work in isolated areas. **l** Work in forming voids and holes = 1.00m^2.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Covering > 500mm wide Covering ≤ 500mm wide	m^2	**1** Horizontal **2** Sloping, pitch stated **3** Vertical **4** Curved; radii stated	**1** Underlays **2** Insulation **3** Finish to exposed surface	**1** Nature of base	**M2** No deduction is made for voids ≤ 1.00m^2		**C2** Finishes include solar reflective paint, chippings and the like. They exclude tiles, paving, slabs, grass roofs and the like.	
3 Extra over for forming	m	**1** Drips	**1** Height stated			**D2** Length shall be net length of each labour	**C3** All additional sheet material required to form the labour item shall be deemed included. **C4** All additional underlay, insulation and surface finish shall be deemed included.	
		2 Welts	**1** Height stated	**1** Type stated				
		3 Rolls	**1** Width and height stated	**1** Wood core **2** Hollow				
		4 Seams	**1** Length stated					
		5 Laps	**1** Length stated					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
4 Boundary work, location and method of forming described	m	**1** Net girth stated **2** Average net girth stated, number of lengths stated	**1** Abutments. **2** Eaves. **3** Ridges. **4** Verges. **5** Valleys. **6** Hips. **7** Vertical angles. **8** Upstands = 500mm high. **9** Downstands = 500mm high.	**1** Horizontal. **2** Vertical. **3** Raking. **4** Curved, radius stated. **5** Stepped. **6** Preformed. **7** Sloping.	**M3** Boundary work to voids is only measured where the void is >1.00m^2. **M4** Where several items of the same type of boundary work have slightly differing net girths these girths may be averaged and the lengths aggregated. **M5** The number of separate lengths aggregated must be stated	**D3** Boundary work is work associated with closing off or finishing off sheet roofing at the external perimeter, at the abutment with different materials or the perimeter of openings and voids.	**C5** Boundary work is deemed to include undercloaks, insulation, strip ventilators, rough and fair cutting, bedding, pointing, laps, seams, ends, angles, intersections, rolls, upstands, downstands, welted edges, dressings and wedgings and additional covering material needed to form the detail and all associated labours.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Boundary work, location and method of forming described (Cont'd)					**M6** An upstand or downstand over 500mm high is measured as vertical work.		**C6** Valleys include everything necessary to form and line the valley.	
5 Flashings	m	**1** Net girth stated	**1** Flashings. **2** Aprons. **3** Sills. **4** Weatherings. **5** Cappings. **6** Hips. **7** Kerbs. **8** Ridges. **9** Linings to openings.	**1** Horizontal. **2** Sloping **3** Vertical. **4** Raking. **5** Curved, radius stated. **6** Stepped. **7** Preformed.			**C7** Flashings are deemed to include bedding, pointing, ends, angles, undercloaks, rough and fair cutting, intersections, welted, beaded or shaped edges and all dressings.	

SHEET ROOF COVERING (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
6 Gutters **7** Valleys	m	**1** Net girth stated	**1** Sloping. **2** Stepped. **3** Curved, radius stated. **4** Secret. **5** Tapered.	**1** Nature of base. **2** Spacing of structural supports. **3** Preformed.		**D4** The length is the mean length measured over all fittings.	**C8** Gutter and valley work is deemed to include all dressings required to form the profile complete with all joints, laps, seams, brackets, undercloaks and other associated linings, outlets, overflows, ends, angles, intersections, bedding, pointing, fixings and the like.	**S1** Maximum and minimum girth of tapered gutters or valleys shall be given.
8 Spot items	nr	**1** Dimensioned diagram or dimensioned description.	**1** Catchpits. **2** Sumps. **3** Outlets. **4** Hatch covers. **5** Canopy covers. **6** Collars or sleeves around pipes and the like. **7** Others: type stated.	**1** Nature of base. **2** Spacing of structural supports. **3** Preformed.			**C9** Spot item work is deemed to include joints, dressing and bonding to surrounding work, undercloaks and other associated linings, ends, angles, bedding, pointing, fixings and the like.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
9 Fittings	nr	**1** Dimensioned description	**1** Ventilators. **2** Finials. **3** Gas terminals. **4** Hip irons. **5** Soakers. **6** Saddles. **7** Rooflights. **8** Other: type stated.	**1** Nature of base. **2** Method of fixing where not at the discretion of the contractor.			**C10** Fittings are deemed to include joints, dressing and bonding to surrounding work, undercloaks and other associated linings, ends, angles, bedding, pointing, fixings and the like.	**S2** Proprietary references may be given in lieu of a dimensioned description.

1.18 TILE AND SLATE ROOF AND WALL

Plain Tiling
Interlocking Tiling
Fibre Cement Slating
Natural Slating
Natural Or Artificial Stone Slating
Timber or Bituminous Felt Shingles
Any other type of Tile, Slate, Slab or Block Roof or Wall Covering

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement 1 Plans of each roof **2** Principal sections of roof **3** External elevations. **B Mandatory information to be provided 1** Method of fixing **2** Minimum laps. **3** Spacing of battens and counter battens. **4** Composition and mix of mortar **5** Type of pointing **6** Bonding to other work **7** Radius of curved work **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Major horizontal and vertical dimensions. **2** Types of materials used to construct the roof and its structures **3** Position of main structural frame members **4** Height of work above ground level stating whether working platform level or finished level.			**C1** Works and materials deemed included: **a** All rough and fair square, raking and curved cutting. **b** All ends and angles either formed or proprietary **c** Extra material for curved work. **d** Forming all rough and fair grooves, throats, mortices, chases, rebates, holes, stops, mitres and the like labours. **e** Raking out joints to form key.	**S1** Kind, quality and size of materials.

TILE AND SLATE ROOF AND WALL COVERINGS (cont'd)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							f Raking out joints to insert flashings. **g** Labours in returns, ends and angles. **h** All extra materials required for bonding **j** Additional material in laps. **k** Work in forming voids and holes = 1.00m^2.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Roof covering **2** Wall covering	m^2	**1** Pitch stated **2** Vertical	**1** Underlays and battens	**1** Curved, radii stated **2** Conical: maximum and minimum radii stated	**M1** No deduction is made for voids $\leq$ 1.00 m^2		**C2** Coverings are deemed to include underlays, battens and work in forming voids = 1.00m^2.	

TILE AND SLATE ROOF AND WALL COVERINGS (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
3 Boundary work, location and method of forming described	m	**1** Dimensioned description stating net girth	**1** Abutments. **2** Eaves. **3** Ridges. **4** Verges. **5** Valleys. **6** Hips. **7** Vertical angles.	**1** Horizontal. **2** Sloping **3** Vertical. **4** Raking. **5** Curved, radius stated. **6** Stepped. **7** Preformed.	**M2** Boundary work to voids is only measured where the void is >1.00m^2.	**D1** Boundary work is work associated with closing off or finishing off sheet roofing at the external perimeter, at the abutment with different materials or the perimeter of openings and voids.	**C3** Boundary work is deemed to include undercloaks, insulation, strip ventilators, rough and fair cutting, bedding, pointing, seams, ends, angles, intersections, rolls, upstands, downstands, welted edges, dressings and wedgings and additional covering material needed to form the detail and all associated **C4** Valleys include everything necessary to form and line the valley.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
4 Fittings	nr	**1** Dimensioned description	**1** Ventilators. **2** Finials. **3** Gas terminals. **4** Hip irons. **5** Soakers. **6** Saddles. **7** Rooflights. **8** Others: type stated.	**1** Nature of base. **2** Method of fixing where not at the discretion of the contractor.				**S2** Proprietary references may be given in lieu of a dimensioned description.

1.19 WATERPROOFING

Mastic Asphalt Roofing
Applied Liquid
Asphalt Tanking or Damp Proofing Applied
Liquid Tanking or Damp Proofing Flexible Sheet
Tanking or Damp Proofing
Other Proprietary Systems of Tanking or Damp Proofing

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Plans of each level where work is to be carried out. **2** Sections showing extent of work **B Mandatory information to be provided** **1** Restrictions on sitting of plant and materials. **2** Minimum laps. **3** Thickness and number of coats. **4** Surface finishes or treatments. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Extent of work. **2** Height of work above ground level.			The following are deemed to be included:- **C1** Cutting to line. **C2** All labour and material associated with cutting, notching, bending, lapping and reinforcement. **C3** Working into recesses such as duct covers, shaped insets, manhole covers, mat wells, outlet pipes, dished gullies and the like. **C4** Work to falls and cross falls **C5** All boundary work to openings = 1.00m^2	**S1** Kind, quality of all materials.

TO BE MEASURED	UNIT	FIRST DIVISION	SECOND	THIRD DIVISION	MEASUREMENT RULES	DEFINITION	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C6 All preparation to background necessary to form a key, including raking out joints, scrabbling or the application of a bonding agent	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Coverings > 500mm wide	m^2	**1** Horizontal. **2** Sloping: pitch stated. **3** Vertical. **4** Curved: radii stated	**1** Underlays **2** Insulation. **3** Finish to exposed **4** Protection.	**1** Nature of base. **2** Number of coats or layers.	**M1** The area measured is that in contact with base. **M2** No deduction is made for voids $\leq$ 1.00 m^2			
2 Coverings = 500mm wide	m							
3 Skirtings **4** Fascias **5** Aprons	m	**1** Net girth on face.	**1** Horizontal. **2** Sloping **3** Vertical. **4** Raking **5** Curved: radii stated **6** Stepped.		**M3** Boundary work to voids is only measured where the void is > 1.00m^2.	**D1** Boundary work is work associated with closing off or finishing off the waterproofing at the external perimeter, at the abutment with different materials or the perimeter of openings and	**C7** Boundary work is deemed to include undercloaks, insulation, strip ventilators, rough and fair cutting, bedding, rough and fair cutting, bedding, pointing, laps, seams, ends, angles, intersections,	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
6 Gutters **7** Channels **8** Valleys **9** Kerbs	m	**1** Net girth on face.	**1** Sloping **2** Stepped. **3** Curved **4** Secret **5** Tapered	**1** Nature of base. **2** Number of coats or layers. **3** Spacing of structural supports.	**M4** Maximum and minimum girth of tapered gutters shall be given.	**D2** The length is the mean length measured over all fittings.	**C8** Lining and covering work is deemed to include all dressings required to form the profile, joints, brackets, undercloaks and other associated linings, outlets, overflows, ends, angles, intersections, bedding, pointing, fixings and the like	
10 Spot items	nr	**1** Dimensioned diagram or dimensioned description.	**1** Catch pits. **2** Sumps **3** Outlets. **4** Hatch covers. **5** Canopy covers. **6** Collars or sleeves around pipes and the like. **7** Others: type stated.				**C9** Spot item work is deemed to include joints, dressing and bonding to surrounding work, undercloaks and other associated linings, ends, angles bedding, pointing, fixings and the like.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
11 Fittings	nr	**1** Dimensioned description.	**1** Ventilators. **2** Others: type stated	**1** Nature of base. **2** Number of coats or layers. **3** Method of fixing where not at the discretion of the contractor			**C10** Fittings are deemed to include joints, dressing and bonding to surrounding work, undercloaks and other associated linings, ends, angles bedding, pointing, fixings and the like.	**S2** Proprietary references may be given in lieu of a dimensioned description.
12 Edge trim	m							

1.20 PROPRIETARY LININGS AND PARTITIONS

Metal Framed Systems to Walls and Ceilings

Drylining and Partitioning Systems to Walls and Ceilings

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Plans, sections and elevations to show scope and location of work **B Mandatory information to be provided** **1** Method of fixing framing and linings. **2** Layout and treatment of joints. **3** Method of jointing. **4** Surface preparation. **5** Finish. **6** Radii of curved work. **7** Nature of base. **8** Location of wall or ceiling mounted fittings requiring additional support or framing **9** Sealants to joints and perimeters. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Structural floor to ceiling heights. **2** The services located within the ceiling, partition or lining where the work includes complex integral services.		**D1** The finish is that which is applied as part of the normal proprietary system. Any finish to be subsequently made by a different trade shall be measured separately under the relevant rules.	**C1** All work is deemed internal unless otherwise stated. **C2** All rough and fair cutting. **C3** Extra material for curved work **C4** Additional framing to support fittings. **C5** Ends, fair ends and abutments with adjoining work **C6** Working around columns, beams and services where the outer face of the work is continuous.	**S1** This refers to the final location of the work. **S2** Kind, quality and size or thickness of materials and components. **S3** Examples of proprietary finishes are taped joints, slurry coats, and the like. **S4** Skim coats or other wet trade finishes are measured in Work Section 28: Floor, wall, ceiling and roof coverings

PROPRIETARY LININGS AND PARTITIONS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Proprietary metal framed system to form walls	m^2	**1** Finished thickness stated **2** Height or average height stated in 1.00m increments. **3** Total length stated measured along centre line.	**1** Insulation. **2** Vapour barriers **3** Sub linings. **4** Finish. **5** Glazing **6** Curved: radius stated		**M1** No deduction is made for voids < 1.00 m^2 **M2** The average height will be calculated for each length of partition with a sloping head measured between junctions.			
2 Proprietary metal framed system to form ceilings	m^2	**1** Over 300mm wide on face	**1** Insulation. **2** Vapour barriers **3** Sub linings. **4** Finish. **5** Curved: radii stated **6** Sloping **7** Convex or concave: radius stated	**1** Fixed direct to structural soffit **2** Supported on adjacent structure, span stated in 1.00m increments	**M3** The adjacent structure such as secondary steel framing or timber framing is measured elsewhere			
	m	**2** Not exceeding 300mm wide on face						

PROPRIETARY LININGS AND PARTITIONS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
3 Extra over for different	m^2	**1** Lining: details stated. **2** Finish: details stated			**M4** This will apply to partitions that have areas of different linings or finishes than that specified in the general description heading. It will not apply to different forms of construction or components. These differences will require separate items measured			
4 Extra over for forming openings	nr	**1** Not exceeding $2.50m^2$. **2** $2.50–5.00m^2$.	**1** Lined: details stated **2** Unlined.			**D2** Lined openings are those lined as part of this work.	**C7** Non-proprietary materials are not deemed included (e.g. timber grounds or inserts would be measured in Work Section 16: Carpentry).	

PROPRIETARY LININGS AND PARTITIONS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Extra over for non-standard perimeter details	m	**1** Dimensioned or description proprietary reference.	**1** Heads: details stated **2** Soles: details stated. **3** Abutments: details stated	**1** Nature of base.		**D3** Non-standard details are those that involve the use of components that are not used within the main body of work such as deflection heads, acoustic seals, fire seals and the like.		
6 Extra over for angles	m							
7 Extra over for junctions	m		**1** Tee **2** Cross.					
8 Extra over for access panels	nr	**1** Dimensioned description or proprietary reference.	**1** Details stated.		**M5** Fair ends are only measured where the exposed end is finished with the same finish as the face(s) or with a trim which is an integral part of the partition system.		**C8** Details include type of frame, ironmongery, finish, insulation, fire rating, and proprietary reference number where applicable. **C9** Fair ends are deemed to include all extra work and materials involved.	

PROPRIETARY LININGS AND PARTITIONS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
9 Fair ends to partitions	m	**1** Thickness of partition stated.	**1** Finish and/or trims					
10 Proprietary linings to walls	m^2		**1** Insulation. **2** Vapour barriers. **3** Sub linings. **4** Finish **5** Curved, radius stated **6** Sloping **7** Convex or concave: radius stated.	**1** Method of fixing **2** Nature of base				
11 Proprietary linings to ceilings	m^2							
12 Proprietary linings to columns	m	**1** Girth ≤ 300mm, nr of faces stated. **2** Girth 300–600mm, nr of faces stated. **3** Girth 600–900mm, nr of faces stated. **4** Thereafter in 300mm stages, nr of faces stated.						
13 Proprietary linings to beams	m							

PROPRIETARY LININGS AND PARTITIONS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
14 Proprietary linings to bulkheads	m							
15 Extra over for forming openings	nr	**1** Not exceeding 2.50m^2. **2** 2.50–5.00m^2 **3** Exceeding 5.00m^2 in further increments of 2.50m^2	**1** Lined, details stated **2** Unlined.				**C10** This includes openings for doors, windows, screens and the like.	
16 Extra over for non-standard perimeter details	m	**1** Dimensioned description or proprietary reference.	**1** Heads: details stated **2** Soles: details stated. **3** Abutments: details stated	**1** Nature of base.		**D4** Non-standard details are those that involve the use of components that are not used within the main body of work such as deflection heads, acoustic seals, fire seals and the		
17 Extra over for angles	m							
18 Extra over for junctions	m		**1** Tee **2** Cross.					

PROPRIETARY LININGS AND PARTITIONS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
19 Extra over for access panels	nr	**1** Dimensioned description or proprietary reference.	**1** Details stated.				**C11** Details include type of frame, ironmongery, finish, insulation, fire rating, and proprietary reference number where applicable.	
20 Beads, function stated	m		**1** Method of fixing.	**1** Nature of base.			**C12** Function of beads include angle beads, stop beads, shadow gap beads, casing beads and the like.	

1.21 CLADDING AND COVERING

Patent Glazing, Curtain Walling, Rainscreen Cladding, Glazed Vaulting and Rigid Sheet Weatherboarding
Profiled Sheet Cladding or Roofing
Panel Or Slab Cladding or Roofing
Sheet Claddings or Coverings
Any other type of Cladding, Lining or Covering

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Plans, sections and elevations sufficient to show scope and location of the works **2** Component drawings. **B Mandatory information to be provided** **1** Type, finish and spacing of framing members. **2** Nature, thickness and spacing of structural supports **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Construction of each installation.	**M1** Secondary steel support work is measured in accordance with the rules of Work Section 1.15: Structural Metalwork		The following are deemed to be included:- **C1** Ironmongery where supplied with the component. **C2** Integral insulation, membranes and the like where part of the system. **C3** Mastics and sealants. **C4** Cleats and brackets. **C5** Fixings and fastenings. **C6** part of the installation **C7** Mechanical and electrical operating equipment where supplied with the component.	**S1** Kind, quality of materials.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Walls **2** Floors **3** Ceilings **4** Roofs **5** Sides and tops of dormers **6** Sides and soffits of beams **7** Sides of columns	m	**1** Not exceeding 600 wide.	**1** Laid diagonally. **2** Sloping: pitch stated **3** Vertical. **4** Curved: radius stated. **5** To soffits.	**1** Internal. **2** External.				
1 Walls **2** Floors **3** Ceilings **4** Roofs **5** Sides and tops of dormers **6** Sides and soffits of **7** Sides of columns	m^2	**1** Exceeding 600 wide.						
	nr	**1** Isolated area not exceeding $1m^2$, irrespective of width.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Items extra over the work in which they occur	nr	**1** Dimensioned description **2** Proprietary reference.	**1** Opening lights. **2** Doors. **3** Rooflights. **4** Ventilator panels. **5** Forming openings for other trades.	**1** Electrical requirements.				**S2** These items are only for components that are part of the system to which they are installed.
9 Boundary work	m	**1** Dimensioned description **2** Proprietary reference.	**1** Heads. **2** Ridges. **3** Valleys. **4** Hips. **5** Bottom edges. **6** Eaves. **7** Verges. **8** Abutments. **9** Flashing pieces.	**1** Horizontal. **2** Vertical. **3** Sloping **4** Raking **5** Curved: radii stated **6** Stepped. **7** Preformed.		**D1** Boundary work is work associated with closing off or finishing off claddings or coverings at the external perimeter or at the abutment with different materials.	**C8** All edge and intermediate trims, cover pieces, mastics and sealants are deemed included with the associated boundary or opening perimeter work.	

CLADDING AND COVERING (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
10 Opening perimeters	m		**1** Heads. **2** Jambs. **3** Sills. **4** Flashing pieces.	**1** Horizontal. **2** Vertical. **3** Sloping **4** Raking **5** Curved: radii stated **6** Stepped. **7** Preformed.		**D2** Opening work is associated with closing off or finishing off claddings and coverings at the perimeter of an opening or a void.		
11 Angles	m		**1** Internal. **2** External.	**1** Regular. **2** Irregular: angle stated		**D3** Irregular angles are any that are not 90°.		
12 Closers	m		**1** Fire stops. **2** Others: details stated.					

1.22 GENERAL JOINERY

Unframed Isolated Trims, Skirtings, or Sundry Joinery Items In-Fill Panels
Sealant Joints
General Ironmongery Not Associated With Windows and D o o r s

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** General arrangement plans. **B Mandatory information to be provided** **1** Kind and quality of materials **2** Finish of timber if not wrot. **3** Preservation treatments. **4** Surface treatments as part of the production process. **5** Selection and protection for subsequent treatment or clear finish. **6** Matching grain or colour. **7** Limits on planning margins **8** Method of jointing and construction where not at the discretion of the contractor. **9** Form of construction and jointing where not at the discretion of the contractor. **10** Method of fixing where not at the discretion of the contractor. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Scope and location of work.			**C1** All timbers are deemed to be wrot finish unless described otherwise. **C2** All timber sizes are nominal unless stated as finished size. **C3** Ends, angles, mitres and intersections irrespective of cross section area of timber. **C4** Work is deemed internal unless described as external	**S1** Kind and quality of materials.

GENERAL JOINERY (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Skirtings, picture rails. **2** Architraves and the like. **3** Cover fillets, stops, trims, beads, nosings and the like **4** Isolated shelves and worktops **5** Window boards **6** Isolated handrails and grab rails	m	**1** Dimensioned overall cross section stated. **2** Number and type of labours described.	**1** Built up members, size of components stated. **2** Timber components tongued on. **3** Different cross-section shapes (nr).	**1** Fixing through vulnerable materials. **2** Curved work to be described stating the radius.				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
7 Duct covers **8** Pipe casings **9** Shelves	m	**1** Thickness and width or girth stated. **2** Thickness and width or girth not exceeding 300mm stated. **3** Thickness and width or girth over 300mm but not exceeding 600mm stated. **4** Thickness and width or girth thereafter in 300mm stages.	**1** Nature and method of forming joints.		**M1** Usually the exact width or girth shall be given for these items but where there are many differing but similar widths these may be grouped into 300mm wide or girth bands.			
10 Pinboards, backboards, plinth blocks and the like.	nr	**1** Dimensioned description		**1** Method of fixing. **2** Nature of base.				
Floor, Wall and Ceiling Boarding, Sheeting, Panelling, Linings and Casings								
11 Boarding, sheeting, panelling over 600mm wide	m^2	**1** Thickness.	**1** Floors **2** Walls **3** Ceilings	**1** Method of fixing. **2** Method of jointing **3** Nature of base.				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Proprietary Partitions, Panels and Cubicles								
12 Boarding, sheeting, panelling not exceeding 600mm wide	m	**1** Width and thickness.						
13 Partitions	m	**1** Height and thickness stated **2** Proprietary name or reference stated	**1** Off site applied finish. **2** On site applied finish.	**1** Method of fixing and jointing **2** Curved: radii stated **3** Nature of background. **4** Integral	**M2** The length is measured through and over all obstructions and openings.	**D1** The linear measurement is the mean length measured along the centre line of the partition.	**C5** Partitions are deemed to include all integral components, fixings, joints, factory applied trims, holes and mortices.	
14 Items extra over the partition they occur in	nr	**1** Openings: size stated	**1** Blanks. **2** Doors. **3** Windows. **4** Glazed panels. **5** Access panels.				**C6** Openings are deemed to include everything necessary to form the opening together with their associated integral components, glass, doors, ironmongery, linings and factory-applied	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
	m	**2** Trims: dimensioned description			**M3** Trims are only measured as separate items if fixed on site.			
15 Duct panels, Sanitary ware back panels and the like	nr	**1** Overall size stated. **2** Thickness stated.	**1** Sub frames, details stated.	**1** Method of fixing. **2** Nature of background.	**M4** Associated sanitary ware items are not included in these items but shall be measured in accordance with the rules of Work Section 32: Furniture, fittings		**C7** All ancillary items needed to connect the sanitary appliances to the M and E installations are deemed included.	
16 Cubicle partition sets	nr	**1** Dimensioned diagram or dimensioned description.	**1** Complete cubicles: number of bays stated	**1** Nature of background. **2** Method of fixing to background.			**C8** Cubicle sets are deemed to include all integral frames, panels, doors, factory applied trims, connections, fixings and fastenings complete with support legs, brackets and standard ironmongery.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Infill panels								
17 Items extra over the partition they	m	**1** Trims: dimensioned description			**M5** Trims are only measured as separate items if fixed on site.	**D2** Infill panels are non-glass and non-glass plastics rigid sheet spandrel and infill panels of all kinds fixed with beads, gaskets and the like to wood, metal, plastic and concrete surrounds. They exclude panels or sheets forming an integral part of a component or proprietary cladding system.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
18 Infill panels and sheets, number stated	m^2	**1** Thickness stated.	**1** Curved: stating radius. **2** Panels exceeding size of normal manufactured unit. **3** Panels requiring special treatment to edges.				**C9** Work is deemed internal unless described as external. **C10** Infill panels are deemed to include bedding compounds, sealants, in tumescent compounds and strips, distance pieces, location and setting blocks and fixings.	
Sealant joints								
19 Joints: contact surfaces stated	m	**1** Type and size of components stated.	**1** Vertical. **2** Sloping **3** Soffit. **4** Horizontal.		**M6** Lengths are measured on face		**C11** Work is deemed to include preparation, cleaners, primers, sealers, backing strips and fillers.	
20 Pointing: contact surfaces stated	m	**1** One side. **2** Both sides						
21 Raking out existing joints	m	**1** Width , depth and type of material stated	**1** Vertical. **2** Sloping **3** Soffit. **4** Horizontal.				**C12** Deemed to include disposal of all debris.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ironmongery								**S2** Kind and quality of materials and fixings. **S3** Surface finish. **S4** Constituent parts of the units or sets. **S5** Fixing through vulnerable materials.
22 Type of item, unit or set stated	nr	**1** Method of fixing. **2** Nature of base.			**M7** Shelf brackets and associated fittings will be measured here.		**C13** Ironmongery is deemed to include fixing with screws to match and preparing the base to receive it.	

1.23 WINDOWS, SCREENS AND LIGHTS

Timber, Metal, Plastic:
Windows and Shop Fronts
Rooflights
Screens
Louvres, Shutters, Canopies and Blinds Associated Glass and Glazing Associated Ironmongery

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** General arrangement plans. **2** Door and window schedule **3** Glazing schedule. **4** Ironmongery schedule **B Mandatory information to be provided 1** Preservation **2** Surface **3** Selection and **4** Matching grain **5** Limits on planning margins **6** Method of jointing and construction where not at the discretion of the contractor. **7** Form of construction and jointing where not at the discretion of the contractor. **8** Method of fixing where not at the discretion of the contractor. **9** Background where vulnerable **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Shape and size of units. **2** Methods of fixing units. **3** Glazing			The following are deemed to be included: **C1** Bedding and pointing frames. **C2** Glass and the like to be installed on site unless stated otherwise.	**S1** Kind and quality of materials and if timber describing if wrot or sawn.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Windows and window frames **2** Window shutters **3** Sun shields **4** Rooflights, skylights and lantern lights **5** Screens, borrowed lights and	nr	**1** Dimensioned description or diagram.	**1** Curved work: radius stated	**1** Method of fixing if not left to discretion of contractor. **2** Factory glazed.				**S2** Glass may be incorporated into the associated work following the rules set out below.
6 Shop fronts **7** Louvers and frames								
Glazing								
8 Glass, type stated **9** Louvre blades	nr	**1** Thickness of glass or overall thickness of sealed unit stated **2** Pane size.	**1** Shape if other than square or rectangular. **2** Airspace(s) width(s) of double/triple glazed units **3** Gas filling of airspaces.	**1** Bent: direction stated. **3** Edge treatment.		**D1** The size given for irregular panes shall be the smallest rectangular size from which the pane can be obtained	**C3** Glass includes plastic, or any type of material glazed into openings except glass blocks	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ironmongery								
10 Type of item, unit or set stated	nr	**1** Method of fixing. **2** Nature of base.	**1** Kind and quality of materials and fixings. **2** Surface finish. **3** Constituent parts of the units or sets.				**C4** Ironmongery is deemed to include fixing with screws to match and preparing the base to receive it.	

1.24 DOORS, SHUTTERS AND HATCHES

Timber, Metal, Plastic:
Doors and Frames
Shutters and Hatches
Sliding or Folding Doors and Partitions Grilles
Associated Glass and Glazing
Associated Ironmongery

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Floor plans **2** Door schedule. **3** Ironmongery schedule. **B Mandatory information to be provided** **1** Kind and quality of materials and, if timber, describing if wrot or sawn. **2** Preservation treatments. **3** Surface treatments applied as part of the production process. **4** Selection and protection for subsequent treatment or clear finish. **5** Matching grain or colour **6** Limits on planning margins. **7** Method of jointing and construction where not at the discretion of the contractor. **8** Form of construction and jointing where not at the discretion of the contractor. **9** Method of fixing where not at the discretion of the contractor **10** Background where vulnerable. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Location of all doors. **2** Door reference number.			The following are deemed to be included: **C1** Bedding and pointing frames deemed to be included. **C2** Glass and the like to be installed on site unless stated otherwise.	**S1** Kind and quality of materials and, if timber, describing if wrot or sawn.

DOORS, SHUTTERS AND HATCHES (Cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Door sets	nr	**1** Detailed description of **2** Structural opening size.	**1** Background	**1** Smoke stops: details stated. **2** Fire stops: details stated.	**M1** Any trims and the like fixed to the set after installation must be measured in accordance with the rules of Work Section 1.22: General joinery.	**D1** Door sets comprise the door or doors complete with associated frame, stops, architraves and trims.		**S2** Glass maybe incorporated into the associated work following the rules set out in coverage rules
2 Doors **3** Roller shutters **4** Collapsible gates **5** Sliding folding partitions **6** Hatches **7** Strong room doors **8** Grilles	nr	**1** Dimensioned description or diagram.	**1** Fire resistance performance	**1** Method of fixing if not left to discretion of contractor **2** Factory glazed	**M2** Each leaf of a multi-leafed door is counted as one door.			**S3** Glass may be incorporated into the associated work following the rules set out in coverage rules
9 Door frames **10** Door linings **11** Door stops **12** Associated fire stops **13** Associated smoke stops	m	**1** Dimensioned overall cross section description.	**1** Labours described			**D2** Frames and linings includes jambs, heads, sills, mullions and transoms.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Glazing								
14 Glass, type stated **15** Louvre blades	nr	**1** Thickness of glass or overall thickness of sealed unit stated **2** Pane size.	**1** Shape if other than square or rectangular. **2** Airspace(s) width(s) of double /triple glazed units. **3** Gas filling of air spaces.	**1** Bent: direction stated. **2** Method of glazing or securing. **3** Edge treatment.		**D3** Glass includes plastic, or any type of material glazed into openings except glass blocks. **D4** The size given for irregular panes shall be the smallest rectangular size from which the pane can be obtained.		
Ironmongery								
16 Type of item, unit or set stated	nr	**1** Method of fixing. **2** Nature of base.	**1** Kind and quality of materials and fixings. **2** Surface finish. **3** Constituent parts of the units or sets.				**C3** Ironmongery is deemed to include fixing with screws to match and preparing the base to receive it.	

1.25 STAIRS, WALKWAYS AND BALUSTRADES

Timber, Metal, Plastic:
Staircases
Walkways and Gantries
Balustrades
Barriers
Guardrails

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement 1 Plans and sections showing location and scope of each unit. **B Mandatory information to be provided** **1** Kind and quality of materials and if timber describing if wrot or sawn. **2** Preservation treatments. **3** Surface treatments applied as part of the production process. **4** Selection and protection for subsequent treatment or clear finish. **5** Matching grain or colour **6** Limits on planning margins. **7** Method of jointing and construction where not at the discretion of the contractor. **8** Form of construction and jointing where not at the discretion of the contractor. **9** Method of fixing where not at the discretion of the contractor **10** Background where vulnerable. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Principle dimensions of each staircase or unit. **2** Height from structural floor to structural floor level.			The following are deemed to be included: **C1** All ramps, wreaths, bends, plain ends, ornamental ends and the like labours. **C2** Linings, nosings, cover moulds, trims and the like where integral part of the unit. **C3** Soffit linings, spandrel panels and the like where an integral part of the unit. **C4** Ironmongery where an integral part of the unit. **C5** Finishes applied off site.	

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C6 Fixings, fastenings, blockings, wedges, bolts, brackets, cleats and the like.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Staircase: type stated	nr	**1** Dimensioned description **2** Component drawing		**1** Hatch doors where part of the loft ladder component.	**M1** Any trims and the like fixed to the unit after installation must be measured in accordance with the rules of Work Section 1.22: General joinery.		**C7** Staircases are deemed to include attached balustrades and newel posts.	
2 Loft ladders								
3 Ladders	nr							
4 Extra over for:		**1** Quarter landing **2** Half landing	**1** Dimensioned description.					
5 Cat walks	m	**1** Dimensioned description.	**1** Curved; radius stated	**1** Method of fixing or support **2** Background.				
6 Walkways								

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
7 Balustrades	m	**1** Dimensioned description.	**1** Curved: radius stated	**1** Infill panels :detail stated. **2** Method of fixing or support. **3** Background.	**M2** These items are measured when not forming an integral part of a staircase unit.		**C8** These items include all integral rails, infill panels, ironmongery, factory applied finishes, fixings and fastenings.	
	nr	**2** Component drawing reference.						
8 Handrails								
9 Barriers								
10 Guardrails								
11 Balcony units								
13 Extra over for	nr	**1** Opening portions: size and details stated.						

1.26 METAL WORK

Isolated Metal Members
General Metal Work
General Metalwork Fittings and Fixtures

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement **1** Plans, sections, elevations or details sufficient to show location, scope and construction of each item measured. **B Mandatory information to be provided** **1** Types and grades of materials. **2** Details of connections. **3** Details of welding tests and x-rays. **4** Details of performance tests. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Position of the work. **2** Types and sizes of structural members and their position relative to each other.							**C1** Works and materials deemed included. **(a)** All fabrication and erection. **(b)** All bolts, nuts, washers, fixings and fastenings.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Isolated metal members	m	**1** Dimensioned description: cross section profile stated	**1** Built up: details of construction stated. **2** Tapered. **3** Curved: radius stated. **4** Hollow: shape stated.	**1** Method of fixing in position. **2** Fixing plates, brackets and the like: dimensioned description **3** Protective coating applied			**C2** Forming holes, mortices and the like in the item and background for fixing purposes are deemed included.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 General metal work members	m nr	**1** Dimensioned description: cross section profile stated.	**1** Protective coating applied off site.	**1** Method of fixing. **2** Nature of background. **3** Surface pattern or finish.				
3 Sheet metal	m^2 nr	**1** Thickness stated. **2** Dimensioned description.						
4 Wire mesh	m^2 nr	**1** Mesh size and thickness stated. **2** Dimensioned description.	**1** Protective coating applied off site.	**1** Method of fixing. **2** Nature of background. **3** Surface pattern or finish.			**C3** Forming holes, mortices and the like in the item and background for fixing purposes are deemed included.	
5 Composite items	m nr	**1** Dimensioned description. **2** Drawing reference .	**1** Full description of all components including integral non-metallic items.	**1** Method of fixing. **2** Nature of background.	**M1** Any non-metallic item applied after the composite item has been installed on site shall be measured in accordance with the rules of the relevant section.		**C4** Non-metallic integral items forming part of the composite item mayb e included in the composite item provided they are clearly described as such.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
6 Filling holes in	item	**1** Water. **2** Concrete. **3** Other material: type stated	**1** Details stated.					
7 Surface treatments	m^2	**1** Galvanising **2** Sprayed coating. **3** Painting. **4** Other treatment: type	**1** On site. **2** Off site.	**1** Surface preparation described	**M2** Any finish applied after installation is measured in accordance with the rules of the 'Painting and decorating' section.		**C5** All preparation is deemed included.	
8 Isolated protective coatings	nr	**1** Approximate size or area stated	**1** Type of protective coat stated. **2** Preparation described.	**1** On site. **2** Off site.				

1.27 GLAZING

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Elevations. **2** Component drawings. **B Mandatory information to be provided. 1** Kind, quality and thickness of glass **2** Kind and quality of glazing material. **3** Method of glazing including details of gaskets where required. **4** Nature of frame or background. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** General scope and location of work.					**M1** Glazing supplied as part of a window, door or other component is measured elsewhere with those components.		**C1** All raking and curved cutting. **C2** All polished and bevelled edges unless given in the description. **C3** Drilling holes.	**S1** This section covers all types of glass and glass plastics and glazing with putty, beads.
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Glass, type stated **2** Sealed glazed units, type of glass stated **3** Louvre blades, type of glass stated	nr	**1** Thickness of glass or overall thickness of sealed unit. **2** Pane size.	**1** Shape if other than square or rectangular. **2** Airspace(s) width(s) of double/triple glazed units. **3** Gas filling of airspaces.	**1** Bent: direction stated. **2** Method of glazing or securing: details stated **3** Edge treatment.	**M2** Glass blocks are measured in accordance with the rules of Work Section 1.14: Masonry.		**C4** Glass includes laminated, toughened, double or triple glazing units, lead, acrylic, polycarbonate and the like, or any other type of material glazed into any other type of material glazed into openings except glass blocks.	**S2** The size given for irregular panes shall be the smallest rectangular size from which the pane can be obtained.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
4 Extra for:	nr	**1** Grinding. **2** Sandblasting **3** Acid etching. **4** Embossing **5** Engraving.	**1** Plain work; pane size stated. **2** Design work: pane size stated.	**1** Detailed description of work including extent of work over the pane.	**M3** This information may be given by reference to detailed drawings.			
5 Lead light glazing, type of glass stated	nr	**1** Thickness of glass. 2 Overall window or light size.	**1** Pane sizes where regular shape. **2** Pattern of lead cames. **3** Spacings of lead cames.	**1** Bent: direction stated. **2** Method of glazing or securing: details stated. **3** Edge treatment.		**D1** Cames are the H-shaped lead strips that are soldered together to form the lead glazed panes.		**S3** The pattern will be described as regular, diamond or irregular.
6 Saddle bars	m	**1** Diameter and length stated.	**1** Method of fixing to window surround.	**1** Method of attaching to lead light glazing.		**D2** Saddle bars are metal rods used to strengthen and provide support to large lead glazed panes. These bars are usually built in to the window surround.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
7 Mirrors	nr	**1** Dimensioned description.	**1** Edge description. **2** Pattern: method of forming described.	**1** Nature of background. **2** Method of fixing: details stated.		**D3** Edges can be plain, rounded, bevelled and the like.		
8 Removing existing glass and preparing frame or surround to receive new glass	nr	**1** Type of frame or surround. **2** Type of glass to be removed.	**1** Panes = 1.00m^2. **2** Panes 1.00–2.00m^2. **3** Panes 2.00–3.00m^2. **4** Panes 3.00–4.00m^2. **5** Panes = 4.00m^2 : size stated.	**1** Method of disposal of debris where not at the discretion of the **2** Materials to be kept for re-use.				

1.28 FLOOR, WALL, CEILING AND ROOF FINISHINGS

Insitu, Tiled, Block, Mosaic, Sheet, Applied Liquid or Planted finishes

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement 1 Plans of each floor. **2** Principal sections through building. **3** External elevations. **4** Finishes schedules. **B Mandatory information to be provided.** **1** Kind, quality and size of materials. **2** Number of coats of Insitu work. **3** Types of underlays or linings. **4** Bedding or other method of fixing **5** Type of pointing **6** Laps **7** Layout and width of joints and bays. **8** Patterns **9** Spacing of battens and counter battens. **10** Composition and mix of mortars. **11** Bonding to other work. **12** Radius of curved work. **13** Nature of base. **14** Laid in one operation with base. **15** Applied finishes/sealers/polishes. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** All floor-to-floor or ceiling heights. **2** Layout of patterned work. **3** Details of moulded work.	**M1** Area measured is net in contact with base. **M2** Widths stated are the width of each finished face. **M3** No deduction for voids = 1.00m^2. **M4** No allowance to be made in net area for laps, dressings or any other labour. **M5** All work is deemed internal unless stated as external.		**C1** All rough and fair square, raking and curved cutting and waste. **C2** All ends and angles either formed or proprietary. **C3** Extra material for curved work. **C4** Forming all rough and fair grooves, throats, mortices, chases, edges, rebates, holes, stops, mitres and the like labours. **C5** Raking out joints to form a key. **C6** Raking out joints to insert skirtings or the like	

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
			C7 All work in forming returns, ends, internal and external angles. **C8** All extra material required for bonding. **C9** Additional material in laps and dressings. **C10** All work on attached columns and beams **C11** Work in forming voids and holes = 1.00 m^2 **C12** Formwork or any other form of temporary support. **C13** Working finishes up to and around all accessories.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Screeds, beds and toppings, thickness and number of coats stated.	m	**1** = 600mm wide.	**1** Level and to falls only = 15° from horizontal.	**1** Surface finish **2** Nature of background. **3** Backings and beddings: thickness stated. **4** Underlays: type and thickness stated. **5** Insulation: type and thickness stated. **6** Laid in one operation with its base. **7** Pattern of joints. **8** Laid in one operation with base monolithically	**M6** Thicknesses stated are the thickness exclusive of adhesives, keys, grooves and the like.		**C14** Underlays include plasterboard or other rigid sheet lathing.	
	m^2	**2** > 600mm wide	**2** To falls, cross falls and slopes = 15° from horizontal **3** To falls, cross falls and slopes > 15° from horizontal.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Finish to floors, type of finish and overall thickness stated.	m	**1** = 600mm wide.					**C15** Tiles are deemed to be laid with their long edges vertical or parallel to the long axis of the floor or ceiling.	
	m^2	**2** > 600mm wide.						
3 Raised access floors, type of finish and thickness of panels stated.			**1** Height of cavity stated.					
4 Ramps to raised acces s floors.	nr	**1** Length, width and height.					**C16** The height stated is the height at the top end.	
5 Fire barriers within void below raised floor.	m^2	**1** Thickness stated.	**1** Fire rating. **2** Method of fixing in position. **3** Obstructed by services.					
	m	**2** Thickness and height stated.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
6 Finish to roofs, type of finish and overall thickness stated.	m m^2	**1** = 600mm wide. **2** > 600mm wide.	**1** Level and to falls only = 15° from horizontal. **2** To falls, cross falls and slopes > 15° from horizontal. **3** To falls, cross falls and slopes > 15° from horizontal.		**M7** They exclude solar reflective paint, chippings and the like that will be measured with their associated roof covering.	**D1** Width is the width of each face.	**C17** Roof finishes include grass, sedum and the like live finishes.	
7 Finish to walls, type of finish and overall thickness	m m^2	**1** = 600mm wide. **2** > 600mm wide.	**1** Curved, radius stated.		**M8** Height of wall finishes is measured to finished ceiling height or specified height past suspended ceilings.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
					M9 Wall finishes are measured behind skirtings unless the skirting is installed prior to the wall finish. **M10** Width is the width of each face.			
8 Finish to isolated columns, type of finish and overall thickness stated.								
9 Finish to ceilings, type of finish and overall thickness stated.	m m^2	**1** = 600mm wide. **2** > 600mm wide.	**1** Over 3.50m above structural floor level.		**M11** Height of ceiling is measured from structural floor level to soffit of finished ceiling or isolated beam. **M12** Width is the width of each face.			
10 Finish to isolated beams, type of finish and overall thickness stated.								

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
11 Finish to treads	m	**1** Net width stated.	**1** Curved: radius stated.	**1** Nature of background. **2** Insets described.			**C18** Ends, angles and outlets deemed included.	
12 Finish to risers	m	**1** Net width stated.	**1** Curved: radius stated. **2** Undercut.	**1** Nature of background.				
13 Finish to strings and aprons.	m	**1** Net width or girth on face stated.	**1** Raking. **2** Sloping **3** Curved: radius stated.	**1** Nature of background.				
14 Skirtings, net height stated.	m	**1** Net width stated.	**1** Raking. **2** Sloping **3** Curved: radius stated.	**1** Nature of background. **2** Cove formers.				
15 Linings to channels.	m	**1** Net girth on face stated.	**1** Raking. **2** Sloping **3** Curved: radius stated.	**1** Nature of background.			**C19** Ends, angles and outlets deemed included.	
16 Kerbs and cappings.	m	**1** Net girth on face stated.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
17 Coves **18** Mouldings **19** Cornices **20** Architraves **21** Ceiling ribs **22** Bands	m	**1** Girth and shape stated. **2** Dimensioned description.	**1** Patterned: details stated. **2** Raking . **3** Vertical. **4** Sloping **5** Vertical. **6** Undercut **7** Flush. **8** Raised. **9** Sunk.	**1** Method of fixing. **2** Nature of background.	**M13** Measure the length in contact with the base.		**C20** All ends, angles, intersections are deemed included.	
23 Special tiles, slabs or blocks	nr m^2	**1** Dimensioned description. **2** Manufacturers' reference.						**S1** Special tiles are only measured here if not already measured under previous rules 11–22 inclusive.
24 Surface dressings, sealers or polishes.	m^2	**1** Type stated. **2** Rate of coverage stated.	**1** Horizontal. **2** Sloping. **3** Vertical. **4** Soffits	**1** Nature of finish being treated.			**C21** Dressings include carborundum grains, stone chippings and the like. **C22** Sealers include waterproofers, hardeners, dust-proofers, polishes and the like.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
25 Movement joints. **26** Cover strips. **27** Dividing strips. **28** Beads, function stated. **29** Nosings.	m	**1** Dimensioned description.		**1** Method of fixing. **2** Nature of background.			**C23** Movement joints include expansion joints.	**S2** Function of beads include angle beads, stop beads, shadow gap beads, casing beads and the like.
30 Reinforcement, details stated. **31** Metal mesh lathing, details stated. **32** Board insulation, thickness stated. **33** Quilt insulation, thickness stated. **34** Isolation membranes, thickness stated.	m m^2	**1** To walls. **2** To ceilings. **3** To floors. **4** To roofs. **5** To isolated beams. **6** To isolated columns.	**1** Depth of suspension.	**1** Method of fixing. **2** Nature of background.				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
35 Accessories.	nr	**1** Dimensioned description.	**1** Method of fixing.	**1** Nature of background.			**C24** Accessories include access panels, special panels in raised access floors, vent grilles, ornaments and the like.	
36 Precast plaster components.	m	**1** Dimensioned description or proprietary reference.	**1** Method of fixing.	**1** Nature of background.				**S3** The dimensioned description shall fully describe the component including stating the end use such as cornice, moulded band, architrave and the like.

1.29 DECORATION

Painting and Clear Finishes.

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Plans and elevation drawings showing scope and location of the work. **2** Painting/decorating schedule(s). **B Mandatory information to be provided. 1** Kind and quality of materials. **2** Nature of base. **3** Description of surface if not smooth. **4** Preparatory work to base. **5** Type and number of priming or sealing coats. **6** Type and number of undercoats. **7** Type and number of finishing coats. **8** Method of application. **9** Type of treatment applied between coats. **10** Pattern of decorative papers. **11** Method of fixing and jointing papers or fabrics. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Ceilings over 3.5m above finished floor level.			**C1** All cutting to line. **C2** Multi-coloured work on differing surfaces. **C3** All cutting to papers and fabrics. **C4** Nature of non-smooth base includes description of base texture and profile such as corrugated, fluted, moulded or carved.	**S1** Method of application left to contractor unless stated otherwise.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Painting to general surfaces	m	**1** = 300mm girth.	**1** Internal.	**1** Work to ceilings or beams over 3.5m but not exceeding 5m above finished floor level and thereafter in 3.00m stages.	**M1** The area or girth measured is that covered.	**D1** Examples of radiator types are flat, panelled, column, tubular or the like.	**C5** The girth of frames and the like is calculated from one edge to the other over all trims, architraves, stops and the like and assume doors have been removed prior to painting.	
2 Painting to glazed surfaces irrespective of pane sizes.	m^2	**2** > 300mm girth.	**2** External.	**2** Surfaces to remain unpainted.				
3 Painting structural metalwork.	nr	**3** Isolated areas = $1.00m^2$ irrespective of location or girth.		**3** Multi-coloured on one surface.				
4 Painting radiators, type stated.				**4** Patterned: details stated.			**C6** The girth designated as external on door frames and the like is that part of the frame visible when the door is closed.	
5 Painting gutters.				**5** Fire rating.				
6 Painting pipes.				**6** Application on site prior to fixing.				
7 Painting services, type stated.				**7** Application off site prior to fixing.				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Painting railings, fences and gates.	m m^2 nr	**1** = 300mm girth. **2** > 300mm girth. **3** Isolated areas = $1.00m^2$ irrespective of location or girth.	**1** Closed. **2** Open. **3** Ornamental.			**D2** Closed means no gaps whatsoever.		
Decorative papers or fabrics								
9 Walls and columns. **10** Ceilings and beams.	m m^2	**1** Areas = $1.00m^2$. **2** Areas > $1.00m^2$.	**1** Curved surfaces. radii stated.	**1** Lining paper. **2** Work to ceilings or beams over 3.5m but not exceeding 5m above finished floor level and thereafter in 3.00m stages.			**C7** No deduction made for voids = $1.00m^2$.	
11 Borders.	m	**1** Dimensioned description.					**C8** Mitres, ends angles, scribing or cutting deemed included.	
12 Motifs.	nr	**1** Dimensioned description or diagram.						

1.30 SUSPENDED CEILINGS

Demountable Suspended Ceilings

Solid Suspended Ceilings

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Plans showing location of the work **2** Reflected ceiling plans showing scope and complexity of the work. **B Mandatory information to beprovided. 1** Kind and quality of materials. **2** Size(s) of panels and strips. **3** Construction of suspension framing and systems. **4** Method of fixing. **5** Nature of background. **6** Nature of services located in the ceiling void. **7** Nature of integral services and fittings **C Minimum information that must be shown on the drawings that accompany this section ofmeasurement.** **1** Services located within the ceiling void. **2** Location of integral services and fittings.	**M1** Secondary steel or timber support work is measured elsewhere.		**C1** All work is deemed internal unless described as external. **C2** Working over and around obstructions. **C3** All cutting. **C4** Forming openings and holes. **C5** All additional support work including bridging for fittings. **C6** All extra work required for work described as patterned.	**S1** Integral fittings are those designed to be incorporated into the ceiling structure.

SUSPENDED CEILINGS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Ceilings **2** Plenum ceilings **3** Beams **4** Bulkheads	m^2	**1** Depth of suspension ≤ 150 mm. **2** Depth of suspension 150–500mm. **3** Depth thereafter in 500mm stages.	**1** Type and thickness of lining. **2** Method of fixing lining to suspension. **3** Integral insulation. **4** Integral vapour barrier. **5** Height of work exceeding 3.50m above finished floor level in 1.50m	**1** Patterned: details stated. **2** Sloping. **3** Curved: radius and plane of curve stated **4** Suspension obstructed by services. services. **5** Trims at regular intervals within area of ceiling: details stated.	**M2** Area measured is that on the exposed face. **M3** Depth of suspension is measured from underside of main structural soffit or secondary support work to back surface of lining. **M4** Height of work is measured from finished floor level to face of ceiling.			**S2** Where ceilings are suspended from sloping, curved or irregular structures an average depth of suspension shall be stated.

SUSPENDED CEILINGS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
6 Upstands	m	**1** Height = 600mm **2** Height thereafter in 300mm stages						
7 Access panels	nr	**1** Dimensioned description. **2** Proprietary reference.	**1** Composition **2** Method of fixing.	**1** Method of locking.			**C7** Access panels are deemed to include all additional support work, framing, edge trim and fixings.	
8 Edge trims **9** Angle trims **10** Fire barriers	m m m^2	**1** Dimensioned description. **1** Dimensioned description. **2** Thickness stated.	**1** Plain. **2** Floating **1** Fire rating where required.	**1** Centres of fixing. **2** Nature of background. **1** Method of fixing in **2** Obstructed by services stated.		**D1** Plain trims are those fixed to the structure. **D2** Floating trims are those fixed to the ceiling system	**C8** Trims are deemed to include ends and angles. **C9** Fire barriers are deemed to include all support work, scribing or forming to fit, angles, ends and working around structures, support work and services.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
11 Collars for services passing through fire barriers	nr	**1** Diameter of pipe. **2** Size of trunking	**1** Pipes. **2** Trunking.	**1** Length of sleeve each side of barrier stated. **2** Fire rating	**M5** Collars are only measured here where they are integral with the fire barrier. If they are not integral they shall be measured in accordance with the rules of Work Section 1.40: connection with mechanical, electrical Builder's work in and transportation installations.			
12 Fittings.	nr m	**1** Dimensioned description.		**1** Nature of background stated.				
13 Shadow gap battens.	m	**1** Dimensioned description.		**1** Centres of fixing. **2** Nature of background.			**C10** Ends, angles and the like deemed included.	

1.31 INSULATION, FIRE STOPPING AND FIRE PROTECTION

Board, Sheet, Quilt, Sprayed, Loose Fill or Foamed Insulation and Fire Protection Installations.

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Plans, sections and details sufficient to show the scope and location of the various works relating to this section. **B Mandatory information to beprovided.** **1** Type, quality and thickness of materials. **2** Fire rating where required. **3** Extent of laps. **4** Method of fixing, laying or applying where not at the discretion of the contractor.							**C1** All cutting. **C2** Working around or over all members and services.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Boards **2** Sheets **3** Quilts **4** Loose	m^2	**1** Thickness stated.	**1** Plain areas. **2** Laid across joists, rafters, partition framing or like members: centres of members stated. **3** Laid between joists, rafters, partition framing or like members: centres of members stated.	**1** Horizontal. **2** Vertical. **3** Sloping. **4** Soffit.	**M1** No deduction is made for voids = $1.00m^2$. **M2** The area of joists, rafters, partition framing or like members is deducted when the material is laid between such members.	**D1** Sloping is the upper surface not horizontal. **D2** Soffits are the underside of any horizontal or sloping structure.	**C3** The area measured is that covered.	
	m	**2** Purpose and dimensioned description.	**4** Location stated.					
5 Sprayed	m^2	**1** Thickness stated.		**1** Nature of background.				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
6 Filling cavities	m^2	**1** Thickness stated.	**1** Mineral fibre. **2** Plastic beads. **3** Cellulose fibre. **4** Expanding foam. **5** Other type of blown or injected material.	**1** Internal. **2** External.			**C4** Drilling holes in structure to allow injection of material and subsequent making good is deemed included.	
7 Fire stops, type stated.	m	**1** Dimensioned description. **2** Fire rating.	**1** Horizontal. **2** Vertical. **3** Raking. **4** Stepped. **5** Curved, radius stated.	**1** Method of fixing. **2** Background when mechanically fixed.				
8 Fire sleeves, collars and the like.	nr	**1** Dimensioned description. **2** Fire rating.		**1** Method of fixing. **2** Background when mechanically fixed.				

1.32 FURNITURE, FITTINGS AND EQUIPMENT

General Fixtures, Furnishings and Equipment. Kitchen Fittings.
Catering Equipment.
Sanitary Appliances and Fittings.
Notices and Signs.
Site and Street Furniture.
Bird/Vermin Control.

<table>
<tr><th colspan="5">INFORMATION PROVIDED</th><th>MEASUREMENT RULES</th><th>DEFINITION RULES</th><th>COVERAGE RULES</th><th>SUPPLEMENTARY INFORMATION</th></tr>
<tr><td colspan="5">A Drawings that must accompany this section of measurement.

1 Location drawings.
B Mandatory information to be provided.

1 Sufficient information to design, procure or manufacture the item.
2 Kind and quality of materials.
3 Details of all associated building work.
4 Tests with which materials and equipment must comply.

C Minimum information that must be shown on the drawings that accompany this section of measurement.
1 Components.</td><td rowspan="2"></td><td rowspan="2"></td><td rowspan="2">C1 Marking positions.
C2 Connecting to services.

C3 Commissioning
C4 All associated excavations and concrete foundations or bases unless stated as measured elsewhere.</td><td rowspan="2"></td></tr>
<tr><td>Item or work to be measured</td><td>Unit</td><td>First Division</td><td>Second Division</td><td>Third Division</td></tr>
<tr><td>1 Fixtures, fittings or equipment without services.</td><td>nr</td><td>1 Component drawing reference.

2 Dimensioned diagram.</td><td>1 Ancillary items provided with the equipment.</td><td>1 Method of fixing.

2 Nature of background.</td><td></td><td></td><td></td><td></td></tr>
</table>

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Fixtures, fittings or equipment with services		**3** Manufacturers reference.	**2** Integral controls and indicators. **3** Remote controls and indicators. **4** Supports, mountings and brackets.					
3 Ancillary items not provided with the item of equipment.	nr	**1** Type, size and method of jointing stated.						
4 Fixtures, fittings or equipment supplied by the employer	nr	**1** Type and size.	**1** Collecting from location off site: details stated.				**C5** Accepting delivery, unloading, transporting about site, storing and handling and disposing of all packing materials is deemed included.	
5 Sign writing	nr	**1** Dimensioned description.	**1** Font.	**1** Nature of background.				

1.33 DRAINAGE ABOVE GROUND

Rainwater installations
Foul drainage installations

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. 1 Roof plan(s). **2** Floor plans of any floors that have drainage installation **3** Principal sections and elevations **B Mandatory information to be provided.** **1** Location of installation. **2** Nature of background. **3** Method and spacing of fixing. **4** Method and spacing of joints. **5** Description of material. **6** Type of brackets or supports. **7** Finish. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Scope and location of work.							**C1** All joints. **C2** All brackets and supports.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Pipework	m	**1** Nominal diameter.	**1** Straight, curved, flexible. **2** Extendable. **3** Method of jointing.	**1** Method of fixing to background.	**M1** Measured over all fittings.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Pipework ancillaries.	nr	**1** Dimensioned description.					**C3** Ancillaries include valves, non-return flaps, reducers, tapers, bends, hoppers, gullies, tundishes dishes, rodding eyes, traps, access doors, angles, offsets, shoes,	**S1** The list is not exhaustive. **S2** Fittings which are reducing are measured extra over the largest pipe in which they occur.
3 Items extra over the pipe in which they occur.	nr	**1** Fittings, pipe size = 65mm.	**1** One end. **2** Two ends. **3** Three ends. **4** Other, details stated.	**1** With inspection door. **2** Method of jointing stated where different from pipe in which the fitting occurs.			**C4** Cutting and jointing pipes to fittings is deemed to be included.	
4 Pipe sleeves through walls, floors and ceilings.	nr	**1** Type and nominal diameter of pipe **2** Length of sleeve or thickness of structure being passed through.	**1** Type of structure being passed through described.	**1** Method of fixing and packing.			**C5** All making good including fire stopping is deemed included.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
		3 Fire rating.						
5 Gutters	m	**1** Nominal size.	**1** Straight, curved, flexible. **2** Extendable. **3** Method of jointing.	**1** Method of fixing to background.	**M2** Measured over all fittings.			
6 Gutter ancillaries	nr	**1** Dimensioned description.					**C6** Ancillaries include stop ends, angles, outlets, overflows, tapers, reducers and the like.	**S3** The list is not exhaustive.
7 Items extra over the gutter in which they occur.	nr	**1** Dimensioned description.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Marking, position of and leaving or forming all holes, mortices, chases and forming all holes, mortices, chases and the like required in the structure.	item	**1** Number and type of installations.						
9 Identification	nr	**1** Plates. **2** Discs **3** Labels **4** Tapes. **5** Symbols or numbers. **6** Bands **7** Charts or diagrams. **8** Others: type stated.	**1** Details stated.	**1** Method of fixing.				

DRAINAGE ABOVE GROUND (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
10 Testing and commissioning	item	**1** Installation stated.	**1** Attendance required. **2** Insurances required by employer.	**1** Preparatory operations stated. **2** Stage tests (nr) listed and purpose stated.			**C7** All fuel and power used is deemed included. **C8** Provision of test certificates is deemed included.	
11 Preparing drawings	item	**1** Number of copies stated.	**1** Method stated. **2** Information to be shown stated.	**1** Working drawings. **2** As fitted drawings.				
12 Operating manuals and instructions	item							

1.34 DRAINAGE BELOW GROUND

Storm Water Drain Systems
Foul Drain Systems
Pumped Drain Systems
Land Drainage

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Drainage layout(s). **B Mandatory information to be provided** **1** Kind and quality of materials. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Layout of drainage showing scope of work. **2** Invert depths. **3** Cover levels. **4** Pipe sizes. **5** Details of manholes, inspection chambers and other pits, tanks, chambers or the like. **6** Work outside site boundary.			**C1** Earthwork support. **C2** Compacting bottoms of excavations. **C3** Trimming excavations. **C4** Backfilling with excavated materials. **C5** Compacting backfill. **C6** Disposal of surplus excavated materials. **C7** Disposal of all water. **C8** Those lengths of pipes within manhole walls. **C9** Building in ends of pipes.	**S1** Work outside site boundary shall be measured separately. **S2** Work below buildings shall be measured separately

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Drain runs	m	**1** Average trench depth in 500mm increments. **2** Type and nominal diameter of **3** Multiple pipes stating number and nominal diameter of pipes.	**1** Method of jointing pipes. **2** Pipe bedding and or surround: details stated **3** Type of backfill if not obtained from the excavations.	**1** Vertical. **2** Curved. **3** Below ground water level. **4** Next to existing roadway or path. **5** Next to existing building. **6** Specified multiple handling: details stated **7** Disposal of excavated material where not at the discretion of the contractor: details stated.	**M1** Drain runs are measured from external face of manhole to external face of manhole or accessory. **M2** Average depth is calculated for each run irrespective of maximum depth.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Items extra over drain runs irrespective of depth or pipe size.	m^2	**1** Breaking up hard surface pavings: thickness stated. **2** Lifting and preserving turf: thickness	**1** Reinstating to match existing: details stated. **2** Relaying turf previously set aside.		**M3** The measurement of these extra over items shall be based on the designed width of beds in the trench. In the absence of a bed the width shall be calculated as the nominal size of the service plus 300mm subject to a minimum width of 500mm.	**D1** Hard material is any material which is of such a size, position or consistency that it can only be removed by special plant or explosives.		
	m^3	**3** Breaking out hard materials. **4** Excavating in and removing hazardous material.	**1** Details stated.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
		5 Excavating in running silt, running sand or other unstable ground. **6** Excavating below ground water level.						
	m	**7** Next to existing live services.			**M4** To be measured where precautions are specifically required. The method of protection is left to the discretion of the contractor.			**S3** If in doubt the surveyor must measure an item giving the nature of the service.
	nr	**8** Around existing live services crossing trench.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
3 Pipe fittings	nr	**1** Dimensioned description.	**1** Type stated. **2** Method of jointing to pipes.				**C11** Pipe fittings include bends, junctions, inspection pipes, rodding eyes and the like.	
4 Accessories	nr						**C12** Accessories include gullies, traps, inspection shoes, fresh air inlets, non-return flaps, valves, rodding eyes and the like and are deemed to include gratings, covers, frames, baskets, filters and all other integral and or associated fittings.	**S4** The list is not exhaustive.

DRAINAGE BELOW GROUND (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Pumps	nr	**1** Size and capacity stated.	**1** Integral controls. **2** Remote controls. **3** Indicators. **4** Supports.	**1** Method of fixing. **2** Nature of background.				
6 Manholes **7** Inspection chambers **8** Soak-aways- pits **9** Cesspits **10** Septic tanks, **11** Other tanks and pits, type stated.	nr	**1** Detailed description stating maximum internal size of chamber. **2** Depth from top surface of cover to invert level in 250 mm stages. **3** Proprietary system, details stated.	**1** Base slab thickness. **2** Wall thickness. **3** Cover slab dimensions. **4** Intermediate slab dimensions. **5** Benching dimensions. **6** Main channel diameter and configuration. **7** Number and diameter of branch channels **8** Internal finish. **9** External finish.		**M5** Size stated is the maximum internal size of the chamber.		**C13** Rocker joints are deemed included. **C14** Other tanks and pits would include catch pits, service chambers, stopcock pits and the like.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
12 Extra over the excavation for:	m^2	**1** Breaking up hard surface pavings: thickness stated. **2** Lifting and preserving turf: thickness stated.					**C15** Making good around edges of surfaces is deemed included	
	m^3	**3** Breaking out hard materials. **4** Excavating in and removing hazardous material. **5** Excavating in running silt, running sand or other unstable ground. **6** Excavating below ground water level.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
13 Sundries	nr	**1** Detailed dimensioned description.	**1** Step irons. **2** Intercepting traps. **3** Backdrops **4** Any other associated item: type stated.				**C16** Bedding, jointing and building in deemed included. **C17** These items may be included with their associated manhole or chamber if part of a proprietary system.	
14 Covers and frames	nr	**1** Dimensioned description.	**1** Manufacturers reference.	**1** Method of fixing frame.			**C18** Bedding covers in grease is deemed included.	
15 Marker posts	nr	**1** Dimensioned description.	**1** Identification plates: details stated.	**1** Method of fixing: details stated.			**C19** Excavation, disposal of spoil and concrete for base deemed included.	
16 Connections	item	**1** Details stated.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
17 Testing and commissioning	item	**1** Detailed description. **2** Type of test and standard to be achieved.	**1** Attendance required. **2** Instruments and equipment to be provided.	**1** Preparatory operations stated. **2** Stage tests (nr) listed.			**C20** Provision of all fuel, power, water and other supplies is deemed included. **C21** Provision of test certificates is deemed included.	

1.35 SITE WORKS

Road and Path Pavings
Hard Landscaping
Sports Surfacing

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Site plans and sections. **2** Component details. **B Mandatory information to be provided.** **1** Kind, quality, shape and size of materials and components. **2** Treatment and layout of joints. **3** Nature of base. **4** Preparatory work. **5** Bedding and/or methods of fixing. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Scope and location of the work.	**M1** Unless stated as deemed included all associated excavations, disposals and fillings are measured in accordance with Work Section 1.5: Excavating and **M2** The areas measured are those in contact with the base. **M3** No deductions made in superficial items for voids = 1.00m^2.		**C1** Kind, quality, shape and size of materials and components. **C2** Treatment and layout of joints. **C3** Nature of base. **C4** Preparatory work. **C5** Bedding and/or methods of fixing. **C6** All work is deemed external unless described as internal. **C7** Formwork and all other temporary support. **C8** Fair joints. **C9** Fair edges. **C10** Working over and around obstructions and into recesses and shaped inserts.	**S1** All thickness stated are nominal or finished after laying and compacting.

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							C11 Forming shallow channels. **C12** Cutting. **C13** Disposal of surplus excavated material off site.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Kerbs **2** Edgings **3** Channels **4** Paving accessories	m	**1** Dimensioned description.	**1** Curved: radius stated.	**1** Foundation: size and details stated. **2** Reinforcement: details stated. **3** Method of fixing or			**C14** Excavation and disposal is deemed included. **C15** Ends, angles, outlets and intersections deemed included. **C16** Edgings are deemed to include pegs and supports.	
5 Extra over for:	nr	**1** Special units. **2** Accessories.	**1** Dimensioned description.					
6 Insitu concrete **7** Formwork **8** Reinforcement **9** Joints **10** Worked **11** Accessories cast in		**1** All measured in accordance with rules of Work Section 11: Insitu concrete works.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
12 Coated macadam and asphalt	m^2	**1** Over 300mm wide: thickness stated.	**1** Roads. **2** Pavings.	**1** Level and to falls only. **2** To falls and crossfalls and slopes = 15° from horizontal. **3** To slopes > 15° from horizontal. **4** Method of application **5** Surface treatment or finish. **6** Special curing of finished work.				
	m	**2** Not exceeding 300mm wide: thickness and width stated.						
	m	**3** Thickness and girth on face stated.	**3** Linings to channels.	**7** Horizontal. **8** To falls.			**C17** Arrises, coves, ends, angles and outlets deemed included.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
13 Gravel, hoggin and woodchip	m^2	**1** Over 300mm wide: thickness stated.	**1** Roads **2** Pavings.	**1** Level and to falls only. **2** To falls and crossfalls and slopes = 15° from horizontal. **3** To slopes > 15° from horizontal. **4** Method of application. **5** Surface treatment or finish. **6** Special curing of finished work.	**M4** The area measured is that in contact with the base.			
	m	**2** Not exceeding 300mm wide: thickness and width stated.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
14 Interlocking brick and blocks, slabs, bricks, blocks, setts and cobbles	m^2 m	**1** Over 300mm wide. **2** Not exceeding 300mm wide: thickness and width stated.	**1** Roads. **2** Pavings. **3** Treads. **4** Risers. **5** Margins or bands. **6** Channels.	**1** Bedding: thickness stated. **2** Level and to falls only. **3** To falls, crossfalls and slopes = 15° from horizontal. **4** To slopes > 15° from horizontal. **5** Laid in bays: average size of bays stated. **6** Joint pattern stated. **7** Curved: radius stated. **8** Foundation and haunching.			**C18** Setting pavings into recessed manhole covers is deemed included. **C19** All cutting and fitting is deemed included. **C20** Excavation and disposal is deemed included. **C21** Ends, angles, outlets and intersections deemed included.	
	m	**3** Thickness and girth on face stated.	**7** Linings to channels.	**9** Horizontal **10** To falls.			**C22** Arrises, coves, ends, angles and outlets deemed	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
15 Extra over for:	nr	**1** Special units. **2** Accessories.	**1** Dimensioned description.			**D1** Special units include gulley surrounds, hoppers and the like.		
16 Accessories	m^2	**1** Thickness stated.	**1** Membranes.					
	m	**2** Dimensioned description.	**2** Movement joints. **3** Expansion joints.		**M5** Joints located at the discretion of the contractor are not measurable.			
17 Site furniture	nr	**1** Dimensioned description. **2** Manufacturers reference.	**1** Tree grilles. **2** Bollards. **3** Litter bins. **4** Seats **5** Other: type stated.	**1** Method of fixing.				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
18 Liquid applied surfacings	m^2	**1** Thickness stated.	**1** Number of coats or layers.	**1** Level and to falls only.				
19 Sheet surfacings				**2** To falls and crossfalls and slopes = 15° from horizontal.				
20 Tufted surfacings								
21 Proprietary coloured sports surfacings				To slopes > 15° from horizontal.				
22 Surface dressings				**3** Method of application or fixing.				
				4				
				5 Surface treatment or finish.				
				6 Special curing of finished work.				
				7 Extent of laps.				
				8 Type of seams.				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
23 Line markings, width stated	m	**1** Width = 300mm. **2** Width > 300mm: width stated	**1** Straight, continuous Straight, broken **2** **3** Curved, continuous: radius stated. **4** Curved, broken: radius stated.	**1** Number of coats. **2** Method of application. **3** Treatment applied between coats. **4** Colour.				
24 Letters, figures and symbols	nr	**1** Dimensioned description.						
25 Sheet linings to pools, lakes, ponds, waterways and the like.	m m^2	**1** Width = 500mm **2** Width $\geq$ 500mm	**1** Horizontal. **2** Sloping, slope stated. **3** Curved, radius stated. **4** Vertical.	**1** Underlays.	**M6** The area measured is net and is that in contact with base. **M7** No deduction is made for voids = 1.00m^2. **M8** No allowance is made for laps and the like.			**S2** All other requirements of Work Section 19:Waterproofing will apply to the measurement of this work.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
26 Spot items	nr	**1** Dimensioned diagram. Or **2** Dimensioned description.	**1** Sumps. **2** Outlets. **3** Collars or sleeves. **4** Others, type stated.				**C23** Spot item work is deemed to include joints, dressing and bonding to surrounding work, undercloaks and other associated linings, ends, angles, bedding, pointing, fixings and the like.	

1.36 FENCING

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Site plans and sections. **2** Component details. **B Mandatory information to be provided.** **1** Kind and quality of materials. **2** Method of construction. **3** Surface treatments or finishes applied as part of the manufacturing process or applied before delivery to site. **4** Nature and size of backfilling. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Scope and location of the work. **2** Location of fencing designed for sloping ground. **3** Pre-contract ground water level(s) and date(s) established.					**M1** Unless stated as deemed included all associated excavations, disposals and fillings are measured in accordance with Work Section 1.5: Excavating and filling.		**C1** All excavations, backfilling and disposal off site of surplus materials. **C2** Earthwork support. **C3** Disposal of ground and surface water. **C4** Insitu concrete for post bases and the like **C5** Temporary supports. **C6** Formwork.	
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Fencing, type stated		**1** Height of fence.	**1** Spacing, height and depth of supports.	**1** Set to curve but straight between posts. **2** Curved: radius stated. **3** Ground sloping > 15° from **4** Lengths = 3m.	**M2** Fencing is measured over all supports. **M3** The height of fencing is measured from the finished surface of the ground to the top of the infilling, or where there is no infilling, to the top wire or rail unless otherwise stated.	**D1** Supports are posts, struts or the like occurring at regular intervals. **D2** The depth of supports and special supports is the depth below the ground surface or other stated base.	**C7** The height of supports and special supports is the height above the ground **C8** Integral and independent gate posts are deemed to include slamming stops and hanging pins or fillets.	**S1** Curved fencing is fencing curved between supports.

FENCING (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Extra over for special supports.	nr	**1** Size, height and depth stated.	**1** End post. **2** Angle or corner post. **3** Integral gate post. **4** Straining post. **5** Others: type stated.	**1** Method of fixing. **2** Background stated. **3** Details of backstays or struts stated.				
3 Independent gate posts.	nr							
4 Items extra over fencing, supports and special supports and independent gate posts irrespective of type.	m^3	**1** Excavating below ground water level. **2** Excavating in hazardous materials details stated.					**C9** If the post-contract ground water level differs from the pre-contract ground water level then the measurements are adjusted accordingly.	
	m^3 m^2	**3** Breaking out existing. **4** Breaking up existing hard pavings: thickness stated.	**1** Rock. **2** Concrete. **3** Reinforced concrete. **4** Brickwork, blockwork or stonework. **5** Tarmacadam or asphalt.			**D3** Rock is any hard material which is of such size or location that it can only be removed by the use of wedges, rock hammers, special plant or explosives.	**C10** Making good existing hard pavings is deemed included.	**S2** A boulder = $5m^3$ in volume or one that can be lifted out in the bucket of an excavator will not constitute rock. **S3** Degraded or friable rock that can be scraped out by the excavator bucket does not constitute rock.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Gates	nr	**1** Height and width stated.	**1** Type stated.	**1** Power supply			**C11** Gates are deemed to include stops, catches, independent stays, sounders, warning lights and all associated work.	
6 Ironmongery	nr	**1** Type of item, unit or set stated	**1** Kind and quality of materials and fixings. **2** Surface finish. **3** Constituent parts of the units or sets.	**1** Method of fixing. **2** Nature of base.			**C12** Ironmongery is deemed to include fixing with screws or bolts to match and preparing the base to receive it.	

1.37 SOFT LANDSCAPING

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Site plans and sections. **2** Component details. **3** Planting schedule. **B Mandatory information to be provided.** **1** Kind, quality, size and composition of materials. **2** Preparatory work. **3** Timing of operations. **4** Size and type of pits, holes and trenches, either excavated or formed. **5** Types of supports and ties. **6** Special filling materials. **7** Method of labelling. **8** Work on roofs and the like stated. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Scope and location of the work.					**M1** No deductions made in superficial items for voids = 1.00m^2.	**D1** Unless stated as deemed included all associated excavations, disposals and fillings are measured in accordance with Work Section 1.5: Excavating and filling. **D2** The areas measured are those in contact with the base.	**C1** All work is deemed external unless described as internal. **C2** All excavations and backfilling. **C3** All necessary multiple handling of excavated material. **C4** Disposal of surplus excavated material off site. **C5** Removal of stones and rubbish **C6** Watering. **C7** Labelling.	**S1** All thickness stated are nominal or finished after laying and compacting
Item or work to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Cultivating	m^2	**1** Depth stated.	**1** Method and degree of tilth.					
2 Surface applications.	m^2	**1** Type and rate stated. **2** Method of application state.	**1** Herbicides **2** Weedkillers **3** Peat. **4** Manure. **5** Compost.	**1** Before sowing or planting. **2** After planting. **3** Around individual plants.			**C8** Working in is deemed included.	

SOFT LANDSCAPING

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
			6 Mulch. **7** Fertiliser. **8** Sand. **9** Soil ameliorates **10** Other: details stated.	**4** To general areas. **5** To beds. **6** To planters. **7** To pots.				
3 Seeding	m^2	**1** Rate stated.	**1** Grass seed. **2** Cultivated plant seed. **3** Wild flower seed. **4** Other: type stated.				**C9** Raking in, harrowing and rolling is deemed included.	
4 Turfing	m^2	**1** Type and thickness stated.	**1** Cutting is deemed to include trimming edges.				**C10** Cutting is deemed to include trimming edges.	
	m	**2** Type, width and thickness stated.						

SOFT LANDSCAPING

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Trees	nr	**1** Botanical name.	**1** BS size designation and root system stated. **2** Girth, height and clear stem and root system stated.	**1** Planting in cultivated or grassed areas prepared by others to be stated including all necessary reinstatement. **2** Planting indoors to be stated. **3** Details of initial cut back stated. **4** Details of watering stated. **5** Backfill type stated if not the material arising from excavations.		**D3** Young nursery stock includes seedlings, transplants and whips.	**C11** Planting in cultivated or grassed areas prepared by others to be stated including all necessary reinstatement. **C12** Planting indoors to be stated. **C14** Supports and ties deemed included.	**S2** Backfill type stated if not the material arising from excavations. **S3** BS size include standard, advanced nursery stock or semi-mature types.
6 Young nursery stock trees	nr		**3** Height and root system stated.					
7 Shrubs	nr							

SOFT LANDSCAPING

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
8 Hedge plants	nr m		**4** Height stated. **5** Height, spacing, number of rows and layout stated.					
9 Plants	nr m^2		**6** Size stated. **7** Size and number per m^2 stated.					
10 Bulbs, corms and tubers	nr m2		**8** Size stated. **9** Size and weight per m^2 stated					
11 Plant containers	nr	**1** Dimensioned description.	**1** Linings: type stated.	**1** Method of fixing.				

SOFT LANDSCAPING

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
12 Protection	m	**1** Temporary fencing: type and duration stated.	**1** Ultimate ownership: details stated.		**M2** Permanent fencing is measured in accordance with the rules of Work Section 1.36: Fencing.			
	nr	**2** Tree guards, dimensioned description.	**2** Type stated.	**1** Method of fixing stated.				
	nr	**3** Wrappings: height of wrapping and girth of tree stated.	**3** Chemical application stated.	**2** Method of fixing stated.				
	nr	**4** Anti-desiccant sprays, height and girth of tree or spread of plant stated.	**4** Type stated.	**3** Rate of spray stated.				
	item	**5** General: details stated.	**5** Duration.					
13 Maintenance	item	**1** Details stated.						

1.38 MECHANICAL SERVICES

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Drawings that must accompany this section of measurement include the following: **1** Floor and site plans detailing layout of systems. **2** System schematics. **3** Cross sections and elevations. **4** Detail layouts for plant rooms and principal services installation areas. **5** Specific installation details for equipment and fittings. **6** Equipment schedules. **Mandatory** **1** Type, quality, size or thickness of the materials **2** Method of fixing **3** Location	**M1** Work Sections are classified in accordance with Appendix A **M2** Work in plant rooms is identified separately **M3** Where the mandatory information is available, measurement of pipework and/or ductwork will be fully detailed with the measurement of fittings identified and measured separately, in accordance with Alternative 1. However, where the mandatory information is not available then pipework/ductwork and associated insulation and fire protection is to be measured inclusive of fittngs.	**D1** Finishes and surface treatments exclude insulation and decorative finishes which are measured under Section 28.	**C1** All work is deemed internal unless stated as external. **C2** Works and materials are deemed to include the following: a) All waste. b) Extra material for labour made fittings. c) All couplings, brackets, supports, fixings and cast in channels. d) All labelling tagging, identification and charts. e) Marking of all holes chases and mortices. f) Earth connectors, straps and links.	**S1** Specified codes of practice and regulations **S2** Kind and quality of materials **S3** Gauge, thickness or substance of materials **S4** Tests with which materials must comply **S5** Finishes or surface treatments applied on site **S6** Finishes or surface treatments applied off site stating whether applied before or after fabrication or assembly.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Primary equipment	nr	**1** System	**1** Type, size, capacity, load, rating, special finishes, casings, in-fills, associated integrated or remote ancillaries, controls, indicators or components, anti-vibration mountings, acoustic performance/treatment and method of fixing.	**1** Location of installation - Roofs, high or low level in plant rooms, risers or service ducts, high or low level on floors.	fire protection is to be measured inclusive of fittings.	**D2** Primary equipment is defined as plant or equipment from which a system originates, e.g. boiler, mains to storage tank, air handling unit, fan and the like. **D3** Cross reference to drawings and/or a full specification. **D4** High and low level to be in accordance with National Health and Safety recommendations and National Working Rule Agreements.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
	nr	Off-load and position of primary equipment.						
	nr	Assembly of composite items of primary equipment.						
2 Terminal equipment and fittings		**1** System.	**1** Type, size, capacity, load, rating, special finishes, casings, in-fills, associated integrated or remote ancillaries, controls, indicators or components, anti-vibration mountings, acoustic performance/ treatment and method of fixing.	**1** Location of installation–roofs, high or low level in plant rooms, risers or service ducts, high or low level on floors.		**D5** Terminal equipment and fittings are defined as any item to which a system is distributed, e.g. heat emitters, grilles, fan coil units and the like.		

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
3 Pipework	m	**1** Straight **2** Curved, radii stated **3** Flexible **4** Extendable	**1** Material, finish, nominal diameter, method of jointing, fixing and background for fixing.	**1** Location of installation–Roofs, high or low level in plant rooms, risers or service ducts, high or low level on floors, in trenches.	**M4** Unless measured separately, all fittings are deemed to be included.		**C3** Cutting and jointing pipes to the fittings, loops and compensators is deemed to be included.	
4 Alternative 1 – pipe fittings	nr	**1** unions **2** connectors **3** flanges, **4** bends, **5** tees, **6** sets, **7** junctions, **8** reducers, **9** test points, **10** bosses, **11** sockets, **12** tappings and the like.	**2** Type, material, finish, nominal diameter, method of jointing.		**M5** Fittings which are reducing are measured extra over the largest pipe in which they occur.			

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
5 Pipe ancillaries	nr	**1** valves, **2** strainers, **3** expansion bellows, **4** anchors, **5** guides and rollers, **6** gullies, **7** outlets, **8** rainwater heads, **9** tundishes, **10** traps, **11** pipe sleeves **12** wall, floor and ceiling plates and the like.	**1** Type, material, finish, nominal, diameter, method of jointing.					
6 Ventilation ducts	m^2 m	Material, dimensions, method of jointing, finish, section shape, fixings and background for fixing.	Location of installation – roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors		**M6** Generally all ductwork to be measured inclusive of all fittings, i.e. connectors, flanges, bends, tees, junctions, reducers, spigots, test holes and the like.			**S7** Unless measured separately, all fittings are deemed to be fittings deemed to be included.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
7 Alternative1 – duct fittings	nr	**1** Joints, flanges, bends, tees, sets, junctions, reducers, spigots, test holes and the like.	Type, material, finish, shape, dimensions, method of jointing.					
8 Duct ancillaries		Dampers, in duct heater/cooling coils.	Type, material, finish, shape, Type, material, finish, shape, dimensions, method of jointing.					
9 Insulation and fire protection	nr	**1** To equipment and fittings	Type, material, finish	Location of installation – roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors, in trenches.			**C4** Screwed sockets, tappings and bosses are deemed to include perforating the pipe	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
	m	**2** To pipework	Thickness, material, finish, nominal diameter of pipe.		**M7** Generally all insulation to pipework to be measured inclusive of all fittings, i.e. unions, connectors, flanges, bends, tees, bosses, sockets, tappings and the like.		**C5** Unless measured separately, all fittings are deemed to be included.	
10 Alternative 1 – Insulation and fire protection to pipe fittings	nr	unions connectors flanges, bends, tees, sets, junctions, reducers, test points, bosses, sockets, tappings and the like.	**1** Type, material, finish, nominal diameter, method of jointing.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
11 Insulation and fire protection to pipe ancillaries	nr	**1** valves, **2** strainers, **3** expansion bellows, **4** anchors, **5** guides and rollers, **6** gullies, **7** outlets, **8** rainwater heads, **9** tundishes, **10** traps, **11** pipe sleeves and the like	**1** Type, material, finish, nominal diameter, method of jointing.	**1** Location of installation– roofs, high or low level in plant rooms, risers or service ducts, high or low level on floors, in trenches.				
12 Insulation and fire protection to pipe ancillaries	m^2	**1** Ventilation ducts	**1** Thickness, material, finish, duct dimensions.	Location of installation – Roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors, in trenches.	**M8** Generally all insulation to ductwork to be measured inclusive of all fittings, i.e. connectors, flanges, bends, tees, sets, junctions, reducers, spigots, test holes and the like, on the external girth of the duct to be Insulated.		**C6** Unless measured separately, insulation and fire protection to all fittings are deemed to be included.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
13 Alternative 1 – Insulation and fire protection to duct fittings		Joints, flanges, bends, tees, sets, junctions, reducers, spigots, test holes and the like.	Type, material, finish, nominal diameter, method of jointing.					
14 Insulation and fire protection to equipment		Thickness, material, finish, nominal area, performance rating.			**M9** Insulation to equipment measured separately where not given in the description of the items to which it relates.			
15 Fire stopping			Material, location, size of opening to be stopped, size of service stopping to be installed around, depth and fire rating.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
16 Identification	item	**1** Plates. **2** Discs. **3** Labels. **4** Tapes. **5** Symbols or numbers. **6** Bands. **7** Charts or diagrams. **8** Other: type stated.						
17 Testing	item						**C7** To include chemical treatment, purging, sterilisation, pressure testing.	
18 Commissioning	item						**C8** To include all final balancing, calibration and setting to work.	
19 System validation	item							
20 Operation and maintenance manuals	item							

MECHANICAL SERVICES (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
21 Drawing preparation	item	Installation, co-ordination, as installed, and record drawings.						
22 Training	item							
23 Loose ancillaries	item	Keys, tools, spares, chemicals and the like.						
24 Post-completion services	item	Maintenance, adjustments, servicing and the like.				**D6** See glossary of terms for definition of completion.		

1.39 ELECTRICAL SERVICES

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Drawings that must accompany this section of measurement include the following: **1** Floor and site plans detailing layout of systems. **2** System schematics. **3** Cross sections and elevations. **4** Detail layouts for plantrooms **5** Specific installation details for equipment and fittings. **6** Equipment schedules. **Mandatory information to beprovided** **1** Type, quality, size or thickness of materials **2** Method of fixing **3** Location	**M1** Where the mandatory information is available, measurement of cable containment and busbars will be fully detailed with the measurement of fittings identified and separately, in accordance with Alternative 1. However, where the mandatory information is not available then cable containment and busbars are to be measured inclusive of fittings.		**C1** All work is deemed internal unless stated as external. **C2** All waste **C3** Extra material for labour made fittings. **C4** All brackets, supports, fixings and cast in channels. **C5** All labelling, tagging, identification and charts. **C6** Marking of all holes chases and mortices. **C7** Earth connectors, straps and links. **C8** The allowances for cables entering fittings, luminaries, equipment and the like	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Primary equipment	nr	**1** System	**1** Location of installation - Roofs, high or low level in plant rooms, risers or service ducts, high or low level on floors.	**1** Type, size, capacity, load, rating, special finishes, casings, in-fills, associated integrated or remote ancillaries, controls, indicators or components, anti-vibration mountings, acoustic performance/treatment and method of fixing.		**D1** Primary equipment is defined as plant or equipment from which a system originates, e.g. main switch board, main control box and the like.		
	nr	**2** Off-load and position of primary equipment.						
	nr	**3** Assembly of composite items of primary equipment.						

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
2 Terminal Equipment	nr	System.	Type, size, capacity, load, rating, special finishes, casings, in-fills, associated integrated or remote ancillaries, controls, indicators or components and method of fixing.	Location of installation – Roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors.		**D2** Terminal equipment and fittings are defined as any item to which a system is distributed, e.g. luminaires, switches, actuators and the like.		
3 Cable Containment (Excluding those in final circuits)	m		Type, material, finish, section shape, dimensions, number of compartments, method of jointing, fixings and background for fixing.	**1** Location of installation – roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors, in trenches.	**M2** Generally all cable containment to be measured inclusive of all fittings, i.e. joint boxes, connectors, flanges, bends, tees, junctions, reducers, spigots, fire barriers and the like.		**C9** Unless measured separately, all fittings are deemed to be included	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
4 Alternative 1 – cable containment fittings	nr	Joint boxes, connectors, flanges, bends, tees, sets, junctions, reducers, spigots, fire barriers and the like.	**1** Type, size, overall length and type of adaptors stated					
5 Cables	m		**1** Type, rating, size, number of cores, material, armouring, sheathing, method of jointing, fixings and background for fixing.	**1** Location of installation – roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors, in trenches.				
6 Cable terminations and joints	nr		**1** Type, rating, size, number of cores, material, armouring, sheathing.				**C10** Includes all pots, seals, glands, lugs, connector blocks and shrouds.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
7 Final circuits	nr		**1** Cable type, rating, size, material, sheathing, number and type of points.	**1** Location of installation – roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors.			**C11** Includes all containment not measured separately, junction boxes, terminations, pots, seals, glands, lugs, connector blocks and shrouds. **C12** Includes fixing containment or cables in chases, surface or suspended from soffits.	
8 Modular wiring systems	nr		**1** Cable type, rating, size, material, sheathing, number and type of points.				**C13** Includes all containment not measured separately, junction boxes, terminations, pots, seals, glands, lugs, connector blocks and shrouds.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Modular wiring systems (Cont'd)							**C14** Includes fixing containment or cables in chases, surface or suspended from soffits.	
9 Busbar	m		**1** Type, rating, material, number of bars, method of jointing, fixings and background for fixing.	**1** Location of installation – Roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors, in trenches.	**M3** Generally all busbar to be measured inclusive of all fittings, i.e. connectors, flanges, bends, tees, sets, junctions, feeder units, tap-off boxes fire barriers and the like.		**C15** Unless measured separately, all fittings are deemed to be included	
10 Alternative 1 – busbar fittings	nr	**1** Connectors, flanges, bends, tees, sets, junctions, feeder units, tap off boxes fire barriers and the like.	**1** Type, material, finish, shape, dimensions, method of jointing.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Alternative 1 – busbar fittings (Cont'd)		**2** Joints, flanges, bends, tees, sets, junctions, reducers, spigots, test holes and the like.						
11 Tapes	m		**1** Type, rating, size, material, sheathing, method of jointing, fixings and background for fixing.	**1** Location of installation – Roofs, high or low level in plantrooms, risers or service ducts, high or low level on floors, in trenches.			**C16** Includes all connections, joints, test clamps.	
12 Electrodes, earth rods, air terminations, bars	nr		**1** Type, rating, size, material, method of jointing, fixings and background for fixing.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
13 Fire stopping and other associated fire protect work.	nr		**1** Material, location, size of opening to be stopped, size of service stopping to be installed around, depth and fire rating.					
14 Identification	item	**1** Plates. **2** Discs. **3** Labels. **4** Tapes. **5** Symbols or numbers. **6** Bands. **7** Charts or diagrams. **8** Other: type stated.						
15 Testing	item							
16 Commissioning	item							
17 System validation	item							

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
18 Operation and maintenance manuals	item							
19 Drawing preparation	item	**1** Installation, co-ordination, as installed and record drawings.	**1** Classification of drawings.					
20 Training	item							
21 Loose ancillaries	item	**1** Keys, tools, spares, chemicals.						
22 Post practical completion Services	item		**1** Maintenance, adjustments and servicing.					

1.40 TRANSPORTATION

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
A Drawings that must accompany this section of measurement. **1** Location plans **B Mandatory information to be provided.** **1** Kind, quality and size or thickness of materials. **2** Location. **3** Earth connectors, straps and links. **C Minimum information that must be shown on the drawings that accompany this section of measurement.** **1** Overall dimensions of items or systems			**C1** All work is deemed internal unless stated as external. **C2** All waste. **C3** Extra material for labour made fittings. **C4** All brackets, supports, fixings and cast in channels. **C5** All labelling, tagging, identification and charts. **C6** Marking of all holes chases and mortices.	

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 System	nr	**1** Type, size, capacity, load, rating, length, number of stops, storey height, associated integrated or remote ancillaries, controls, indicators or components.					**C7** Cross referenced to drawings and/or a full specification detailing finishes.	
	nr	**2** Off-load and position of primary equipment						
	nr	**3** Assembly of composite items of primary						
	item	**4** Free issue fixing steelwork and other components for installation by others.	**1** Type of free issue materials.					

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Systems (Cont'd)	item	**5** Interface with and connection to systems supplied and installed by others.	**1** Type of system.	**1** Nature of interface.			**C8** Systems supplied and installed by others will include telephones, intercoms, fire alarm and the like.	
2 Fire stopping and other associated fire protect work	nr	**1** Material, location, size of opening to be stopped, size of service stopping to be installed around, depth and fire rating.						
3 Identification	item	**1** Plates. **2** Discs. **3** Labels. **4** Tapes. **5** Symbols or numbers. **6** Bands. **7** Charts or diagrams. **8** Other: type stated.						

TRANSPORTATION (cont'd)

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
4 Testing and commissioning	item							
5 System validation	item							
6 Operation and maintenance manuals	item							
7 Drawing preparation	item	**1** Classification of drawings.					**C9** Installation, co-ordination, as installed and record drawings.	
8 Training	item							
9 Loose ancillaries	item	**1** Keys, tools, spares, chemicals.						
10 Post practical completion services	item	**1** Maintenance, adjustments and servicing.						

SECTION TWO (CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKE)

2.0 CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKES

2A GENERAL ITEMS

Includes: **General Conditions, Site Services and Facilities, Temporary Works, Testing of Materials and Work, Provisional Sums and Prime Cost Sum**.

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Location drawings as defined in General Rule 5.1 **2** Information to facilitate visiting site and addresses where drawings or other information additional to that required by these rules or listed information may be inspected					**M1** The unit of measurement for general items may be the sum, except where another unit of measurement is used in accordance with rule M2			
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 General items	item sum	**1** Contractual requirements	**1** Performance bond **2** Insurance of the works **3** Parent company guarantee		**M2** An item may be given for a Parent Company Guarantee where it is envisaged a tendered may be a subsidiary of a larger company or group	General items include general conditions, site services and facilities, temporary works, testing of materials and work, provisional sums and prime cost sum.	**C1** Items for insurance classed as contractual requirements shall be deemed to include only provision of insurance in accordance with the relevant clauses of contract in use.	

2.0 CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKES

GENERAL ITEMS

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
General items (Cont'd)		**2** Specified requirements	**1** Accommodation for the contract administrators	**1** Offices **2** Laboratories **3** Cabins	**M3** A quantity shall be given against items for specified requirements of which the value is to be ascertained and determined by measurement in accordance with the Conditions of contract. A unit of measurement shall be stated for each item	**D1** All works other than the permanent works which is expressly stated in the Contract to be carried out by the Contractor and of which the nature and extent is expressly stated in the Contract shall be classed as specified requirements		**S1** Item description for work classed as *specified requirement* which is to be carried out after the date for completion shall so state.
			2 Services for the contract administrators	**1** Transport vehicles **2** Telephones				
			3 Equipment for use by the contract administrators	**1** Office equipment **2** Laboratory equipment **3** Surveying equipment				

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
General items (Cont'd)	item	Specified requirements (Cont'd)	**4** Attendance upon the contract administrators	**1** Drivers **2** Chainmen **3** Laboratory assistants				**S2** Item descriptions for work classed as specified requirements shall allow for the pricing of the services as follows: **a)** establishment **b)** continued operation or maintenance **c)** removal
	sum		**5** Testing of materials the **6** Testing of works					
			7 Temporary works	**1** Traffic diversion **2** Traffic regulation **3** Access roads **4** Bridges **5** Cofferdams **6** Pumping **7** De-watering **8** Compressed air tunnelling	**M4** Items shall be given in this class for all testing for which items are not given separately as set out in other classes			**S3** Item description for testing of materials and testing of the works shall include particulars of samples and methods of testing.

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
General items (Cont'd)	item	**3** Method related charges	**1** Accommodation and buildings	**1** Offices **2** Laboratories **3** Cabins **4** Stores **5** Canteens and messroom	**M5** Items for Method-Related Charges, if any shall be inserted by the Tenderer			**S4** An item description for Method-Related Charges shall distinguish between Fixed and Time-Related charges
			2 Services	**1** Electricity **2** Water **3** Security **4** Hoardings **5** Site transport **6** Personnel transport **7** Welfare				
	sum		**3** Plant	**1** Cranes **2** Transport **3** Earthmoving **4** Compaction **5** Concrete mixing **6** Concrete transport **7** Pile driving **8** Pile boring				
			4 Plant	**1** Pipe laying **2** Paving **3** Tunnelling **4** Crushing and screening **5** Boring and drilling				

CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKES

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
General items (Cont'd)			**5** Temporary works	**1** Traffic diversion				
				2 Traffic regulation				
				3 Access roads				
				4 Bridges				
				5 Cofferdams				
				6 Pumping				
				7 De-watering				
				8 Compressed air for tunnelling				
			6 Temporary works	**1** Access scaffolding				
				2 Support for scaffolding and propping				
				3 Piling				
				4 Formwork				
				5 Shafts and pits				
				6 Hardstandings				
			7 Supervision and labour	**1** Supervision				
				2 Administration				
				3 Labour teams				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
General items (Cont'd)		**4** Provisional sums	**1** Daywork	**1** Labour **2** Percentage adjustment to Provisional Sum for Daywork labour **3** Materials **4** Percentage adjustment to Provisional Sum for Daywork materials **5** Plant **6** Percentage adjustment to Provisional Sum for Daywork plant **7** Other charges **8** Percentage adjustment to Provisional Sum for Daywork other charges	**M6** Items for percentage adjustment of to Provisional Sums for Daywork shall be given only where a Daywork Schedule is given in the Bill of Quantities. Adjustments shall be inserted against such items to correspond with adjustments if any inserted by the Tenderer in the Daywork Schedule			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
General items (Cont'd)			**2** Provisional Sums -Defined		**M7** Provisional Sums –Defined Work shall be given where the scope of the Work cannot be completely designed but where the scope can be defined in rule C2	**D2** Provisional Sums –Defined Work where the scope of the work cannot be completely designed but where the scope can be defined.	**C2** Where Provisional Sums –Defined Work are included the Tenderer is deemed to have made due allowances for the programming, planning and pricing of the Works within his general Items	**S5** Item descriptions for Provisional Sums –Defined Work shall state the nature of the work, indicative quantities of the scope and extent of the work

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
General items (Cont'd)		**5** Nominated Sub contractors which include work on site **6** Nominated Sub contractors which do not include work on site	**1** Prime Cost Item **2** Labour **3** Special labour **4** Other charges and profit		**M8** Each Prime Cost item shall be followed by an item for labours and an item for other charges and profit. Where labours other than or in addition to those defined in the Definition rule, are to be provided, the item for labours shall be designated as for special labours			**S6** Item descriptions for Prime Cost items shall identify the work involved **S7** The labours shall be stated in item descriptions for special labours

2B GROUND INVESTIGATION

Includes: Trial Pits, Trenches, Boreholes, Samples, Site and Laboratory Tests, Instrumental Observations and Professional Services in Connection with Ground Investigation.
Excludes: Excavation not Carried out for the Purpose of Ground Investigation for Ground Work, Piling and Pile Ancillaries (2E, 2P and 2Q)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS			**C1** Items for ground investigation shall be deemed to include Trial pits and trenches, boreholes, samples, site and laboratory tests, instrumental observations and professional services in connection with ground investigation **C2** Items for ground investigation shall be deemed to include disposal of excavated material and removal of dead services.	

2B GROUND INVESTIGATION

SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Ground investigation	nr	**1** Trial pits and trenches	**1** Number in material other than rock **2** Number in material which included rock	**1** Maximum depth: not exceeding 1.00 m **2** 1.00-2.00 m **3** 2.00-3.00 m **4** 3.00-5.00 m **5** 5.00-10.00 m **6** 10.00-15.00 m **7** 15.00-20.00 m **8** stated exceeding 20.00 m				**S1** Item description for the number and depth of trial pits and trenches shall state the minimum plan area at the bottom of the pit or trench, or where the work is undertaken to locate services the maximum length of the trench.
	m		**3** Depth in material other than rock **4** Depth in rock **5** Depth supported **6** Depth backfilled, material stated					**S2** Item description for the number of depth of trial pits and trenches (2B1:1-4) shall identify separately those which are expressly required to be excavated by hand.

SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground investigation (Cont′d)	h		**7** Removal of obstructions **8** Pumping at a stated minimum extraction rate					**S3** Item description for pumping shall state any special de-watering methods which are expressly required.
	nr	**2** Light cable percussion boreholes	**1** Number				**C3** Items for the depth of light percussion boreholes shall be deemed to include casings.	**S4** Items descriptions for the number and depth of light cables percussion boreholes, shall state the nominal diameter of the bases of the boreholes.
			2 Depth	**1** In holes maximum depth: not exceeding 5.00 m **2** 5.00-10.00 m **3** 10.00-20.00 m **4** 20.00-30.00 m **5** 30.00-40.00 m **6** stated exceeding 40.00 m				
	m		**3** Depth backfilled, material stated					

SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)	h		**4** Chiselling to prove rock or to penetrate obstructions		**M1** Chiselling to prove rock or to penetrate obstructions, shall be measured only where it is specifically required			
	nr	**3** Rotary drill boreholes	**1** Number			**D1** Rotary drilled boreholes shall be classed as such whether the boring is carried out by a rig or an attachment		**S5** Item descriptions for rotary drilled boreholes shall state the nominal minimum core diameter
	m		**2** Depth without core recovery **3** Depth with recovery	**1** In holes maximum depth: not exceeding 5.00 m **2** 5.00-10.00 m **3** 10.00-20.00 m **4** 20.00-30.00 m **5** 30.00-40.00 m **6** stated exceeding 40.00 m		**D2** Core boxes shall be deemed to become the property of the Employer unless otherwise stated		**S6** Item descriptions for the number of rotary drilled boreholes shall state those which are continuations of light percussion boreholes
	m		**4** Depth cased **5** Depth backfilled, material stated					

SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)	nr		**6** Core boxes, length core stated			**D3** The maximum depth used for calculation or rotary drilled boreholes which are continuations of light cable percussion boreholes shall be measured from the Commencing Surface of the light cable percussion borehole		**S7** Item descriptions for rotary boreholes which are inclined shall state the angle of inclination
	nr	**4** Samples	**1** From the surface or from trial pits and trenches	**1** Undisturbed soft material **2** Disturbed soft material **3** Rock **4** Ground water				**S8** Item descriptions for samples shall state their size, type and class in accordance with BS 5930 or equal approved standard

ITEM OR WORK TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)			**2** From boreholes	**1** Open tube **2** disturbed **3** Ground water **4** Stationary piston **5** Swedish foil **6** Delft **7** Bishop sand				
	h	**5** Site tests and observations	**1**	**1** Permeability				**S9** Item descriptions for permeability tests shall state the type and give particulars of the tests.
	nr			**2** Ground water level				**S10** Item descriptions for ground water level observations shall state when the measurements are to be taken.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground investigation (Cont'd)				**3** Standard penetration				**S11** Item s for standard penetration tests shall state whether they are in light cable percussion or rotary drilled boreholes
				4 Penetration vane **5** Vane in borehole **6** Pressure meter				
			2	**1** Plate bearing				**S12** Item descriptions for plate bearing tests shall state whether they are in pits and trenches or boreholes or at the surface.
				2 Self- boring Pressure meter **3** California bearing ratio				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)	nr			**4** Static cone sounding				**S13** Item descriptions for static cone sounding shall state the maximum depth of the cone and, where electric cones are used, the maximum capacity of the machine
				5 In situ density **6** Mackintosh probe				
				7 Hand auger borehole				**S14** Item descriptions for hand auger borehole tests and observations shall state the maximum diameter and maximum depth of the boreholes
	m	**6** Instrumental observations	**1** Pressure head	**1** Stand pipes				**S15** Item description for instrumental observations shall state details of the type of observations and the type of protective fences
	m			**2** Piezometers				
	nr			**3** Install covers **4** Readings				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)	m		**2** Inclinometers	**1** Installations				
	nr			**2** Readings			**C4** Item descriptions for inclinometers and settlement gauges shall be deemed to include provision of special boreholes.	**S16** Item descriptions for inclinometers and settlement gauges shall state whether the instruments are in boreholes.
	nr		**3** Settlement gauge **4** Resistivity **5** Seismic **6** Magnetometer **7** self-potential **8** Gravimetric	**1** Installations **2** Readings				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)	nr	**7** Laboratory tests	**1** Classification	**1** Moisture content **2** Atterberg limits **3** Specific gravity **4** Particle size analysis by sieve **5** Particle size analysis by pipette or hydrometer **6** Frost susceptibility				
			2 Chemical content	**1** Original organic matter **2** Sulphate **3** pH value **4** Contaminants				**S17** Item descriptions for tests should state the standards required and the contaminants to be analysed.
			3 Compaction	**1** Standard **2** Heavy **3** Vibratory				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)		Laboratory tests (Cont'd)	**4** Consolidation	**1** Odometer cell **2** Triaxial cell **3** Rowe cell				**S18** Item descriptions for triaxial cell and Rowe cell tests shall state the number of increments and the effective pressures.
			5 Permeability	**1** Constant head **2** Falling head				
			6 Soil strength	**1** Quick undrained triaxial **2** Consolidated undrained triaxial with water pressure measurement				**S19** Item descriptions for quick undrained triaxial test shall state the diameter and whether a single, multistage or set of three specimens is required.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)		Laboratory tests (Cont'd)	Soil strength (Cont'd	**3** Consolidated drained triaxial with volume change measurement **4** Shearbox: peak only **5** peak and residual **6** residual only **7** ring shear **8** California bearing ratio				**S20** Item description for consolidated triaxial tests shall state the diameter and the effective pressures. Multistage tests shall be identified in item descriptions **S21** Item descriptions for shearbox tests shall state the normal pressures and the size of the shearbox **S22** Item descriptions for California bearing ratio tests shall state the compactive effort, surcharge and whether soaking is required.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)		Laboratory tests (Cont'd)	**7** Rock strength	**1** Unconfined compressive strength of core samples. **2** Consolidated drained triaxial with volume change measurement **3** Brazilian **4** Ring shear **5** Point load				**S23** Item descriptions for unconfined compressive strength tests shall state the diameter and height of the samples. **S24** Item descriptions for consolidated drained triaxial tests shall state the diameter and height of the samples and the effective pressures. **S25** Item descriptions for Brazilian tests shall state the diameter and length of the samples.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)		Laboratory tests (Cont'd)	Rock strength (Cont'd)					**S26** Item descriptions for ring shear tests shall state the normal pressures and the diameter of the samples. **S27** Item descriptions for load tests shall state the type of test required and the minimum dimensions of the samples.
	h	**8** Professional services	**1** Technician **2** Technician engineer		**M2** Professional services shall be measured only where they are expressly required for analysis or records and results		**C5** Items for professional services shall be deemed to include preparation and submission of reports and keeping records of time spent	
			3 Engineer or Geologist	**1** Graduate **2** Registered **3** Principal or Consultant				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Ground Investigation (Cont'd)			**4** Visits to the site **5** Overnight stays in connection with visits to the sites		**M3** The hours measurement shall be working hours and shall exclude hours occupied in travel, meals, etc		**C6** Items for visits to the site and overnight stays in connection with visits to the site shall be deemed to include travelling, meals, accommodation and other incidental expenses	

2C GEOTECHNICAL AND OTHER SPECIALIST PROCESSES

Includes: Geotechnical processes for altering the properties of soils and rocks and other specialist processes as listed

Excludes: Compaction (included in 2E), Grouting carried out from within tunnels, shafts and other subterranean cavities (included in 2T), Grouting carried out from within sewers (included in 2Y)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** The Commencing Surface adopted in the preparation of the Bill of Quantities, shall be adopted for the measurement of the completed work		**C1** Items for work in this Section shall be deemed to include disposal of excavated material and removal of dead services	
			C2 Drilling and excavation for work in this Section, shall be deemed to include material other than rock or artificial hard material unless otherwise stated in the descriptions	
	M2 The depth of grout holes for ground anchorages and drains shall be measured along the holes irrespective of the inclination			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Geotechnical processes	m	**1** Drilling for grout holes through materials other than rock or artificial hard material **2** Drilling for grout holes through rock or artificial hard material **3** Driving injection pipes	**1** Vertically downwards **2** Downwards at an angle 0^0 -45 to the vertical **3** Horizontally or downward at an angle less than 45^0 to the horizontal **4** Upwards at an angle 0^0-45^0 to the horizontal **5** Upwards at an angle less than 45^0 to the	**1** In holes of depth: not exceeding 5 m **2** 5-10 m **3** 10-20 m **4** 20-30 m **5** 30-40 m **6** stated exceeding 40m	**M3** Drilling through previously grouted holes in the course of stage grouting shall not be measured. Where holes are expressly required to be extended, the number of holes shall be measured and drilling through previously grouted holes shall be measured as drilling through rock or artificial hard material.			**S1** The diameter of holes shall be stated in item descriptions for drilling and driving for grout holes and the likes.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Geotechnical and Other Specialist Processes (Cont'd)	nr	**4** Grout holes materials and injection	**1** Number of holes **2** Number of stages **3** Single water pressure tests **4** Multiple water pressure tests	**1** Cement **2** Pulverized fuel ash **3** Sand **4** Pea gravel **5** Bentonite **6** Chemicals	**M4** The number of stages measured shall be the total number of grouting stages expressly required.			
	t							
	nr		**6** Injection	**1** Number of injections **2** Neat cement grout **3** Cement and stated filler **4** Chemical grout **5** Other stated grout	**M5** The mass of grout materials measured shall not include the mass of mixing water			**S2** The type of materials shall be stated in item descriptions for grout materials
	t							
	nr			**6** Single packer settings				
	nr			**7** Double packer settings				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Geotechnical and Other Specialist Processes (Cont'd)					**M6** The number of injections measured shall be total number of injections expressly required **M7** The mass of grout injection measured shall not include the mass of mixing water			**S3** Item descriptions for number of injections shall identify those which are in stages distinguishing between those which are in ascending and descending stages
	m^3	**5** Diaphragm walls	**1** Excavation in material other than rock or artificial hard **2** Excavation in rock **3** Excavation in artificial hard material	**1** Maximum depth: not **2** 5-10 m **3** 10-15 m **4** 15-20 m **5** 20-25 m **6** 25-30 m **7** stated exceeding 30m	**M8** Formwork for voids, rebates and fillets in diaphragm walls shall be classified as concrete accillaries (class 2G)	**D2** Diaphragm walls, are walls constructed using bentonite slurry or other support fluids	**C2** Items for excavation for diaphragm walls shall be deemed to include preparation and upholding sides of excavation	**S4** The thicknesses of diaphragm walls shall be stated in item descriptions for excavation and concrete for diaphragm walls

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Geotechnical and Other Specialist Processes (Cont'd)	m^3 t		**4** Concrete **5** Plain round steel bars reinforcement **6** Deformed high yield steel bars reinforcement	**1** Nominal size: 6 mm 8 mm 10 mm 12 mm 16 mm 20 mm 25 mm 32 mm or greater	**M9** The depths of concrete in diaphragm walls shall be measured from the cut-off levels expressly required. The volume of concrete shall be calculated as set out in the relevant rules M1 & M2 in class 2F **M10** The mass measured for reinforcement in diaphragm walls shall include that of stiffening, lifting and supporting steel cast in	**D3** The nominal size stated in item descriptions for bar reinforcement in diaphragm walls shall be the cross section size	**C3** Items for concrete in diaphragm walls shall be deemed to include trimming the faces of diaphragm walls and preparing their tops to receive other work **C4** Items for reinforcement in diaphragm walls shall be deemed to include supporting reinforcement and preparing protruding reinforcement to receive other work	**S5** The nature of the material shall be stated in item descriptions for excavation in artificial hard material **S6** The mix specification or strengths shall be stated in item descriptions for concrete in diaphragm walls
	sum m		**7** Waterproofed joints **8** Guide walls					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Geotechnical and Other Specialist Processes (Cont'd)					**M11** The mass of steel reinforcement shall be taken as o.785kg/100 mm^2 of cross section (7.85 t/m^3). The mass of other reinforcing materials shall be taken as stated in the contract. **M12** Guide wall shall be measured each side of the diaphragm wall			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Geotechnical and Other Specialist Processes (Cont'd)	nr	**6** Anchorages/Ground Reinforcement	**1** Number in material other than rock or artificial hard material to stated depth	**1** Temporary	**M13** The lengths of tendons for ground anchorages shall be measured between the outer ends of anchorages	**D4** Ground anchorages and soil nailing shall be classed as ground reinforcement		**S7** The composition, location and working load and details of water and grout testing, pregrouting and grouting shall be stated in item descriptions for ground anchorages
	m		**2** Total length of tendons in material other than rock or artificial hard material	**2** Temporary with single corrosion protection **3** Temporary with double corrosion protection **4** Permanent				
	nr		**3** Number in material which includes rock or artificial hard material to a stated maximum depth	**5** Permanent with single corrosion protection **6** Permanent with double corrosion protection				**S8** The composition, location and details of testing shall be stated in item descriptions for soil nailing

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Geotechnical and Other Specialist Processes (Cont'd)	m	**7** Sand bands and wick drains	**1** Number of drains	**1** Cross sectional dimension: not exceeding 100mm **2** 100-200mm **3** 200-300mm **4** 300-400mm **5** 400-500mm **6** stated exceeding 500mm	**M14** The number of pre drilled holes measured for sand, band and wick drains shall be the number expressly required to be predrilled through overlaying material			**S9** Sand drains, band drains and wick drains shall be separately identified in item descriptions and the materials of which they are composed stated
	nr		**2** Number of pre drilled holes					
	m		**3** Depth of overlaying material **4** Depth of drains of maximum depth: not exceeding 10 m **5** 10-15 m **6** 15-20 m **7** 20-25 m **8** stated exceeding 25m					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Geotechnical and Other Specialist Processes (Cont'd)	nr	**8** Vibro flotation	**1** Number of probes **2** Depth of probes	**1** Depth: not exceeding 5 m **2** Depth: 5-10 m **3** Depth: 10-20 m **4** Depth: 20-30 m **5** Depth: 30-40 m **6** Depth: stated exceeding 40m				

2.D DEMOLITION AND SITE CLEARANCE

Includes: Demolition and removal of natural and artificial articles, objects and obstructions which are above the original surface

Excludes: removal of articles, objects and obstructions and materials (other than tree roots) at or below the original surface (included in 44, 46, 50, 51, 52, 59, 61, 65, 66)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE							**C1** Items for demolition and site clearance shall be deemed to include disposal of the materials arising	**S1** Item description of work from which the materials arising remain property of the Employer shall so state
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
Demolition and Site Clearance	ha m^2	**1** Site clearance	**1** General clearance **2** Invasive plant species			**D1** General clearance shall include the demolition and removal of all articles, objects and obstructions which are expressly required to be cleared, except those for which separate items are given as set out in this class	**C2** Items for general clearance which include the removal of hedges shall be deemed to include the removal of hedges stumps of any diameter where these are also required to be removed	**S2** Item description for general clearance shall identify the area included unless it is the total area of the site **S3** Item description for invasive species shall state the method of treatment and type of vegetation

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Demolition and Site Clearance (Cont'd)		Site clearance (Cont'd)				**D2** Invasive plant species shall include any plant whose control is governed by legislation and which is treated by herbicidal or chemical process.	**C3** Items for invasive plant species shall include for the disposal of any vegetable matter remaining after treatment	**S4** Where holes left by stump removal are to be backfilled, item description for general clearance, trees and stumps shall state the nature of the backfilling material.
	nr	**2** Trees	**1** Girth: 500mm-2m **2** exceeding 2m			**D3** Girths of trees shall be measured 1m above ground level	**C4** Items for clearance of trees shall be deemed to include removal of the stumps where they are also required to be removed	
	nr	**3** Stumps	**1** Diameter: less **2** exceeding 1m					

DEMOLITION AND SITE CLEARANCE (cont'd)

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Demolition and Site Clearance (Cont'd)	sum	**4** Buildings **5** Other structures	**1** Brickwork **2** Concrete **3** Masonry **4** Metal **5** Timber **6** No predominant material	**1** Volume: not exceeding $50m^3$ **2** $50\text{-}100m^3$ **3** $100\text{-}250m^3$ **4** $250\text{-}500m^3$ **5** $500\text{-}1000m^3$ **6** $1000\text{-}2500m^3$ **7** $2500\text{-}5000m^3$ **8** stated exceeding $5000m^3$		**D4** The volume used in the classification of building and other structures shall be their approximate value occupied, excluding any volume below the Original surface.		**S5** Buildings and other structures shall be identified in item descriptions
	m	**6** Pipelines	**1** Nominal bore: 100-300mm **2** 300-500mm **3** exceeding 500mm		**M1** Pipelines within buildings and other structures shall be measured only where their nominal bore exceeds 300mm		**C5** Items for demolition of pipelines shall be deemed to include demolition and removal of	

2E EARTHWORKS

Includes: Excavation, Dredging, Filling, Compaction, Disposal and Landscaping

Excludes: Investigation, Diaphragm Walls and Ground Anchorages, Piles, Tunnels Shafts

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** In accordance with Section 3.1 General rules, the quantities of earthworks shall be computed net using dimensions from the drawings with no allowance for bulking, shrinkage or waste. Where boundaries between different materials are not shown on drawings, measurement shall be made on site			

CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKES

EARTHWORKS (cont'd)

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks	m^3	**1** Excavation by dredging **2** Excavation of cuttings	**1** Top soil **2** Material other than topsoil, rock or artificial hard material		**M2** The Commencing surface in the preparation of the Bill of Quantities shall be adopted for the measurement of the completed work	**D1** Excavated material shall be deemed to be made of material other than top soil, rock or artificial hard material unless otherwise stated in the item descriptions	**C1** Items for excavation shall be deemed to include upholding of excavation, additional excavation to provide working space and removal of dead services	**S1** The location and limits of excavation by dredging shall be stated in item descriptions where its extent would otherwise be uncertain
		3 Excavation for foundation	**3** Rock	**1** Maximum depth: not exceeding 0.25 m				
			4 Stated artificial hard material exposed at the commencing surface	**2** 0.25-0.5m	**M3** Excavation classed as by dredging in the Bill of Quantities, shall be measured as by dredging, irrespective of the method of excavation adopted by the Contractor			**S2** Item descriptions for excavation below a body of open water identified in the Preamble or General items shall identify the body of water
		4 General excavations		**3** 0.5-1m		**D2** Items for excavation within borrow pits shall be deemed to include removal and replacement of overburden and unsuitable material	**C2** Item descriptions for excavation shall be deemed to include removal of existing pipes of any material of any material or diameter.	
				4 1-2m				
				5 2-5m				
			5 Stated artificial hard material not exposed at the commencing surface	**6** 5-10m				
				7 10-15m				
				8 stated exceeding 15m				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)			**6** Controlled and hazardous material exposed at the commencing surface **7** Controlled and hazardous material not exposed at the commencing surface		**M4** Measurement of excavation by dredging shall be made from soundings unless otherwise stated **M5** An item shall be given for each separate stage of excavation where separate stages in the conduct of the work are expressly required	**D3** Excavation from within borrow pits shall be classed as general excavation **D4** Controlled hazardous materials shall include any material whose excavation and disposal is governed by legislation including invasive plant species	**C3** Items for excavation within borrow pits shall be deemed to include removal and replacement of overburden and unsuitable material	**S3** The location and limits of excavation for foundations shall be stated in item descriptions, where the limits would otherwise be uncertain. Excavation around pile shafts and for underpinning shall be so described and classed as excavation for foundations

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)					**M6** The volume measured for the excavation of a structure or foundation shall be the volume which is to be either occupied by or vertically above any part of the structure or foundation **M7** The volume measured for excavation below a body of open water shall be the volume below water when the water surface is at the level (or the higher level of fluctuation applicable) shown on the drawing to which reference is given			**S4** The Commencing Surface shall be identified in the description of each item for work involving excavation for which the Commencing Surface is not the Original Surface. The Excavated Surface shall be identified in the description of each item for which the Excavated surface is not the Final Surface S5 Item descriptions for excavation within borrow pits shall so state

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)					**M8** An isolated volume of artificial hard material or rock occurring within other materials to be excavated shall not be measured separately unless its volume exceeds 1 m^3 except that the minimum volume **M9** The volume measured for excavation within borrow pits shall be the net volume measured for filling			**S6** Item descriptions shall identify separately, excavation which is expressly required to be executed by hand **S7** Item description for controlled and hazardous materials shall state the nature of the material.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)	m^2	**5** Excavation ancillaries	**1** Trimming of excavated surfaces **2** Preparation of excavated surfaces	**1** Topsoil **2** Material other than topsoil, rock or artificial hard material	**M10** Trimming of excavated surfaces shall be to surfaces which are to receive no Permanent Works, whether trimming is expressly required or not	**D5** Disposal of excavated material shall be deemed to be disposal off the site unless otherwise stated in item description		**S8** Item descriptions for excavation ancillaries in connection with excavation by dredging shall be described **S9** Item descriptions for trimming of excavated surfaces and preparation of excavated surface shall identify surfaces which are: (a) inclined at an angle of 10-45^0 to the horizontal (b) inclined at an angle of 45-90^0 to the horizontal (c) vertical
	m^3		**3** Disposal of excavated material **4** Double handling of excavated material	**3** Rock **4** Stated artificial hard material **5** Controlled and hazardous materials				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)	m^3		**5** Dredging to remove silts **6** Excavation of material below the Final Surface and replacement with stated material		**M11** Preparation of excavated surfaces shall be to surfaces which are to receive Permanent Works whether preparation is expressly required or not except surfaces which are to receive filling or landscaping and surfaces for which formwork is measured	**D6** Trimming, preparation, disposal and double handling shall be deemed to be carried out upon material other than topsoil, rock or artificial hard material, unless otherwise stated in item descriptions		**S10** Where material is for disposal on the site, the location shall be stated in the descriptions for disposal of excavated material **S11** Item description for disposal of controlled and hazardous materials shall state the nature of the material
	m^2		**7** Timber supports left in **8** Metal supports left in					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
					M12 The volume of disposal of excavated material measured shall be the difference between the total net volume of excavation and the net volume of excavated material used for filling			
					M13 Double handling of excavated material shall be measured only to the extent that it is expressly required. The volume measured for double handling shall be that of the void formed in temporary stockpile from which the material is removed			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)					**M14** Dredgingto remove silt shall be measured only to the extent that is expressly required that silt which accumulates after the Final Surface has been reached, shall be removed **M15** The area measured for timber or metal supports left in shall be the area of supported surface surfaces for which the supports are expressly required to be left-in			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)	m^3	**6** Filling	**1** To structures **2** Embankment **3** General	**1** Excavated top soil **2** Imported top soil **3** Non-selected excavated material other than rock or top soil **4** Selected excavated material other than rock or top soil **5** Imported natural material other than rock or top soil **6** Excavated rock **7** Imported rock **8** Imported artificial material	**M16** The filling of excavations around completed structures, shall be measured only to the extent that the volume filled is also measured as excavation in accordance with rule M6 **M17** Where filling to form temporary roads is subsequently approved by the Engineer for incorporation into permanent filling, the volume placed shall not be deducted from the measurement of filling	**D7** Filling material shall be deemed to be non-selected excavated material other than topsoil or rock, unless otherwise stated in the descriptions. **D8** Filling material shall be deemed classed as excavated rock only where the use of rock as filling at stated locations is expressly required.	**C4** Items for filling shall be deemed to include compaction	**S12** The material shall be identified in item descriptions for filling with imported material **S13** Where different compaction requirements are specified for the same filling, they shall be stated in item descriptions for the filling **S14** Where the rate of deposition of filling material is limited, the limitation shall be stated in item descriptions for filling.
	m^2		**4** To stated depth or thickness					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)		Filling (Cont'd)			**M18** Additional filling necessitated by settlement of or penetration into underlying material, shall be measured only to the extent that its depth exceeds 75 mm **M19** The volume of imported filling material measured, shall be the difference between the net volume of filling and net volume of excavated material derived from work within classes 2E and 2T used for filling.	**D9** Filling shall be classed as to stated depth or thickness where material is provided of uniform total compacted depth or thickness such as in drainage blankets, topsoiling, pitching and beaching. Bulk filling shall not be classed as to stated depth or thickness notwithstanding that it may be compacted in separate layers of material of stated thickness.		**S15** The materials shall be identified in item descriptions for filling to stated depth or thickness. **S16** Item descriptions for filling to stated depth or thickness shall identify work upon surfaces which are: (a) inclined at an angle of 10-45^0 to the horizontal (b) inclined at an angle of 45-90^0 to the horizontal (c) vertical

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)					**M20** Where rock filling is deposited into soft areas, the volume shall be measured in the transport vehicles at the place of deposition			**S17** Item descriptions for high energy impact compaction shall state the type of compaction.
					M21 Where filling is to be deposited below water, and the quantity cannot be measured satisfactorily by any other means, its volume shall be measured in the transport vehicles at the place of deposition			**S17** Item descriptions for high energy impact compaction shall state the type of compaction.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)	m^3		**1** High energy impact compaction general fill	**1** Selected excavated material other than topsoil or rock **2** Imported natural material other than topsoil or rock	**M22** High energy impact compaction general fill existing ground shall be measured to the surface area to be treated			
	m^2			**3** Existing ground				
	m^2	**7** Filling ancillaries	**1** Trimming of filled surfaces **2** Preparation of filled surfaces	**1** Topsoil **2** Material other than topsoil, rock or artificial hard material **3** Rock **4** Stated artificial hard material	**M23** Trimming of filled surfaces shall be measured to surfaces which are to receive Permanent Works whether trimming is expressly required or not **M24**	**D10** Trimming and preparation shall be deemed to be carried out upon material other than topsoil, rock or artificial hard material unless otherwise stated in the description		**S18** Item descriptions for filling ancillaries stated depth or thickness shall identify work upon surfaces which are: (a) inclined at an angle of 10-45^0 to the horizontal (b) inclined at an angle of 45-90^0 to the horizontal (c) vertical

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)					Preparation of filled surfaces shall be measured to the surfaces which are to receive Permanent Works whether preparation is expressly required or not, except surfaces which are to receive filling or landscaping and surfaces for which formwork is measured			**S19** The type and grade of material shall be stated in item descriptions for geotextiles
	m^2		**3** Geotextiles		**M25** The area of additional geotextiles in laps shall not be measured			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Earthworks (cont'd)	m^2	**8** Landscaping	**1** Turfing **2** Hydraulic mulch grass seeding **3** Other grass seeding		**M26** The lengths of hedges measured shall be their developed lengths along the centre lines		**C5** Items for landscaping shall be deemed to include fertilizing, trimming and preparation of surfaces	**S20** Where turfing is pegged or wired item, descriptions shall so state **S21** Item descriptions for turfing and grass seeding shall identify work upon surfaces which are inclined at an angle exceeding 10^0 to the horizontal
	nr		**4** Plants, stated species and size **5** Shrubs, stated species and size **6** Trees, stated species and size					
	m		**7** Hedges, stated species size and spacing	**1** Single row **2** Double row				

2F INSITU CONCRETE

Excludes: Capping of Boreholes, Diaphragm Walls, Excavation Ancillaries, Granolithic and Other Applied Finishes, Drainage and Pipework, Piles, Tunnel and Shaft Linings Roads, Pavings and Kerbs, Foundations for Fences and Gates

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** The volume of concrete shall include that occupied by **(a)** reinforcement and other metal sections **(b)** prestressing components **(c)** cast-in components each not exceeding $0.1m^3$ in volume **(d)** rebates, grooves, throats, fillets, chamfers or internal splays each not exceeding $0.01m^2$ in cross-sectional area			

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
	(e) pockets and holes which are defined as large or small voids in accordance with rule D3 of class G			
	(f) joints or joint components between adjacent volumes of Insitu concrete			
	M2 The volume of concrete measured shall exclude that of nibs or external splays each not exceeding $0.01m^2$ in cross-sectional area			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu concrete (cont'd)	m^3	Provision of concrete **1** Designed concrete	**1** Strength C8/10 **2** C12/15 **3** C16/20 **4** C20/25 **5** C25/30 **6** C28/35 **7** C30/37 **8** C32/40	**1** Maximum aggregate size 20mm 40mm		**D1** Items for provision of concrete shall be classified in accordance with BS 8500 and BS EN 206-1 **D2** Concrete shall be classed as Designed concrete where the designer is responsible for ensuring the requirement for the concrete are in accordance with the standards		**S1** The specification of concrete in accordance with BS 8500 shall be stated in term description for Provision of concrete unless the concrete reference is stated for which the specification is given elsewhere in the contract **S2** Where BS 8500 is not used the specification or item description shall stated the details of the concrete to be provided

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu Concrete (Cont'd)	m^3	**2** Designated concrete	**1** Strength RC20/25 **2** RC25/30 **3** RC28/35 **4** RC30/37 **5** RC32/40 **6** RC35/40 **7** RC40/50			**D3** Concrete shall be classed as Designated concrete where the design of the concrete is generated by the concrete producer		**S3** The specified design exposure condition shall be stated in the item description for provision of concrete unless the concrete reference is stated for which the specification is given elsewhere in the contract
		3 Standardised prescribed	**1** ST1 **2** ST2 **3** ST3 **4** ST4 **5** ST5			**D4** Concrete shall be classed as Standardised prescribed concrete where it is limited to simple structural non reinforced applications		

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu Concrete (cont'd)		**4** Prescribed mix				**D5** Concrete shall be classed as Prescribed concrete where the designer provides full details of the concrete such that it will produce concrete that will achieve the required performance		**S4** Any specified requirements for admixtures shall be stated in the item description for Provision of concrete unless the concrete reference is stated for which the specification is given elsewhere in the contract
	m^3	**5** Proprietary concrete				**D6** Concrete shall be classed as Proprietary concrete where the application is to achieve a particular performance outside normal performance criteria.		**S5** Any specified requirements for 'Consistence categories' shall be stated in the item description for Provision of concrete unless the concrete reference is stated for which the

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu Concrete (cont'd)								**S5 (Cont'd)** specification is given elsewhere in the contract **S6** Any specified requirements for 'Maximum water to cement ratios' shall be stated in the item description for Provision of concrete unless the concrete reference is stated for which the specification is given elsewhere in the contract

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu Concrete (cont'd)								**S7** Any specified requirements for 'Minimum cement content kg/m^3 shall be stated in the item description for Provision of concrete unless the concrete reference is stated for which the specification is given elsewhere in the contract
	m^3	Placing of concrete **6** Mass **7** Reinforced **8** Prestressed	**1** Blinding **2** Bases, footings, pile caps and ground slabs **3** Suspended slabs **4** Walls	**1** Thickness not exceeding **2** 150-300mm **3** 3000-500mm **4** exceeding 500mm	**M3** Column and piers integral with a wall shall be measured as part of the wall, except where expressly required to be cast separately	**D7** Prestressed concrete which is also reinforced shall be classed as prestressed concrete.		**S8** Item descriptions for placing of concrete which is expressly required to be placed against an excavated surface(other than blinding) shall so be stated

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu Concrete (cont'd)			**5** Columns and piers **6** Beams **7** Casing to metal sections	**1** Cross-sectional area: not exceeding $0.03m^2$ **2** $0.03\text{-}0.1m^2$ **3** $0.1\text{-}0.25m^2$ **4** $0.25\text{-}1m^2$ **5** exceeding $1m^2$ **6** Special beam sections	**M4** Beams integral with a slab shall be measured as part of the slab, except where expressly required to be cast separately	**D8** The thickness used for classification of blinding shall be the minimum thickness. **D9** The thickness used for classification of ground slabs, suspended slabs and walls shall exclude the additional thickness of integral beams, columns, piers and other projections		**S9** The cross-sectional dimensions of special beam sections shall be stated in item descriptions, except where a beam type or mark number is stated for which dimensions are given on drawings. **S10** Item descriptions for components classed as other concrete forms shall identify the work and include one of the following: **(a)** the principal dimensions

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu Concrete (cont'd)						**D10** Concrete in suspended slabs and walls less than 1m wide or long shall be classed as concrete in beams and column respectively **D11** Beams shall be classed as special beam sections where their cross-sections profiles are rectangular or approximately rectangular less than 4/5 of their length or where they are of box or other composite section.		**(b)** a type of mark number for which principal dimensions are given on drawings **(c)** a statement locating the work for which principal dimensions are given on the **S11** Item description for sprayed concrete support shall state the specification of the concrete and whether it is reinforced and the minimum thickness

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu Concrete (cont'd)			**8** Other concrete forms			**D12** Sprayed concrete shall be designated as other concrete forms **D13** Reinforcing materials added to the mix for sprayed concrete shall not be classed as reinforcement		

2G CONCRETE ANCILLARIES

Includes: Formwork for Insitu concrete; Reinforcement for Insitu concrete; Joints in Insitu concrete; Post-tensioned prestressing; Accessories for Insitu concrete

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** Formwork shall be measured for surfaces of Insitu concrete which require temporary support during casting except where otherwise stated **M2** Formwork shall not be measured for the following **(a)** edges of blinding concrete not exceeding 0.2m **(b)** joints and associated rebates and groove **(c)** temporary surfaces formed at the discretion of the contractor			**S1** Formwork left in shall be so described in item description for formwork **S2** Item descriptions for formwork which is to upper surfaces shall so state, except where the surfaces are inclined at an angle not exceeding 10^0 to the vertical **S3** Item descriptions for formwork shall state where the formwork is to blinding concrete

CONCRETE ANCILLARIES (cont'd)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
					(d) surfaces of concrete which are expressly required to be cast against an excavated surface **(e)** surfaces of concrete which are cast against an excavated surface inclined at an angle less than 45^0 to the horizontal			**S4** Radii of curved formwork shall be stated in item description as follows: **(a)** to one radius in one plane (cylindrical), radius stated **(b)** to one radius in two planes (cylindrical), radius stated **(c)** varying result (conical), maximum and minimum radii stated.
WORK SECTION TO BE MEASURED	**UNIT**	**FIRST DIVISION**	**SECOND DIVISION**	**THIRD DIVISION**				
	m m^2	**1** Formwork: rough finish **2** Fair finish **3** Other stated finish **4** Stated surface features	**1** Plane horizontal **2** Plane sloping **3** Plane battered **4** Plane vertical **5** Curved to one radius in one plane **6** Other curved	**1** Width: not exceeding 0.1m **2** 0.1-0.2m **3** 0.2-0.4m **4** 0.4-1.22m **5** exceeding 1.22m		**D1** Plane formwork shall be classified according to its angle of inclination as follows: Class and Angle of inclination to the vertical		

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Concrete Ancillaries (cont'd)	nr		**7** For voids	**1** small void depth: not exceeding 0.5m **2** 0.5-1m **3** 1-2m **4** stated exceeding 2m **5** Large void depth: not exceeding 0.5m **6** 0.5-1m **7** 1-2m **8** stated exceeding 2m		**(a)** Horizontal, 85^0 90^0 **(b)** Sloping, 10^0-85^0 **(c)** Battered, 0^0-10^0 **(d)** Vertical, 0^0 **D2** Formwork shall be deemed to be for plane areas and to exceed 1.22m wide, unless otherwise stated.		
	m		**8** For concrete components of constant cross section	**1** Beams **2** Columns **3** Walls **4** Other members **5** Projections **6** Intrusions		The **D3** classification of large and small voids shall be as follows: Class, Maximum cross-section(circular voids (diameter), other voids(area)) **(a)** Large, circular voids, 0.35-0.7m		

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Concrete Ancillaries (cont'd)					**M3** Formwork to upper surfaces of concrete shall be measured to surfaces inclined at an angle exceeding 15^0 to the horizontal and to other required upper surfaces for which formwork is expressly required. **M4** Formwork to the surfaces of voids larger than those classed as large voids in accordance with rule D3 shall be measured as set out in this class for formwork generally.	**(b)** Large, other voids, 0.1-0.5m^2 **(c)** Small, circular voids, not exceeding 0.35m^2 **(d)** Small, other voids, not exceeding 0.1m^2 The depth of voids shall be measured perpendicularly to the adjacent surface of concrete. **D4** Nibs and external splays not exceeding 0.01m^2 in cross-sectional area shall be classed as projections		**S5** Item description for formwork for concrete components of constant cross-section, other than projections and intrusions, shall state the principal cross-sectional dimensions of the component and its mark number, location or other unique identifying feature. **S6** Formwork for curved concrete components of constant cross-section shall be so described stating the radii

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Concrete Ancillaries (cont'd)					**M5** Formwork for the surfaces of projections and intrusions exceeding 0.01m^2 in cross-sectional area shall be measured as set out in this class for formwork generally. **M6** The area of formwork measured shall include the area of formwork obscured by: **(a)** forms of large and small voids **(b)** forms of projection and intrusions	**D5** Rebates, grooves, internal splays, throats, fillets and chamfers not exceeding 0.01m^2 in cross-sectional area shall be classed intrusions		

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Concrete Ancillaries (cont'd)	t	**5** Reinforcement	**1** Plain round steel bars **2** Deformed high yield steel bars **3** Stainless steel bar of stated quality **4** Reinforcing bars of other stated material	**1** Nominal size:6mm **2** 8mm **3** 10mm **4** 112mm **5** 16mm **6** 20mm **7** 25mm **8** 32mm or greater	**M7** The mass of steel reinforcement shall be taken as 0.785kg/m per 100mm^2 0f cross-section (7.85t/m^3). The mass of other reinforcing materials shall be taken as stated in the contract	**D6** Welded, swaged or screwed sleeve joints in reinforcing bars shall be classed as special joints	**C1** Items for reinforcement shall be deemed to include supporting reinforcement other than steel supports to top reinforcement	**S7** Item description for bar reinforcement shall state the lengths of bars to the next higher multiple of 3m where they exceed 12m before bending
	nr		**5** Special joints					**S8** Item description for special joints shall state the type of joint and type and size of reinforcing bar.
	m^2		**6** Steel fabric **7** Fabric of other stated material	**1** Nominal mass: not exceeding 2kg/m^2 **2** 2-3kg/m^2 **3** 3-4kg/m^2 **4** 4-5kg/m^2 **5** 5-6kg/m^2 **6** 6-7kg/m^2 **7** 7-8kg/m^2 **8** stated exceeding 8kg/m^2	**M8** The mass of reinforcement measured shall include the mass of steel supports to top reinforcement **M9** The area of additional fabric in laps shall not be measured			**S9** Item description for steel fabric shall state the fabric reference size and mass or mass per square metre.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Insitu Concrete (cont'd)								**S10** Item description for fabric of other stated materials shall state the fabric sizes and nominal mass per square metre
	m^2	**6** Joints	**1** Open surface plain **2** Open surface with filter **3** Formed surface plain **4** Formed surface with filter	**1** Average width: not exceeding 0.5m **2** 0.5-1m **3** stated exceeding 1m	**M10** Joints shall be measured only where they are at locations where joints are expressly required. **M11** The widths of joints shall be measured between the outer surfaces of concrete with no deduction or addition for widths or depths occupied by rebates, grooves, fillets or	**D7** Joints for which temporary support of the whole surface area of concrete is required during casting shall be classed as formed surface joints. Other joints shall be classed as open surface joints	**C2** Items for open surface and formed surface joints shall be deemed to include intermediate surface treatment where expressly required. **C3** Items for joints shall be deemed to include formwork	**S11** The dimension and nature of components shall be stated in item descriptions for joints

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Concrete Ancillaries (cont'd)	m	Joints (Cont'd)	**5** Plastics or rubber **6** Metal waterstops	**1** Average width: not exceeding **2** 150-200mm **3** 200-300mm **4** stated exceeding 300mm			**C4** Item for waterstops shall be deemed to include cutting and joining of waterstops and provision of special fittings at angles and junctions	
	m		**7** Sealed rebates or grooves					
	nr		**8** Dowels	**1** Plain or greased **2** Sleeved or capped				
	nr	**7** Post-tensioned prestressing	**1** Horizontal internal tendons in in situ concrete **2** Inclined or vertical tendons in in situ concrete **3** Horizontal internal tendons in precast concrete **4** Inclined or vertical tendons in precast concrete	**1** Length: not exceeding 5m **2** 5-7m **3** 7-10m **4** 10-15m **5** 15-20m **6** 20-25m **7** 25-30m **8** stated exceeding 30m	**M12** Prestressing shall be measured by the number of tendons where tendons are used and by the number of external jacking operations where stress is induced by jacking only	**D8** Profiled tendons in horizontal components shall be classed as horizontal tendons **D9** The lengths of tendons used for classification shall be their developed lengths between the outer faces of anchorage	**C5** Items for prestressing shall be deemed to include ducts, grouting and other components and tasks ancillary to prestressing	**S12** Item description for prestressing shall identify the e concrete component to be stressed and state the composition of the tendon and particulars of the anchorage.
			5 External jacking operations					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Concrete Ancillaries (cont'd)	m^2	**8** Concrete accessories	**1** Finishing of top surfaces	**1** Wood float **2** Steel trowel **3** Other stated surface treatment **4** Granolithic finish **5** Other stated applied finish	**M13** Surface treatment shall not be measured to surfaces formed at the contractor's discretion **M14** The top areas of tops of wall and other surfaces which are not given separate finishing treatment shall not be measured as finishing of top surfaces **M15** No deduction from the areas measured for finishing shall be made for holes and openings in the finished surfaces each not exceeding $0.5m^2$		**C6** Items for granolithic and other stated applied finish shall be deemed to include materials, surface treatment, joints and formwork	**S13** The materials, thickness and surface treatments of granolithic and other stated applied finish shall be stated in item description
			2 Finish of formed surface	**1** Aggregate exposed using retarder **2** Bush hammering **3** Other stated surface treatment carried out after striking formwork				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Concrete Ancillaries (cont'd)	m nr		**3** Inserts	**1** Linear inserts **2** Other inserts	**M16** where inserts are expressly required to be grouted into performed openings the formwork shall be measured	**D10** Components cast or grouted into in situ concrete except reinforcement, prestressing and jointing materials shall be classed as	**C7** Items for inserts shall be deemed to include their supply unless otherwise stated	**S14** Item description for inserts shall identify the components to be cast or grouted in and state their principal dimension **S15** Item description for inserts shall identify: **(a)** those which project from one surface of the concrete **(b)** those which project from two surface of the concrete **(c)** those which are totally within the concrete volume

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Concrete Ancillaries (cont'd)		Concrete accessories (Cont'd)	Inserts (Cont'd)					**S16** where inserts are expressly required to be grouted into performed openings in concrete item descriptions shall so state. Materials for grouting and sizes of openings shall be stated.
	nr		**4** Grouting under plates	**1** Area: not exceeding $0.1m^2$ **2** $0.1\text{-}0.5m^2$ **3** $0.5\text{-}1m^2$ **4** stated exceeding $1m^2$				**S17** Materials shall be stated in item descriptions for grouting under plates

2H PRECAST CONCRETE

Excluded:
Pre-tensioned prestressing (included in 2G)
Precast concrete pipework (included in 2J and 2K)
Precast concrete manholes, catchpit and gullies (included in 2L)
Precast concrete piles (included in 2Q and 2R)
Precast concrete paving, kerbs and traffic sign supports (included in 2R and 2S)
Precast concrete tunnel linings (included in 2V)
Precast concrete blockwork (included in 2W)
Precast concrete fencing (included in 2Z)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE							Precast concrete is deemed to include manufacture, erection, joining and fixing of precast concrete units	
WORK SECTION TO BE MEASURED	**UNIT**	**FIRST DIVISION**	**SECOND DIVISION**	**THIRD DIVISION**				
Precast Concrete	nr	**1** Beams **2** Prestressed pre-tensioned **3** Prestressed post tensioned **4** Columns	**1** Length: not exceeding 5m **2** 5-7m **3** 7-10m **4** 10-15m **5** 15-20m **6** 20-30m **7** exceeding 30m	**1** Mass: not exceeding 250kg **2** 250-500kg **3** 500kg-1t **4** 1-2t **5** 2-5t **6** 5-10t **7** 10-20t **8** stated exceeding 20t		**D1** The mass used for classification in the second division shall be the mass of each unit	**C1** Items for precast concrete shall be deemed to include reinforcement, formwork, joints and finishes	**S1** The position in the works and specification of the concrete to be used in each type of precast unit shall be stated in item descriptions

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Precast Concrete (cont'd)	nr	**5** Slabs	**1** Area: not exceeding $1m^2$ **2** $1\text{-}4m^2$ **3** $4\text{-}15m^2$ **4** $15\text{-}50m^2$ **5** exceeding $50m^2$			**D2** Concrete components which are cast other than in their final position shall generally be classed as precast concrete unit **D3** Where site precasting of units is adopted for reasons other than to obtain multiple use of formwork and the nature of the work is characteristic of in situ concrete, but involves the movement of the cast units into their final positions, the units shall be classed as in situ concrete and items given in class A for the Temporary		**S2** Item descriptions shall state the mark or type number of each precast concrete unit. Units with different dimensions shall be given different mark or type numbers.
	nr	**6** Segmental units						
	m	**7** Units for subways, culverts and ducts			**M1** The length measured for units for subways, culverts and ducts and for copings, sills and weir blocks shall be the total length of identical units			**S3** Particulars of tendons and prestressing shall be stated in item description for prestressed pre-tensioned units.
	m	**8** Copings, sills and weir blocks	**1** Cross-sectional area: not exceeding $0.1m^2$ **2** $0.1\text{-}0.5m^2$ **3** $0.5\text{-}1m^2$ **4** exceeding $1m^2$					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Precast Concrete (cont'd)								**S4** The cross-section type and principal dimensions shall be stated in item description for beams, columns, segmental units, units for subways, culverts, ducts, copings, sills and weir blocks **S5** The average thickness shall be stated in item description for slabs **S6** The mass per metre shall be stated in item descriptions for units for subways, culverts, ducts, copings, sills and weir blocks

2J PIPEWORKS - PIPES, FITTINGS AND VALVE, MANHOLE AND ANCILLIARIES

Includes: Provision, Laying and Jointing of Pipes, Excavating and Backfilling Pipe Trenches

Excludes: Piped Building Services

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Pipes	m	**1** Clay pipes **2** Concrete pipes **3** Iron pipes **4** Steel pipes **5** Polyvinyl chloride pipes **6** Glass reinforced plastic pipes **7** High density polyethylene pipes **8** Medium density	**1** Nominal bore: not exceeding 200mm **2** 200-300mm **3** 300-600mm **4** 600-900mm **5** 900-1200mm **6** 1200-1500mm **7** 1500-1800mm **8** exceeding 1800mm	**1** Not in trenches **2** In trenches, depth: not exceeding 1.5m **3** 1.5-2m **4** 2-2.5m **5** 2.5-3m **6** 3-3.5m **7** 3.5-4m **8** exceeding 4m	**M1** The Commencing Surface adopted in the preparation of the Bill of Quantities shall be adopted for the measurement of the completed work. **M2** Backfilling of trenches shall not be measured except as set out in class K for filling of French and rubble drains and in class L for backfilling with material other than that excavated from the trenches.	**D1** Pipes not in trenches shall include pipes suspended or supported above the ground or other surface, pipes in headings, tunnels or shafts, pipes jacking and pipes laid within volumes measured separately for excavation. **D2** Pipes not in trenches shall be classed as such only where pipes are expressly required not to be laid in trenches.		**S1** The location or type of pipework in each item or group of items shall be stated in item description so that the pipe runs included can be identified by reference to the Drawings. **S2** The materials, joints types, nominal bores and lining requirement of pipes shall be stated in item descriptions. **S3** Item descriptions for pipes not in trenches shall distinguish between the different categories of pipes listed in rule 2D1.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipes (cont'd)						**D3** Depths used for classification of pipes in trenches shall be measured from the commencing surface to the inverts of the pipes.		**S4** The Commencing Surface shall be identified in the description of each item for work involving excavation for which the commencing surface is not also the Original Surface.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipes (cont'd)					**M3** Lengths of pipes shall be measured along their centre lines. Lengths of pipes in trenches shall include lengths occupied by fittings and valves and exclude lengths occupied by pipes and fittings comprising backdrops to manholes. Lengths of pipes and fittings comprising backdrops to manholes. Lengths of pipes not in trenches shall exclude lengths occupied by fittings and valves. Where more than one pipe is expressly required to be laid in one trench, the lengths measured shall be the length			**S5** Where more than one pipe is expressly required to be laid in one trench the item description for each pipe shall so state and also identify the pipe run. Where pipes are laid in French or rubble drains item descriptions shall so state. **S6** Trench depths exceeding 4m shall be stated in item descriptions to the next higher multiple of 0.5m **S7** Item descriptions shall identify separately excavation which is expressly required to be carried out by hand.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipes (cont'd)					**M4** Additional items shall be given in class 2K and 2L for work in connection with pipes not in trenches other than provision, laying and jointing of pipes. **M5** Lengths of pipes entering manholes and other chambers shall be measured to the inside surfaces of the chambers except that pipes and fittings comprising backdrops to manholes shall be included in items for manholes measured in 2L.			

2K PIPEWORKS - FITTINGS AND VALVES

Includes : Fittings and valves for pipework
Excludes : Work included in 2J, 2L, 2M and 2AA
Piped building services (included in 2AA)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE			**C1** Items for fittings and valves shall be deemed to include the supply of materials by the Contractor, unless otherwise stated. **C2** Items for fittings and valves shall be deemed to include excavation, preparation of surfaces, disposal of excavated material, upholding sides of excavation, backfilling and removal of dead services except	

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
							to the extent that such work is included in 2J, 2K and 2L.	
Fittings and valve	nr	**1** Clay pipe fittings **2** Concrete pipe fittings **3** Iron or steel pipe fittings **4** Polyvinyl chloride pipe fittings **5** Glass reinforced plastic pipe fittings **6** High density polythene	**1** Bends **2** Junctions and branches **3** Tapers **4** Double collars **5** Adaptors **6** Glands **7** Bellmouths **8** Straight specials	**1** Nominal bore : not exceeding 200mm **2** 200 - 300 mm **3** 300 - 600 mm **4** 600 - 900 mm **5** 900 - 1200 mm **6** 1200 - 1500 mm **7** **8** exceeding 1800 mm	**M1** Pipes fittings comprising backdrops to manholes shall be included in the items for manholes measured in 2K.	**D1** Pipes fittings on pipes of different nominal bores shall be classified in the third division according to the nominal bore of the largest pipe.	**C3** Items for straight specials shall be deemed to include cutting.	**S1** The materials, joints types, nominal bores and lining requirements of pipe fittings shall be stated in item. Fittings with puddle flanges shall be so described.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Fittings and valve (cont'd)		**7** Medium density polythene pipe fittings			**M2** Straight specials shall be measured only where they are expressly required. Straight specials shall not be measured where they are expressly required. Straight specials shall not be measured where they are necessitated only by the layout of the work. Pipe runs between manholes, chambers and other structures whose lengths are not an exact multiples of a standard pipe length shall be deemed to require straight specials only if they are expressly required as rocker pipes.	**D2** A straight special is a length or made to order.		**S2** Item descriptions for pipe fittings to cast iron or spun iron pipework of nominal bore exceeding 300mm and to all steel pipework shall state the principal dimensions of each fitting. **S3** Vertical bends in pipework of which the nominal bore exceeds 300mm shall be so described. **S4** Fittings to pipework not in trenches shall be so described. **S5** Fittings to relined water mains measured in class 2Y shall be so described.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Fittings and valve (cont'd)		**8** Valves and penstocks	**1** Gate valves: hand operated **2** Power operated **3** Non-return operated **4** Butterfly valves : hand operated **5** Power operated **6** Air valves **7** Pressure reducing valves **8** Penstocks					**A6** The materials, nominal bore and any additional requirements such as joints, draincocks, extension spindles and brackets shall be stated in item descriptions for valves and penstocks. **A7** Valves and penstocks to relined water mains measured in class 2Y shall be so described.

2L PIPEWORK - MANHOLES AND PIPEWORK ANCILLARIES

Includes: Manholes and Other Chambers, Ducts, Culverts, Crossings and Reinstatement, Other Ancillaries as Listed

Excludes: Work Included in Class 2J, 2K, 2M and 2AA; Ducted Building Services (Included in Class 2AB)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE		**D1** The centre line for multiple pipes, ducts or culverts shall be the line equidistance between the inside faces of the outer pipe walls.	**C1** Items for work in this class shall be deemed to include excavation, preparation of surfaces, disposal of excavated material, upholding sides of excavation backfilling and removal of dead services, except to the extent that such work is included in classes 2J, 2K and 2L. **C2** Items for work in this class shall be deemed to include concrete, reinforcement, formwork, joints and finishes.	

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Manhole and pipework ancillaries (Cont'd)	nr	**1** Manholes	**1** Brick **2** Brick with backdrop **3** In situ concrete **4** In situ concrete with backdrop **5** Pre cast concrete **6** Pre cast concrete with backdrop	**1** Depth: not exceeding 1.5 m **2** 1.5 - 2 m **3** 2 - 2.5 m **4** 2.5 - 3 m **5** 3 - 3.5 m **6** 3.5 - 4 m **7** stated exceeding		**D2** The depths of manholes and other chambers shall be measured from the tops of covers to channel inverts or tops of base slabs, whichever is the lower.	**C3** Item for manholes, other stated chambers and gullies shall be deemed to include all items of metalworks and pipework, other than valves, which occur within or at the surface of the item.	**S1** Type or mark numbers shall be stated in item descriptions for manholes, other stated chambers and gullies of which details are given elsewhere in the Contract.
	nr	**2** Other stated chambers	**1** Brick **3** In situ concrete **5** Precast concrete			**D3** Drawpits shall be classed as other stated chambers.	**C4** Items for manholes with backdrops shall be deemed to include the pipework and associated fittings comprising the backdrop.	**S2** Types and loading duties of covers shall be stated in item descriptions for manholes, other stated **S3** Items descriptions for manholes, other stated chambers and gullies shall identify separately those which are expressly required to be excavated by hand.
	nr	**3** Gullies	**1** Clay **2** Clay trapped **3** In situ concrete **4** In situ concrete trapped **5** Precast concrete **6** Precast concrete trapped **7** Plastics **8** Plastics trapped					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES		SUPPLEMENTARY INFORMATION
Manhole and pipework ancillaries (Cont'd)	m3	**4** French drains, rubble drains, ditches and trenches	**1** Filling French and rubble drains with graded material **2** Filing French and rubble drains rubble		**M1** Excavation and pipe laying for piped French and rubble drains are measured in class 2J.				**S4** The nature of the filling material shall be stated in item descriptions for filling French and rubble drains.
	m		**3** Trenches for unpiped rubble drains **4** Rectangular section ditches **5** unlined and lined **6** Vee section ditched: unlined **7** lined **8** Trenches for pipes or cable not to be laid by the contractor	**1** Cross-sectional area: not exceeding 0.25m^2 **2** 0.25 - 0.5 m^2 **3** 0.5 -0.75 m^2 **4** 0.75 - 1m^2 **5** 1 - 1.5 m^2 **6** 1.5- 2 m^2 **7** 2 - 3 m^2 **8** stated exceeding 3 m^2	**M2** The cross-sectional areas of lined ditches shall be measured to the Excavated Surface.				**S5** Materials and dimensions of linings to ditches shall be stated in item descriptions.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Manhole and pipework ancillaries (Cont'd)	m	**5** Ducts and metal culverts	**1** Cable ducts: 1 way **2** 2 way **3** 3 way **4** stated number of ways exceeding 3m **5** Sectional corrugated metal culverts nominal internal diameter: not exceeding 0.5m **6** 0.5 - 1m **7** 1-1.5m	**1** Not in trenches **2** In trenches, depth: not exceeding 1.5m **3** 1.5 - 2m 4 2 -2.5 m **5** 2.5 -3 m **6** 3- 3.5m **7** 3.5 - 4m **8** exceeding 4m	**M3** The rules in class 2J for pipes shall also apply to ducts and metal culverts in this class except that lengths measured for ducts and metal culverts not in trenches shall include lengths occupied by fittings.	**D4** Non-circular metal culverts shall be classified by their maximum nominal internal cross-sectional dimension. **D5** The rules in class 2J for pipes shall also apply to ducts and metal culverts in this class.	**C5** Items for ducts and metal culverts shall be deemed to include cutting and fittings.	**S6** The rules in class 2J for pipes shall also apply to ducts and metal culverts in this class.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Manhole and pipework ancillaries (Cont'd)	nr	**6** Crossings	**1** River, stream or canal, width: 1-3 m **2** 3- 10 m **3** stated exceeding 10m **4** Hedge **5** Wall **6** Fence **7** Sewer or drain **8** Other stated underground service	**1** Pipe bore: not exceeding 300 mm **2** 300- 900 mm **3** 900- 1800 mm **4** stated exceeding 1800 mm	**M4** Crossings shall be measured for pipes, ducts and metal culverts. **M5** Crossings of streams shall be measured only where their width exceeds 1 m.	**D6** River, stream and canal crossings shall be classified by their widths measured along pipes, duct or culvert centre lines when the water surface is at the level (or the higher level of fluctuation if applicable) shown on the drawing to which reference is given in the Preamble in accordance with paragraph 5.20.	**C6** Items for crossings shall be deemed to include reinstatement unless otherwise stated.	**S7** Where linings to rivers, streams or canals are to be broken through and reinstated the type of lining shall be stated in item description.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Manhole and pipework ancillaries (Cont'd)						**D7** The dimension used for classification of bore in the third division shall be the maximum nominal distance between the inside faces of the outer walls of the pipe, duct or culvert to be installed.		
	m	**7** Reinstatement	**1** Breaking up and temporary reinstatement of roads **2** Breaking up and temporary reinstatement of foothpaths **3** Breaking up, temporary and permanent reinstatement of roads.	**1** Pipe bore: not exceeding 300 **2** 300- 900 mm **3** 900- 1800 mm **4** stated exceeding 1800mm	**M6** Reinstatement shall be measured for pipes, ducts and metal culverts. **M7** Lengths of reinstatement shall be measured along centre lines and shall include lengths occupied by manholes and other chambers.	**D7** The dimension used for classification of bore in the third division shall be the maximum nominal distance between the inside faces of the outer walls of the pipe, duct or culvert to be installed.	**C7** Additional reinstatement shall be deemed to be included in the items for manholes and other chambers. **C8** Removal and reinstatement of kerbs and channels shall be deemed to be included in the items for breaking up and reinstatement of roads and	**S8** Types and depths of surfacing, including base and subbase courses, shall be stated in item descriptions for breaking up and reinstatement of roads and foothpaths.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Manhole and pipework ancillaries (Cont'd)			**4** Breaking up, temporary and permanent reinstatement of foothpaths. **5** Reinstatement of land		**M8** Strip topsoil from easement and reinstate shall be measured only where it is expressly required that a width of ground greater than the nominal trench width defined in accordance with rule D1 in class L is to be stripped of topsoil before trench excavation and subsequently reinstated.		**C9** Items for strip topsoil from easement and reinstate shall be deemed to include storing and protecting topsoil and reinstatement of land.	**S9** Item descriptions for strip topsoil from easement and reinstate shall state any limitations on the width to be stripped and reinstated. **S10** Item descriptions for reinstatement of land and for strip topsoil from easement and reinstate shall distinguish between grassland, gardens, sports fields and cultivated land.
			6 Strip topsoil from easement and reinstate					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Manhole and pipework ancillaries (Cont'd)	m	**8** Other pipework ancillaries	**1** Reinstatement of field drains		**M9** Other pipework ancillaries shall be measured for pipes, ducts and metals culverts.	**D7** The dimension used for classification of bore in the third division shall be the maximum nominal distance between the inside faces of the outer walls of the pipe, duct or culvert to be installed.	**C10** Itemsfor reinstatement of field drains shall be deemed to include connections to existing field drains.	**S11** Sizes and types of marker posts shall be stated in item descriptions.
	nr		**2** Marker posts					
	m		**3** Timber supports left in					**S12** Item descriptions for connections to existing manholes and other chambers and to existing pipes, ducts and culverts shall identify the nature of the existing service and the extent of the work to be included.
	m^2		**4** Metal supports left in		**M10** The lengths measured for reinstatement of field drains shall be the nominal trench width defined in accordance with rule 2D in class 2M.			
	nr		**5** Connections to existing manholes and other chambers.	**1** Pipe bore: not exceeding 200mm				
	nr		**6** Connections to existing pipes, ducts and culverts	**2** 200- 300 mm **3** 300- 600 mm **4** 600- 900 mm **5** 900- 1200 mm **6** 1200- 1500 mm **7** 1500- 1800 mm **8** stated exceeding 1800mm				

2M PIPEWORK - SUPPORTS AND PROTECTION, ANCILLARIES TO LAYING AND EXCAVATION

Includes: Extras to excavation and backfilling of trenches for pipework, ducts and metal culverts, manholes and other chambers, headings, thrust boring and pipe jacking

Pipe laying in headings and thrust boring and pipe j a c k i n g

Provisions of supports and protection to pipework, ducts and metal culverts

Excludes: Work included in classes 2J, 2K, 2L and 2 Z

Insulation to building services (included in Class 2 A A)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** Items shall be given in this class in addition to the items for provision, laying and jointing of pipes, ducts and culverts and for the excavation and backfilling of trenches in 2J and 2L. Items shall be given in this class in addition to the items for manholes and other chambers in 2L and 2Z.		**C1** Items for work in this class shall be deemed to include excavation, preparation of surfaces, disposal of excavated materials, upholding sides of excavation, backfilling and removal of dead services, except to the extent that such work is included in classes 2J, 2K and 2L or in the items for extras to excavation and backfilling in this class.	

PIPEWORK - SUPPORTS AND PROTECTION, ANCILLARIES TO LAYING AND EXCAVATION (cont'd)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
	M2 Work in this class associated with ducts and metal culverts shall be measured as set out for work associated with pipes . The dimension used for classification in the third division (2M 2-8) shall be the maximum nominal distance between the inside faces of the outer duct or culvert walls.		**C2** Items for work in this class shall be deemed to include concrete, reinforcement, formwork, joints and finishes.	
	M3 Breaking up and reinstatement of roads and pavings shall be included in class 2L.			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipework -	m^3	**1** Extras to excavation and backfilling	**1** In pipe trenches	**1** Excavation of rock	**M4** The volume of extras to excavation and backfilling in pipe trenches shall be calculated by multiplying together the average depth and length of the material removed or backfilled and the nominal trench width.	**D1** The nominal trench width if not stated in the contract shall be taken as 500 mm greater than the maximum nominal distance between the inside faces of the outer pipe walls where this distance does not exceed 1m and as 750 mm greater than this distance where it exceeds 1 m .		**S1** Item descriptions shall separately excavation which is expressly required to be carried out by hand.
Supports and protection, Ancillaries to laying and excavation (cont′d)			**2** In manholes and other chambers	**2** Excavation of mass concrete				
			3 In headings	**3** Excavation of reinforced concrete				
			4 In thrust boring	**4** Excavation of other artificial hard materials				
			5 In pipe jacking	**5** Backfilling above the final surface with concrete				
				6 Backfilling above the final surface with stated material other than	**M5** No volume of extras to excavation and backfilling in manholes and other chambers shall be measured outside the maximum plan area of the manhole or other chamber.			
				7 Excavation of natural material below the final surface and backfilling with concrete				
				8 Excavation of natural material below the final surface and backfilling with stated material				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipework - Supports and protection, Ancillaries to laying and excavation (cont'd)		Extras to excavation and backfilling (Cont'd)			**M6** The volume of extras to excavation and backfilling for pipe laying in headings, in thrust boring and in pipe jacking shall be measured by multiplying together the internal cross-sectional area of pipe and the length of the material excavated or backfilled. Packing in headings shall not be measured.			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipework - Supports and protection, Ancillaries to laying and excavation (cont'd)					**M7** Backfilling above the final surface (L 1* 5-6) shall be measured only where it is expressly required that the material excavated shall not be used for backfilling (L 1 * 7-8) shall be measured only where it is expressly required. **M8** An isolated volume of rock concrete or other artificial hard material occurring within other material to be excavated shall not be measured separately unless its volume exceeds 0.25 m^3			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipework - Supports and protection, Ancillaries to laying and excavation (cont'd)	m	**2** Special pipe laying methods	**1** In heading **2** Thrust boring **3** Pipe jacking	**1** Nominal bore: not exceeding 200mm **2** 200-300mm **3** 300-600mm **4** 600-900mm **5** 900-1200mm **6** 1200-1500mm **7** 1500-1800mm **8** stated exceeding 1800mm	**M9** Special pipe laying methods shall be measured only where they are expressly required **M10** Access pits, shafts and jacking blocks, where expressly stated in the contracts to be execute by the contractor and of which the nature and extent are expressly stated in the contract, shall be measured as specified requirement		**C3** Items for special type of pipe laying methods shall be deemed to include crossings, provision and removal of access pits, shafts and jacking blocks unless otherwise stated and other work associated with special pipe laying methods not included in the items for provision, laying and jointing of pipes given in 2J	**S2** Item descriptions fro special pipe laying method shall identify the run of pipe or pipes. The type of packing shall be stated in item descriptions for pipes in headings.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipework - Supports and protection, Ancillaries to laying and excavation (cont'd)	m	**3** Beds **4** Haunches **5** Surrounds	**1** Sand **2** Selected excavated granular material **3** Imported granular material **4** Mass concrete **5** Reinforced concrete		**M11** Lengths of beds; haunches and surround shall be measured along pipe centre line including lengths occupied by fittings and valves but not including lengths occupied by manholes and other chambers through which they are not continued	**D2** Items for haunches and surrounds shall include beds of same material. **D3** Items for beds, haunches and surrounds to multiple pipes shall be classified in the third division according to the nominal distance between the inside faces of the outer pipe walls		**S3** Materials used for beds, haunches and surrounds and the depths of beds shall be stated in item descriptions. **S4** Beds, haunches and surrounds to multiple pipes shall be so described stating the number of pipes and the maximum nominal distance between the inside faces of the outer pipe walls

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipework - Supports and protection, Ancillaries to laying and excavation (cont'd)	m	**6** Wrapping and lagging			**M12** Lengths of wrapping and lagging shall be measured along each pipe centre line including lengths occupied by fittings and valves but not including lengths occupied by manholes and other chambers through which they are not continued		**C4** Items for wrapping and lagging of pipes shall be deemed to include wrapping and lagging of fittings, valves and joints	**S5** Materials used for wrapping and lagging of pipes shall be stated in item descriptions
	nr	**7** Concrete stools and thrust blocks	**1** Volume: not exceeding $0.1m^3$ **2** $0.1\text{-}0.2m^3$ **3** $0.2\text{-}0.5m^3$ **4** $0.5\text{-}1m^3$ **5** $1\text{-}2m^3$ **6** $2\text{-}4m^3$ **7** $4\text{-}6m^3$ **8** stated exceeding $6m^3$			**D4** The volumes used for classification of concrete stools and thrust blocks shall exclude the volumes occupied by pipes.	**C5** Item for concrete stools and thrust block shall be deemed to include pipe fixings	**S6** Item description of concrete stools and thrust blocks shall state the specification of the concrete and whether it is reinforced.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pipework - Supports and protection, Ancillaries to laying and excavation (cont'd)	nr	**8** Other isolated pipe supports	**1** Height: not exceeding 1m **2** 1-1.5m **3** 1.5-2m **4** 2-3m **5** 3-4m **6** 4-5m **7** 5-6m **8** stated exceeding 6m			**D5** The height of pipe supports used for classification shall be measured from the ground or other supporting surface to the invert of the highest pipe where pipes are supported from below and of the lowest pipe where pipes are supported from above **D6** Where two or more pipes are carried by one support, the item for the support shall be classified in the third division by the aggregate bore of the pipes supported		**S7** Principal dimensions and materials shall be stated in item descriptions for pipe supports

2N STRUCTURAL METALWORK AND MISCELLANEOUS METALWORK

Excludes: Metalwork in concrete; Metalwork in pipework; Miscellaneous (non-structural) metalwork; Metalwork in piles; Metal Work in Fence

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE					**M1** Items shall be included in miscellaneous metalwork, for metal components not included in this class but associated with metal structures.			
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
Structural metalwork	t	**1** Fabrication of main members for bridges	**1** Rolled sections **2** Plates or flats **3** Built-up or hollow sections	**1** Straight on Plan **2** Curved on Plan **3** Straight on Plan and cambered **4** Curved on Plan and cambered.	**M2** The mass of members, other than plates or flats, shall be calculated from the overall lengths of the members with no deductions for splay-cut or mitred ends. **M3** The mass of members, measured shall be that of plates, rolled sections, shear connectors, stiffeners, cleats, packs, splice plates and other fittings.		**C1** Items for fabrication of metalwork shall be deemed to include delivery of fabricated metalwork to the Site	**S1** The materials and grades of materials used shall be stated in item descriptions for fabrication of members. **S2** item descriptions for fabrication of members other than for portal frames shall identify cranked members.
	t	**2** Fabrication of subsidiary members for bridges	**1** Deck panels **2** Bracing **3** External diaphragms					
	t	**3** Fabrication of members for frames **4** Fabrication of other members	**1** Columns **2** Beams **3** Portal frames **4** Trestles, towers and built-up columns	**1** Straight on plan **2** Curved on plan **3** Straight on plan and cambered				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Structural metalwork (cont'd)			**5** Trusses and built-up girders **6** Bracings, purlins and cladding rails	**4** Curved on plan and cambered	**M4** No allowance shall be made in the measurements for rolling margin and other permissible deviations. The mass of weld fillets, bolts, nuts washers, rivets and protective coatings shall not be measured. **M5** No deductions shall be made for the mass of metal removed to form notches and holes each not exceeding 0.1m2 in area measured in plane			**S3** Details of the members comprising boom and infill construction shall be stated in item descriptions for trestles, towers and built-up columns and trusses and built-up girders. **A5** Item descriptions for anchorages and holding down bolt assemblies shall state particulars of the type of anchorage or assembly.
	nr		**7** Grillages **8** Anchorages and holding down bolt assemblies					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Structural metalwork (cont'd)					**M6** The mass of steel shall be taken for measurement as 785Kg/m^2 per 100mm thickness (7.85t/m^3). The masses of other metals shall be taken as stated in the Specification or, where not so stated, as stated in the supplier's catalogue. **M7** Anchorages and holding down bolt assemblies shall be measured by the number of complete assemblies.			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Structural metalwork (cont'd)	t	**5** Erection of members for bridges	**1** Trial erection **2** Permanent erection				**C2** Items for erection of members shall be deemed to include work carried out after delivery of fabricated metalwork to the site	**S6** Item descriptions for erection shall separately identify and locate separate bridges and structural frames and, where appropriate, parts of bridges or frames.
	nr	**6** Erection of members for frames	**3** Site bolts: Black	**1** Diameter: not exceeding 16mm				
			4 HFSG general grade	**2** 16-20mm				
		7 Erection of other members	**5** HSFG higher grade	**3** 20-24mm				
				4 24-30mm			**C3** Items for site bolts shall be deemed to include supply and delivery to	**S7** Where fixing clips and resilient pads are used to secure overhead crane rails, item descriptions shall so state.
			6 SFG load indicating or load limit types,	**5** 30-36mm				
				6 36-42mm				
			7 SFG load indicating or load limit types, higher grade	**7** stated exceeding 42mm				
	m^2	**8** Off site surface treatment	**1** Blast Cleaning		**M8** Surface treatment carried out on the Site shall be classed as painting.			**S8** Materials and number of applications shall be stated in item descriptions for metal spraying, galvanizing and painting
			2 Picking					
			3 Flame cleaning					
			4 Wire brushing					
			5 Metal spraying					
			6 Galvanizing					
			7 Painting					

2P MISCELLANEOUS METALWORK

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** Painting carried out on the site shall be classed as painting **M2** Masses calculated for miscellaneous metalwork assemblies shall include the mass of all metal components and attached pieces. **M3** No deductions from the masses or areas measured for miscellaneous metalwork shall be made for openings and holes each not exceeding 0.5m^2 in area		**C1** Items for miscellaneous metalwork shall be deemed to include fixing components and drilling or cutting of other work	**S1** Item descriptions shall state the specification and thickness of metal , surface treatments carried out prior to delivery to site and principal dimensions of miscellaneous metalwork assemblies.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Miscellaneous metalwork	t	**1**	**1** Stairways and landings **2** walkways and platforms					
	m		**3** Ladders **4** Handrails **5** Bridge parapets		**M4** The lengths of ladders shall be measured along lengths of stringers. The lengths of handrails and bridge parapets shall be measured			**S2** Where ladders include safety loops, rest platforms or returned stringers, item descriptions shall so state
	m		**6** Miscellaneous framing	**1** Angle section **2** Channel section **3** I section **4** Tubular section	**M5** The lengths of miscellaneous framing shall be measured along the external perimeter of framing			
	m^2		**7** Plate flooring **8** Open grid flooring				**C2** Item for plate and open grid flooring shall be deemed to include supporting metalwork unless otherwise stated	

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Miscellaneous metalwork (cont'd)	m^2	**2**	**1** Cladding **2** **3** Duct covers				**C3** Item for welded mesh panelling and duct covers shall be deemed to include supporting metalwork unless otherwise stated	
	nr		**4** Tie rods				**C4** Item for tie rods shall be deemed to include concrete, reinforcement and joints	
	m		**5** Walings					
	nr		**6** Bridge bearings	**1** Roller **2** Slide **3** Rocker **4** Cylindrical **5** Spherical **6** Elastomeric **7** Rubber pot **8** Proprietary			**C5** Item for bridge bearings shall be deemed to include grouting beneath plates.	**S3** Item description for bridge bearings shall state the composition and material of the bearing

CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKES

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Miscellaneous metalwork (cont'd)	nr		**7** Uncovered tanks **8** Covered tanks	**1** Volume: not exceeding 1m^3 **2** 1-3m^3 **3** 3-10m^3 **4** 10-30m^3 **5** 30-100m^3 **6** 100-300m^3 **7** 300-1000m^3 **8** stated exceeding1000 m^3				**S4** Item description for tanks shall state the principal dimensions

2Q Timber

Includes: Timber components and fittings, Timber decking fittings and fastenings to timber components and decking

Excludes: Formwork to concrete; Timber piles; Timber sleepers; Timber supports in tunnels; Timber fencing; Carpentry and joinery in simple building works incidental to civil engineering works

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE							**C1** Items for timber shall be deemed to include fixing, boring, cutting and jointing.	
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
Timber	m	**1** Hardwood components	**1** Cross-sectional area: not exceeding $0.01m^2$	**1** Length: not exceeding 1.5m	**M1** The length of timber components measured shall be their overall lengths with no allowance for scarf or other joints	**D1** The cross-sectional area stated for classification of timber components shall be their nominal gross cross-sectional areas		**S1** The nominal gross cross-sectional area dimensions, thickness, grade or species and any impregnation requirements or special surface finishes shall be stated in item descriptions for timber components. **S2** The structural use and location of timber components shall be stated in item descriptions for components longer than 3m
		2 Softwood components	**2** $0.01-0.02m^2$ **3** $0.02-0.04m^2$ **4** $0.04-0.1m^2$ **5** $0.1-0.2m^2$ **6** $0.2-0.4m^2$ **7** exceeding $0.4m^2$	**2** 1.5-3m **3** 3-5m **4** 5-8m **5** 8-12m **6** 12-20m **7** exceeding 20m				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Timber (cont'd)	m^2	**3** Hardwood decking **4** Softwood decking	**1** Thickness: not exceeding 25mm **2** 25-50mm **3** 50-75mm **4** 75-100mm **5** 100-125mm **6** 125-150mm **7** exceeding 150mm		**M2** No deductions from the areas measured for timber decking shall be made for openings and holes each not exceeding $0.5m^2$ in area	**D2** The thickness stated for classification of timber decking shall be the nominal gross thickness.		**S3** The nominal gross cross-sectional area dimensions, thickness, grade or species and any impregnation requirements or special surface finishes shall be stated in item descriptions for timber decking.
	nr	**5** Fittings and fastenings	**1** Straps **2** Spikes **3** Coach screws **4** Bolts **5** Plates					**S4** Materials, types and sizes of fittings and fastenings shall be stated in item description

2R PILES

Excludes: Boring for site investigation (included in 42); Ground anchors (included in 43); Walings and Tie Rods (included in 55)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** Bored and driven depths shall be measured along the axes of piles from the Commencing Surface to the toe levels of bored piles, to the bottom of the casing of driven cast in place piles and to the bottom of the toes of other driven piles. The commencing surface adopted in the preparation of the bill of Quantities as the surface at which boring or driving is expected to begin shall be adopted for the measurement of the completed work.		**C1** Items for piles and stone columns shall be deemed to include disposal of excavated material.	**S1** Materials of which piles are composed **S2** Preliminary piles shall be identified in items description Raked piles shall be identified in item descriptions and their inclination ratios stated **S3** The structure to be supported and the commencing surface shall be identified in item descriptions for piles.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Piles (cont'd)	nr m	**1** Bored cast in place concrete piles **2** Driven cast in place concrete piles	**1** Diameter: 300mm or 350mm **2** 400mm or 450mm **3** 500mm or 550mm	**1** Number of piles **2** Concrete length **3** Depth bored or driven to stated maximum depth	**M2** Each group of items for cast in place concrete piles shall comprise **(a)** an item for the number of piles (57.1.2*1)	**D1** The maximum depth stated in item descriptions for the depth of cast in place concrete piles shall be the depth which is not exceeded by any pile included in the item.		**S4** The diameter shall be stated in item description for cast in place concrete piles. **S5** Contiguous bored piles shall be identified in item descriptions.
Piles (cont'd)			**4** 600mm or 750mm **5** 900mm or 1050mm **6** 120mm or 1350mm **7** 1500mm		**(b)** an item for the total concrete length of piles (57.1.2*2) **(c)** an item for total depth bored or driven (57.1.2*3) **M3** The concreted lengths of cast in place concrete piles shall be measured from the cut-off level expressly required to the toe levels expressly required.	**D2** Piles comprising a permanent steel casing which is filled with concrete shall be classed as driven cast in place concrete piles where the piles are designed for the load to be carried on the concrete.		

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Piles (cont'd)	nr	**3** Performed concrete piles	**1** Cross-sectional area: not exceeding $0.025m^2$	**1** Number of piles of stated length	**M4** Each group of items for performed concrete and timber piles shall comprise **(b)** an items for the total depth driven	**D3** The length of performed concrete and timber piles shall be lengths expressly required to be supplied excluding extensions but including heads and shoes.		**S6** The cross-section type and the cross-sectional dimensions or diameter shall be stated in item description for performed concrete and timber piles. **S7** Items description for performed concrete piles shall state if they are prestressed. **S8** Details of treatments and coating shall be stated in item descriptions for the number of piles. **S9** Details of driving heads and shoes shall be stated in item descriptions for the number of piles.
	m	**4** Performed concrete sheet piles	**2** $0.025\text{-}0.05\ m^2$	**2** Depth driven				
		5 Timber piles	**3** $0.05\text{-}0.1\ m^2$					
			4 $0.1\text{-}0.15\ m^2$					
			5 $0.15\text{-}0.25\ m^2$					
			6 $0.25\text{-}0.5\ m^2$					
			7 $0.5\text{-}1\ m^2$					
			8 exceeding $1m^2$					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Piles (cont'd)	nr	**6** Isolated steel piles	**1** Mass: not exceeding 15kg/m	**1** Number of piles of stated length	**M5** Each group of items for isolated steel piles shall comprise:	**D4** The length of isolated steel piles shall be lengths expressly required to be supplied excluding extensions.		**S10** The mass per metre and cross-sectional dimensions shall be stated in item descriptions for isolated steel piles.
	m		**2** 15-30 kg/m **3** 30-60 kg/m **4** 60-120 kg/m **5** 120-250 kg/m **6** 250-500kg/m **7** 500kg/m- 1t/m **8** exceeding 1t/m	**2** Depth driven	**(a)** one or more items for the number of piles of stated length. **(b)** an items for the total depth driven.	**D5** Piles comprising a driven permanent steel casing which is filled with concrete shall be classed as isolated steel pipes where the piles are designed for the load to be carried on the casing. Filling such piles shall be classed as filling hollow piles with concrete.		**S11** Details of treatments and coating shall be stated in item descriptions for the number of piles.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Piles (cont'd)	m m^2	**7** Interlocking steel piles	**1** Section modulus: not exceeding $500cm^3/m$ **2** $500\text{-}800cm^3/m$ **3** $800\text{-}1200cm^3/m$ **4** 1200-$2000cm^3/m$ **5** 2000-$3000cm^3/m$ **6** 3000-$4000cm^3/m$ **7** 4000-$5000cm^3/m$ **8** exceeding $5000cm^3/m$	**1** length of special piles **2** Driven area **3** Area of piles length: not exceeding 14m **4** 14-24m **5** exceeding 24m	**M6** Each group of items for interlocking steel piles shall comprise: **(a)** one or more items for the total length for each type of special pile, if any **(b)** an item for the total driven area of piles **(c)** one or more items for the total area of piles divided into ranges of length given in the third division	**D6** The lengths of interlocking steel piles shall be the lengths expressly required to be supplied excluding extensions. **D7** Interlocking steel corner, junction, closure and taper piles shall be classed as special piles.		**S12** The section reference or mass per metre and section modulus shall be stated in item descriptions for interlocking steel piles. **S13** Details treatment and coatings shall be stated in item descriptions for area of piles

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Piles (cont'd)					**M7** The areas of interlocking steel piles shall be calculated by multiplying the mean undeveloped horizontal lengths of the pile walls formed (including lengths occupied by special piles) by the depths measured in accordance with rule M1 in the case of items for the driven areas and by the lengths defined in accordance with rule D6 in the case of items for the areas of piles			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Piles (cont'd)					**M8** Closure and taper piles classed as special piles shall be measured only where they are expressly required.			
	nr m	**8** Stone columns	**1** Diameter: not exceeding 300mm **2** 300mm-450mm **3** 450mm - 600mm **4** 600mm - 750mm **5** 750mm - 900mm **6** 900mm - 1050mm **7** 1050mm-1200mm **8** exceeding 1200mm	**1** Number of columns **2** Length of columns			**C2** Items for stone columns shall include for boring	**S15** The diameter shall be stated in item descriptions for stone columns.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Piles (cont'd)					**M9** Work in this class, other than backfilling empty bore for cast in place concrete piles, shall be measured only where it is expressly required		**C1** Items for piling ancillaries shall be deemed to include disposal of surplus materials unless otherwise stated.	

CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKES

2S PILING ANCILLARIES

Includes:

Excludes: Ground anchors (included in class 2C), Piles (included in class 2R), Walings and Tie Rods (included in class in 2P)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE					**M1** The lengths of permanent casing shall be measured from the commencing surface to the bottom of the casing.		**C1** Items for permanent casing shall be deemed to include driving head and shoes.	**S1** The diameter of enlarged bases for bored piles shall be stated in item descriptions.
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
Pile ancillaries	m	**1** Cast in place concrete piles	**1** Pre-boring **2** Backfilling empty bore with the stated material **3** Permanent casing each length: not exceeding 13m **4** exceeding 13m	**1** Diameter: 300mm or **2** 400mm or 450mm **3** 500mm or 550mm **4** 600mm or750mm **5** 900mm or 1050mm **6** 1200mm or 1350mm **7** 1500mm		**D1** The diameter used for classification in the third division shall be the diameter of the piles		**S2** Materials, thickness and details of treatments and coatings shall be stated in item descriptions for permanent casings. Item descriptions for cutting off surplus lengths which include permanent casings shall so state.
	nr		**5** Enlarged bases					
	m		**6** Cutting off surplus lengths					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pile ancillaries (cont'd)	nr		**7** Preparing heads					**S3** Materials shall be stated in item descriptions for reinforcement
	t	**2** Cast in place concrete piles	**1** Reinforcement	**1** Straight bars, nominal size: not exceeding **2** exceeding 25mm **3** Helical bars of stated nominal	**M2** The mass measured for reinforcement shall include the mass of reinforcements in **M3** The mass of steel reinforcement shall be taken as 0.785 kg/m per $100mm^2$ of cross-section ($7.85t/m^3$). The mass of other reinforcing materials shall be taken as stated in the contract.	**D2** The nominal size used for classification in item description for bar reinforcement shall be the cross sectional size.	**C3** Items for reinforcement shall be deemed to include supporting reinforcement.	**S4** Details of couplers for high tensile steel reinforcement which are expressly required shall be stated in item descriptions for reinforcement.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pile ancillaries (cont'd)	m	**3** Preformed concrete piles **4** Timber piles	**1** Pre-boring **2** Jetting **3** Filling hollow piles with concrete	**1** Cross-sectional area: not exceeding $0.025m^2$ **2** $0.025m^2$-$0.05m^2$ **3** $0.05m^2$-$0.1m^2$	**M5** Driving extended piles shall included in the measurement of the items for driven depth in Section 57.1	**D3** The cross-sectional area used for classification in the third division shall be the cross sectional area of the piles	**C4** Items for pre-boring shall be deemed to include grouting voids between the pile and the bore	**S5** Items description for filling hollow piles with concrete shall state the specification of the concrete
	nr		**4** Number of pile extensions	**4** $0.1m^2$-$0.15m^2$ **5** $0.15m^2$-$0.25m^2$ **6** $0.25m^2$-$0.5m^2$	**M6** Each group of items for pile extensions shall comprise		**C5** Items for pile extensions shall be deemed to include the work necessary to attach the extensions to the pile	**S6** Materials of which pile extension are composed shall be stated in item descriptions for their length
	m		**5** Length of pile extensions, each length: not exceeding 3m **6** exceeding 3m **7** Cutting of surplus length **8** Preparing heads	**7** $0.5m^2$-$1m^2$ **8** stated exceeding $1m^2$	**(a)** an item for the number of pile extensions (57.2. 3-4 .4*). **(b)** one or more items for the length of pile extensions divided into ranges of length given in the second division .		**C6** Items for filling hollow piles shall be deemed to include removal of material from within the pile before concreting.	

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pile ancillaries (cont'd)					**M7** The lengths of pile extensions measured shall not include lengths formed from material arising from cutting off surplus length of other piles. **M8** The length measured for timber pile extensions shall include lengths occupied by scarfed or other joints.			
	m nr	**5** Isolated steel piles	**1** Pre-boring **2** Jetting **3** Filling hollow piles with concrete **4** Number of pile extensions **5** Length of pile extensions, each length: not exceeding 3m	**1** Mass: not exceeding 15kg/m **2** 15-30 kg/m **3** 30-60 kg/m **4** 60-120 kg/m **5** 120-250 kg/m **6** 250-500kg/m **7** 500kg/m- 1t/m **8** exceeding 1t/m	**M5** Driving extended piles shall included in the measurement of the items for driven depth in Section 57.1 **M9** Each group of items for pile extensions shall comprise	**D4** The mass used for classification in the third division shall be the mass of the		

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pile ancillaries (cont'd)			**6** exceeding 3m **7** Cutting of surplus length **8** Preparing heads		**M9 (Cont'd)** **(a)** an item for the number of pile extensions (57.2.5 .4*). **(b)** one or more items for the length pile extensions divided into ranges of length given in the second division (57. 2. 5. 5-6*). **M7** The lengths of pile extensions measured shall not include lengths formed from material arising from cutting off surplus length of other piles.			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pile ancillaries (cont'd)	m	**6** Interlocking steel piles	**1** Pre-boring **2** Jetting **3** Filling hollow piles with concrete **4** Number of pile extensions **5** Length of pile extensions, each length: not exceeding 3m **6** exceeding 3m **7** Cutting of surplus length **8** Preparing heads	**1** Section modulus: not exceeding 500cm^3/m **2** 500-800cm^3/m **3** 800-1200cm^3/m **4** 1200-2000cm^3/m **5** 2000-3000cm^3/m **6** 3000-4000cm^3/m **7** 4000-5000cm^3/m **8** stated exceeding 5000cm^3/m	**M5** Driving extended piles shall included in the measurement of the items for driven depth in Section 57.1 **M10** Each group of items for pile extensions shall comprise **(a)** an item for the number of pile extensions. **(b)** one or more items for the length pile extensions divided into ranges of length given in the second division. **M7** The lengths of pile extensions measured shall not include lengths formed from material arising from cutting off surplus length of other piles.	**D5** The section modulus used for classification in the third division shall be the section modulus of piles	**C5** Items for pile extensions shall be deemed to include the work necessary to attach the extensions to the pile	**S6** Materials of which pile extensions are composed shall be stated in item description for their length.
	m							
	nr							

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pile ancillaries (cont'd)					**M11** The lengths measured for cutting off surplus lengths of interlocking steel piles shall be the mean undeveloped horizontal lengths to be cut (including lengths occupied by special piles).			
	h	**7** Obstructions			**M12** Obstructions shall be measured only for breaking out rock or artificial hard material encountered above the founding stratum of bored piles.			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Pile ancillaries (cont'd)	nr	**8** Pile tests	**1** Maintained loading with various reactions **2** Constant rate of penetration **3** Horizontal loading	**1** Test load: not exceeding 100t **2** 100-200 t **3** 200-300 t **4** 300-400 t **5** 400-600 t **6** 600-800 t **7** 800-1000 t **8** exceeding 1000 t				**S7** item descriptions for pile tests shall identify those which are preliminary piles. **S8** Item descriptions for loading tests shall state the load. Where the load is applied to raking piles, item descriptions shall so state.
			4 Non-destructive integrity **5** Inclinometer installations					

2T ROADS AND PAVINGS

Includes: Sub-base, base and surfacing of roads, runways and other paved areas; Kerbing and light duty pavements, footways and cycle tracks; Traffic signs and markings

Excludes: Earthworks (included in 2E); Drainage (included in 2J, 2K, 2L and 2M); Fence and gates (included in 65); Maintenance of road and pavings

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE		**D1** The expression FMW shall mean as specified in the specification for roads and bridge works or similarly approved by the Federal Ministry of Works (Transport) or such body responsible for providing such road **D2** Types, categorized and mixtures refer to the manual of Contract Documents for Highway Works published by The Federal Ministry of Works.	**C1** Items for work in this class involving Insitu concrete shall be deemed to include formwork and integral finishes to concrete	**S1** Item descriptions for courses of paving, road making materials and pavements slabs, shall identify the material and state the depth of each course or slab and the spread rate of applied surface finish **S2** Item descriptions for work in this class which is applied to surfaces inclined at an angle exceeding 10^0 to the horizontal, shall so be stated.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)	m^2	**1** Unbound sub-base	**1** Type 1 unbound mixtures **2** Type 2 unbound mixtures **3** Type 3 (open graded) unbound **4** Category B (close graded) unbound mixtures	**1** Depth: not exceeding 30mm **2** 30-60mm **3** 60-100mm **4** 100-150mm **5** 150-200mm **6** 200-250mm **7** 250-300mm **8** exceeding 300mm	**M1** The width of each course of material shall be measured at the top surface of that course. The area of manhole covers and other intrusions into a course shall not be deducted where the area of the intrusion is less than $1m^2$			
			5 Type 4 (asphalt arrisings) unbound mixtures		**M2** The area of additional geotextiles in laps shall not be measured			
	m^2		**6** Geotextiles					**S3** The type and grade of materials shall be stated in item descriptions for geotextiles
	m^3		**7** Additional depth of stated material					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)	m^2	**2** Cement and other hydraulically bound pavements	**1** Cement bound granular mixture **2** Fly ash bound mixture 1and hydraulic road binder bound mixture 1 **3** Slag bound mixture B2, fly ash bound mixture 2 and hydraulic road binder bond mixture 2 **4** Slag bound mixture B3, fly ash bound mixture 3 and hydraulic road binder bond mixture 3 **5** Fly ash bound mixture 5	**1** Depth: not exceeding 30mm **2** 30-60mm **3** 60-100mm **4** 100-150mm **5** 150-200mm **6** 200-250mm **7** 250-300mm **8** exceeding 300mm				**S4** The type of mixtures shall be stated in item descriptions for cement and other hydraulically bound pavements.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)			**6** Slag bound mixture B1-1, B1-2, B1-3. B1-4 **7** Soil treated by cement **8** Soil treated by other stated material					
	m^2	**3** Bituminous bound pavements	**1** Hot rolled asphalt base **2** Hot rolled asphalt binder **3** Dense base and binder course asphalt concrete paving grade bitumen (recipe mixtures) **4** Dense asphalt concrete surface course **5** Hot rolled asphalt concrete surface **6** Close graded asphalt concrete surface course	**1** Depth: not exceeding 30mm **2** 30-60mm **3** 60-100mm **4** 100-150mm **5** 150-200mm **6** 200-250mm **7** 250-300mm **8** exceeding 300mm				**S5** The type of mixtures shall be stated for bituminous bound pavements.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)			**7** Fine graded asphalt concrete surface course **8** Open graded asphalt concrete surface course					
	m^2	**4** Bituminous bound pavements	**1** Dense base and binder course asphalt concrete (design mixtures) **2** EME2 base and binder course asphalt concrete **3** Stone mastic asphalt binder course and regulating course **4** Porous asphalt				**C2** Items for cold milling / planning shall be deemed to include for disposal	**S6** Item descriptions for cold recycled bound materials shall state whether it is Insitu or ex-situ

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)			**5** Hot rolled asphalt surface course and binder course **6** Cold recycled bound material **7** Cold milling/ planning					
	t		**8** Regulating course of stated material					
	m^2	**5** Concrete Pavements	**1** Unreinforced concrete surface slabs **2** Jointed reinforced concrete surface slabs **3** Continuously reinforced concrete surface **4** Continuously reinforced concrete base slabs **5** Wet lean concrete	**1** Depth: not exceeding 30mm **2** 30-60mm **3** 60-100mm **4** 100-150mm **5** 150-200mm **6** 200-250mm **7** 250-300mm **8** exceeding 300mm				**S7** Item descriptions for wet lean concrete shall state the mix

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)	m^2		**6** Steel fabric reinforcement to BSS4483 **7** Other fabric reinforcement of stated material	**1** Nominal mass not exceeding 2 kg/m2 **2** 2-3 kg/m^2 **3** 3-4 kg/m^2 **4** 4-5 kg/m^2 **5** 5-6 kg/m^2 **6** 6-7 kg/m^2 **7** 7-8 kg/m^2 **8** stated exceeding 8kg/m^2	**M3** The areas of additional fabric reinforcement in laps shall not be measured **M4** The mass of steel reinforcement shall be taken as 0.785kg/m per 100mm of cross section (7.85kg/m3). The mass of other reinforcing materials shall be taken as stated in the contract.		**C3** Items for reinforcement shall be deemed to include supporting reinforcement other than steel supports to top reinforcement	**S8** Item description for reinforcement for steel fabric reinforcement to BS4483 shall state the fabric reference in accordance with BS4483 or the wire and mesh arrangement in accordance with BS4466. Item descriptions for other fabric reinforcement shall state the sizes and nominal mass per square meter.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)	t		**8** Plain round steelbars reinforcement to BSS 4449 **9** Deformed high yield steel reinforcement bars to BSS 4449	**1** Nominal size: 6mm **2** 8mm **3** 10mm **4** 12mm **5** 16mm **6** 20mm **7** 25mm **8** 32mm or greater	**M5** The mass of reinforcement measurement shall include the mass of steel supports to top reinforcement			**S9** Item descriptions for steel bar reinforcement shall state the type of reinforcement
	m^2		**10** Separation and waterproof membranes		**M6** The areas of additional membranes in laps shall not be measured			**S10** Item descriptions for separation and waterproof membranes shall state their material and thickness.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)	m	**6** Joints in concrete pavements	**1** Longitudinal joints **2** Expansion joints **3** Contraction joints **4** Wrapping joints **5** Butt joints **6** Construction joints	**1** Depth: not exceeding 30mm **2** 30-60mm **3** 60-100mm **4** 100-150mm **5** 150-200mm **6** 200-250mm **7** 250-300mm **8** exceeding 300mm	**M7** Construction joints shall be measured only where they are at locations where construction joints are expressly required			**S11** The dimensions, spacing and nature of sealed grooves and rebates, water stops, dowels and other components shall be stated in item descriptions for joints in concrete pavements
	m	**7** Kerbs, channels, edgings, footways and paved areas	**1** Precast concrete kerbs to BS 7263 Part1 figure reference stated	**1** Straight or curved to radius exceeding 12 m **2** Curved to radius not exceeding 12 m **3** Quadrants	**M8** Excavation and filling for kerbs, channels, edgings shall be measured in accordance with the rules of Section 2E Earthworks (civil and industrial engineering works and the like)		**C4** Items for kerbs, channels, and edgings shall be deemed to included beds, backings, reinforcement joints and cuttings.	**S12** Materials and cross-sectional dimensions of kerbs , channels, edgings and their beds and backings shall be stated in item descriptions
	nr						**C5** Items descriptions for kerbs, channels, and edgings shall be deemed to included for earthworks and	**S13** The material, size and thickness for precast concrete natural stone block, clay slabs and pavers shall be
			2 Precast concrete edgings to BS 7263 Part1 figure reference stated	**4** Drops **5** Transitions				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)			**3** Other precast concrete kerbs, channels, edgings, dimensioned description **4** In situ concrete kerbs, channels, edgings, dimensioned description				**C6** Items for precast concrete natural stone block, clay slab and pavers shall be deemed to included beds, backings, joints and cuttings.	**S14** The material and thickness for flexible surfacing, insitu concrete surfacing and grass concrete paving shall be stated in item description.
	m^2		**5** Asphalt kerbs, dimensioned description **6** Asphalt channels dimensioned description **7** Precast concrete, natural stone, block and clay slabs and pavers **8** Flexible surfacing **9** Insitu concrete surfacing **10** Grass concrete paving				**C7** Items for flexible surfacing and insitu concrete surfacing shall be deemed to included beds and joints **C8** Items for grass concrete pavings shall be deemed to include for beds, joints, soiling and seeding.	

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)	m^2	**8** Light duty pavements	**1** Granular base **2** Hardcore base **3** Tarmacadam **4** Rolled asphalt **5** Bitumen macadam **6** Dense tar **7** In situ concrete of stated **8** Precast concrete flags to stated specification		**M9** The width of each course material shall be measured at the top of that surface of that course. The areas on manhole covers and other intrusion into a course shall not be deducted where the area of the intrusion is less than $1m^2$			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Roads and Pavings (cont'd)	nr	**9** Ancillaries	**1** Traffic signs	**1** Non-illuminated **2** Illuminated	**M10** Lengths measured for linear surfaced markings shall take account of the gaps in intermittent markings **M11** Items for support gantries and other substantial structures associated with traffic signs which are constructed in concrete or structural metalwork, or other materials, shall be measured under the appropriate work Section in part 2 (civil, industrial engineering works and the likes).		**C9** Items for traffic signs other than traffic signs measured in accordance with the rule M10, shall be deemed to include foundations, supporting posts excavation preparation of surfaces,, disposal of excavated material, removal of existing services, upholding sides of excavation, backfilling, concrete , reinforcement and joints	**S15** The material, size and diagram number taken from Traffic signs regulation and general directions, issued by the FMW, or Road Safety Traffic Division of the Police Force, shall be stated in item descriptions for traffic signs and surface markings **S16** The shape and colour of aspects shall be stated in item descriptions for reflecting road studs.
			2 Surface markings	**1** Non reflecting road studs **2** Reflecting road studs **3** Letters and shapes				
	m			**4** Continuous lines **5** Intermittent (broken) lines **6** Raised rib				

2U RAILWAY TRACK CONSTRUCTION

Includes: Track foundations, rail, sleepers, fittings, switches and crossings
Excludes: Overhead crane rails (included in 2N); Concrete track foundations (included in 2F and 2G)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE								
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
Rail Track Construction	m^3	**1** Track bases/foundation	**1** Bottom ballast **2** Top ballast		**M1** The volume of the top ballast measured shall include the volume of the sleepers	**D1** Bottom ballast shall be placed before the track is laid **D2** Top ballast shall be ballast placed after the track is laid	**C1** Item for ballast cleaning shall be deemed to include the provision of ballast to fill voids	**S1** Item descriptions for track foundations shall state the material
	m^2		**3** Blinding bed **4** Blankets **5** Water proof membranes		**M2** Area of additional waterproof membrane in laps shall not be measured			**S2** Item descriptions for blinding, blankets and waterproof membranes shall state their materials and thickness
	m^3		**6** Ballast cleaning		**M3** Item description for ballast cleaning shall state the depth of material to be cleaned and the cleaning material to be used to fill			
	m		**7** Tamping **8** Pneumatic ballast		**M4** Item description for tamping shall state the width of the ballast to be tamped			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)					**M5** Item description for pneumatic ballast injection shall state the width of the ballast to be treated			
	nr	**2** Taking up	**1** Bull head rails **2** Flat bottom rails **3** Dock and crane rails	**1** Plain track **2** Turnout **3** Diamond crossing	**M6** The length of taking up plain track shall be measured along the centre line of the track (two rail) and shall exclude lengths occupied by turnouts and diamond		**C2** Items for taking up turnout and diamond crossing shall be deemed to include check, guard, and conductor rails	**S3** Item description for taking up track shall state the amount of dismantling, details of disposal of the track and the type of rail, sleeper and joint
	m		**4** Check and guard rails **5** Conductor rails		**M7** The lengths of taking up check, guard and conductor rails shall be measured along the lengths of the rail (one rail) shall exclude lengths within turnouts and diamond crossings			**S4** Item description for taking up buffer stops shall state their approximate weight and type of
	nr		**6** Sundries	**1** Buffer stops **2** Retarders **3** Wheel stops **4** Lubricants **5** Switch heaters **6** Switch levers **7** Eutectic strip				
	m							

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)	nr	**3** Lifting, packing and slewing and work to existing track	**1** Bull head rail track **2** Flat bottom rail track **4** Existing rail turning and fixing **5** Rerailing **6** Stressing rail	**1** Plane track **2** Plane track: with conductor **3** Plane track: with turnout **4** Plane track: with conductor rail and turnout **5** Buffer stops	**M8** The length of track stated in item description for liftings, packing and slewing track shall be the length measured along the centre line of the track(two rails) and shall be taken over all roads, switch roads shall be measured from the toe of switches **M9** The length track for rerailing stressing rail shall be the length of a single rail and exclude length of turnouts and crossings.	**D3** Spot replacement of sleepers shall cover the replacement of sleepers in track neither supplied nor laid by the contractor	**C3** Items for lifting, packing and slewing tracks shall be deemed to include opening out, packing and boxing in with ballast and insertion of closure rails **C4** Item for spot replacement of sleepers shall be deemed to include opening out of ballast,	**S5** Item descriptions for lifting, packing and slewing track, shall state the length of track, the maximum distance of slew and the minimum lift **S6** Where extra ballast is required, item descriptions for lifting, packing and slewing, shall so state **S7** Items for spot replacement of sleepers shall state the type of sleepers and type of fixing
	nr							

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)					**M10** Item description s for spot replacement for sleepers shall state the type of rail fittings required and the method of fixing to track and to the sleeper.		**C5** Item for rerailing shall be deemed to include for grinding, milling and planning. **C6** Item for spot replacement of sleepers shall be deemed to include opening out of ballast, disposal of existing sleepers and replacing ballast removing old sleeper, the supply and laying of new sleepers and repacking ballast	**S8** Items for existing rail turning and refixing shall state the type of rail and type of fixing **S9** Items for rerailing shall state the type of joints **S10** Items for stressing rail shall state when the work is restressing
	t	**4** Supplying	**1** Bullhead rails **2** Flat bottom rails	**1** Mass not exceeding 20 **2** 20 - 30 kg/m	**M11** The mass measured for supplying rails shall include the mass of twist rails		**C7** Items for supplying shall be deemed to include delivery of components to site	**S11** Item descriptions for supplying sleepers and fittings shall state the type.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)			**3** Dock and crane rails **4** Check and guard rails **5** Conductor rails **6** Twist rails	**3** 30-40 kg/m **4** 40-50 kg/m **5** exceeding 50kg/m	**M12** Fish plates shall be measured by the number of pairs		**C8** Items for supplying sleepers shall be deemed to include fittings **C9** Items for supplying fittings shall be deemed to include fixings, keys, clips, bolts, nuts, screws, spikes, ferrules, track circuit insulators, pads and conductor rail insulator	**S12** Item for supplying sleepers, shall state the size of the sleepers and identify the fittings which are attached by the supplier. Sleeper with integral ducts and services shall be so described **13** Hollow sleepers shall be identified in item description for sleepers
	nr		**7** Sleepers	**1** Timber **2** Concrete **3** Steel				
	nr		**8** Fittings	**1** Chairs **2** Base plates **3** Pandrol rail fastenings **4** Plain fish plates **5** Insulated fish plates **6** Switch heaters **7** Switch levers **8** Conductor rail insulator **9** Conductor rail side ramp				**14** Items for supplying sleepers shall state the size of the sleepers and identify the fittings which are attached by the supplier. Sleepers with integral ducts for services shall be so described

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)		**5** Supplying	**1** Turnouts and crossing	**1** Trap points **2** Catch points **3** Turnouts **4** Diamond crossings	**M13** Conductor rail and guard boards shall be measured each side of the rail		**C10** Items for supplying shall be deemed to include delivery of components to site **C11** Items for supplying turnouts and diamond crossings shall be deemed to include timber, fittings and check rails **C12** Items for supplying conductor rail guard board shall be deemed to include fixings	**S15** Item descriptions for supplying prefabricated track panels shall state length of the panels **S16** Item description for supplying turnout and crossing shall state the type **S17** Item descriptions for supplying sundries shall state the type **S18** Item descriptions for supplying buffer stops shall state their approximate weight **S19** Item descriptions for supplying buffer stops shall state the type and approximate weight.
	m		**2** Sundries	**1** Buffer stops **2** Retarders **3** Wheel stops **4** Lubricator **5** Switch heaters **6** Switch levers **7** Conductor rail guard board				
	m		**3** Prefabricated track panel 18.288m long with rails and bearers	**1** Bull head track: timber sleepers **2** Concrete sleepers **3** Steel sleepers **4** Flat bottom track: timber sleepers **5** Concrete sleepers **6** Steel sleepers				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)	nr		**4** Prefabricated turnouts and crossings	**1** Turnouts: type stated timber bearers **2** Turnouts: type stated concrete bearers **3** Turnouts: type stated steel bearers **4** Diamond crossings: type stated timber bearers **5** Diamond crossings: type stated concrete bearers **6** Diamond crossings: type stated steel bearers				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)	m nr	**6** Laying with bearers on ballast	**1** Bullhead rails **2** Flat bottom rails **3** Dock **4** Crane rails	**1** Plain track **2** Form curve in plain track radius not exceeding 300m **3** Form curve in plain track radius exceeding **4** Turnouts **5** Diamond crossings **6** Welded joints	**M14** The length of laying plain track shall be measured along centre line of the track(two rail) and shall include length occupied by turnout and diamond crossings **M15** The length of laying checks, guard and conductor rails shall measured along the length of rail (one rail). **M16** Conductor rail board shall be measured each side of the rail **M17** Laying rails direct to concrete base shall be measured separately	**D4** Laying rails direct to concrete base shall be classed as slab track	**C13** Items for laying shall be deemed to include work carried out after delivery of components to the site, where track is not to be supplied by the contractor , to the location stated in accordance to rule S14 **C14** Laying rails direct to concrete, slab track, shall be deemed to include fixings and pads.	**S20** Item description for laying rail track not supplied by contractor shall identify the form in which its to be supplied **S21** Item descriptions fro layin g prefabricated track shall identify pre-fabricated lengths. **S22** Items description for laying rails shall state the type and mass per metre of rail and the type of joint and sleeper **S23** Item descriptions for prefabricated track panel shall state the panel

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)	m nr nr	Laying with bearers on ballast (Cont'd)	**5** Check rails **6** Guard rails **7** Conductor rails	**1** Rail **2** Length end **3** Side ramps **4** Welded joints				**24** Item description for laying turnout and diamond crossing shall state their type and weight **S25** Item description for welded joint shall state the rail section and the type of weld. **S26** Item descriptions for laying buffer stops shall state their approximate weight
	m	**7** Laying	**1** Bullhead rails: slab track **2** Flat bottom rails: slab track	**1** Plain track **2** From curve in plain track not exceeding 300m **3** From curve in plain rack exceeding 300m	**M18** The length of laying plain track shall be measured along the centre line of the track (two rail) and shall include the lengths occupied by the turnouts and diamond crossings	**D4** Laying rails direct to concrete bases shall be classed as slab track	**C15** Items for laying shall be deemed to include work carried out after delivery of components to the site, where track is not to be supplied by the contractor , to the location stated in accordance to rule S14	**S27** Item description for laying rail track not supplied by contractor shall identify the form in which its to be supplied **S28** Item descriptions fro layin g fabricated lengths.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)	nr	Laying (Cont'd)		**4** Turnouts **5** Diamond crossing **6** Welded joints **7** Spotre- sleeping	**M19** The lengths of laying check, guard and conductor rails shall be measured along the lengths of the rail (one		**C16** Laying rails direct to concrete, slab track, shall be deemed to include fixings and pads.	**S29** Items description for laying rails shall state the type and mass per metre of rail and the type of joint and sleeper **S30** Item descriptions for prefabricated track panel shall state the panel length **S31** Item description for laying turnout and diamond crossing shall state their type and weight **S32** Item description for welded joint shall state the rail section and the
	m		**3** Prefabricated track panel	**1** Bullhead rail: timber sleepers **2** Bullhead rails: concrete sleepers **3** Flat bottom rails: timber **4** Flat bottom rails: concrete sleepers **5** Flat bottom rails: steel sleepers	**M20** Conductor rail guard boards shall be measured each side of the rail **M21** Laying rails direct to concrete base shall be measured separately			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Railway track construction (cont'd)	nr	Laying (Cont'd)	**4** Prefabricated turnouts and crossings: with bearers	**1** Turnouts: timber bearers **2** Turnouts: concrete bearers **3** Turnouts: steel bearers **4** Diamond crossings: **5** Diamond crossings: **6** Diamond crossings: steel bearers				**S33** Item descriptions for laying buffer stops shall state their approximate weight
	nr		**5** Sundries	**1** Buffer stops **2** Retarders **3** Wheel stops **4** Lubricators **5** Switch heaters **6** Switch levers **7** Conductor rail guard board **8** Static sander **9** Eutectic strip				
	m							
	nr							
	m							

2V TUNNEL

Includes: Excavation, lining and securing of tunnels, shafts and other subterranean cavities

Excludes: Geotechnical processes carried out from he ground; Filling within tunnels; Reinforcement in in situ concrete lining, Cut and cover tunnels

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** Tunnels constructed by cut and cover are excluded from this Section. The earthworks, in situ concrete, and other components of tunnels constructed by cut and cover method shall be measured under the relevant Sections in Part 2 (Civil, Industrial works and the like)	**D1** Transitions, breakaways and intersections between tunnels and shafts which include work outside the normal profile of the tunnels shall be measured as other cavities		**S1** Where tunnelling work is expressly required to be executed under compressed air, items shall be so described. Item descriptions shall state the gauge pressure in stages of increments of 0.4 bars after the first 1 bar. The provision and operation of plant and services associated with the use of compressed air shall be measured as specified requirements in Section 2A **S2** Item descriptions for excavation and lining of other cavities shall identify the cavity

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
1 Tunnels	m^3	**1** Excavation	**1** Tunnels in rock **2** Tunnels in other stated material **3** Shafts in rock **4** Shafts in other stated material **5** Other cavities in rock **6** Other cavities in other stated material	**1** Stated diameter : not exceeding 2 m **2** 2-3 m **3** 3-4 m **4** 4-5 m **5** 5-6 m **6** 6-7 m **7** 7-8 m **8** stated exceeding 8 m	**M2** The volume measured for excavation shall be calculated to the payment lines shown on Drawings or where no payment lines are shown, to the net dimensions of the volumes to be excavated (other than overbreak) outside the normal cross sectional profile of tunnels and shafts shall be classes as excavation of other cavities.	**D2** The diameter used for classification and stated in the item descriptions shall be the external diameter of the excavation cross section of tunnels and other cavities and shaft	**C1** Item descriptions for excavation shall be deemed to include disposal of excavated material off the site and removal of dead services unless otherwise stated in the item descriptions	**S3** Item descriptions for excavations shall state whether tunnels and shafts are straight, curved or tapered. Item descriptions for excavations shall state the gradient of tunnels sloping at a gradient of 1 in 25 or steeper and the inclination to the vertical of inclined shafts
	m^2		**7** Excavated surfaces in rock **8** Excavated surfaces in other stated material		**M3** An isolated volume of rock occurring within other material to be excavated shall not be measured separately not be measured separately unless its volume exceeds 0.25 m^3			**S4** Where the whole material is for disposal on the site, the location of the disposal areas shall be stated in item descriptions for excavations.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Tunnels (cont'd)		Excavation (Cont'd)			**M4** The area measured for excavated surfaces shall be the area of payment surfaces shown on the drawings or where no payment surfaces are shown, the net areas of the surfaces of the volumes to be excavated			Where excavated material is to be used as filling, item descriptions shall so state. **S5** Details of filling for voids caused by overbreak shall be stated in item description for excavated surfaces. **S6** Where tunnels, shafts and other cavities are not of circular cross-section, their maximum external dimension of cross section shall be substituted for the diameter and their external cross-sectional dimensions shall be stated in item descriptions.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Tunnels (cont'd)	m^2	**2** In situ lining to tunnels **3** In situ lining to shafts	**1** Sprayed concrete **2** Sprayed concrete **3** Cast concrete primary **4** Cast concrete secondary **5** Formwork to stated finish	**1** Stated diameter : not exceeding **2** 2-3 m **3** 3-4 m **4** 4-5 m **5** 5-6 m **6** 6-7 m **7** 7-8 m **8** stated exceeding 8 m	**M5** The thickness of in situ concrete shall be measured to the payment lines shown on the drawings or, where no payment lines are shown, to the net dimensions of the volumes to be lined (see rule M2)	**D3** Reinforcing materials added to the mix of sprayed concrete shall not be treated as **D4** The diameter of lining used for measurement and stated in item descriptions shall be the internal diameter	**C2** Items for lining shall be deemed to include joints and finishes **C3** Items for performing lining shall be deemed to include reinforcement and formwork	**S7** Item descriptions for lining shall state whether tunnels and shafts are straight, curved or tapered, and the gradient of tunnels sloping at the rate of 1 in 25 or steeper shall be stated. The inclination to the vertical in the case of inclines shall be stated
	m^3	**4** In situ lining to other cavities						
	m^2							
	nr	**5** Preformed segmental lining to Preformed segmental lining to shafts **7** Preformed segmental lining to other	**1** Precast concrete bolted rings **2** Precast concrete expanded rings **3** Cast iron bolted rings **4** Cast iron expanded rings **5** Nodular iron rings **6** Fabricated steel rings		**M6** The volumes of an in situ cast concrete lining shall be calculated as set out in Section E65/E66 In situ concrete **M7** The measurement unit for packing shall be the number of rings of segments packed			**S8** Items for in situ concrete lining shall state the specification of concrete and whether it is reinforced and identify those linings which are to form head walls, shaft bottoms and other similar components. Item descriptions for

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Tunnels (cont'd)								sprayed concrete lining shall state the minimum thickness
	m		**4** Caulking of stated material					**S9** Item descriptions for preformed segmental lining rings shall identify the components of each ring and state the nominal ring width and maximum piece weight **S10** Item descriptions for preformed segmental lining in pilot tunnels or shafts shall so state. Where the materials used in preformed segmental lining to pilot tunnels and shafts are to remain the property of the Employer it shall be so stated.
	nr	**7** Lining ancillaries	**1** Parallel circumferential packing **2** Tapered circumferential packing **3** Stepped junctions					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Tunnels (cont'd)		Lining ancillaries						**S11** Precast concrete segmental lining rings which are flanged or solid shall be kept separately **S12** The internal diameter of the lining shall be stated in item descriptions for lining ancillaries **S13** Where the inner lining section is not circular, their maximum internal dimensions of cross section shall be substituted for the diameter and the dimensions shall be stated in the descriptions

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Tunnels (cont'd)	m	**8** Support and stabilization	**1** Rock bolts	**1** Mechanical **2** Mechanical grouted **3** Pre-grouted impacted **4** Chemical end anchor **5** Chemical grouted **6** Chemically filled	**M8** Both temporary and permanent support and stabilization shall be measured **M9** The mass measured for steel arches shall be calculated as set out in Section 2G. the volume of timber support shall be measured in accordance with the rules in Section 2G. the area of sprayed concrete support shall be measured at the payment lines show on the drawings or, where no payment lines are shown, to the net dimensions of the support to be provided.	**D4** Reinforcing materials added to the mix of sprayed concrete shall not be treated as reinforcement. **D5** The mesh or link reinforcement in sprayed concrete support shall be measured as mesh or link support	**C4** Items for face packers shall be deemed to include collating, securing and making good linings on completion	**S14** Item descriptions for rock bolts, shall state their size, type, shank detail and the maximum length. **S15** The materials used for lagging and for packing or grouting behind lagging shall be stated in item descriptions **S16** Item descriptions for sprayed concrete support shall state the specification of concrete and whether it is reinforced and the minimum thickness. Item descriptions for mesh or link support shall state the size and mass of mesh or link fabric
	t		**2** Internal support	**1** Steel arches: supply **2** Steel arches: erection				
	m^3			**3** Timber supports: supply				
	m^2			**4** Timber supports: **5** Lagging **6** Sprayed concrete **7** Mesh or link				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Tunnels (cont'd)	nr m t		**3** Pressure grouting	**1** Sets of drilling and grouting plant **2** Face packers **3** Deep packers of stated size **4** Drilling and flushing to stated diameter **5** Re-drilling and flushing **6** Injection of grout materials of stated composition	**M10** The number of face packers and deep packers measured shall be the number of injections **M11** The mass measured for injection of grout materials shall not include the mass of mixing water			**S17** The lengths of holes shall be stated in stages of 5 m in item descriptions for drilling and re-drilling holes for pressure grouting and in item descriptions for forward probing

2W BRICKWORK, BLOCKWORK AND MASONRY

Excludes: Brickwork in manholes and other brickwork incidental to pipeworks (included in class K); Brickwork in sewer renovation (included in class Y)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** Each Skin of brickwork, brickwork or masonry which is in cavity or composite construction shall be measured. **M2** Volumes and areas measured for brickwork, blockwork and masonry shall include the volumes and areas of joints, and exclude those of copings and sills. No deduction from or addition to volumes and areas shall be made for intruding or projecting surface features. No deduction from the volumes and areas measured shall be made for holes and openings in walls or			

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
					surfaces each not exceeding 0.25m^2 in cross-sectional area			
WORK SECTION TO BE MEASURED		**FIRST DIVISION**	**SECOND DIVISION**	**THIRD DIVISION**				
1 Masonry	m2	**1** Common brickwork **2** Facing brickwork **3** Engineering brickwork **4** Lightweight blockwork **5** Dense concrete blockwork **6** Artificial stone blockwork **7** Ashlar masonry **8** Rubble masonry	**1** Thickness: not exceeding 150mm **2** 150-250mm **3** 250-500mm **4** 500mm - 1m **5** exceeding 1m	**1** Vertical straight walls **2** Vertical curved walls **3** Battered straight walls **4** Battered curved walls **5** Vertical facing to concrete **6** Battered facing to concrete **7** Casing to metal sections	**M3** Mean dimensions shall be used to calculate the areas and volumes of concrete and casing to metal sections and the heights of columns and piers.	**D1** Walls or facing battered on one or both sides shall be classed as battered walls or battered facing. **D2** Isolated walls having a length on plan not exceeding four times their thickness shall be classed as piers. **D4** In determining the thickness of walls, facing and casing, the presence of surface features shall be ignored.	**C1** Items for masonry shall be deemed to include fair facing	**S1** Item descriptions for walls, Facing to concrete, casing to metal sections, columns and piers shall either state the materials, nominal dimensions and types of brick, block and stone or give equivalent references or applicable British Standard specifications.
	m		**6** Columns and piers of stated cross-sectional dimensions					**S2** Items descriptions for masonry walls, facing to concrete, casing to metal sections, columns and piers shall state the surface finish.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Masonry (cont'd)								**S3** The bonding pattern, type of mortar and type of jointing pointing shall be stated in item description for walls, facing to concrete, casing to metal sections, columns and piers. **S4** Item descriptions for walls, facing to concrete, casing to metal sections, columns and piers which are in cavity or composite construction shall so state. **S5** The nominal thickness of walls, facing to concrete and casing to metal sections shall be stated in item descriptions.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Masonry (cont'd)	m		**7** Surface features	**1** Copings and sills, material **2** Rebates and chases **3** Cornices **4** Band courses **5** Corbels **6** Pilasters **7** Plinths **8** Fair facing	**M4** The lengths of surface features measured shall be mean lengths. The areas measured for fair facing shall be those expressly required and shall be measured at the face.	**D5** Columns and piers attached to walls or facing of the same material shall be classed as		**S6** Item descriptions for surface features shall include sufficient detail to identify special masonry and special or cut bricks and blocks. The spacing of intermittent surface shall be stated in item descriptions. **S7** The cross-sectional dimensions of surface features shall be stated in item descriptions where the cross-sectional area of the surface features exceeds $0.05m^2$.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Masonry (cont'd)	m		**8** Ancillaries	**1** Joint reinforcement **2** Damp proof courses **4** Bonds to existing work **5** Infills of stated thickness **6** Fixings and ties **7** Built-in pipes and ducts, cross-sectional area; not exceeding $0.05m^2$ **8** stated exceeding $0.05m^2$	**M5** The of additional material in laps of joint reinforcement and damp proof courses shall not be measured. **M6** The areas of fixings and ties measured shall be the areas of brickwork, blockwork or masonry fixed or tied. Where two areas of brickwork, blockwork or masonry are fixed or tied to each other, only the smaller of the two areas shall be measured.		**C2** Items for built-in pipes and ducts shall be deemed to include their supply unless otherwise stated.	**S8** The materials and dimensions of joint reinforcement and damp proof courses, the materials of infills and the type and spacing of fixings and ties shall be stated in item descriptions. **A9** The dimensions and nature of components including face or internal details shall be stated in item descriptions for movement joints. **A10** The lengths of built-in pipes and ducts shall be stated in item descriptions where they exceed 1m.
	m^2							
	nr							

2X PAINTING

Includes: In situ painting

Exclude: Painting carried out of prior to delivery of components to the site

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE					**M1** No deduction from the areas measured shall be made for holes and openings in the painted surfaces each not exceeding 0.5m2 in area.		**C1** Items for painting shall be deemed to include preparations of surface before painting.	**A1** Item descriptions for painting shall state the material used and either the number of coats or the film thickness. **A2** Preparation of surface shall be identified in item descriptions where more than one type of preparation is specified for the same surface.
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
Painting	m^2	**1** Iron or zinc based primer **2** Etch primer paint **3** Oil paint **4** Alkyd gloss paint **5** Emulsion paint **6** Cement paint **7** Epoxy or polyurethane paint **8** Bituminous or coal tar paint	**1** Metal, other than metal sections and pipework **2** Timber **3** Smooth concrete **4** Rough concrete **5** Masonry **6** Brickwork and blockwork	**1** Upper surfaces inclined at an angle not exceeding 30^0 to the horizontal **2** Upper surfaces inclined at an angle not exceeding 30-60^0 to the horizontal **3** Surfaces inclined at an angle exceeding 60^0 to the horizontal	**M2** Surfaces of width not exceeding 1m shall not be distinguished by inclination			

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Painting (cont'd)				**4** Soffit surface and lower surfaces inclined at an angle not exceeding 60^0 to the horizontal				
	m			**5** Surfaces of width not exceeding 300mm **6** Surfaces of width 300mm-1m				
	nr			**7** Isolated groups of surfaces	**M3** Isolated groups of surfaces of different shapes or dimensions shall be measured as separate items.	**D1** Isolated groups of surfaces shall be classed as such only where the total surface area of each group does not exceed $6m^2$		**A3** Item descriptions for isolated groups of surfaces shall identify the work to be painted and its locations
	m^2		**7** Metal sections **8** Pipework		**M4** In calculating the painted are of metal sections the presence of connecting plates, bracelets, rivets, bolts, nuts and similar projections shall be ignored.		**C2** Painting of metal sections shall be deemed to include painting the surfaces of connecting plates, brackets, rivets, bolts, nuts and similar projections.	

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Painting (cont'd)					**M5** The area measured for painting pipework shall be the length multiplied by the girth of each length of pipe or lagged pipe with no deductions or additions for flanges, valves, other projecting fittings and hangers.		**C3** Painting of pipework shall be deemed to include painting the surfaces of flanges, valves, other projecting fittings and hangers.	

2Y WATERPROOFING

Includes: Damp proofing, tanking and roofing

Excludes: Waterproofed joints (included in classes 2C, 2G, 2H, 2J, 2K, 2L, 2T, 2W, 2Z, and 2AA); Damp proof courses in brickwork, blocking and masonry (included in class 2AA)

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE					**M1** The areas measured shall be those of the surface covered. No deduction from the areas measured shall be made for holes and openings in the waterproofed surfaces each not exceeding 0.5m^2 in area.		**C1** Items for waterproofing shall be deemed to include preparing surfaces, forming joints, overlaps, mitres, angle, fillets, built-up edges and laying to falls or cambers.	**S1** Item descriptions for waterproofing shall state the materials used and the number and thickness of coatings or layers.
Work Section to be measured	**Unit**	**First Division**	**Second Division**	**Third Division**				
1 Water proofing	m^2	**1** Damp proofing **2** Tanking **3** Roofing	**1** Asphalt **2** Sheet metal **3** Waterproof sheeting **4** Waterproof coating **5** Rendering in ordinary cement **6** Rendering in waterproof cement mortar	**1** Upper surfaces inclined at an angle not exceeding 30° to the horizontal **2** Upper surfaces inclined at 30° 60° to the horizontal **3** Surfaces inclined at an angle exceeding 60° to the **4** Curved surfaces	**M2** Surfaces of width not exceeding 1m shall not be distinguished by inclination or by curvature. **M3** Waterproofing classed as to curved or doomed surfaces shall not be distinguished by inclination.	**D1** Waterproofing shall be classed as to curved or domed surfaces only where a radius of curvature of the surface is less than 10m.	
	m							

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Water proofing (cont'd)			**7** Tiles	**5** Domed surfaces **6** Surfaces of width not exceeding **7** Surface of width 300mm -1m				
	nr			**8** Isolated groups of surfaces	**M4** Isolated groups of surfaces of different shape or dimensions shall be measured as separate items.	**D2** Isolated groups of surfaces shall be classed as such only where the total surface area of each group does not exceed $6m^2$		**S2** Item descriptions for isolated groups of surfaces shall identify the work to be waterproofed and state its location.
	m^2 m	**4** Protective layers	**1** Sand asphalt **2** Flexible **3** Sand **4** Sand and cement screed **5** Tiles	**1** Upper surfaces inclined at 30^0 to the horizontal **2** Upper surfaces inclined at 30^0–60^0 to the horizontal **3** Surfaces inclined at an angle exceeding 60^0 to the **4** Curved surfaces **5** Domed surfaces **6** Surfaces of width not exceeding 300mm	**M5** Surfaces of width not exceeding 1m shall not be distinguished by inclination or by **M6** Waterproofing classed as to curved or doomed surfaces shall not be distinguished by inclination.	**D3** Waterproofing shall be classed as to curved or domed surfaces only where a radius of curvature of the surface is less than 10m.		

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Water proofing (cont'd)				**7** Surfaces of width 300mm - 1m				
	nr			**8** Isolated groups of surfaces	**M7** Isolated groups of surfaces of different shape or dimensions shall be measured as separate items.	**D4** Isolated groups of surfaces shall be classed as such only where the total surface area of each group does not exceed $6m^2$		**S3** Item description for isolated groups of surfaces shall identify the work to be waterproofed and state its location.
	m^2	**5** Sprayed or brushed waterproofing						
	m^2	**6** Sheet linings membrane	**1** Butyl rubber **2** EPDM rubber **3** PVC **4** Polypropylene **5** Polyethylene **6** Polyurethane **7** Proprietary	**1** Upper surfaces inclined at an angle not exceeding 30^0 to the horizontal **2** Upper surfaces inclined at 30^0 60^0 to the horizontal **3** Surface inclined at an angle exceeding 60 to the horizontal **4** Curved surfaces **5** Domed surfaces				
	m							

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Water proofing (cont'd)				**6** Surfaces of width not exceeding 300mm **7** Surfaces of width 300 mm - 1m				
	nr			**8** Isolated groups of surfaces				

2Z MISCELLANEOUS WORK

Includes: Fences, gates and their foundations; Drainage to structures above ground; Rock filled gabions

INFORMATION PROVIDED					MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE								
WORK SECTION TO BE MEASURED	**UNIT**	**FIRST DIVISION**	**SECOND DIVISION**	**THIRD DIVISION**				
Miscellaneous work	m	**1** Fences	**1** Timber post and rail **2** Timber post and wire **3** Concrete post and wire **4** Metal post and wire **5** Coated metal post and wire **6** Timber close boarded **7** Metal guard system	**1** Height: not exceeding 1 m **2** 1-1.25 m **3** 1.25 - 1.5 m **4** 1.5 - 2 m **5** 2 - 2.5 m **6** 2.5 - 3 m **7** exceeding 3m	**M1** Length of fences shall exclude lengths occupied by gates and stiles	**D1** The heights used for classification of fences shall be measured from Commencing Surface	**C1** Items for fences shall be deemed to include excavation, preparation of surfaces, disposal of excavated material, upholding sides of excavation, backfilling, removal of existing services, concrete, formwork and reinforcement. **C2** Items for fences shall be deemed to include end posts, angle posts, straining posts and gate posts.	**S1** Item descriptions for fences which are erected to a curve of radius not exceeding 100m or on a surface inclined at an angle exceeding 10^0 shall so state **S2** The types and principal dimensions of fences and of their foundations shall be stated in item descriptions.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Miscellaneous work	nr	**2** Gates and stiles	**1** Timber gates **2** Metal gates **3** Stiles	**1** Width: not exceeding 1.5 m **2** 1.5 - 2 m **3** 2 - 2.5 m **4** 2.5 - 3 m **5** 3 - 4 m **6** 4 - 5 m **7** exceeding 5m		**D2** The widths used for classification of gates and stiles shall be measured between the inside faces of posts.		**S3** The types and principal dimensions of gates and stiles shall be stated in item descriptions. **S4** Item descriptions for gates composed of more than one leaf shall identify the number of leaves in item descriptions.
	m nr	**3** Drainage to structures above ground	**1** Mild steel **2** Cast iron **3** Plastics	**1** Gutters **2** Downpipes **3** Fittings to gutters **4** Fittings to downpipes		**D3** Fittings to gutters shall include bends, angles, stop ends and outlets. Fittings to downpipes shall include bends, swan necks, shoes and roof outlets fixed directly to downpipes.		**S5** Item descriptions for drainage to structures above ground shall state the type, principal dimensions and materials of the components.

CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKES

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Miscellaneous work	nr	**4** Rock filled gabions	**1** Box of stated size			**D4** Rock filled gabions exceeding 300mm thick shall be class as boxed gabions and those not exceeding 300mm thick as mattress gabions. Filling shall be deemed to be imported unless otherwise stated.		**S6** Item descriptions for rock filled gabions shall state the type and grading of filling, the size of mesh and diameter of mesh wire. Details of protective coatings shall be stated.
	m^2		**2** Mattress of stated thickness					
	nr	**5** Open cell block systems						**S7** The material, cross section type and principal dimensions or type shall be stated for open cell block systems.

2AA SEWER AND WATER MAIN RENOVATION AND ANCILLARY WORKS

Includes: Preparation and renovation of existing sewers and water mains; New manholes within the length of existing sewers; Work to existing manholes

Excludes: Grouting carried out from outside the sewer (included in 44); New pipework (included in 50-53); New fittings and valves used in water main renovation (included in 51); Extras to excavation and backfilling for new manholes and other chambers (included in 53)

INFORMATION PROVIDED	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
SEE GENERAL ITEMS (2A) ABOVE	**M1** Length of sewer shall be measured along their centre lines between the inside surfaces of manholes but shall exclude length occupied by pipes and fittings comprising backdrops to manholes.		**C1** Items for work which is expressly required to be carried out by excavation shall be deemed to include preparation of surfaces, disposal of excavated material, upholding sides of excavation, backfilling and removal of dead services.	**S1** The location of the work in each item or group of items shall be stated so that the work can be identified by reference to the Drawings **S2** Principal dimensions and profiles of sewers shall be stated in item descriptions.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Sewer and Water Main Renovation and Ancillary Works (cont'd)					**M2** Where work is expressly required to be carried out by excavation, crossings, reinstatement and other pipework ancillaries shall be measured in 52 and extras to excavation and backfilling shall be measured in class 53.			**S3** Work which is expressly required to be carried out manually or by remotely controlled methods shall each be so stated in item descriptions. **S4** Item description shall identify work which is expressly required to be carried out by excavation and (except for manholes) shall state the maximum depth of excavation in stages of 1m measured to the invert of the sewer or water main.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Sewer and Water Main Renovation and Ancillary Works (cont'd)								**S5** Item description for preparation, stabilization, renovation and laterals shall state the material forming the existing sewer or water main.
	m	**1** Preparation of existing sewers	**1** Cleaning				**C2** Items for cleaning shall be deemed to include making good resultant damage.	
	nr		**2** Removing Intrusions	**1** Laterals, bore not exceeding 150mm **2** Laterals, stated profile and size exceeding 150mm in one or more dimension **3** Other stated artificial		**D1** Items shall be classed as removing intrusions where intrusions into the bores of existing sewers are to be removed prior to renovation.	**C3** Items for removing intrusions shall be deemed to include making good.	**S6** Item description for removing intrusions shall state the material forming the intrusions.

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Sewer and Water main renovation and ancillary works (cont'd)	m		**3** Closed circuit television surveys					
	nr		**4** Plugging laterals, material stated	**1** Bore not exceeding 300mm				
	m^3		**5** Filling laterals and other pipes, materials stated	**2** Stated profile and size exceeding 300mm in one or more dimension				
	nr		**6** Local internal repairs	**1** Area: not exceeding $0.1m^2$	**M3** No deductions shall be made from the areas of sewer surface pointing for openings or voids each not exceeding $0.5m^2$ in area	**D2** The areas stated in item description for local internal repairs shall be the finished surface areas.	**C4** Items for local internal repairs shall be deemed to include cutting out and repointing **C5** Items for pointing and pipe joint sealing shall be deemed to include preparation of joints.	
	m^2	**2** Stabilization of existing sewers	**1** Pointing, materials stated	**2** $0.1\text{-}0.25\ m^2$				
	nr		**2** Pipe joint sealing, materials stated	**3** stated exceeding $0.25m^2$				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Sewer and Water main renovation and ancillary works (cont'd)	m^3		**3** External grouting	**1** Number holes **2** Injection of grout, materials stated	**M4** External grouting shall be measured only where grouting is expressly required to be carried out as seperate operations from annulus grouting (67.3.6.0)	**D3** External grouting shall be grouting of voids outside the existing sewer from within the existing sewer other than voids grouted in the course of annulus grouting.		**A7** Where external grouting is carried out through pipe joints, description of items for the number of holes shall so state.
	m	**3** Renovation of existing sewers	**1** Sliplining	**1** Polyethylene **2** Polypropelene				**S8** Item description for slip-lining, in situ jointed pipe lining, segmental lining and stated propreitary lining shall state the type of lining, its minimum finished internal size and its thickness or grade.
			2 In situ jointed pipe lining	**1** Polyethylene **2** Polypropelene **3** Glass reinforced plastic				
			3 Segmental lining	**3** Glass reinforced plastic **4** Glass reinforced concrete **5** Cast Gunite **6** Resin concrete				
			4 Stated proprietary linning					

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Sewer and Water main renovation and ancillary works (cont'd)	m^3		**5** Gunite coating of stated		**M5** The volume measured for annulus grouting shall not include the volume measured for external grouting (Y 2 3 2)	**D4** Annulus grouting shall be grouting of the annular voids between new linings and existing sewer and of other voids grouted in the course of grouting annular voids		**S9** Item description for in situ jointed pipe lining and segmental lining shall state the offset where lining is curved to an offset which exceeds 35mm per metre.
			6 Annulus grouting, materials stated					
	nr	**4** Lateral of renovated sewers	**1** Jointing	**1** Bore: not exceeding 150mm **2** 150-300mm **3** Stated profile and size exceeding 300mm in one or more dimension			**C6** Items for lateralsmshall be deemed to include the work involved in connecting to the lining within 1m from the inside face of the lined sewer	**S10** Items description for jointing laterals shall state the type of lining to which the laterals are to be connected and identify those laterals which are to be regarded.
			2 Flap valves of stated size	**1** Remove existing **2** Replace existing **3** New flap valve of stated type				

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Sewer and Water main renovation and ancillary works (cont'd)	m nr m	**5** Water main renovation and ancillary works	**1** Cleaning **2** Removing intruisions **3** Pipe sample inspection **4** Closed-circuit television surveys **5** Cement mortar lining **6** Epoxy lining	**1** Nominal bore: not exceeding 200mm **2** 200-300mm **3** 300-600mm **4** 600-900mm **5** 900-1200mm **6** stated exceeding 1200mm	**M6** Length of water mains shall be measured along centre lines and shall include lengths occupied by fittings and valves	**D5** Pipe sample inspections and closed-circuit television surveys shall include work carried out either before or after cleaning and lining	**C7** Items for pipe sample inspections shall be deemed to include replacing the length removed by new pipework.	**S11** Item descriptions for lining shall state the materials, nominal bores and thicknesses of the lining.
	nr	**6** New manholes	**1** Brick **2** Brick with backdrop **3** Insitu concrete **4** Insitu concrete with backdrop **5** Precast concrete **6** Precast concrete with backdrop	**1** Depth: not exceeding 1.5m **2** 1.5 - 2m **3** 2 - 2.5m **4** 2.5 - 3m **5** 3 - 3.5m **6** 3.5 - 4m **7** stated exceeding 4m		**D6** The depth of manholes shall be measured from the tops of covers to channel inverts or tops of base slabs whichever is below	**C8** Items for new manholes shall be deemed to include excavation, decoration of surfaces, disposal of material, upholding sides of excavation, backfilling, concrete, reinforcement, formwork, joints, finishes and reinstatement	**S12** Type or mark numbers shall be stated in item description for manholes of which details are given elsewhere in the contract. Items description shall identify different configurations of manholes **S13** Types and loading duties of covers shall be stated item description for new manholes

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Sewer and Water main renovation and ancillary works (cont'd)	nr	**7** Existing manholes	**1** Abandonment				**C9** Items for new manholes shall be deemed to include all items of metalwork and pipe work which occur within or at the surface of the manhole	**S14** Item description shall separately identify new manholes which replace existing manholes
			2 Alteration				**C10** Items of new manholes with backdrops shall be deemed to include the pipe and fittings comprising the backdrop	**S15** item description for existing manhole shall state the details of the work required

WORK SECTION TO BE MEASURED	UNIT	FIRST DIVISION	SECOND DIVISION	THIRD DIVISION	MEASUREMENT RULES	DEFINITION RULES	COVERAGE RULES	SUPPLEMENTARY INFORMATION
Sewer and Water main renovation and ancillary works (cont'd)	h	**8** Interruptions	**1** Preparation of existing sewers **2** Stabilization of existing sewers		**M7** Interruptions shall be measured only where a minimum pumping capacity is expressly required and for periods of time during normal working hours when the flow in the sewer exceeds the installed pumping capacity and work is interrupted.			
			3 Renovation of existing sewers	**1** Sliplining **2** In situ jointed pipe lining **3** Segmenta l lining **4** Stated proprietary linning **5** Gunite coating **6** Annulus grouting				
			4 Work on lateral to renovated sewers **5** Work on manholes					

CIVIL/INDUSTRIAL ENGINEERING WORKS AND THE LIKES

2AB SIMPLE BUILDING WORKS INCIDENTAL TO CIVIL ENGINEERING WORKS

Includes: Carpentry and joinery; Insulation; Windows, doors and glazing; Surface finishes, linings and partitions; Piped building services; Ducted building services; Cabled building services etc. are to be measured in accordance with the appropriate Sections in Part 1

Excludes: Drainage below ground (included in 2J-2M); Duct and trenches for electrical services below ground or outside the building (included in 2J-2M); Metal work (included in 2N); Civil engineering timber works (included in 2Q); Brickwork, blockwork and masonry (included in 2V); Painting (included in 2W); Asphalt work (included in 2X); Roofing, cladding and coverings (included in class W); Drainage to structures above ground (included in class X)

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